What Is Valuable to Know?

A Statistical Design Document for Investment-Worth Modeling on PitchBook/FactSet Deal Data

hawkes-sim design document — master reference for model and code decisions

Alberto Gennaro[1]

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[1]Document drafted with model-assisted tooling; all modeling decisions reviewed by the author.

How to read this document

This document is the master reference for the hawkes-sim modeling effort: every future code modification in hawkes_calibration/ should be traceable to a numbered decision, assumption, or protocol stated here. It supersedes the earlier working notes (THEORY.md, MATH_NOTES.md, STRATEGY.md) as the single source of truth, while carrying forward their results wherever those results survived scrutiny; the numerical verdicts inherited from the synthetic rehearsal campaigns are collected in Appendix C so that the main text can cite them without re-deriving them.

The document is organized around one organizing question — *what is valuable to know?* — posed in Chapter 1 and answered differently by every subsequent part. Chapter 2 fixes, once, what the PitchBook/FactSet data actually delivers and the six ways observation corrupts it. Chapters 3 to 7 build the continuous-time machinery: point-process foundations, Poisson and exponential-Hawkes models with covariate-driven excitation and inhibition, the treatment of untrusted and censored deal times, and the goodness-of-fit discipline. Chapter 8 confronts the covariate staleness that PitchBook imposes. Chapters 9 and 10 develop the discrete-time bucket formulation solved with gradient boosting and the ranking view evaluated by recall at K. Chapter 11 handles deals arriving from companies outside the tracked universe. Chapters 12 and 13 close the loop: from model outputs to investment decisions, and from decisions to code.

Three reading conventions. First, *assumptions are contracts*: every model result is stated under numbered assumptions, and the text says explicitly which assumptions are believed, which are convenient fictions, and which are known to be false in the data (with the consequence of the violation). Second, *Design decision* boxes record commitments — a choice among alternatives, with the rationale and the trigger that would reverse it. Third, *Code hooks* boxes at the end of each chapter map the mathematics onto modules of hawkes_calibration/, existing or to be written; these boxes, together with Chapter 13, are the implementation contract.

Proofs are given in the running text when they are short and instructive; longer arguments are deferred to the ends of chapters. Citations are restricted to work we have actually relied on — either for a theorem, a method we implement, or a warning we heed — and the bibliography is annotated in that spirit throughout the text rather than in a separate rubric.

Contents

Part I

The Question and the Data

Chapter 1

The Central Question: What Is Valuable to Know?

1.1 Worth, formally

The business objective is to understand which private-market investments are "worth it." That phrase hides a decision problem, and the first discipline this document imposes is to make the decision problem explicit *before* any model is chosen. An investment thesis about a private company is a bet on a small set of future observables: whether the company raises again, when it raises, at what valuation, and whether the eventual exit repays the capital deployed today. A model earns its keep only by improving one of those bets. So the central question is not "which model is best" but:

What is valuable to know?

Formally, let $\mathcal{D}$ be a space of actions available to the investor (allocate attention to a company, enter a diligence process, price a secondary position, pass), let ω denote the future evolution of the market — the marked point process of deals defined in Chapter 2 — and let $u(d, \omega)$ be the utility of action $d \in \mathcal{D}$ under outcome ω. A predictive object π (a probability, an intensity, a ranking) is *valuable* exactly to the extent that

$$\mathrm{VoI}(\pi) \;=\; \mathbb{E}[\,u(d_\pi^*,\,\omega)] \;-\; \mathbb{E}[\,u(d_0^*,\,\omega)]\,, \qquad d_\pi^* = \arg\max_{d \in \mathcal{D}} \mathbb{E}[u(d,\omega) \mid \pi]\,, \tag{1.1}$$

is positive and large, where d_0^* is the action taken without the model. This is the standard value-of-information decomposition, and its practical force here is negative: it rules out building models whose outputs no action consumes. Every model family developed in this document is admitted only because a concrete action consumes its output, and the evaluation metric of each family is chosen to be a proxy for Equation (1.1) under that action, not a generic accuracy score.

1.2 The four answers and the targets they induce

Interrogating the sourcing and monitoring workflows that would consume model output yields four distinct answers to "what is valuable to know," each inducing a different mathematical target on the same underlying data.

T1 — Identity: *which* company gets funded next. The sourcing action is a short list: given finite diligence capacity, which K companies should be looked at now because they are most likely to be the next ones to raise? The natural target is the conditional distribution of the next deal's identity,

$$\mathbb{P}(\text{next deal in sector } s \text{ goes to firm } i \mid \mathcal{F}_{t^-}), \tag{1.2}$$

and the natural metric is ranking quality — recall at K against the next deals actually observed (Chapter 10). Crucially, Equation (1.2) conditions on *a deal occurring*: the timing of deals cancels from the target, which is why identity models can survive data defects that destroy timing models (Section 1.3).

T2 — Timing: *when* the deals arrive. Pricing, portfolio pacing, and capital-deployment planning consume arrival times: how many deals will sector s produce next quarter, and how is firm i's next round distributed in time? The target is the conditional intensity $\lambda(t)$ of the deal process (Chapter 3) and functionals of it (expected counts, waiting-time distributions), evaluated by likelihood on held-out windows and by calibration diagnostics (Chapter 7). This is the target the Poisson and Hawkes machinery of Chapters 4 and 5 serves.

T3 — Horizon events: will firm i raise within Δ. Many consuming actions are binary and horizon-bound: enter diligence now if the company is likely to raise within the next six months; deprioritize if no event is plausible within two years. The target is the fixed-horizon event probability

$$p_i(t, \Delta) \;=\; \mathbb{P}\big(N_i(t+\Delta) - N_i(t) \geq 1 \mid \mathcal{F}_t\big), \qquad \Delta \in \{6\text{M}, 1\text{Y}, 2\text{Y}\}, \tag{1.3}$$

a discrete-time object that can be attacked directly with supervised learning on a bucketed panel (Chapter 9) — at the price of label construction subtleties that Chapter 9 treats as first-class mathematics, not plumbing.

T4 — Marks: size, valuation, terms. Finally, worth depends on *what* the deal is, not only that it happens: round size, pre/post valuation, up/down/flat classification. These are mark distributions conditional on an event, $\mathbb{P}(m \in \cdot \mid \text{deal at } t \text{ by } i, \mathcal{F}_{t^-})$. This document develops T4 only to the extent needed by the decision layer (Chapter 12); its estimation reuses the regression machinery of Chapter 9 and inherits every covariate caveat of Chapter 8.

These four targets are not competing models of the same quantity; they are different marginals and conditionals of one object — the marked point process — consumed by different actions. The recurring failure mode this document is designed to prevent is answering one target's question with another target's model and metric (for instance, selling a horizon classifier's probability, T3, as a timing model, T2; or judging a ranking model, T1, by likelihood alone).

1.3 One process, two factorizations

The unifying object is the marked point process of deals: ground times t_m (when a deal happens), marks (s_m, i_m, m_m) (which sector, which firm, what terms). Its likelihood factorizes *exactly* as

$$p(\text{times, sectors, firms, terms}) \;=\; \underbrace{p(\text{times, sectors})}_{\text{ground process}} \times \underbrace{p(\text{firm} \mid \text{sector, time, history})}_{\text{mark factor}} \times \underbrace{p(\text{terms} \mid \text{firm, time})}_{\text{deal marks}}. \tag{1.4}$$

This identity — established as Theorem 3.7 and inherited from the repository's earlier theory notes, where it is the decomposition labeled M1 — is the load-bearing wall of the whole design. It is an identity of likelihoods, not an approximation: all approximations live *inside* one factor at a time, and defects of the data can be assigned to the factor they damage. Three consequences organize the document.

First, *the targets map onto factors.* T2 lives in the ground process; T1 lives in the mark factor; T4 lives in the deal marks; T3 is a horizon functional of ground × mark. A team can therefore improve one factor without touching the others, and Chapter 13 assigns code modules factor by factor.

Second, *the mark factor is robust where the ground process is fragile.* The mark factor is a conditional-choice likelihood in which the sector-time baseline cancels (Chapter 10); it survives untrusted deal dates, weekly aggregation, and even moderate intensity misspecification. The ground process absorbs precisely those defects. When the data cannot support a trustworthy ground model, the mark factor still delivers T1 — this asymmetry, proved and quantified in Chapters 6 and 10, is why ranking is the most defensible deliverable on current data quality.

Third, *discrete-time models are the same object at coarser resolution.* The bucketed panel of Chapter 9 is the ground+mark object aggregated to intervals: Theorem 9.4 shows the bucket-probability model is exactly a discrete-time hazard model, and recovers the continuous-time intensity as bucket width shrinks. XGBoost on buckets and Hawkes in continuous time are therefore rungs of one ladder, not rival ideologies; which rung to stand on is decided by the target and by what the data defects of Chapter 2 leave standing.

1.4 From questions to models: the decision map

Table 1.1 is the document in one table. Each row is an answer to "what is valuable to know"; the columns give the mathematical target, the model family this document commits to for that target, the evaluation metric that proxies Equation (1.1), and the chapter that develops it.

Valuable to know	**Target**	**Committed model**	**Metric**	**Ch.**
Which firm raises next (identity)	$\mathbb{P}(i \mid \text{deal}, s, t, \mathcal{H})$	risk-set conditional logit → gradient-boosted ranker	recall@K, MRR, held-out conditional NLL	10
When/how many deals (timing)	ground intensity $\lambda_s(t)$	covariate NHPP → exponential Hawkes with engineered excitation/inhibition	held-out NLL, time-rescaling, PIT	4–7
Will i raise within Δ (horizon)	$p_i(t, \Delta)$, $\Delta \in$ {6M, 1Y, 2Y}	discrete-hazard panel + XGBoost with staleness features	calibrated probabilities; PR-AUC; lift@K	9
Deal terms (marks)	$p(m \mid i, t)$	mark regressions on deal covariates	per-mark scores	12

Table 1.1: The decision map: answers to "what is valuable to know," the targets they induce, and the models and metrics this document commits to.

Design decision (D1.1 — target before model).

No model is built, modified, or benchmarked in this repository without first naming which target (T1–T4) it serves and which action consumes its output. Pull requests that improve a metric not attached to a target are rejected by policy. Rationale: Equation (1.1);

the failure mode it prevents is optimizing likelihood in a factor whose output nothing consumes. Reversal trigger: none anticipated — this is the document's charter.

Design decision (D1.2 — both stacks, one object).

We maintain *both* the continuous-time stack (NHPP/Hawkes ground process + risk-set mark factor) and the discrete-time stack (bucketed panel + XGBoost), as views of the same marked process at different resolutions, rather than choosing one ideology. Rationale: the synthetic rehearsals recorded in Appendix C show the covariate-driven baseline carries most of the predictive mass (nats-scale gains) while excitation contributes decimal points — but also show a parity XGBoost beating the linear conditional logit on identity (the fired B1 test). Each stack dominates somewhere. Reversal trigger: if, after the stale-covariate remedies of Chapter 8, one stack dominates on *every* consuming action for two consecutive evaluation campaigns, the other is demoted to benchmark-only status.

1.5 What "worth it" will mean operationally

The decision layer of Chapter 12 composes the four targets into the quantities the investment workflow actually consumes: a ranked sourcing list with horizon-qualified event probabilities attached (T1 + T3), sector pacing forecasts with uncertainty bands (T2), and expected-terms context (T4). Until that chapter, "worth it" is used in the narrow, precise sense of Equation (1.1): a model output is worth producing if and only if the action that consumes it gains utility, and the document's evaluation protocols are designed so that offline metrics are monotone proxies for that gain. The chapters in between are the machinery that makes those outputs trustworthy on the data we actually have — which is the subject of the next chapter, and the source of every hard problem in this document.

Code hooks.

The decision map of Table 1.1 is mirrored in code by keeping the factors of Equation (1.4) in separate modules with separate evaluation harnesses: ground-process fitters (`hawkes_calibration/sector_stability.py`, `hawkes_calibration/eventtime/`), mark-factor fitters (`hawkes_calibration/sector_ranker.py`, `hawkes_calibration/models/hawkes_ranker.py`), and (to be written) the bucketed-panel builder and horizon classifier of Chapter 9 (`hawkes_calibration/panel/`), the staleness state layer of Chapter 8 (`hawkes_calibration/covstate/`), and the decision layer of Chapter 12 (`hawkes_calibration/decision/`). Chapter 13 gives the full mapping with migration steps.

Chapter 2

The Data: PitchBook Events, FactSet Macro, and What Observation Does to Them

2.1 Sources and window

Two data sources feed every model in this document.

PitchBook: the event-driven layer. The deal history of the private market, delivered as three relational tables whose column-level dictionary is given in Appendix B. The *deals* table is the event log: one row per financing event, keyed by `deal_id` and `company_id`, carrying the deal date, sector taxonomy (`primary_industry_sector / group / code`), round descriptors (`vc_round`, `deal_type`, `deal_class`, up/down/flat flags), economics (`deal_size` with a status flag distinguishing `Actual/Estimated/Undisclosed`, `pre_money_valuation`, `post_valuation`, `raised_to_date`, invested capital and ownership fields), firm financials as reported around the deal (`revenue`, `ebitda`, `gross_profit`, `net_income`, employees, growth since last deal), and geography. The *companies* table is a snapshot per tracked firm: identity, sector, founding year, business status, and last-observed-deal summaries (`last_financing_date`, `_size`, `_valuation`, `_deal_type`). The *investors* table links firms to their investors with first-investment dates. The universe flag `company_universe` marks whether a deal belongs to a firm PitchBook tracks; deals do occur with `company_universe = 'Out of Universe'`, whose firms have *no row* in the companies table — defect (C6) below.

FactSet: the market-clock layer. A small panel of macro series in FactSet Standardized Economics form: policy and effective rates (`FFUNDT`, `FFUND`, daily), inflation (`CPI_RAW` level and `CPI_YOY`, monthly), unemployment (`RUNEMP`, monthly), each published with a lag of days to weeks after its reference period. These series are the exogenous covariates of the ground process: they move on calendar time regardless of deal activity, which makes them the only covariates in the system that are *never stale* in the sense of Chapter 8 — but they are *lagged*, and the lag matters for point-in-time discipline (Section 2.4).

Window. The joined history runs from January 2011 through June 2024: $T \approx 13.5$ years, i.e. roughly 704 weeks of joint deal-and-macro history. All development described in this document uses a temporal split: a training window, a validation window for model selection, and a strictly held-out test window at the end (the split protocol is fixed in Chapter 7). The 2011 start postdates the crisis regime shift of 2008–2010 by design; the June 2024 end is an administrative censoring time in the precise sense of defect (C1).

2.2 The formal observation model

We now fix, once, the probabilistic description of what the data is. Every later chapter states its assumptions relative to this model.

Definition 2.1 (Market process)**.** Work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$. The market is a marked point process

$$\{(t_m, s_m, i_m, m_m)\}_{m\geq 1}, \qquad t_1 \leq t_2 \leq \cdots,$$

where $t_m \in [0, T]$ is the (true) time of the m-th deal, $s_m \in \mathcal{S}$ its sector, $i_m \in \mathcal{I} \cup \mathcal{I}^\dagger$ the receiving firm — $\mathcal{I}$ the tracked universe, $\mathcal{I}^\dagger$ the untracked remainder — and m_m the deal marks (size, valuation, round type, ...). For each sector s, $N_s(t)$ counts deals in s up to t; for each tracked firm i, $N_i(t)$ counts firm i's deals. The filtration $\mathcal{F}_t$ contains the internal history of the marked process and the paths of all exogenous covariates up to t.

Definition 2.2 (Covariate processes)**.** Three covariate processes enter the models:

(i) *Macro covariates* $x^{\mathrm{mac}}(t) \in \mathbb{R}^p$: exogenous, càdlàg, observed for all t but only through their *published* version $x^{\mathrm{mac}}_{\mathrm{pub}}(t) = x^{\mathrm{mac}}(t - L(t))$, where $L(t)$ is the publication lag process (Section 2.4).

(ii) *Firm covariates* $x_i(t) \in \mathbb{R}^q$ for $i \in \mathcal{I}$: the true, latent, evolving state of firm i (scale, financials, capitalization). The data does *not* deliver $x_i(t)$; it delivers the last value recorded at firm i's most recent deal,

$$\tilde{x}_i(t) \;=\; x_i(\tau_i(t)), \qquad \tau_i(t) \;=\; \sup\{t_m < t : i_m = i\}, \tag{2.1}$$

the last-observation-carried-forward (LOCF) covariate, whose observation times are themselves event times of the modeled process. This is defect (C5), the subject of Chapter 8.

(iii) *Engineered history covariates*: deterministic functionals of the observed history $\mathcal{H}_{t-}$ chosen by us — time since firm i's last deal, deal counts in trailing windows, sector activity summaries, staleness ages. These are always available and always point-in-time correct; Chapter 5 uses them to build excitation *and* inhibition into intensity models, and Chapter 9 feeds them to gradient boosting.

Definition 2.3 (Conditional intensity and at-risk indicator)**.** For a counting process N_e (e a sector, firm, or firm-transition index), a *conditional intensity* $\lambda_e(t)$ is a nonnegative predictable process such that

$$M_e(t) \;=\; N_e(t) - \int_0^t Y_e(u)\,\lambda_e(u)\,\mathrm{d}u \tag{2.2}$$

is a local martingale, where $Y_e(t) \in \{0, 1\}$ is the at-risk indicator: the unit is under observation and eligible for the event at t^- (Andersen et al., 1993). All likelihoods in this document are instances of the point-process (Jacod) likelihood

$$\ell(\vartheta) \;=\; \sum_m \log \lambda_{e_m}(t_m; \vartheta) \;-\; \sum_e \int_0^T Y_e(u)\,\lambda_e(u; \vartheta)\,\mathrm{d}u, \tag{2.3}$$

whose martingale estimation theory (unbiased score at the truth, Fisher information as predictable variation, asymptotic normality) is developed in Chapter 3 following Andersen et al. (1993) and Andersen and Gill (1982).

The at-risk indicator is where most of the data's pathology concentrates: nearly every defect below is, mechanically, a corruption of either Y (who is really at risk, and when did we start watching them) or of the event times t_m (when did it really happen), or of the covariates the intensity conditions on.

2.3 The defect ledger

Six defects, (C1)–(C6), exhaust the ways this dataset deviates from the ideal of Definition 2.1. For each: the formal statement, the consequence if ignored, and the chapter that owns the treatment. The first four are inherited from the repository's earlier theory notes and restated here in final form; (C5) and (C6) are promoted to first-class defects in this rewrite because the new-data review showed they bind hardest.

(C1) Administrative right censoring at the window end

Every firm alive in June 2024 has its next round unobserved. Formally, censoring is the predictable process $C_i(t) = \mathbf{1}\{t \le T\}$; under independent censoring the observed process $\int C_i \, \mathrm{d}N_i$ has intensity $C_i(t)\lambda_i(t)$ and the likelihood Equation (2.3) computed on observed data is a genuine partial likelihood (Andersen et al., 1993, III.2). Administrative censoring at fixed T is independent by construction. The censored spell contributes exactly its survival factor $\exp(-\int \lambda)$: no term dropped, no term invented. This is the benign defect — *provided open spells are kept.*

Warning 2.4. Dropping open (censored) spells converts right censoring into right *truncation*: selection on the event occurring in-window, which oversamples short inter-deal gaps and biases every fitted hazard upward. This is not classical length bias (which oversamples *long* spells in prevalent-cohort designs); the correct treatment of sampling schemes and truncation is Kalbfleisch and Prentice (2002). Policy: every dataset builder in the repository must emit censored spells with an explicit censoring flag, and every likelihood must consume them. Chapters 4 and 6 implement this for Poisson and Hawkes calibration respectively; Chapter 9 implements the bucketed analogue (censored buckets are labeled, not silently dropped).

(C2) Left truncation: firms enter observation by raising

A firm is invisible to PitchBook until its first recorded financing: it enters observation at V_i = first-deal date, and enters *because of an event.* Mechanically this is delayed entry (Keiding, 1990): risk sets are $Y_i(t) = \mathbf{1}\{V_i \le t < U_i\}$ and every integral in Equation (2.3) starts at V_i. Statistically it is a selection: the observed universe is "firms conditioned on having raised at least once," so unconditional statements about companies are not identified from tracked data alone. Chapter 11 is where this selection is modeled rather than ignored.

(C3) Unlabeled exits: the immortal-firm problem

Failures and quiet wind-downs are mostly unrecorded (`business_status` is coarse and laggy). A dead firm keeps $Y_i(t) = 1$ in any naive risk set. Three facts must be kept distinct. (i) The contamination bias — dead firms held at risk deflate every fitted hazard — arises under *any* unlabeled-exit mechanism, even exit independent of funding propensity: it is a misrecorded Y_i, not a dependence effect. (ii) Because exit also correlates with the latent quality that drives funding, the defect is not correctable from observables: the synthetic rehearsal (Appendix C) showed that an exit classifier with AUC 0.877 was *not* sufficient to reconstruct truthful risk sets, and that risk-set truthfulness was the single largest sensitivity in the whole system (observed-vs-oracle top-5 hit rate 0.14 vs 0.29). (iii) If exits *were* labeled, removing firms at exit would be valid for cause-specific funding intensities with no independence assumption at all — the standard competing-risks fact (Putter et al., 2007). The problem is that exits are unobserved, not that they are dependent. Treatments — exposure windows, mover–stayer mixtures (Farewell, 1982), and ranking-layer mitigations — are developed where they bind: Chapters 4, 9 and 10.

(C4) Untrusted deal dates and aggregation

Two distinct corruptions of t_m. First, *date jitter*: PitchBook records announced vs. closed dates that can differ by days, and a nontrivial fraction of dates are back-filled or rounded; the recorded $\hat{t}_m$ is best modeled as t_m observed through a small-window error. Second, *deliberate aggregation*: for modeling at weekly resolution we replace times by bin counts, which is interval censoring by construction. Both are instances of one statement: the event count over an interval is trusted, the position within the interval is not. Chapter 6 builds the corresponding likelihoods: interval-censored NHPP calibration in closed form, and interval-censored Hawkes via the mean-intensity (mean-Poisson-process) approximation of Rizoiu et al. (2022), with the information loss quantified — branching-scale parameters survive aggregation, timescale parameters do not (the κ–θ ridge of Chapter 7; Fisher information in the decay $\to 0$ as bin width grows past the kernel half-life).

(C5) Stale firm covariates, updated only at own deals

Defect Equation (2.1): firm covariates are observed only at the firm's own deals, so at any t the available $\tilde{x}_i(t)$ has age $a_i(t) = t - \tau_i(t)$, and the age is largest exactly for the firms that have gone longest without raising — the observation times are *informative*, being event times of the process under study. This is simultaneously a measurement-error problem (attenuation of covariate effects; Carroll et al., 2006; Prentice, 1982) and an internal-covariate problem in the sense of Kalbfleisch and Prentice (2002). Chapter 8 formalizes both faces, quantifies the damage (the synthetic covariate-noise rehearsal, exp22, measured the attenuation mechanics), and commits to a layered remedy: staleness-age features always; an explicit covariate-evolution state layer where it pays; joint-model escalation (Rizopoulos, 2012; Tsiatis and Davidian, 2004) only if a trigger fires.

(C6) Out-of-universe deals: events with no covariates

Deals occur from firms not in the tracked universe (`company_universe = 'Out of Universe'`, no companies-table row, no covariates, no history). Consequences: sector deal counts are complete but firm-attributable counts are thinned; any firm-level risk set is missing members who *do* compete for deals; and the mark factor must own an outside option or it will overstate every tracked firm's probability. Chapter 11 develops the treatment: an explicit newcomer/outside component in the mark factor (with the hard-won discipline that *newcomer probability* — first appearance in the data — must never be conflated with *entrant probability* — economic entry; the rehearsal measured the conflation at a negligible likelihood cost but a permanent block on compositional claims), a ghost/external component in the ground process, and unseen-mass estimation in the Good–Turing spirit (Good, 1953; Gale and Sampson, 1995) as the model-free sanity check on how much deal mass the tracked universe can even see.

2.4 Point-in-time discipline

A model that consumes information not available at prediction time is worthless regardless of its metrics. The repository already enforces, and this document elevates to policy, the following discipline (implemented in `synthetic/loaders.py` and to be preserved verbatim by every real-data loader):

Protocol 2.5 (Point-in-time construction)**.**

(a) *Macro covariates are as-of joined on publish date*: the value of $x^{\mathrm{mac}}_{\mathrm{pub}}(t)$ is the last value whose *publication* date (not reference date) precedes t.

Defect	What is corrupted	If ignored	Owned by
(C1)	spells open at June 2024	upward-biased hazards if open spells dropped	Chapters 4, 6 and 9
(C2)	entry into observation (Y_i before first deal)	selection mistaken for dynamics	Chapter 11
(C3)	Y_i after unrecorded death	deflated hazards, inverted rankings	Chapters 4, 9 and 10
(C4)	event times t_m	spurious timing estimates; broken θ	Chapters 6 and 7
(C5)	firm covariates $x_i(t)$	attenuated, staleness-confounded effects	Chapter 8
(C6)	firm identity i_m, risk-set membership	overstated tracked-firm probabilities	Chapter 11

Table 2.1: The defect ledger. Every modeling chapter states which defects it is robust to and which assumptions it needs about the rest.

(b) *Firm features are LOCF from past deals only*: $\tilde{x}_i(t)$ uses deals strictly before t; the staleness age $a_i(t)$ is always carried alongside as a feature, never discarded.

(c) *Risk-set membership is inferred from observables*: founding year and first-deal date determine entry; any exit proxy used to close spells must itself be point-in-time.

(d) *Deduplication and hygiene run before everything*: duplicate deal rows (same firm within a few days, jittered size) are collapsed by the documented rule; sector mislabels and undisclosed sizes are handled by the imputation rules of Appendix B, never ad hoc.

(e) *Evaluation splits are temporal*, with model selection strictly inside the training era. No hyperparameter, feature transform, or calibration map may see post-split data.

2.5 The synthetic rehearsal world

Because the real data's defects cannot be switched off, the repository maintains a synthetic world — generated PitchBook-style tables (deals/companies/investors) and a FactSet-style macro feed with publication lags, materialized through an observation layer that reproduces defects (C1)–(C6) with known ground truth (`synthetic/`: 12 sectors, 8,000 firms of which 7,031 tracked, 520 weeks, ∼49k events, ∼27% newcomer deal share). Two regimes matter: regime A, generated by the exact two-stage model family we fit (well-specified rehearsal), and regime B, a daily firm-level block-Hawkes world with self-inhibition and latent quality, deliberately *not* in the fitted family (misspecification stress test). Every methodological claim this document inherits from the rehearsal campaigns — the covariates-dominate verdict, the risk-set sensitivity, the fired B1 test, the κ–θ aggregation ridge, attenuation under covariate noise — is catalogued with its numbers in Appendix C, and the transfer convention is fixed there: rehearsal results set *priors and vetoes* for real-data work, never substitutes for real-data evaluation.

Design decision (D2.1 — defects are assumptions, stated per model).

Every model chapter opens by declaring, defect by defect against Table 2.1, whether the model (i) is robust to the defect, (ii) corrects it under stated assumptions, or (iii) is vulnerable and accepts the bias with a documented sign. No silent vulnerability is permitted. Rationale: the rehearsal showed the largest errors came not from wrong models but from unstated exposure to (C3) and (C5).

Code hooks.

Real-data ingestion will mirror `synthetic/loaders.py`: an adapter `hawkes_calibration/data/pitchbook.py` (to be written) emitting the same fitter-ready arrays — weekly sector counts, as-of macro matrix, LOCF firm features *plus staleness ages*, at-risk masks, and the event list — so that every fitter downstream is source-agnostic. The defect ledger of Table 2.1 becomes a validation suite (`hawkes_calibration/data/audits.py`): open-spell retention check, entry-date consistency, duplicate-collapse idempotence, as-of-join look-ahead test, and universe-share reporting (`off_universe_share` is already a first-class output of the synthetic loader and must be one for the real loader).

Part II

Continuous-Time Machinery

Chapter 3

Temporal Point Processes: Foundations

3.1 One machinery, many models

Every probabilistic model in this document is an instance of one object: a counting process characterized by its conditional intensity, fitted by the likelihood that the intensity induces, checked by the residuals that its compensator induces, and simulated by algorithms whose correctness is the intensity calculus read backwards. This chapter builds that machinery once, at the level of rigor the rest of the document consumes, so that Chapters 4 to 7 and 10 can say "instantiate Theorem 3.5 with this λ" and be done.

Five results carry the weight. First, the point-process likelihood (Theorem 3.5): one formula, an event term plus an exposure term, that every fitter in `hawkes_calibration/` implements. Second, the ground/mark factorization (Theorem 3.7): the exact identity, promised in Section 1.3, that splits the marked deal process into a timing factor and an identity factor and thereby licenses the repository's two-layer sector→firm architecture. Third, martingale estimation theory (Section 3.6): why maximum likelihood works for dependent event data, and which standard errors to trust when the model is wrong. Fourth, the time-rescaling theorem (Theorem 3.16): the goodness-of-fit engine of Chapter 7. Fifth, simulation correctness (Proposition 3.18): why the generators used in the synthetic rehearsal world of Section 2.5 produce the processes they claim to. The chapter closes, as Chapter 2's decision D2.1 requires, with an explicit account of which data defects (C1)–(C6) this machinery absorbs and which it cannot.

3.2 Counting processes, histories, and conditional intensity

Definition 3.1 (Counting process)**.** On the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ of Definition 2.1, a *counting process* N is an adapted, right-continuous process with $N(0) = 0$, nondecreasing piecewise-constant paths, jumps of size $+1$, and $N(t) < \infty$ a.s. for every $t \leq T$. A *multivariate counting process* is a finite family $(N_e)_{e \in \mathcal{E}}$ of counting processes, no two of which jump at the same time.

The index e will range over sectors, firms, or firm–transition pairs as each chapter requires; the filtration $\mathcal{F}_t$ contains the internal history $\mathcal{H}_t$ of the marked deal process *and* the observed covariate paths, as fixed in Definition 2.1. Two pieces of structure make the theory work, and both deserve to be named precisely because their violation is a software bug before it is a mathematical one.

Predictability. A process H is *predictable* if it is measurable with respect to the σ-algebra on $\Omega \times [0, T]$ generated by left-continuous adapted processes. The operational content: the value $H(t)$ must be computable from information available strictly before t, i.e. from $\mathcal{F}_{t^-}$. Left-continuity in t is the working test — an adapted process with left-continuous paths is predictable, and every intensity model in this document is built from such pieces (LOCF covariates held constant between deals, kernel sums evaluated as left limits, at-risk indicators of the form $\mathbf{1}\{V_i \le t < U_i\}$ which are left-continuous in t).

Warning 3.2 (Predictability is point-in-time discipline)*.* Predictability of λ is the continuous-time statement of Protocol 2.5. If any input to $\lambda(t)$ carries a timestamp not strictly before t — a covariate as-of joined on reference date instead of publish date, a kernel sum that includes the event at t itself, a risk-set flag closed using information revealed at t — then λ is not predictable, $N - \int Y\lambda$ is not a martingale with respect to the filtration actually used, and everything downstream fails *silently*: the score of Proposition 3.12 acquires a bias, the information identities break, and held-out likelihoods flatter the model. This is why `hawkes_calibration/eventtime/likelihood.py` decays its excitation state to *just before* each event when evaluating $\log \lambda$ at that event.

Compensators. The second piece of structure is the decomposition of a counting process into signal and noise.

Theorem 3.3 (Doob–Meyer decomposition for counting processes; stated, not proved)**.** *Let N be a counting process with $\mathbb{E}N(T) < \infty$. There exists an a.s. unique predictable, nondecreasing process Λ with $\Lambda(0) = 0$, the* compensator*, such that $M = N - \Lambda$ is a martingale. Under local integrability the same holds with "local martingale" (Andersen et al., 1993, Ch. II).*

Definition 2.3 specializes this to the absolutely continuous case that this document lives in: a *conditional intensity* for N_e is a nonnegative predictable process λ_e such that

$$M_e(t) \;=\; N_e(t) - \int_0^t Y_e(u)\,\lambda_e(u)\,\mathrm{d}u \tag{3.1}$$

is a local martingale, with Y_e the at-risk indicator. The heuristic that gives the definition its meaning, and that every proof in this chapter leans on, is the infinitesimal-bin reading of the martingale property:

$$\lambda_e(t)\,\mathrm{d}t \;=\; \mathbb{P}\big(N_e \text{ jumps in } [t, t+\mathrm{d}t) \mid \mathcal{F}_{t^-}\big) \qquad \text{on } \{Y_e(t) = 1\}. \tag{3.2}$$

The intensity is the instantaneous event rate given *everything known just before now*: internal history, covariate paths, risk-set composition. It is a modeling object (we write λ down) and simultaneously a probabilistic one (Equation (3.1) pins down what writing it down commits us to).

Assumption 3.4 (Standing regularity)**.** Throughout this chapter $(N_e)_{e\in\mathcal{E}}$ is a multivariate counting process such that (i) each N_e admits the intensity $Y_e\lambda_e(\cdot;\vartheta)$ of Equation (3.1), with λ_e predictable, locally bounded, and strictly positive at the jump times of N_e; (ii) $\mathbb{E}\int_0^T Y_e\lambda_e\,\mathrm{d}u < \infty$; (iii) $\lambda_e(t;\cdot)$ is twice continuously differentiable in ϑ on a neighborhood of the truth, with predictable, locally bounded derivative processes, and differentiation under the integral sign is valid; (iv) no two components jump simultaneously. All models fitted in this repository satisfy (i)–(iv) by construction (exponential-family intensities, piecewise-constant covariates, finite windows).

3.3 The point-process likelihood

Theorem 3.5 (Point-process (Jacod) likelihood)**.** *Under Assumption 3.4, let the data on $[0, T]$ be the event times and types (t_m, e_m), $m = 1, \ldots, n$. Then, up to an additive term not depending on ϑ (the dominating measure), the log-likelihood is*

$$\ell_T(\vartheta) \;=\; \sum_{m=1}^{n} \log \lambda_{e_m}(t_m; \vartheta) \;-\; \sum_{e \in \mathcal{E}} \int_0^T Y_e(u)\, \lambda_e(u; \vartheta)\, \mathrm{d}u, \tag{3.3}$$

which is Equation (2.3) of Definition 2.3.

Proof. We give the product-integral derivation over infinitesimal bins; it is the argument to remember, and it becomes rigorous as a product-integral limit (Daley and Vere-Jones, 2008; Andersen et al., 1993). Partition $[0, T]$ into K bins $B_k = ((k-1)\delta, k\delta]$ with $\delta = T/K$, and K large enough that no bin contains two events. By the chain rule of conditional probability, the probability of the observed configuration is the product over bins of the conditional probability of what happened in B_k given $\mathcal{F}_{(k-1)\delta}$. By Equation (3.2) and Assumption 3.4, a bin containing the event (t_m, e_m) contributes $\lambda_{e_m}(t_m)\, \delta + o(\delta)$; an empty bin contributes $1 - \sum_e Y_e \lambda_e\, \delta + o(\delta)$; two or more events in one bin have probability $o(\delta)$. Hence

$$\log \mathbb{P}(\text{data}) - n \log \delta \;=\; \sum_{m=1}^{n} \log \lambda_{e_m}(t_m) \;+\; \sum_{k:\, B_k \text{ empty}} \log \Big(1 - \textstyle\sum_e Y_e \lambda_e\, \delta \Big) \;+\; o(1).$$

Since $\log(1 - x) = -x + O(x^2)$, the middle sum is a Riemann sum of $-\sum_e \int_0^T Y_e \lambda_e\, \mathrm{d}u$ and converges to it as $\delta \to 0$. The diverging term $n \log \delta$ is the same for every ϑ: it cancels in every likelihood ratio and belongs to the dominating measure (concretely, Equation (3.3) exponentiated and multiplied by e^T is the density of the observed process with respect to a unit-rate Poisson process). This yields Equation (3.3). $\square$

The two terms have distinct jobs, and keeping them distinct organizes half of this document. The *event term* $\sum_m \log \lambda_{e_m}(t_m)$ rewards the model for placing intensity where events actually occurred — evaluated exactly at the jumps, on the log scale, so that assigning near-zero intensity to a realized event is catastrophic. The *compensator term* $\sum_e \int Y_e \lambda_e$ charges the model for exposure: every unit of intensity claimed over every firm-week at risk is paid for linearly, whether or not an event arrives. A model can only win the event term by spending in the compensator term, and the optimal spending pattern is the true intensity — that is the content of the score identity in Section 3.6. Two corollaries of the shape of Equation (3.3) recur constantly. An event-free spell $(a, b]$ contributes exactly the survival factor $\exp(-\int_a^b Y \lambda)$ — this is why censored spells carry information and must never be dropped (Warning 2.4, Proposition 3.20). And the event term is the only place event *times* enter; corrupt the times and the compensator term survives while the event term degrades — the germ of the interval-censored treatment of Chapter 6.

3.4 The ground/mark factorization

The market process of Definition 2.1 is a *marked* point process: ground times t_m with marks m_m (in our data the mark is the triple sector–firm–terms). The likelihood of a marked process factorizes exactly into a timing part and a mark part. This is the single most consequential piece of mathematics in the document — it is the formal license for the two-layer architecture — so we set it up carefully and prove it.

Definition 3.6 (Marked intensity kernel)**.** Let marks live in a measurable space $(\mathcal{M}, \nu)$ with ν a σ-finite reference measure (counting measure for discrete marks such as sector and firm identity, Lebesgue for continuous terms). A *marked point process* with intensity kernel $\lambda(t, m)$ is one for which, for every measurable $A \subseteq \mathcal{M}$, the counting process $N(t, A) = \#\{m : t_m \le t,\ m_m \in A\}$ has conditional intensity $\int_A \lambda(t, m)\, \nu(\,\mathrm{d}m)$. The *ground process* is $N_\mathrm{g}(t) = N(t, \mathcal{M})$ with *ground intensity* $\lambda_\mathrm{g}(t) = \int_\mathcal{M} \lambda(t, m)\, \nu(\,\mathrm{d}m)$, and the *mark density* is

$$p(m \mid t, \mathcal{H}_{t^-}) \;=\; \frac{\lambda(t, m)}{\lambda_\mathrm{g}(t)} \qquad \text{whenever } \lambda_\mathrm{g}(t) > 0, \tag{3.4}$$

a probability density in m with respect to ν, predictable in t.

Theorem 3.7 (Ground/mark factorization)**.** *Under Assumption 3.4 applied to the family $\{N(\cdot, A)\}$, the log-likelihood of the marked data $\{(t_m, m_m)\}_{m=1}^n$ on $[0, T]$ satisfies, exactly and for every parameter value and every realization,*

$$\ell_T \;=\; \underbrace{\sum_{m=1}^{n} \log \lambda_\mathrm{g}(t_m) \;-\; \int_0^T \lambda_\mathrm{g}(u)\,\mathrm{d}u}_{\textit{ground factor: timing}} \;+\; \underbrace{\sum_{m=1}^{n} \log p(m_m \mid t_m, \mathcal{H}_{t_m^-})}_{\textit{mark factor: identity}}\;. \tag{3.5}$$

Proof. Take first the case of finitely many mark values, which covers the sector–firm part of our data. The marked process is then the multivariate counting process $(N(\cdot, \{m\}))_{m \in \mathcal{M}}$ with intensities $\lambda(t, m)$, and Theorem 3.5 gives

$$\ell_T \;=\; \sum_{m=1}^{n} \log \lambda(t_m, m_m) \;-\; \sum_{m \in \mathcal{M}} \int_0^T \lambda(u, m)\,\mathrm{d}u.$$

Substitute $\lambda(t, m) = \lambda_\mathrm{g}(t)\, p(m \mid t, \mathcal{H}_{t^-})$ from Equation (3.4). The event term splits into $\sum_m \log \lambda_\mathrm{g}(t_m) + \sum_m \log p(m_m \mid t_m, \mathcal{H}_{t_m^-})$. The compensator term collapses:

$$\sum_m \int_0^T \lambda_\mathrm{g}(u)\, p(m \mid u, \mathcal{H}_{u^-})\,\mathrm{d}u \;=\; \int_0^T \lambda_\mathrm{g}(u) \underbrace{\sum_m p(m \mid u, \mathcal{H}_{u^-})}_{=1}\,\mathrm{d}u \;=\; \int_0^T \lambda_\mathrm{g}(u)\,\mathrm{d}u,$$

because the mark density integrates to one at every u. For general mark spaces, replace the sum over m by $\int \cdot\, \nu(\,\mathrm{d}m)$; the computation is identical and the underlying likelihood formula is Jacod's formula for marked processes (Daley and Vere-Jones, 2008). This proves Equation (3.5). $\square$

The crux is worth stating in words: *the mark density leaves no trace in the compensator*, because it integrates to one under the exposure integral. The mark factor therefore contributes only at event times — it is a product of genuine conditional probabilities "given a deal at t_m, which mark" — and involves intensities only through the ratio Equation (3.4), in which any common timing factor cancels. That is the mathematical root of the robustness asymmetry announced in Section 1.3: distortions that rescale or warp λ_g (untrusted dates, aggregation, baseline misspecification) damage the ground factor and cancel from the mark factor. Chapters 6 and 10 quantify this.

Remark 3.8 (Identity, not approximation)*.* Equation (3.5) assumes no independence, no Poisson structure, and no correct specification; it is algebra performed on the likelihood of Theorem 3.5, valid realization by realization. Every approximation made in this document lives *inside* one factor: the weekly MBPP approximation of Chapter 6 approximates the ground factor; the linear-index softmax of Chapter 10 approximates the mark factor; neither touches the other factor, and a data defect can be assigned to the factor whose term it corrupts. This bookkeeping — which factor is damaged — is how the defect ledger of Table 2.1 becomes actionable.

Corollary 3.9 (The two-layer sector→firm architecture). *Specialize the mark to $m = (s, i, m')$: sector, receiving firm, deal terms. Applying Theorem 3.7 and factoring the mark density by the chain rule, $p(s, i, m' \mid t) = p(s \mid t)\, p(i \mid s, t)\, p(m' \mid s, i, t)$, the likelihood splits into the three factors of Equation (1.4). Equivalently, defining sector processes N_s with intensities $\lambda_s(t) = \lambda_{\mathrm{g}}(t)\, p(s \mid t, \mathcal{H}_{t-})$, every tracked firm's intensity factorizes as*

$$\lambda_i(t) \;=\; \lambda_{s(i)}(t)\, \mathbb{P}(i \mid s(i),\, t,\, \mathcal{H}_{t-}), \qquad i \in \mathcal{I}, \tag{3.6}$$

and the log-likelihood is the sum of a sector-timing part (in λ_s only) and a firm-identity part (in the conditional choice probabilities only).

Proof. Apply the theorem twice: once with mark (s, i, m'), once viewing i as a mark of the sector-s ground process N_s; the chain-rule factorization of the joint mark density is the elementary identity for conditional densities, and Equation (3.6) is Equation (3.4) read at the firm level. □

Design decision (D3.1 — the factorization is the architecture).

The repository's two-layer design — a sector-level ground model (`hawkes_calibration/sector_stability.py`, `hawkes_calibration/eventtime/`) and a firm-level risk-set choice model (`hawkes_calibration/sector_ranker.py`) — is Corollary 3.9 implemented literally, with the two factors parameterized by variation-independent parameter blocks so that maximizing the sum Equation (3.5) decouples into two exact, separate maximizations. Rationale: the factorization is an identity, the factors' metrics align with distinct targets (T2 vs. T1 of Section 1.2), and the factors have opposite robustness profiles under the defect ledger. Reversal trigger: evidence of material parameter sharing across factors — e.g. a jointly estimated timing–identity model beating the separate fits on *both* held-out ground NLL and identity recall@K in two consecutive evaluation campaigns — would move us to joint estimation; the factorized *likelihood* would remain, only the separability of its maximization would be given up.

3.5 Superposition and the identity of the next event

Two small results connect the multivariate and marked views, and both are used constantly — in the ranking chapter as the modeling target, and in the simulation section as the attribution step.

Proposition 3.10 (Superposition). *Under Assumption 3.4, the pooled process $\bar{N} = \sum_e N_e$ is a counting process with conditional intensity $\bar{\lambda}(t) = \sum_e Y_e(t)\lambda_e(t)$.*

Proof. $\bar{N}$ is right-continuous, nondecreasing, and has unit jumps because no two components jump together; and $\bar{N} - \int \bar{\lambda} = \sum_e M_e$ is a finite sum of local martingales, hence a local martingale. Since $\bar{\lambda}$ is a sum of predictable processes it is predictable, so it is the conditional intensity of $\bar{N}$. □

Proposition 3.11 (Which type fires). *Under Assumption 3.4, conditionally on $\mathcal{F}_{t-}$ and on the event that the pooled process jumps at t,*

$$\mathbb{P}(\textit{the jump is of type } e \mid \mathcal{F}_{t-},\ \Delta\bar{N}(t) = 1) \;=\; \frac{Y_e(t)\, \lambda_e(t)}{\sum_{e'} Y_{e'}(t)\, \lambda_{e'}(t)}. \tag{3.7}$$

Proof sketch. By Equation (3.2), $\mathbb{P}(\text{type-}e \text{ jump in } [t, t + \mathrm{d}t) \mid \mathcal{F}_{t-}) = Y_e \lambda_e(t)\,\mathrm{d}t$ and $\mathbb{P}(\text{some jump}) = \bar{\lambda}(t)\,\mathrm{d}t$; the ratio gives Equation (3.7) as $\mathrm{d}t \to 0$. Structurally, the statement is Theorem 3.7 read backwards: view the type e as the mark of the pooled process; then the right-hand side of Equation (3.7) is exactly the mark density Equation (3.4). The full measure-theoretic proof, together with its operational consequences — the risk-set conditional logit, its partial-likelihood justification (Cox, 1975; McFadden, 1974), and the evaluation protocol built on it — is developed in Chapter 10, which restates Equation (3.7) as the definition of the ranking target. □

Equation (3.7) is the entire T1 target of Section 1.2 in one line: ranking firms by $\lambda_i(t)/\sum_j \lambda_j(t)$ is ranking by conditional next-event probability, and by Equation (3.6) the sector timing factor cancels from the ratio within a sector. It is also the attribution rule of the thinning simulator below: accept a proposed event, then label it type e with probability $\lambda_e/\bar{\lambda}$.

3.6 Martingale estimation theory

Why does maximizing Equation (3.3) work, given that events are dependent, non-stationary, and censored? The answer is not the classical i.i.d. theory but the martingale structure of Equation (3.1); this section develops exactly the pieces later chapters invoke.

Differentiating Equation (3.3) in ϑ (legitimate under Assumption 3.4(iii)) and writing $\mathrm{d}N_e - Y_e\lambda_e\,\mathrm{d}u = \mathrm{d}M_e$ gives the *score process*

$$U(\vartheta, t) \;=\; \sum_e \int_0^t \partial_\vartheta \log \lambda_e(u;\vartheta)\,\big[\,\mathrm{d}N_e(u) - Y_e(u)\lambda_e(u;\vartheta)\,\mathrm{d}u\big] \;=\; \sum_e \int_0^t \partial_\vartheta \log \lambda_e \;\mathrm{d}M_e(\vartheta), \tag{3.8}$$

where the second equality holds when ϑ is the true value: the score is a stochastic integral of a predictable process against the compensated noise.

Proposition 3.12 (Unbiased score at the truth)**.** *Let ϑ_0 be the true parameter. Under Assumption 3.4, $U(\vartheta_0,\cdot)$ is a mean-zero local square-integrable martingale; under the integrability of Assumption 3.4(ii)–(iii) it is a martingale, and in particular $\mathbb{E}\,U(\vartheta_0, T) = 0$.*

Proof. At ϑ_0, each M_e in Equation (3.8) is a local martingale by Equation (3.1), and the integrand $H_e = \partial_\vartheta \log \lambda_e$ is predictable and locally bounded by assumption. A stochastic integral of a predictable locally bounded process against a local martingale is a local martingale (Andersen et al., 1993, Ch. II), and $U(\vartheta_0, 0) = 0$. The moment condition upgrades the local martingale to a martingale, whence the expectation is zero at every t, in particular at T. What breaks when the assumptions fail is instructive: if λ peeks at time-t information (Warning 3.2), H_e is not predictable and the integral is no longer a martingale — the score acquires a bias of the same sign as the leak; if the model is misspecified, $M_e(\vartheta)$ is not a martingale for any ϑ in the family, and the score is instead centered at the pseudo-true parameter ϑ_* — the Kullback–Leibler projection of the truth onto the family — which is what the sandwich variance below is built for. □

Proposition 3.13 (Information identities)**.** *Define the observed information $J(\vartheta) = -\partial^2_\vartheta \ell_T(\vartheta)$ and the predictable variation of the score at the truth,*

$$\langle U(\vartheta_0,\cdot)\rangle(T) \;=\; \sum_e \int_0^T \big(\partial_\vartheta \log \lambda_e\big)\big(\partial_\vartheta \log \lambda_e\big)^\top\, Y_e\,\lambda_e\,\mathrm{d}u. \tag{3.9}$$

Then $J(\vartheta_0) - \langle U(\vartheta_0,\cdot)\rangle(T) = -\sum_e \int_0^T \partial^2_\vartheta \log \lambda_e\;\mathrm{d}M_e$ is a mean-zero martingale term, and consequently (the second Bartlett identity) $\mathbb{E}\,J(\vartheta_0) = \mathbb{E}\,\langle U\rangle(T) =: \mathcal{I}(\vartheta_0)$, the expected Fisher information.

Proof. Differentiate Equation (3.3) twice: $-J = \sum_e [\int \partial^2 \log \lambda_e \, \mathrm{d}N_e - \int Y_e \, \partial^2 \lambda_e \, \mathrm{d}u]$. Substitute $\partial^2 \lambda_e = \lambda_e \{\partial^2 \log \lambda_e + (\partial \log \lambda_e)(\partial \log \lambda_e)^\top\}$ in the second integral and collect terms: $J = -\sum_e \int \partial^2 \log \lambda_e \, \mathrm{d}M_e + \langle U \rangle(T)$. The first term is a stochastic integral of a predictable process against a martingale, mean zero at ϑ_0 by the argument of Proposition 3.12; take expectations. That the compensator of the score's quadratic variation is Equation (3.9) is the standard predictable-variation computation for counting-process integrals (Andersen et al., 1993, Ch. II). □

Sandwich variance. The MLE $\hat{\vartheta}$ solves $U(\hat{\vartheta}, T) = 0$; a Taylor expansion around the (pseudo-)truth gives $\hat{\vartheta} - \vartheta_* \approx J(\vartheta_*)^{-1} U(\vartheta_*, T)$, hence the asymptotic variance

$$\mathrm{Var}(\hat{\vartheta}) \;\approx\; A^{-1} B \, A^{-1}, \qquad A = \mathbb{E}\, J(\vartheta_*), \quad B = \mathrm{Var}\; U(\vartheta_*, T). \tag{3.10}$$

When the model is correctly specified, Proposition 3.13 gives $B = A = \mathcal{I}(\vartheta_0)$ and Equation (3.10) collapses to the familiar $\mathcal{I}^{-1}$. When it is not — and regime B of the synthetic rehearsal (Section 2.5) is a standing reminder that our families are misspecified by construction — the equality fails and only the sandwich is honest. In practice B is estimated by the optional-variation analogue of Equation (3.9) (outer products of per-event score contributions), aggregated with clustering at the sector level because latent frailty correlates firms within a sector (Appendix C: the 2.4× branching inflation under latent AR(0.90) frailty is exactly a violation of within-sector independence).

Theorem 3.14 (Rebolledo's martingale CLT; statement only)**.** *Let $U^{(n)}$ be a sequence of mean-zero local square-integrable martingales. If (i) $\langle U^{(n)} \rangle(t) \to V(t)$ in probability for each t, with V deterministic and continuous, and (ii) for each $\varepsilon > 0$ the predictable variation contributed by jumps of magnitude exceeding ε vanishes in probability, then $U^{(n)}$ converges weakly to a continuous Gaussian martingale with variance function V (Andersen et al., 1993; Andersen and Gill, 1982).*

Rebolledo's theorem is the replacement for the classical CLT in all counting-process asymptotics: applied to the normalized score with $V =$ the limit of Equation (3.9), and combined with the Taylor expansion above, it yields asymptotic normality of $\hat{\vartheta}$ with the variance Equation (3.10). Condition (ii) holds here because individual events contribute vanishing shares of the total information as the window or the cross-section grows.

Theorem 3.15 (Consistency and asymptotic normality of the point-process MLE; statement only)**.** *For stationary ergodic point processes with intensity families satisfying identifiability, smoothness, and moment conditions, $\hat{\vartheta}_T \to \vartheta_0$ a.s. and $\sqrt{T}\,(\hat{\vartheta}_T - \vartheta_0) \Rightarrow \mathcal{N}(0, \mathcal{I}_1(\vartheta_0)^{-1})$ as $T \to \infty$, with $\mathcal{I}_1$ the per-unit-time information (Ogata, 1978). The counting-process regime with many units observed on a fixed window — our firm panel — is covered by the same martingale program (Andersen and Gill, 1982; Andersen et al., 1993).*

Both asymptotic regimes are present in this repository and should not be conflated: sector-level timing models see $S = 12$ fixed components over a long window (Ogata's $T \to \infty$ regime), while the mark factor sees thousands of firms per risk set on a fixed window (the Andersen–Gill $n \to \infty$ regime). Standard errors quoted in later chapters state which regime they invoke.

Design decision (D3.2 — martingale-grade inference outputs).

Every likelihood fitter in `hawkes_calibration/` must expose (i) the analytic score Equation (3.8), (ii) an observed-information or numerical Hessian, and (iii) sandwich standard errors Equation (3.10) with sector-level clustering as the default reported uncertainty;

model-based $\mathcal{I}^{-1}$ errors may be shown only alongside. Rationale: misspecification is the operating condition (regime B), under which only the sandwich is valid for the pseudo-true parameter; and exposing the score is what makes the gradient checks and the GOF machinery of Chapter 7 cheap. Reversal trigger: for a model family that the diagnostics of Chapter 7 certify as well specified on real data across two campaigns, model-based errors may be promoted to primary for that family only.

3.7 The time-rescaling theorem

If the fitted intensity is right, the fitted compensator is a clock under which the data become maximally boring. This is the goodness-of-fit engine that Chapter 7 operates; we state and prove it here.

Theorem 3.16 (Time rescaling)**.** *Let N have conditional intensity λ under Assumption 3.4, with compensator $\Lambda(t) = \int_0^t Y\lambda \,\mathrm{d}u$ continuous and strictly increasing on the observation window, and let $t_1 < t_2 < \cdots < t_n$ be the event times on $[0, T]$. Then the rescaled times $\tau_m = \Lambda(t_m)$ form a unit-rate Poisson process on $[0, \Lambda(T)]$; equivalently, the rescaled gaps*

$$\Delta_m \;=\; \Lambda(t_m) - \Lambda(t_{m-1}) \;=\; \int_{t_{m-1}}^{t_m} Y(u)\lambda(u)\,\mathrm{d}u, \qquad t_0 = 0, \tag{3.11}$$

are i.i.d. Exp(1)*, and $\Lambda(T) - \Lambda(t_n)$ is an independent right-censored* Exp(1) *variable (Brown et al., 2002; Ogata, 1988).*

Proof. We give the hazard argument in the case where, between events, the intensity path is determined by the history at the last event together with the observed covariate paths — true of every model fitted in this document, whose $\lambda(u)$ for $u \in (t_{m-1}, t_m]$ is an explicit functional of $\mathcal{H}_{t_{m-1}}$, the no-further-event continuation, and covariates that $\mathcal{F}$ carries. Condition on $\mathcal{F}_{t_{m-1}}$ and let $g(u)$ denote that determined intensity path. By Equation (3.2), the waiting time to the next event has hazard g: $\mathbb{P}(t_m \in [u, u + \mathrm{d}u) \mid t_m \geq u, \mathcal{F}_{t_{m-1}}) = g(u)\,\mathrm{d}u$, hence the conditional survivor function $\mathbb{P}(t_m > u \mid \mathcal{F}_{t_{m-1}}) = \exp(-\int_{t_{m-1}}^u g)$. Now fix $x \geq 0$ and let $u_x = \inf\{u : \int_{t_{m-1}}^u g = x\}$, finite and uniquely attained because Λ is continuous and strictly increasing. The events $\{\Delta_m > x\}$ and $\{t_m > u_x\}$ coincide, so

$$\mathbb{P}(\Delta_m > x \mid \mathcal{F}_{t_{m-1}}) \;=\; \exp\Big(-\int_{t_{m-1}}^{u_x} g(u)\,\mathrm{d}u\Big) \;=\; e^{-x}.$$

The conditional law of Δ_m given the entire past is Exp(1), free of the past; iterating over m makes $(\Delta_1, \ldots, \Delta_n)$ i.i.d. Exp(1), which is the gap characterization of the unit-rate Poisson process. The same computation with x at the censoring boundary gives the final censored gap. In full generality — intensity paths influenced by additional external randomness between events — the result is the random-time-change theorem: $M(\Lambda^{-1}(\tau))$ is a martingale by optional stopping, so the time-changed counting process has deterministic continuous compensator τ, and Watanabe's characterization identifies it as unit-rate Poisson (Daley and Vere-Jones, 2008). □

The practical reading: *a correct model makes its own residuals exponential*, whatever the dependence structure of the original data. The resulting protocol — rescale with the fitted compensator, then test the gaps against Exp(1) by Kolmogorov–Smirnov and QQ plots, as in Ogata's residual analysis (Ogata, 1988) — is fixed as this repository's primary event-time diagnostic in Chapter 7, including the treatment of the censored last gap. Two caveats travel with it. The test consumes exact event times, so under defect (C4) — weekly aggregation — it degrades by construction; the interval-censored regime uses bucket-level Pearson and dispersion residuals instead (Chapter 7, `hawkes_calibration/mbpp/gof.py`). And rescaling with an intensity fitted on the *same* data is optimistic; the split protocol of Chapter 7 rescales held-out windows with frozen parameters.

3.8 Simulation: inversion and thinning

Simulation is estimation's mirror: the rehearsal world of Section 2.5, every parameter-recovery study in Appendix C, and the posterior predictive checks of Chapter 7 all rest on generators whose output provably has the intended intensity. Two algorithms cover our needs.

Corollary 3.17 (Inverse-compensator simulation). *Let $E_1, E_2, \ldots$ be i.i.d.* Exp(1)*. Setting, recursively,*

$$t_m \;=\; \inf\Big\{u > t_{m-1} : \int_{t_{m-1}}^{u} \lambda(v)\,\mathrm{d}v \geq E_m\Big\},$$

with λ evaluated along the history generated so far, produces a point process with conditional intensity λ.

Proof. This is Theorem 3.16 read backwards: the construction forces the rescaled gaps to be the i.i.d. Exp(1) inputs, and the hazard computation in that proof, run in reverse, shows the waiting time generated has exactly the conditional hazard λ. □

Inversion is exact and fast whenever the conditional compensator inverts in closed form — the covariate-driven Poisson models of Chapter 4 with piecewise-constant covariates are the leading case. When the compensator does not invert cheaply, we thin.

Proposition 3.18 (Correctness of Ogata's modified thinning). *Suppose that on each inter-refresh interval (between consecutive accepted events, covariate breakpoints, or bound refreshes) a constant $\bar\lambda$ is available with $\lambda(u) \leq \bar\lambda$ for all u in the interval, where λ is evaluated along the accepted history and its no-new-event continuation. Generate proposals on the interval from a homogeneous Poisson process of rate $\bar\lambda$, and accept a proposal at τ independently with probability $\lambda(\tau)/\bar\lambda$, where $\lambda(\tau)$ is computed from the accepted history up to τ^-. Then the accepted points form a point process with conditional intensity λ (Lewis and Shedler, 1979; Ogata, 1981).*

Proof. Work on the enlarged filtration generated by the accepted history, the proposal process, and the acceptance variables. Given the accepted history $\mathcal{F}^{\mathrm{acc}}_{t-}$, the probability of an accepted point in $[t, t + \mathrm{d}t)$ is the probability of a proposal there, $\bar\lambda\,\mathrm{d}t + o(\mathrm{d}t)$, times the acceptance probability $\lambda(t)/\bar\lambda$ — which is $\mathcal{F}^{\mathrm{acc}}_{t-}$-measurable (predictable), because $\lambda(t)$ depends only on accepted points before t and on covariates; two proposals in the bin have probability $o(\mathrm{d}t)$. The product is $\lambda(t)\,\mathrm{d}t + o(\mathrm{d}t)$, so the accepted counting process minus $\int\lambda$ is a martingale on the enlarged filtration, and since λ is adapted to the accepted history, it is the conditional intensity of the accepted process (Equation (3.1)). The bound being valid on the whole inter-refresh interval is what makes the acceptance probability well defined (≤ 1) throughout; the multivariate version accepts with $\bar\lambda^{-1}\sum_e \lambda_e(\tau)$ and then labels the accepted point type e with probability $\lambda_e(\tau)/\sum_{e'}\lambda_{e'}(\tau)$, which is correct by Proposition 3.11. □

Warning 3.19 (The bound is part of the algorithm's correctness). Thinning is exact only while $\bar\lambda$ truly dominates. For exponential Hawkes with nonnegative kernels ($h(t) = \kappa\theta e^{-\theta t}$), the intensity strictly decreases between events, so the value just after the last event is a valid bound

— *unless the baseline jumps upward inside the interval*, as it does whenever a piecewise-constant covariate steps. A generator that fails to refresh the bound at covariate breakpoints silently under-simulates events after every upward step, biasing rehearsal conclusions in the direction of weaker covariate effects. `hawkes_calibration/eventtime/simulate.py` refreshes at accepted events *and* at covariate breakpoints for exactly this reason, and verifies its recursive intensity against a naive recomputation.

For the exponential kernel there is also an exact, thinning-free simulator that exploits the Markov property of the intensity itself (cluster and shot-noise structure), due to Dassios and Zhao (2013); it and the branching (cluster) representation are developed where they are used, in Chapter 5.

3.9 What the machinery absorbs, and what it cannot

Decision D2.1 of Chapter 2 requires every model chapter to declare its position against the defect ledger of Table 2.1. This chapter contributes the declaration for the machinery itself: which defects the counting-process framework absorbs *by construction*, and which no amount of likelihood theory can repair.

Proposition 3.20 (Independent censoring preserves the likelihood; (C1))**.** *Let C_e be $\{0,1\}$-valued predictable censoring processes, and let $\mathcal{G}_t \supseteq \mathcal{F}_t$ be the filtration enlarged with the censoring information. Assume independent censoring: each N_e has the same intensity $Y_e\lambda_e$ with respect to $\{\mathcal{G}_t\}$ as with respect to $\{\mathcal{F}_t\}$. Then the observed processes $N_e^c(t) = \int_0^t C_e \, \mathrm{d}N_e$ have $\mathcal{G}$-intensity $C_e\, Y_e\, \lambda_e$, and Equation (3.3) computed on the observed events with exposure C_eY_e is a genuine (partial) likelihood: its score retains the martingale property of Proposition 3.12, and all of Section 3.6 applies unchanged (Andersen et al., 1993, III.2).*

Proof sketch. Since C_e is predictable, its value at t is known given $\mathcal{G}_{t-}$, so $\mathbb{E}[\,\mathrm{d}N_e^c \mid \mathcal{G}_{t-}] = C_e(t)\,\mathbb{E}[\,\mathrm{d}N_e \mid \mathcal{G}_{t-}] = C_e\, Y_e\, \lambda_e\, \mathrm{d}t$, the middle equality being the independent-censoring hypothesis. Hence $N_e^c - \int C_eY_e\lambda_e$ is a $\mathcal{G}$-martingale, and Theorem 3.5 applied to (N_e^c) yields Equation (3.3) with Y_e replaced by C_eY_e; at observed events $C_e = 1$, so the event term is untouched. The factor of the joint likelihood governing the censoring mechanism itself contains no ϑ and is dropped — that is the sense in which the result is a *partial* likelihood. Administrative censoring at fixed T has $C_e(t) = \mathbf{1}\{t \le T\}$ deterministic, hence predictable, and trivially preserves the intensity: it is independent by construction. What breaks without predictability is the first equality; what breaks without independence is the second — if censoring anticipates the process (firms leave observation *because* their funding prospects changed), $\mathbb{E}[\,\mathrm{d}N_e \mid \mathcal{G}_{t-}] \ne Y_e\lambda_e\,\mathrm{d}t$ and the observed-data likelihood is biased. That failure mode is exactly defect (C3). □

Operationally, Proposition 3.20 is the license already exercised in Warning 2.4: a spell open at June 2024 contributes its survival factor $\exp(-\int \lambda)$ through the compensator term, no term is dropped and none invented, *provided the open spell is kept in the data.* The machinery is robust to (C1).

(C2) Left truncation. Delayed entry is absorbed by the at-risk indicator: setting $Y_i(t) = \mathbf{1}\{V_i \le t < U_i\}$ makes every compensator integral in Equation (3.3) start at the entry time V_i, and the likelihood is valid *conditionally on the pre-entry history* (Keiding, 1990; Kalbfleisch and Prentice, 2002). The mechanical fix is complete; the statistical selection is not fixed at all: because entry occurs *by raising*, every estimand is conditional on "at least one deal before T," and unconditional statements about companies require the open-universe treatment of Chapter 11. The machinery *corrects (C2) under the stated conditional interpretation*, and silently changes the estimand if that interpretation is forgotten.

(C3) Unlabeled exits. Here the machinery is *vulnerable, with a known sign.* A dead firm with $Y_i \equiv 1$ contributes exposure and no events — indistinguishable, inside the likelihood, from a live quiet firm — so no score, residual, or information identity can detect the corruption: Equation (3.3) is being evaluated with the wrong Y, and every fitted intensity is biased *downward* while ranking risk sets are contaminated with phantom competitors. The rehearsal quantified the damage (observed-vs-oracle top-5 hit rate 0.14 vs 0.29; an exit model with AUC 0.877 insufficient to repair it; Appendix C). Treatments are exposure windows and mover–stayer structure in Chapters 4 and 9 and ranking-layer mitigations in Chapter 10; none is a consequence of the machinery, all are assumptions imported to compensate for it.

(C4) Untrusted times and aggregation. The event term of Equation (3.3) — and only the event term — consumes exact times, and Theorem 3.16 and Proposition 3.18 consume them too. Date jitter perturbs $\log \lambda(t_m)$ where λ moves fastest, i.e. it attacks timescale parameters first; full aggregation deletes the event term and requires the interval-censored likelihoods of Chapter 6 (Rizoiu et al., 2022). The inherited verdict stands: κ survives aggregation, θ does not (Appendix C). The machinery is vulnerable to (C4); Chapter 6 is the correction.

(C5) and (C6). Stale covariates leave the machinery *valid but reinterpreted*: LOCF covariates as constructed under Protocol 2.5 are predictable, so Equation (3.3) is a genuine likelihood for the model that conditions on the *observed* covariates — but its coefficients are attenuated relative to the true-covariate model, which is Chapter 8's subject. Out-of-universe deals leave sector-level ground counts complete but thin the firm-level mark factor: without an explicit outside option, the normalization in Equation (3.7) overstates every tracked firm's probability — Chapter 11 owns the fix.

In summary: the machinery of this chapter is robust to (C1), corrects (C2) under a conditional reading, and is structurally blind to (C3), (C4) on the event term, (C5) in interpretation, and (C6) in the mark normalization. That blindness is the agenda of Chapters 4, 6, 8 and 11.

Code hooks.

Theorem 3.5 is implemented in `hawkes_calibration/eventtime/likelihood.py`: `log_likelihood` assembles the event term by recursive exponential-kernel sums decayed to each event's left limit (Warning 3.2) plus a closed-form compensator term; `log_likelihood_and_grad` returns the analytic score Equation (3.8), satisfying decision D3.2. Proposition 3.18 and Warning 3.19 are implemented in `hawkes_calibration/eventtime/simulate.py` (`simulate_multivariate_hawkes`): Ogata thinning with bound refresh at accepted events and covariate breakpoints, with a naive-recomputation cross-check of the recursive intensity. Theorem 3.16 lives in `hawkes_calibration/mbpp/gof.py` (`time_rescaling_residuals`, `ks_test_exp1`, `qq_exp1`), with bucket-level Pearson/dispersion residuals as the (C4)-regime substitute. Proposition 3.11 is the modeling target of `hawkes_calibration/sector_ranker.py` (Chapter 10). Proposition 3.20 and the (C2) risk-set rule become validation checks in the planned `hawkes_calibration/data/audits.py` of Chapter 2: open-spell retention and entry-date consistency are preconditions for the likelihood to be the one this chapter proved things about.

Chapter 4

Poisson Models: The Baselines That Must Be Beaten

4.1 Why the ladder starts here

The timing target T2 of Chapter 1 asks for the ground intensity $\lambda_s(t)$: how many deals, in which sector, next week, with calibrated uncertainty. This chapter builds the two simplest answers — the homogeneous Poisson process, which asserts that nothing is valuable to know beyond the level of activity, and the inhomogeneous Poisson process (NHPP) with covariates, which asserts that the *market clock* is valuable but the market's own history is not. Their role in the design is deliberately adversarial. Every richer timing model in this document — the history-modulated GLM previewed in Section 4.7, the exponential Hawkes machinery of Chapter 5, the interval-censored variants of Chapter 6 — is admitted into production only by beating a well-covariated NHPP on held-out likelihood under the protocols of Chapter 7. The rehearsal campaigns justify this posture with numbers: in the decomposition benchmark, covariates carried almost the entire held-out gain over a seasonal baseline (+3.5, +5.7, and +1.5 nats per sector-week across the three synthetic worlds), while everything added on top of them — lags, excitation, cross-sector terms — contributed decimal points (Appendix C). A Hawkes fit whose covariate baseline is impoverished will absorb the macro cycle into its kernel and report spurious contagion; the only defense is a baseline that already knows the macro cycle, which is exactly what this chapter calibrates.

The second reason to linger at this rung is that it is the one rung whose likelihood is *exactly right* under the observation scheme of Chapter 2. At sector resolution, right censoring costs nothing (Section 4.5 proves this and thereby settles a standing open question of the repository), interval censoring by weekly aggregation costs nothing (Theorem 4.10), and sectors neither enter late nor die, so (C2) and (C3) do not bind. The Poisson rung is where calibration discipline is cheap; every deviation from it that we later purchase must be priced against a baseline whose own accounting is beyond suspicion.

4.2 The two models and their assumptions

Throughout, units e are sectors $s \in \mathcal{S}$ (the primary case) or, for the event-time path, tracked firms $i \in \mathcal{I}$; N_e is the counting process of Definition 2.3 with at-risk indicator Y_e, and all likelihoods specialize the Jacod likelihood Equation (2.3) whose martingale estimation theory is developed in Chapter 3. The homogeneous model sets $\lambda_e(t) \equiv \mu_e$: conditionally on nothing, events arrive at a constant rate, the MLE is $\hat{\mu}_e = N_e(T)/\int_0^T Y_e$, and the model's only message is the level. It is the null of the whole timing enterprise; its calendar-dummy refinement (a rate per quarter or per season) is the "seasonal baseline" against which the covariate gains above are measured. The inhomogeneous model is the first calibrated one, and we now fix its assumptions as numbered contracts.

Assumption 4.1 (Conditional Poisson structure)**.** Given the covariate paths, the intensity of N_e is a deterministic function of time: $\lambda_e(t;\vartheta) = \lambda_e(t)$ measurable with respect to the covariate history alone, with no dependence on the internal history $\mathcal{H}_{t-}$ of the deal process. Equivalently, conditionally on covariates, N_e is an inhomogeneous Poisson process: counts over disjoint sets are independent, and the count over a set A is Poisson with mean $\int_A Y_e(u)\lambda_e(u)\,\mathrm{d}u$ (Kingman, 1993; Daley and Vere-Jones, 2003).

Assumption 4.2 (Log-linear intensity)**.** $\lambda_s(t) = \exp(\beta^\top x_s(t))$, where $x_s(t) \in \mathbb{R}^d$ stacks a sector intercept (equivalently, sector dummies), the published as-of macro values of Protocol 2.5(a), and deterministic seasonality and calendar terms; per-sector slopes are interactions of the macro block with the sector dummies. The exponential link is the canonical one; Section 4.2.1 defends the choice against its alternatives.

Assumption 4.3 (Piecewise-constant weekly covariates)**.** $x_s(\cdot)$ is constant on the cells of a partition of $[0, T]$ into weekly bins: macro as-of series are step functions by construction (they jump at publication dates, which we align to the weekly grid), calendar terms are constant within a week, and sector dummies are constant always. Consequently $\lambda_s(\cdot)$ is a step function.

Assumption 4.4 (External covariates)**.** The covariates of Assumption 4.2 are *external* in the sense of Kalbfleisch and Prentice (2002): their evolution does not depend on the deal stream. Macro series, calendar terms, and sector dummies qualify. Lagged deal counts, exponentially filtered event streams, and any functional of $\mathcal{H}_{t-}$ do *not*; adding them changes the model class (Section 4.7) and voids Assumption 4.1.

Assumption 4.5 (Spanning)**.** Writing c for the sector-week cells of Assumption 4.3, $w_c = \int_{\text{cell}} Y(u)\,\mathrm{d}u$ for the exposure of cell c and x_c for its covariate vector, the family $\{x_c : w_c > 0\}$ spans $\mathbb{R}^d$; equivalently $\sum_c w_c x_c x_c^\top \succ 0$.

Assumption 4.1 is a known fiction — Section 4.6 measures its failure and upgrades the noise model — while Assumptions 4.3 and 4.4 are believed exactly true for the design of Assumption 4.2, and Assumption 4.5 is checkable from the design matrix before any fit.

4.2.1 Why the exponential link

Three properties recommend exp, and each earns its keep somewhere specific in this document. *Positivity for free*: any $\beta \in \mathbb{R}^d$ yields a valid intensity, so estimation is unconstrained smooth optimization — no boundary, no barrier, no projected gradient. *Concavity*: the log-likelihood is globally concave in β (Theorem 4.6), so the fitted object is a reproducible global optimum rather than a local one an optimizer happened to find. *Multiplicative interpretation*: a unit move in covariate j multiplies the rate by e^{β_j} uniformly across sectors and weeks, so macro effects read as rate ratios — the natural language for "tightening moved weekly late-stage deal flow down 12%" — and sector dummies give sector-proportional baselines. The log link is also *aggregation-coherent* under Assumption 4.3: binning to weeks leaves the functional form intact with the bin width absorbed into the intercept (Theorem 4.10), which is precisely what makes the weekly GLM an exact NHPP calibration rather than a discretization of one.

The alternatives are worth stating because the repository uses one of them. The *identity link* $\lambda = \beta^\top x$ makes effects additive — deals per week per unit of covariate — which is the natural scale for excitation terms with a branching interpretation, and it tolerates within-bin covariate variation exactly (the integral of a linear function is linear in the averaged covariate, so no Jensen gap ever opens). Its price is positivity: the likelihood is concave only on the polyhedron $\{\beta : \beta^\top x_c > 0\ \forall c\}$, the MLE can sit on the boundary where standard asymptotics fail (Self and Liang, 1987), and every fit needs constraints. The *softplus link* $\lambda = \log(1 + e^\eta)$ keeps positivity free and grows only linearly for large η, taming the exponential extrapolation risk that haunts the log link outside the training covariate support; it remains log-concave, but the coefficients lose their rate-ratio reading and the fit leaves standard GLM software. The stabilized sector layer of Chapter 5 is a deliberate hybrid: exponential link for the covariate baseline (this chapter's object, with all three advantages), identity link for the added excitation term (where the branching reading and the stability constraint live).

Design decision (D4.1 — exponential link for the covariate baseline).
The covariate-driven baseline intensity of every ground-process model in this document uses the log-linear form of Assumption 4.2. Rationale: unconstrained concave estimation (Theorem 4.6), exact weekly-GLM equivalence (Theorem 4.10), multiplicative macro interpretation, and sign freedom for later history terms (the exp-link route is one of the three legal ways to express inhibition; Chapter 5). Reversal trigger: if held-out evaluation shows exponential extrapolation failures — predictions pinned at the support-envelope guard of `hawkes_calibration/sector_stability.py` on more than isolated weeks — the baseline link is switched to softplus and this chapter's concavity and equivalence results are re-derived for it (both survive in weakened form; the GLM packaging does not).

4.3 Likelihood, score, and the geometry of the optimum

Under Assumptions 4.1 and 4.2, the Jacod likelihood Equation (2.3) specializes, for the pooled sector process on $[0, T]$, to

$$\ell(\beta) \;=\; \sum_m \beta^\top x_{s_m}(t_m) \;-\; \sum_{s\in\mathcal{S}} \int_0^T Y_s(u)\, e^{\beta^\top x_s(u)}\, \mathrm{d}u, \tag{4.1}$$

the sum running over observed deals m with sector s_m and time t_m. Under Assumption 4.3 the integral collapses over cells: with y_c the event count, w_c the exposure, and x_c the covariate of cell c,

$$\ell(\beta) \;=\; \sum_c \Big(y_c\, \beta^\top x_c \;-\; w_c\, e^{\beta^\top x_c} \Big). \tag{4.2}$$

Differentiating Equation (4.2) (or Equation (4.1); the cell form is the general one with the integral discretized exactly):

$$\nabla \ell(\beta) \;=\; \sum_c \big(y_c - w_c\, e^{\beta^\top x_c}\big)\, x_c, \tag{4.3}$$

$$\nabla^2 \ell(\beta) \;=\; -\sum_c w_c\, e^{\beta^\top x_c}\, x_c x_c^\top. \tag{4.4}$$

The score Equation (4.3) is the familiar "observed minus expected" weighted by covariates; evaluated at the truth it is the terminal value of the martingale $\int x\,(\,\mathrm{d}N - Y\,\lambda\,\mathrm{d}t)$, which is the engine of the asymptotics below. The Hessian Equation (4.4) does not involve the counts y_c at all — a canonical-link property: observed information equals conditional expected information, so Newton's method, Fisher scoring, and iteratively reweighted least squares are the same algorithm here.

Theorem 4.6 (Strict log-concavity under the exponential link)**.** *Under Assumptions 4.1 to 4.3, the map $\beta \mapsto \ell(\beta)$ of Equation (4.2) is concave on all of $\mathbb{R}^d$: $\nabla^2\ell(\beta) \preceq 0$ for every β. If in addition Assumption 4.5 holds, it is strictly concave: $\nabla^2\ell(\beta) \prec 0$ for every β.*

Proof. For any $v \in \mathbb{R}^d$,

$$v^\top \nabla^2 \ell(\beta)\, v \;=\; -\sum_c w_c\, e^{\beta^\top x_c}\, (v^\top x_c)^2 \;\le\; 0,$$

since every summand is a product of nonnegative factors: concavity, with no condition. Equality holds iff $v^\top x_c = 0$ for every cell with $w_c > 0$ (the weights $w_c e^{\beta^\top x_c}$ are strictly positive on exactly those cells). Under Assumption 4.5 no $v \neq 0$ annihilates all exposed x_c, so the quadratic form is strictly negative for all $v \neq 0$ and all β. When Assumption 4.5 fails — say two macro series are exactly collinear over the training window — the likelihood is flat along the offending direction and β is identified only up to it; the fitted *intensity* is still unique, which is why ridge regularization plus inference on well-identified linear combinations, rather than coefficient surgery, is the prescribed response to the near-failure that smooth, correlated macro series actually produce. □

Corollary 4.7 (Global optimum and solver guarantees)**.** *Under the assumptions of Theorem 4.6 with Assumption 4.5, the MLE, when it exists, is the unique global maximizer of ℓ, every stationary point is that maximizer, and any ascent method with standard line-search or trust-region safeguards — Newton, Fisher scoring, L-BFGS — converges to it from any starting point. Two fits of the same data that report different optima are evidence of a software defect, not of multimodality.*

Proof. Strict concavity on $\mathbb{R}^d$ makes the superlevel sets convex and any stationary point the unique global maximum; convergence of safeguarded ascent methods on smooth concave objectives with nonsingular Hessian is standard. The content worth stating is the last sentence: at this rung, optimizer nondeterminism is a bug detector. This guarantee is exactly what is forfeited — first partially (block concavity), then wholly — as excitation parameters with free decays enter in Chapter 5. □

Asymptotics. At the true β_0 the score is a martingale integral with predictable variation $\mathcal{I}(\beta_0) = \sum_c w_c e^{\beta_0^\top x_c} x_c x_c^\top$ — the Fisher information, equal to minus the Hessian Equation (4.4). Rebolledo's martingale CLT then gives $\hat\beta \approx \mathcal{N}(\beta_0, \mathcal{I}^{-1})$ as information accumulates, through a longer window or more sectors; this is the counting-process estimation theory of Andersen et al. (1993) and Andersen and Gill (1982), and the stationary-point-process version of Ogata (1978), specialized to the concave case. What matters practically is that information is *pooled*: a d-dimensional β needs adequate total information $\sum_c w_c e^{\beta^\top x_c} x_c x_c^\top$, not adequate counts in every sector-week cell — sparse cells are a separation risk (Proposition 4.8), not an asymptotics failure per se.

4.3.1 Existence: separation and sparse cells

Concavity guarantees uniqueness, not existence. The Poisson analogue of logistic separation is the one genuine pathology of this rung.

Proposition 4.8 (Nonexistence of the MLE under separation)**.** *Under the assumptions of Theorem 4.6 with Assumption 4.5, the MLE fails to exist iff there is a direction $v \neq 0$ with*

(i) $v^\top x_c \le 0$ *for every cell with* $w_c > 0$*, and*

(ii) $v^\top x_c = 0$ *for every cell with* $y_c > 0$*.*

Along any such v the likelihood increases strictly and monotonically to a finite supremum that is not attained; the fitted intensity on the cells with $v^\top x_c < 0$ degenerates to zero.

Proof. Suppose such v exists. By Assumption 4.5, $v^\top x_c \neq 0$ for at least one exposed cell, which by (ii) has $y_c = 0$ and by (i) has $v^\top x_c < 0$. Along $\beta + \tau v$, the event term of Equation (4.2) is constant in τ by (ii), while each compensator term $-w_c e^{\beta^\top x_c + \tau v^\top x_c}$ is nondecreasing in τ by (i) and strictly increasing on the cell just exhibited. Hence ℓ increases strictly to the finite limit obtained by deleting the vanishing compensator terms, and no finite β attains it. Conversely, if no such v exists, then along every ray $\beta + \tau v$ either some exposed cell has $v^\top x_c > 0$, in which case the compensator term drives $\ell \to -\infty$ exponentially (dominating the linearly growing event term), or all exposed cells have $v^\top x_c \leq 0$ with strict inequality at some event cell, in which case the event term drives $\ell \to -\infty$ linearly. So ℓ is coercive on $\mathbb{R}^d$ and, being continuous, attains its maximum. □

Warning 4.9 (Sparse cells produce infinite coefficients). The separating directions of Proposition 4.8 are not exotic: any indicator covariate whose active cells contain *zero events* — a small sector crossed with a rare calendar dummy, an interaction active only in a few quiet weeks — yields v = minus that coordinate, and the fitted coefficient runs to $-\infty$ (the MLE of a zero rate). At weekly resolution with twelve sectors and rich calendar interactions this *will* occur. Remedies, in the order the repository applies them: the ridge penalty already standard in the fitters (a strictly concave perturbation restores existence unconditionally; the penalty must be reported next to the fit); the Firth adjustment, maximizing $\ell(\beta) + \frac{1}{2}\log\det\mathcal{I}(\beta)$, which removes the leading-order bias of the MLE and keeps estimates finite (Firth, 1993), with the rare-events analysis of King and Zeng (2001) as the companion warning that unpenalized rates estimated from a handful of events are biased as well as fragile; and design hygiene — collapse or drop cells/dummies that the training window cannot inform. Related but distinct is *extrapolation* risk: a log-linear intensity evaluated outside the training covariate support is an exponential extrapolation even when the MLE exists; the support-envelope winsorization in `hawkes_calibration/sector_stability.py` is the implemented guard, and train/test covariate ranges are reported next to every held-out NLL (Chapter 7).

4.4 The binned-GLM equivalence theorem

The repository fits weekly sector counts with a Poisson GLM. The following theorem says this *is* NHPP maximum likelihood — exactly, not approximately — and identifies precisely what binning discards (nothing, under this chapter's assumptions).

Theorem 4.10 (Piecewise-constant NHPP = Poisson GLM with log-exposure offset)**.** *Let Assumptions 4.1 to 4.3 hold, so that λ is constant, equal to $\lambda_c = e^{\beta^\top x_c}$, on each cell c of the weekly partition, and let $w_c = \int_{cell} Y(u)\,du$ be the (possibly partial) exposure of cell c. Then:*

(i) (Distribution.) *Conditionally on covariates and exposures, the cell counts are independent with $y_c \sim \text{Poisson}(w_c\, e^{\beta^\top x_c})$.*

(ii) (Likelihood identity.) *The point-process log-likelihood Equation (4.1) and the log-likelihood of the Poisson GLM with log link, design vectors x_c, and offset $\log w_c$ differ by a constant free of β:*

$$\ell_{\text{pp}}(\beta) \;=\; \sum_c \Big(y_c \log(w_c e^{\beta^\top x_c}) - w_c e^{\beta^\top x_c} - \log y_c! \Big) \;+\; \textit{const.} \tag{4.5}$$

(iii) (Sufficiency.) *The vector of cell counts $(y_c)_c$ is sufficient for β: given the counts, the event times within each cell are distributed as i.i.d. uniform draws on the at-risk subset of the cell, a law free of β.*

Consequently the NHPP MLE, score, observed information, and all likelihood-based inference coincide with those of the binned Poisson GLM; fitting the weekly sector count GLM is an NHPP calibration, not a discretization of one.

Proof. (i) is the independent-increments property of Assumption 4.1 applied to the disjoint at-risk subsets of the cells: the count over the at-risk subset A_c of cell c is Poisson with mean $\int_{A_c} \lambda_c \, \mathrm{d}u = w_c \lambda_c$ (Kingman, 1993).

(ii) Because x is constant on cells, the event term of Equation (4.1) groups as $\sum_m \beta^\top x(t_m) = \sum_c y_c \, \beta^\top x_c$, and the compensator integral collapses to $\sum_c w_c e^{\beta^\top x_c}$, giving Equation (4.2). Expanding the GLM log-likelihood on the right of Equation (4.5):

$$\sum_c \Big(y_c \log w_c + y_c \, \beta^\top x_c - w_c e^{\beta^\top x_c} - \log y_c! \Big) \;=\; \ell_{\mathrm{pp}}(\beta) + \sum_c \left(y_c \log w_c - \log y_c! \right),$$

and the bracketed sum is a statistic of the data only.

(iii) Conditionally on $y_c = n$, the n points of a Poisson process with *constant* rate on A_c are distributed as the order statistics of n i.i.d. uniforms on A_c, for every value of the rate (Kingman, 1993); the conditional law is therefore free of β, and sufficiency follows from the factorization $L(\beta; \text{times}) = L(\beta; \text{counts}) \times p(\text{times} \mid \text{counts})$. The final claims follow by differentiating an identity that holds up to an additive constant. □

Corollary 4.11 (The repository's weekly GLM is an NHPP fit)**.** *With equal-width weekly bins and fully exposed sectors ($w_c \equiv w$), the offset* $\log w$ *is absorbed into the sector intercepts, and the weekly sector Poisson GLM —* `fit_sector_count_model` *of* `hawkes_calibration/sector_stability.py` *with* `n_lags=0`*, and its vectorized rehearsal twin* `fit_sector_glm_fast(n_lags=0)` *— computes exactly the NHPP MLE of Assumptions 4.2 and 4.3. Partial exposure (a unit entering or leaving observation mid-week) is handled exactly by the offset, with no change to the estimator's form.*

Two remarks bound the theorem's reach. First, parts (i) and (iii) never used the log link: the reduction of a piecewise-constant point process to independent bin counts holds for *any* positive rate parameterization; the log link is what makes the reduced problem a canonical-link GLM with the concavity of Theorem 4.6. Second, exactness requires Assumption 4.3. If covariates vary within a bin and the fit uses a bin-representative value $\bar{x}_c$, a Jensen gap opens — the true cell mean $\int e^{\beta^\top x(u)} \, \mathrm{d}u$ exceeds $w_c e^{\beta^\top \bar{x}_c}$ whenever x genuinely varies — and the GLM becomes an approximation. At weekly bins with monthly macro series and calendar terms the gap is identically zero; the exactness is a property of our design, not a general fact about binning.

4.5 Right censoring: the complete treatment at this rung

This section closes a standing open question of the repository: what must Poisson calibration do about right censoring? The answer, proved here in both of the bookkeeping styles used by the codebase, is: *nothing beyond keeping the censored exposure.* There is no correction term, no weighting, no imputation; the survival factor that a censored spell must contribute is already computed by the compensator integral, provided open spells are kept and exposure is integrated to the end of observation.

Proposition 4.12 (Intensity view: censoring is already in the likelihood)**.** *Let unit e be observed on $[V_e, C_e]$ with $C_e = \min(T, U_e)$, where T is the administrative window end and U_e a recorded exit time, and let censoring be independent in the sense of Chapter 3 (the intensity of N_e with respect to the filtration enlarged by censoring information is unchanged; Andersen et al., 1993, III.2). Administrative censoring at the deterministic T is independent* *by construction; removal at a* recorded *exit is valid for the cause-specific funding intensity with no independence assumption at all (Chapter 2, defect (C3)(iii); Putter et al., 2007). Then the observed-data log-likelihood is exactly Equation* (4.1) *with* $Y_e(u) = \mathbf{1}\{V_e \le u \le C_e\}$*; in particular the event-free portion* $(t_{\text{last}}, C_e]$ *of an open spell contributes*

$$-\int_{t_{\text{last}}}^{C_e} \lambda_e(u)\,\mathrm{d}u$$

to ℓ *— the log of its survival factor — and nothing else. In the binned form of Theorem 4.10, the entire treatment is the exposure* $w_c = \int_{\text{cell}} \mathbf{1}\{V_e \le u \le C_e\}\,\mathrm{d}u$*: keeping open spells and integrating exposure to* $\min(T, U_e)$ *is the complete and exact treatment of (C1).*

Proof. Under independent censoring the observed counting process $\int \mathbf{1}\{u \le C_e\}\,\mathrm{d}N_e(u)$ has intensity $Y_e(t)\lambda_e(t)$ with respect to the enlarged filtration (Chapter 3; Andersen et al., 1993, III.2), so the Jacod likelihood of the observed data is Equation (2.3) with this intensity — which is Equation (4.1) restricted to $[V_e, C_e]$. Splitting the compensator integral at the observed event times isolates the term over $(t_{\text{last}}, C_e]$, an interval containing no event: its entire contribution is $-\int \lambda_e$, with no $\log \lambda$ term to drop or invent. The binned statement follows from Theorem 4.10, whose exposure w_c integrates precisely this at-risk indicator. The proposition fails exactly when censoring is *informative* — observation ends for reasons correlated with the intensity given covariates. That is not (C1) but (C3), the unlabeled-exit defect: an unrecorded death is a censoring event we never see and therefore cannot put into C_e; Section 4.8 states the resulting bias. □

Proposition 4.13 (Waiting-time view: the same computation)**.** *Under Assumption 4.1, for a spell starting at a (an entry or an observed event) the waiting time to the next event has survival function*

$$S(c) \;=\; \mathbb{P}(N(a, c] = 0) \;=\; \exp\Big(-\int_a^c \lambda(u)\,\mathrm{d}u\Big), \qquad c \ge a, \tag{4.6}$$

and density $f(c) = \lambda(c)\,S(c)$*. The likelihood assembled spell-by-spell — density factors for completed spells, survival factors* $S(C)$ *for spells censored at* C *— equals the Jacod likelihood of Proposition 4.12 exactly:*

$$\prod_{m=1}^{n} \lambda(t_m)\, e^{-\int_{t_{m-1}}^{t_m} \lambda(u)\,\mathrm{d}u} \;\times\; e^{-\int_{t_n}^{C} \lambda(u)\,\mathrm{d}u} \;=\; \Big(\prod_{m=1}^{n} \lambda(t_m)\Big)\, e^{-\int_{t_0}^{C} \lambda(u)\,\mathrm{d}u}, \qquad t_0 = V.$$

Proof. Equation (4.6) is the Poisson count law of Assumption 4.1 evaluated at zero; differentiating $1 - S(c)$ gives $f(c) = \lambda(c)S(c)$. Taking logs of the displayed product, the survival exponents telescope over the partition $t_0 < t_1 < \cdots < t_n < C$ into the single compensator integral $-\int_{t_0}^{C} \lambda$, and the event factors contribute $\sum_m \log \lambda(t_m)$: this is Equation (4.1) for the unit. The two views are therefore not two treatments to be reconciled but one number computed with two bookkeepings. Practical corollary: a spell-table implementation (one row per spell with a censoring flag, survival factor for flagged rows) and an exposure-table implementation (one row per unit-week cell with exposure w_c) must return identical log-likelihoods on the same data — a cheap and mandatory cross-implementation test. In the homogeneous case Equation (4.6) is the exponential waiting time $e^{-\mu(c-a)}$, and the renewal view is classical survival analysis of exponential spells; for recurrent events the counting-process formulation is the one that generalizes (Cook and Lawless, 2007). □

Corollary 4.14 (Interval censoring is free at this rung)**.** *Suppose deal dates are trusted only at resolution* δ *(the week): the bin of each event is correct, its position within the bin is not. Under the assumptions of Theorem 4.10 with bin width* δ*, the binned Poisson likelihood Equation* (4.5) *is the* exact *likelihood of the trusted data — not an approximation — because by part (iii) the within-bin positions carry no information about β: the corrupted coordinates are exactly the ancillary ones. Boundary cells truncated by entry or censoring retain exactness through their partial exposure w_c. None of this survives history dependence: for a Hawkes process the intensity after t depends on the within-bin positions of earlier events, binning destroys likelihood information, and the interval-censored machinery of Chapter 6 (Rizoiu et al., 2022) becomes necessary. The contrast is the precise sense in which weekly aggregation is free at rung 1 and costly afterwards — and it is why (C4) appears as "robust" in this chapter's defect table and as a first-class problem in Chapters 5 and 6.*

Warning 4.15 (Dropping open spells is right truncation, not censoring). The exactness of Propositions 4.12 and 4.13 holds only if censored spells are *kept.* Discarding them — fitting only spells that end in an observed event — conditions on the event landing in-window, which oversamples short gaps and biases every fitted rate upward; the observation scheme becomes right truncation, a different and harder problem (Kalbfleisch and Prentice, 2002). This restates Warning 2.4 at the point of use. For the weekly sector GLM the trap is nearly impossible to fall into (every sector-week cell exists whether or not it contains events — the zero counts *are* the censored exposure), which is itself an argument for the cell representation; for spell-table firm-level code the trap is one careless `dropna` away, and the open-spell retention audit of `hawkes_calibration/data/audits.py` exists to catch it.

4.6 Overdispersion: from Poisson to negative binomial

Assumption 4.1 forces $\operatorname{Var}(y_c) = \mathbb{E}(y_c)$ cell by cell. Funding counts violate this: latent heterogeneity (unmodeled sector-week quality, vendor coverage drift) and genuine history dependence both inflate variance beyond the mean, and the rehearsal measured quasi-Poisson dispersion 2.72 on the misspecified regime-B weekly sector fit (Appendix C) — weekly variance nearly three times the Poisson prediction. Two upgrades present themselves, one inferential and one distributional, and the design needs both for different purposes.

The inferential upgrade is *quasi-Poisson*: keep the mean model and the estimating equation Equation (4.3), posit $\operatorname{Var}(y_c) = \phi\,\mu_c$, estimate $\hat\phi$ by the Pearson statistic $\hat\phi = \frac{1}{n-d}\sum_c (y_c - \hat\mu_c)^2/\hat\mu_c$, and scale standard errors by $\hat\phi^{1/2}$ (or use the sandwich variance, which is robust to variance misspecification beyond the proportional form). Because the score Equation (4.3) is unbiased whenever the *mean* is correct, $\hat\beta$ keeps its consistency under overdispersion; only its advertised precision was wrong. Quasi-Poisson repairs the advertisement at zero modeling cost, but it supplies no predictive distribution — and the evaluation protocols of Chapter 7 score predictive distributions (PIT, proper scoring rules), not point forecasts. Hence the distributional upgrade.

Proposition 4.16 (Gamma frailty makes a negative binomial)**.** *Let $y \mid Z \sim \operatorname{Poisson}(Z\mu)$ with frailty $Z \sim \operatorname{Gamma}(\alpha, \alpha)$ (shape and rate α, so $\mathbb{E}Z = 1$, $\operatorname{Var} Z = 1/\alpha$). Then y is negative binomial:*

$$\mathbb{P}(y = k) \;=\; \frac{\Gamma(k+\alpha)}{\Gamma(\alpha)\,k!}\left(\frac{\alpha}{\alpha+\mu}\right)^{\alpha}\left(\frac{\mu}{\alpha+\mu}\right)^{k}, \qquad k = 0, 1, 2, \ldots, \tag{4.7}$$

with $\mathbb{E}y = \mu$ and $\operatorname{Var} y = \mu\,(1 + \mu/\alpha)$. As $\alpha \to \infty$ the law converges to $\operatorname{Poisson}(\mu)$. Moreover, with $\mu_c = w_c e^{\beta^\top x_c}$ and α fixed, the negative binomial log-likelihood remains concave in β.

Proof. Integrating the Poisson pmf against the Gamma density,

$$\mathbb{P}(y = k) = \int_0^\infty \frac{e^{-z\mu}(z\mu)^k}{k!}\,\frac{\alpha^\alpha z^{\alpha-1}e^{-\alpha z}}{\Gamma(\alpha)}\,\mathrm{d}z = \frac{\mu^k\alpha^\alpha}{k!\,\Gamma(\alpha)}\int_0^\infty z^{k+\alpha-1}e^{-(\mu+\alpha)z}\,\mathrm{d}z = \frac{\mu^k\alpha^\alpha}{k!\,\Gamma(\alpha)}\,\frac{\Gamma(k+\alpha)}{(\mu+\alpha)^{k+\alpha}},$$

which rearranges to Equation (4.7). The moments follow from the tower rule: $\mathbb{E}y = \mathbb{E}[Z\mu] = \mu$ and $\operatorname{Var} y = \mathbb{E}[Z\mu] + \operatorname{Var}(Z\mu) = \mu + \mu^2/\alpha$. For $\alpha \to \infty$, $Z \to 1$ in probability and the mixture collapses to the Poisson. For concavity, write $\eta_c = \log w_c + \beta^\top x_c$; the β-dependent part of the log-likelihood per cell is $y_c\,\eta_c - (y_c + \alpha)\log(\alpha + e^{\eta_c})$. The first term is linear, and $\log(\alpha + e^{\eta}) = \operatorname{logsumexp}(\log\alpha, \eta)$ is convex in η, hence $-(y_c + \alpha)\log(\alpha + e^{\eta_c})$ is concave in the linear function η_c of β; sums of concave functions are concave. □

The mixture derivation is not a convenience trick; it is the model's meaning. Z is a multiplicative *frailty* — the latent common factor of Vaupel et al. (1979), developed for multivariate event data by Hougaard (2000) — attached here to each sector-week cell to absorb heterogeneity the covariates miss. That reading sets the boundaries of the upgrade honestly. Independent frailties per cell fix the *marginal* variance but assert no dynamics; if the missing structure is autocorrelated (a persistent latent factor, or genuine event-to-event excitation), Pearson residuals will still be serially correlated after the NB upgrade, and that residual autocorrelation — not overdispersion itself — is the escalation signal toward Chapter 5. The distinction matters because overdispersion is produced by frailty *and* by contagion alike; a dispersion statistic can demand escalation but can never certify excitation, and the rehearsal's 2.4× branching-ratio inflation under latent AR frailty (Appendix C) is the standing reminder of what happens when clustered variance is read as contagion. The frailty-vs-contagion identification problem is treated properly in Chapter 5.

Design decision (D4.2 — negative binomial is the default count noise).

The default observation model for sector-week counts is the negative binomial Equation (4.7) with mean $w_c e^{\beta^\top x_c}$ and a fitted dispersion α (per sector, unless pooling is justified by the diagnostics), replacing the pure Poisson wherever a predictive count distribution is consumed — held-out NLL, PIT calibration, decision outputs. Rationale: measured overdispersion (dispersion 2.72 on the stress world, Appendix C); an exact frailty interpretation (Proposition 4.16) that absorbs heterogeneity without asserting dynamics; preserved mean structure and β interpretation; preserved concavity in β at fixed α. Quasi-Poisson/sandwich standard errors remain the reporting standard for coefficient inference in plain-Poisson fits. Dispersion diagnostics — $\hat\phi$, fitted $1/\alpha$, and randomized PIT — are computed with every count fit under the protocols of Chapter 7. Reversal triggers: if real-data fits drive $1/\alpha \to 0$ after the covariate baseline (equidispersion), the upgrade is dropped as needless; if NB residuals stay serially correlated, the deficiency is dynamic and the remedy is history terms (Section 4.7, Chapter 5), not a fatter marginal tail.

4.7 History as covariates: the self-modulated GLM, previewed

One step remains inside GLM-land, and the repository's sector layer already stands on it. Add the recent past to the design: lagged own- and cross-sector counts (or exponentially filtered versions of them) enter the linear predictor,

$$\log \Lambda_{s,t} \;=\; a_s + \beta^\top x_t + \sum_{r\in\mathcal{S}} \sum_{\ell=1}^{L} b_{sr\ell}\, y_{r,t-\ell}, \tag{4.8}$$

which is a *log-linear Poisson autoregression* — the count time series family of Fokianos et al. (2009), log-linear variant Fokianos and Tjøstheim (2011). Everything mechanical from this chapter survives: the parameters sit in a linear predictor, so the likelihood is concave

(Theorem 4.6 verbatim), the solver guarantees of Corollary 4.7 hold, and the exp link grants *sign freedom* — negative b coefficients express inhibition, one of the three legal routes to it catalogued in Chapter 5. This is rung 3 of the ladder: *Hawkes without Hawkes*. Fixed lags (or fixed-decay filters) play the role of a fixed kernel, capturing the predictive content of excitation while refusing to estimate the timescale — the one parameter that weekly data cannot identify anyway (the κ–θ ridge; Chapter 7).

Two prices are paid, and both are conceptual rather than computational. First, lagged counts are *internal* covariates (Assumption 4.4 is voided): the model is a legitimate conditional intensity and one-step-ahead prediction is clean, but multi-step forecasts and any scenario analysis must simulate the system jointly — plugging today's history into a formula for next quarter is not a probability of anything (Kalbfleisch and Prentice, 2002). Second, and this the design must state loudly because the repository's own history includes the failure mode:

Warning 4.17 (Stability of log-link feedback is an operating-point condition)*.* For the *additive* excitation model of Chapter 5, with nonnegative lag matrices summing to G, subcriticality is the coefficient condition $\rho(G) < 1$, enforced in `hawkes_calibration/sector_stability.py` through the row-sum bound $\rho_{\max} = 0.95$. That certificate does *not* transfer to Equation (4.8). Under the log link one past event multiplies next week's rate by e^b, so the marginal effect of an event on the conditional mean is $b\,\Lambda$ — coefficient *times operating level.* A feedback matrix that is harmless at 0.3 events per sector-week can be explosive at 3; no inequality on the b's alone, spectral or otherwise, certifies stability. The applicable theory is the contraction/ergodicity analysis of Fokianos and Tjøstheim (2011), whose conditions constrain the feedback of *bounded or slowly varying* transforms of the past; raw counts in the exponent sit outside the Lipschitz structure that the theory requires, and the rehearsal produced a measured metastable excursion in a configuration whose nominal spectral radius was comfortably subcritical (Appendix C). Design rules that follow: history enters exponents only through bounded transforms (EWMA levels, $\log(1+\cdot)$ counts); any simulation or multi-step use of a fitted self-modulated GLM must verify stability *at the operating point*; and the branching reading of excitation is reserved for the additive parameterization. The full treatment — both parameterizations, their stability certificates, and what each can and cannot claim about contagion — is Chapter 5.

4.8 Defect declaration

Per Design decision D2.1 (Chapter 2), the Poisson models of this chapter declare their position against the defect ledger of Table 2.1. Table 4.1 summarizes; the signs are the chapter's own results.

The asymmetry in Table 4.1 between sector and firm rows is the quantitative reason the ground process is calibrated at sector resolution first: at that resolution the Poisson rung is exactly right under (C1), (C2), (C4) and immune to (C3), (C6), so its held-out likelihood is a trustworthy floor. Firm-level Poisson calibration through the event-time path inherits the (C3) deflation and (C5) attenuation with the stated signs, and every later firm-level model must beat it *while carrying the same declared exposures.*

Code hooks.

Existing. `hawkes_calibration/sector_stability.py` (`fit_sector_count_model`): with `n_lags=0` the additive excitation vanishes and the fit is exactly the weekly NHPP GLM of Theorem 4.10 and Corollary 4.11; its ridge `l2_beta` and baseline support-envelope winsorization implement the remedies of Warning 4.9; `sector_rate_at` is the one-step predictor. `synthetic/fast_fit.py::fit_sector_glm_fast(n_lags=0)`

Defect	Status at this rung	Mechanism and sign	Treatment
(C1)	corrected *exactly*	censored exposure contributes its survival factor through the compensator; no bias (Proposition 4.12)	this chapter
(C2)	corrected mechanically	risk sets start at V_i; exposure w_c integrates from entry. Residual selection ("firms that have raised") is an interpretation limit, not a likelihood error	Chapter 11
(C3)	*vulnerable* (firm level)	unrecorded exits inflate exposure w_c with dead time; fitted hazards *deflated* (downward sign), worst for long-quiet firms. Sector level immune: sectors do not die	exposure windows, Chapter 9
(C4)	robust at weekly resolution	binned likelihood is exact; within-bin positions are ancillary (Corollary 4.14); day-level jitter within weeks is immaterial	Chapter 6
(C5)	*vulnerable*	stale firm covariates attenuate their fitted effects toward zero; macro covariates unaffected (published as-of, never stale). Bites the event-time firm path, not the sector GLM	Chapter 8
(C6)	sector immune / firm vulnerable	sector counts are complete (out-of-universe deals still carry sectors); firm-attributable intensity misses competitors and thins counts	Chapter 11

Table 4.1: Defect declaration for the Poisson rung (D2.1 compliance). "Immune"/"robust" claims hold under this chapter's stated assumptions; signed vulnerabilities are accepted and carried to the chapters that own the remedies.

is the vectorized rehearsal twin used by the exp27-style decomposition benchmarks. `hawkes_calibration/eventtime/estimate.py` (`fit_multivariate_with_covariates`) is the day-resolution NHPP path when excitation is disabled, with `hawkes_calibration/eventtime/covariates.py` (`PiecewiseConstantCovariate.integrate_exp_gamma`) computing the compensator $\int e^{\beta^\top x}$ in closed form cell by cell — the continuous-time twin of the GLM offset. *To be written.* `hawkes_calibration/sector_negbin.py`: the D4.2 negative-binomial layer — NB likelihood Equation (4.7) with log-mean $\log w_c + \beta^\top x_c$, per-sector α, quasi-Poisson $\hat{\phi}$ and PIT diagnostics emitted in the Chapter 7 harness format. `hawkes_calibration/data/audits.py` (exposure integrity, per Chapter 2): open-spell retention (Warning 4.15), entry-date consistency of w_c integration limits, spell-table vs. exposure-table log-likelihood parity (Proposition 4.13), and train/test covariate-support overlap reporting (Warning 4.9).

Chapter 5

Exponential Hawkes: Excitation, Inhibition, and Covariates

5.1 Model and assumptions

The Poisson models of Chapter 4 treat deal flow as exogenously modulated: nothing that happens in the market changes the future of the market. Their residual diagnostics say otherwise — clustering beyond what covariates explain is the recurring finding of the rehearsal campaigns (Appendix C) — and the Hawkes process (Hawkes, 1971) is the minimal extension in which events beget events. This chapter develops the exponential-kernel Hawkes layer end to end: the branching representation that organizes everything (Section 5.2), the likelihood and the recursion that makes it computable (Section 5.3), the EM algorithm on the latent genealogy (Section 5.4), simulation (Section 5.5), and then the section this chapter exists for: what to do about *inhibition*, which a nonnegative-kernel Hawkes process provably cannot express, and which our market exhibits — a firm that has just raised is temporarily *less* likely to raise again (Section 5.6). The chapter closes with the frailty-versus-contagion identifiability problem (Section 5.7) and the defect declaration required by D2.1 (Section 5.8).

Assumption 5.1 (Exponential Hawkes ground process)**.** On the filtered space of Definition 2.1, the univariate model for a single stream is

$$\lambda(t) \;=\; \mu(t) \;+\; \sum_{t_m<t} h(t-t_m), \qquad h(u) \;=\; \kappa\theta\, e^{-\theta u}, \quad \kappa \geq 0,\ \theta > 0, \tag{5.1}$$

and the multivariate, sector-indexed model is, for $s \in \mathcal{S}$,

$$\lambda_s(t) \;=\; \mu_s(t) \;+\; \sum_{r\in\mathcal{S}} \sum_{\substack{t_m<t \\ s_m=r}} \kappa_{sr}\, \theta_{sr}\, e^{-\theta_{sr}(t-t_m)}, \qquad \mu_s(t) \;=\; \exp(\beta_s^\top x_s(t)), \tag{5.2}$$

where the covariate process $x_s(t)$ (macro series, sector activity summaries, seasonal terms) is predictable and locally bounded, constructed under the point-in-time protocol Protocol 2.5. The *branching matrix* is

$$G_{sr} \;=\; \int_0^\infty \kappa_{sr}\theta_{sr}e^{-\theta_{sr}u}\,\mathrm{d}u \;=\; \kappa_{sr}, \tag{5.3}$$

and we impose *subcriticality*: $\rho(G) < 1$.

Two virtues of the (κ, θ) parameterization justify committing to it over the equally common $\alpha e^{-\beta t}$ form. First, $\kappa = \int_0^\infty h$ is a *unitless branching mass*: by Section 5.2 it is the expected number of direct offspring per event, invariant to the time unit, while θ is a pure timescale with half-life $\log 2/\theta$. Second, the two *decouple in aggregation analysis*: binning event times at width Δ (defect (C4)) destroys information about θ once $\Delta\theta$ is large, but the mass κ remains estimable from count balance — the κ–θ ridge quantified in Chapter 7 and the standing verdict "κ survives aggregation, θ does not" (Appendix C). A parameterization in which the kernel height $\alpha = \kappa\theta$ is a free parameter entangles mass and timescale in exactly the direction the data cannot resolve; keeping (κ, θ) explicit keeps the ill-conditioned direction visible. (The event-time fitters currently store $\alpha = \kappa\theta$ and $\beta = \theta$ internally; the mapping is recorded in the code hooks.)

Subcriticality is the standing stability constraint. For constant baselines it is exactly the condition for a stationary version to exist (Hawkes, 1971; Brémaud and Massoulié, 1996); for the covariate-driven baseline of Equation (5.2) the process is defined conditionally on the covariate paths and is non-explosive on finite windows regardless (Remark 5.3), but $\rho(G) < 1$ still governs cluster sizes, forecast amplification, and the sanity of any simulation. Because $G \geq 0$, Perron–Frobenius gives $\rho(G) \leq \max_s \sum_r G_{sr}$, so a row-sum bound is a convex sufficient condition — this is precisely the constraint the repository's discrete-face fitter enforces with $\rho_{\max} = 0.95$ (`hawkes_calibration/sector_stability.py`), and the continuous-time layer inherits the same certificate. Multivariate Hawkes models with spectral-radius stability conditions are standard in financial applications (Embrechts et al., 2011; Bacry et al., 2015).

5.2 The cluster representation

Everything tractable about the linear Hawkes process follows from one theorem: it is secretly a Poisson cluster process (Hawkes and Oakes, 1974).

Theorem 5.2 (Hawkes–Oakes cluster representation)**.** *Let $\mu_s(t) \geq 0$ be locally integrable baselines and $h_{sr} \geq 0$ kernels with $G_{sr} = \int h_{sr} < \infty$. Construct a marked point process as follows:* immigrants *of type s arrive as an inhomogeneous Poisson process with rate $\mu_s(t)$, independently across types; recursively, each event of type r at time t_m (immigrant or offspring) generates* direct offspring *of type s on (t_m, ∞) as an inhomogeneous Poisson process with rate $h_{sr}(t - t_m)$, independently of everything else; the process is the superposition of all generations. Then the superposition is a Hawkes process with intensity Equation (5.2). Conversely, every Hawkes process with these parameters is equal in law to this cluster process.*

Proof of the construction direction. Let $\{\mathcal{G}_t\}$ be the filtration generated by the labeled construction (all events with their genealogy). Fix t and consider arrivals of type s in $(t, t + \mathrm{d}t]$. They are of two kinds. New immigrants form a Poisson process with deterministic rate μ_s, independent of $\mathcal{G}_t$, hence contribute $\mathcal{G}_t$-intensity $\mu_s(t)$. For each event $t_m < t$ of type r already born, its type-s offspring process is inhomogeneous Poisson with rate $h_{sr}(\cdot - t_m)$; by independent increments of the Poisson process, its future beyond t is conditionally independent of $\mathcal{G}_t$ with intensity $h_{sr}(t - t_m)$. Events born after t contribute nothing, since the kernels are supported on positive ages. The streams are independent by construction, and the intensity of a superposition of conditionally independent streams is the sum of the intensities (additivity of compensators), so the type-s superposed counting process has $\mathcal{G}_t$-intensity $\mu_s(t) + \sum_r \sum_{t_m<t,\, s_m=r} h_{sr}(t - t_m)$. Finally, this expression is a functional of the unlabeled event times alone, hence measurable with respect to the internal filtration $\mathcal{F}_t \subseteq \mathcal{G}_t$; by the innovation (projection) theorem the $\mathcal{F}_t$-intensity is the optional projection of the $\mathcal{G}_t$-intensity, which here equals itself (Daley and Vere-Jones, 2003, Ch. 7). So the unlabeled superposition has the Hawkes intensity Equation (5.2). The converse (existence and uniqueness in law of the cluster decomposition for a given Hawkes law) is Hawkes and Oakes (1974); we use only the construction direction, which is what simulation and EM consume. □

Remark 5.3 (Non-explosion does not need subcriticality)*.* On a finite window $[0, T]$ the mean intensity $\xi_s(t) = \mathbb{E}[\lambda_s(t)]$ satisfies the Volterra equation $\xi = \mu + h * \xi$ componentwise; since the exponential kernels are bounded by $\kappa_{sr}\theta_{sr}$, Grönwall's inequality gives $\xi_s(t) \le \bar{\mu}\, e^{ct} < \infty$ for constants depending only on the parameters. The process is therefore well defined and a.s. finite on any finite window for *any* κ; what $\rho(G) < 1$ buys is stationarity, finite cluster sizes in expectation, and bounded long-run amplification — the subject of the corollaries below. This distinction matters later: for the *nonlinear* models of Section 5.6 even finite-window good behavior is no longer automatic.

Corollary 5.4 (Subcriticality and cluster size)**.** *In the cluster construction, the offspring cascade of a single type-r event is a multitype Galton–Watson process with mean-offspring matrix G. The expected total cluster size (all generations, all types, including the ancestor) is finite iff $\rho(G) < 1$, in which case the expected number of type-s events in the cluster of a type-r ancestor is $\left[(I - G)^{-1}\right]_{sr}$. In the univariate case the expected total cluster size is $1/(1 - \kappa)$.*

Proof. The number of direct type-s offspring of a type-r event is Poisson with mean $\int h_{sr} = G_{sr}$, independently across events, so generation sizes form a multitype branching process: the expected type-composition vector of generation n descending from one type-r ancestor is $G^n e_r$ (induction on n, using linearity of expectation and independence of offspring processes). The expected cluster composition is $\sum_{n\ge0} G^n e_r$. For a nonnegative matrix the Neumann series $\sum_n G^n$ converges iff $\rho(G) < 1$ (Gelfand's formula), with limit $(I - G)^{-1}$; if $\rho(G) \ge 1$ the series diverges because $\mathbf{1}^\top G^n e_r \not\to 0$ along the Perron direction. Univariate: $\sum_n \kappa^n = 1/(1 - \kappa)$. □

Corollary 5.5 (Stationary rate and endogenous share)**.** *Let the baselines be constant, $\mu_s(t) \equiv \bar{\mu}_s$, and $\rho(G) < 1$. The stationary mean rate vector is*

$$\bar{\lambda} \;=\; (I - G)^{-1}\bar{\mu}, \qquad \textit{univariate: } \bar{\lambda} = \frac{\bar{\mu}}{1-\kappa}, \tag{5.4}$$

and the stationary fraction of events that are offspring rather than immigrants (the endogenous share*) is $1 - \mathbf{1}^\top\bar{\mu}/\mathbf{1}^\top\bar{\lambda}$; in the univariate case it equals κ.*

Proof. Taking expectations in Equation (5.2) under stationarity, $\bar{\lambda}_s = \bar{\mu}_s + \sum_r G_{sr}\bar{\lambda}_r$ because $\mathbb{E}\sum_{t_m<t, s_m=r} h_{sr}(t-t_m) = \int_0^\infty h_{sr}(u)\,\bar{\lambda}_r\,\mathrm{d}u$ by Campbell's formula; solving gives Equation (5.4). Immigrants arrive at rate $\bar{\mu}_s$ by construction, so the immigrant share of events is $\mathbf{1}^\top\bar{\mu}/\mathbf{1}^\top\bar{\lambda}$ and the endogenous share is its complement; univariate, $1 - \bar{\mu}/\bar{\lambda} = 1 - (1 - \kappa) = \kappa$. Equivalently: each immigrant's cluster has expected size $1/(1 - \kappa)$ (Corollary 5.4), of which exactly one event is the immigrant. □

Remark 5.6 (A common accounting error). It is tempting to reason "rate $\bar{\mu}$ of immigrants, each with κ children, hence total rate $\bar{\mu}(1 + \kappa)$ and endogenous share $\kappa/(1 + \kappa)$." This counts only the first generation: children beget children, the geometric series compounds to $1/(1 - \kappa)$, and the correct endogenous share is κ, not $\kappa/(1 + \kappa)$. At $\kappa = 0.5$ the difference is 50% versus 33% of deal flow attributed to contagion — a headline-sized error. All endogeneity reporting in this repository uses Corollary 5.5, and (better) the posterior parentage diagnostic of Section 5.4, subject to the upper-bound discipline of Section 5.7.

5.3 Likelihood and the Ogata recursion

The Jacod likelihood Equation (2.3) specializes to Equation (5.1) as

$$\ell(\beta,\kappa,\theta) \;=\; \sum_{m=1}^{n} \log\Big(\mu(t_m) + \kappa\theta\, A(m)\Big) \;-\; \int_0^T \mu(u)\,\mathrm{d}u \;-\; \kappa\sum_{m=1}^{n}\Big(1 - e^{-\theta(T-t_m)}\Big), \tag{5.5}$$

where

$$A(m) \;\coloneqq\; \sum_{l<m} e^{-\theta\,(t_m - t_l)} \tag{5.6}$$

is the decayed event count seen by event m. The compensator term follows by exact integration: $\int_{t_m}^{T} \kappa\theta e^{-\theta(u-t_m)}\,du = \kappa(1-e^{-\theta(T-t_m)})$, summed over events; the baseline integral is available in closed form for the piecewise-constant covariates used in the repository (`hawkes_calibration/eventtime/covariates.py`). Computed naively, the $A(m)$ cost $O(n^2)$; the exponential kernel — alone among common kernels — makes them recursive (Ozaki, 1979).

Proposition 5.7 (Ogata recursion). *With $A(1) = 0$, the quantities Equation (5.6) satisfy*

$$A(m) \;=\; e^{-\theta\,(t_m - t_{m-1})}\,(1 + A(m-1)), \qquad m \geq 2, \tag{5.7}$$

so Equation (5.5) is computable in $O(n)$ after sorting.

Proof. By induction on m. Base case $m = 2$: $A(2) = e^{-\theta(t_2-t_1)} = e^{-\theta(t_2-t_1)}(1 + A(1))$ since $A(1) = 0$. Inductive step: assume $A(m-1)$ equals Equation (5.6) at index $m-1$. Splitting off the $l = m-1$ term and factoring the common decay over $[t_{m-1}, t_m]$,

$$A(m) = \sum_{l\leq m-1} e^{-\theta(t_m-t_l)} = e^{-\theta(t_m-t_{m-1})}\Big(e^0 + \sum_{l<m-1} e^{-\theta(t_{m-1}-t_l)}\Big) = e^{-\theta(t_m-t_{m-1})}(1 + A(m-1)),$$

which is Equation (5.7). The recursion works because the exponential kernel satisfies $h(u+v) = h(u)e^{-\theta v}$: the whole past compresses into one number that decays deterministically between events. This is the Markov-embedding property; it fails for power-law kernels, which admit no finite recursion, and it is why this document fixes the exponential family as the parametric default. □

In the multivariate model the same recursion runs per source–target pair: $A_{sr}(m) = e^{-\theta_{sr}(t_m-t_{m-1})}(\mathbf{1}\{s_{m-1} = r\} + A_{sr}(m-1))$ on the merged event stream, giving $O(n\,|\mathcal{S}|^2)$ per likelihood pass; `hawkes_calibration/eventtime/likelihood.py` implements exactly this, in a block-cumsum form that is numerically stable over long horizons, together with analytic gradients. Asymptotic normality of the MLE in the stationary ergodic case is Ogata (1978).

Positivity transforms. The fitters optimize unconstrained by reparameterizing $\kappa = e^a$ (in code, $\alpha = e^a$ with $\alpha = \kappa\theta$), so gradient methods never propose negative branching and standard errors return via the delta method (`hawkes_calibration/eventtime/estimate.py`). Two consequences deserve note. First, the transform is a *modeling* statement, not a numerical trick: it hard-codes the nonnegative-kernel assumption whose consequences Section 5.6 confronts — a very negative a is near-zero excitation, never inhibition. Second, the boundary $\kappa = 0$ is pushed to $a \to -\infty$, so "no excitation" is a flat direction of the transformed likelihood, and testing $H_0\colon \kappa = 0$ is a boundary problem whose LRT null is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Self and Liang, 1987); the testing protocol lives in Chapter 7. When the number of sectors makes the $|\mathcal{S}|^2$ branching entries data-hungry, the ℓ_1-penalized fitter with exact zeros (`hawkes_calibration/eventtime/lasso.py`) implements the penalized multivariate excitation recovery of Hansen et al. (2015).

Warning 5.8 (The likelihood is not concave in (κ, θ)). For *fixed* θ and a linearly parameterized baseline, Equation (5.5) is concave: each event term is the log of an affine function of the parameters and the compensator is linear. Jointly in (κ, θ) it is not — the event terms involve $\kappa\theta e^{-\theta\Delta}$, and the likelihood surface develops the κ–θ ridge analyzed in Chapter 7: data observed at resolution coarser than the kernel half-life constrain roughly the product of mass and effective within-window decay, not each factor. With the exp-link baseline of Equation (5.2) even the fixed-θ problem loses global concavity (the event terms $\log(e^{\beta^\top x} + c)$ are convex in β), retaining concavity only in the excitation block given the baseline. Practice therefore fixes a grid of decays, maximizes at each grid point with multistart, and reports the *profile* likelihood over θ — flagging ridge flatness rather than hiding it. The EM algorithm of the next section is the complementary tool: it restores block-concavity through augmentation and yields the diagnostics no direct MLE provides.

5.4 EM on the latent genealogy

Theorem 5.2 says every event has a latent *parent*: either the baseline (immigrant) or an earlier event. Making the genealogy explicit turns Hawkes estimation into a missing-data problem with an exact E-step and closed-form M-steps — the cascades-of-Poisson-processes formulation of Simma and Jordan (2010), identical in structure to the ETAS EM of Veen and Schoenberg (2008). We derive it in the univariate model; the multivariate bookkeeping is indicated at the end.

Let $u_m \in \{0, 1, \dots, m-1\}$ denote the parent of event m (0 = immigrant, $l \geq 1$ = event t_l).

Lemma 5.9 (Complete-data likelihood)**.** *Given the genealogy $u = (u_1, \dots, u_n)$, the complete-data log-likelihood on $[0, T]$ is*

$$\ell_c(\beta, \kappa, \theta; u) = \sum_{m:u_m=0} \log \mu(t_m) - \int_0^T \mu + \sum_{m:u_m=l\geq 1} \big[\log \kappa + \log \theta - \theta(t_m - t_l)\big] - \kappa \sum_{l=1}^{n} (1 - e^{-\theta(T-t_l)}). \tag{5.8}$$

Proof. Under the cluster construction the labeled data decompose into independent inhomogeneous Poisson processes: the immigrant process with rate μ on $[0, T]$, contributing $\sum_{u_m=0} \log \mu(t_m) - \int_0^T \mu$; and, for each event l, its offspring process with rate $h(\cdot - t_l)$ observed on $(t_l, T]$, contributing $\sum_{m:u_m=l} \log h(t_m - t_l) - \int_0^{T-t_l} h(v)\, dv$. Summing, and evaluating $\int_0^{T-t_l} \kappa\theta e^{-\theta v}\, dv = \kappa(1 - e^{-\theta(T-t_l)})$, gives Equation (5.8). Note the compensator does not depend on u: parents determine only which factor each event's log-intensity lands in. □

The observed-data likelihood Equation (5.5) is recovered by summing e^{ℓ_c} over genealogies — the log-of-sum that destroys concavity. Augmentation splits it back apart.

Proposition 5.10 (E-step: exact posterior parentage)**.** *Under parameters (β, κ, θ), the posterior over genealogies given the event times factorizes across events, $\mathbb{P}(u \mid t_{1:n}) = \prod_m p_{m,u_m}$, with*

$$p_{m0} = \frac{\mu(t_m)}{\mu(t_m) + \kappa\theta A(m)}, \qquad p_{ml} = \frac{\kappa\theta\, e^{-\theta(t_m - t_l)}}{\mu(t_m) + \kappa\theta A(m)}, \quad 1 \leq l < m, \tag{5.9}$$

with $A(m)$ as in Equation (5.6) — so the E-step reuses the Ogata recursion and costs $O(n)$ beyond the pairwise weights.

Proof. By Lemma 5.9, $e^{\ell_c(u)} = C \prod_m w_{m,u_m}$ where $w_{m0} = \mu(t_m)$, $w_{ml} = \kappa\theta e^{-\theta(t_m - t_l)}$, and C collects the compensator factors, which do not involve u. Hence $\mathbb{P}(u \mid t_{1:n}) \propto \prod_m w_{m,u_m}$, which factorizes with per-event normalizers $\mu(t_m) + \kappa\theta A(m)$. Probabilistically this is the thinning identity: given $\mathcal{F}_{t^-}$, the event at t_m belongs to one of the superposed streams with probability proportional to that stream's intensity. □

Proposition 5.11 (M-step)**.** *Let $P_0 = \sum_m p_{m0}$ and $P_1 = n - P_0 = \sum_m \sum_{l \geq 1} p_{ml}$ be the expected immigrant and offspring counts, and $D = \sum_{m, l\geq 1} p_{ml}(t_m - t_l)$ the expected total parent–child gap. Maximizing $Q(\cdot) = \mathbb{E}_{u\sim p}[\ell_c]$ gives, in the interior approximation that replaces the truncated exposures $1 - e^{-\theta(T-t_l)}$ by 1:*

$$\hat{\mu} = \frac{P_0}{T} \quad (\textit{constant baseline}), \qquad \hat{\kappa} = \frac{P_1}{n}, \qquad \hat{\theta} = \frac{P_1}{D}, \tag{5.10}$$

i.e. branching = expected children per event, decay = reciprocal of the p-weighted mean parent–child gap. With exact exposures the κ-update becomes $\hat{\kappa}(\theta) = P_1 / \sum_l (1 - e^{-\theta(T-t_l)})$ and $\hat{\theta}$ solves the scalar equation $P_1/\theta - D = \hat{\kappa}(\theta) \sum_l (T - t_l) e^{-\theta(T-t_l)}$.

Proof. Q inherits the separable form of Equation (5.8) with indicators replaced by posteriors. The κ-part is $P_1 \log\kappa - \kappa\sum_l(1-e^{-\theta(T-t_l)})$, concave with the stated root; the θ-part is $P_1\log\theta - \theta D - \kappa\sum_l(1-e^{-\theta(T-t_l)})$, whose stationarity condition is the displayed scalar equation; setting the exposure terms to 1 (events far from the window edge) decouples them into Equation (5.10). For the exp-link baseline, the β-part of Q is $\sum_m p_{m0}\beta^\top x(t_m) - \int_0^T e^{\beta^\top x}$ — a weighted Poisson GLM, globally concave. The augmentation thus restores concavity to the very block that broke it in Warning 5.8. □

Theorem 5.12 (Monotone ascent)**.** *Let $\vartheta^{(k+1)}$ maximize $Q(\cdot \mid \vartheta^{(k)})$. Then $\ell(\vartheta^{(k+1)}) \geq \ell(\vartheta^{(k)})$, with equality only if $\vartheta^{(k)}$ already maximizes $Q(\cdot \mid \vartheta^{(k)})$.*

Proof. Write $\pi_\vartheta(u) = \mathbb{P}(u \mid t_{1:n};\vartheta)$ for the posterior of Proposition 5.10. For any ϑ', $\ell(\vartheta') = \mathbb{E}_{\pi_\vartheta}[\ell_c(\vartheta';u)] - \mathbb{E}_{\pi_\vartheta}[\log\pi_{\vartheta'}(u)]$, since $\ell_c(\vartheta';u) = \ell(\vartheta') + \log\pi_{\vartheta'}(u)$ holds for every u (Bayes) and the expectation of a constant is itself. Subtracting the same identity at ϑ,

$$\ell(\vartheta') - \ell(\vartheta) = \left[Q(\vartheta' \mid \vartheta) - Q(\vartheta \mid \vartheta)\right] + \mathrm{KL}(\pi_\vartheta \,\|\, \pi_{\vartheta'}) \;\geq\; Q(\vartheta' \mid \vartheta) - Q(\vartheta \mid \vartheta) \;\geq\; 0$$

for $\vartheta' = \vartheta^{(k+1)}$, using nonnegativity of Kullback–Leibler divergence (Jensen) and the definition of the M-step. Equality forces both terms to vanish. □

Three practical points, each load-bearing. *Boundary corrections.* The exact-exposure updates of Proposition 5.11 are not optional refinements: near the right window edge an event's offspring are partially unobserved, and the interior approximation deflates κ while the orphaned early-window events (parents before time 0) inflate μ. The right edge is corrected *exactly* inside the M-step — the offspring process of an observed parent really is Poisson observed on $(t_l, T]$ — while the left edge is only mitigated, by discarding a burn-in of several kernel half-lives from the immigrant accounting (Veen and Schoenberg, 2008). *The E-step is the endogeneity diagnostic.* The honest answer to "how much of deal flow is contagion" is not the plug-in $\hat\kappa$ of Corollary 5.5 but the posterior expected offspring share $\frac{1}{n}\sum_m(1-p_{m0})$, together with the distribution of attributed parent–child gaps: genuine contagion concentrates attributions at the kernel timescale, while a latent common factor inflates $\hat\kappa$ yet produces attributions with no timescale structure (Section 5.7). *Constraints are not automatic.* EM keeps every $\kappa \geq 0$ by construction, but nothing in the parent accounting bounds $\rho\left(\hat G\right)$ below one — a rare source sector attributed many children can push a column past criticality — so the subcriticality certificate must be checked and enforced after each fit, exactly as the discrete-face fitter does. In the multivariate model the updates read $\hat G_{sr} = \sum_{m:s_m=s}\sum_{l:s_l=r} p_{ml} / \sum_{l:s_l=r}\left(1-e^{-\theta(T-t_l)}\right)$, expected type-$s$ children per unit of type-r exposure. If profile likelihoods over the θ-grid show the exponential shape itself misfits (time-rescaling diagnostics of Chapter 7), the escalation is the histogram-kernel nonparametric EM of Lewis and Mohler (2011) — same E-step, kernel M-step replaced by a binned estimator.

Protocol 5.13 (Estimation ladder for the continuous-time sector layer)**.** (a) Fix a grid of decays θ spanning well-separated half-lives from weeks to quarters; do not free θ jointly with κ except on the final reported grid point. (b) At each grid point, fit by EM (Propositions 5.10 and 5.11) with right-edge exposure corrections and a burn-in of at least four half-lives, multistarted; cross-check with the direct MLE of Section 5.3. (c) Report the profile likelihood over θ with its flatness (the ridge diagnostic of Chapter 7), never a single $(\hat\kappa, \hat\theta)$ point alone. (d) Report $\rho\left(\hat G\right)$, the posterior endogenous share, and the parent–child gap histogram; enforce $\rho\left(\hat G\right) \leq \rho_{\max} = 0.95$ as in the discrete face. (e) All branching estimates carry the upper-bound language of Protocol 5.20.

5.5 Simulation

Simulation is not an afterthought: multi-step forecasts require it (history covariates are *internal*, so scenario propagation must draw events and update features jointly (Kalbfleisch and Prentice, 2002)), and the goodness-of-fit machinery of Chapter 7 consumes simulated envelopes. Three routes, all exact in law.

Lemma 5.14 (Thinning bound for exponential kernels)**.** *On any interval containing no events and no baseline increase, $t \mapsto \lambda_s(t)$ of Equation (5.2) is nonincreasing. Hence $\lambda^* = \sum_s \lambda_s(\tau^+)$ evaluated just after the most recent event or baseline breakpoint τ dominates the total intensity until the next event or breakpoint.*

Proof. The excitation part is a nonnegative combination of $e^{-\theta_{sr}(t-t_m)}$, each strictly decreasing in t; the baseline is constant on the interval by hypothesis. A sum of nonincreasing functions is nonincreasing. □

Ogata thinning (Lewis and Shedler, 1979; Ogata, 1981) then specializes cleanly: from the current time propose a candidate gap $\sim \operatorname{Exp}(\lambda^*)$, accept with probability $\sum_s \lambda_s(t_{\text{cand}})/\lambda^*$, attribute the accepted event to sector s with probability proportional to λ_s, refresh the bound at every event *and* at every covariate breakpoint (where the piecewise-constant baseline may jump up). The Markov embedding keeps the per-event cost at $O(|\mathcal{S}|^2)$ state updates; `hawkes_calibration/eventtime/simulate.py` implements exactly this, including the breakpoint refresh. *Exact inversion* (Dassios and Zhao, 2013) removes rejection entirely for the exponential kernel: $(t, \lambda(t))$ is a piecewise-deterministic Markov process, and the next interarrival time is the minimum of a baseline-driven $\operatorname{Exp}(\bar{\mu})$ draw and an excitation-driven time whose survival function $\exp\!\left(-\frac{Y}{\theta}(1-e^{-\theta u})\right)$, Y the current excess intensity, inverts in closed form (defective with probability $e^{-Y/\theta}$: the cluster may die). This is the preferred engine for the long forecast horizons where thinning's bound refreshes dominate cost. *Cluster simulation* (Theorem 5.2: draw immigrants, then recursive Poisson offspring) is the third route; it is the one that exposes the genealogy, which makes it the reference implementation for validating EM attributions and edge corrections on synthetic data. All three must agree in law; the test suite compares them distributionally.

5.6 Inhibition and excitation via engineered covariates

The market has a signed dynamic: sector-level deal flow is *self-exciting* (capital attention, momentum), while a firm's own recent raise is *inhibitory* (a funded firm leaves the market for months). Regime B of the rehearsal world builds in exactly this structure, with a 30-day inhibition half-life against a 7-day contagion half-life (Section 2.5). The linear Hawkes family of Assumption 5.1 cannot represent the inhibitory half, and the proof is one paragraph.

Proposition 5.15 (Monotonicity: nonnegative kernels cannot inhibit)**.** *Let $\lambda(t;\mathcal{H}) = \mu(t) + \sum_{t_m \in \mathcal{H}, t_m < t} h(t-t_m)$ with $h \geq 0$. Then for every history $\mathcal{H}$, every additional event time $t_0 < t$, and every t: $\lambda(t;\mathcal{H} \cup \{t_0\}) \geq \lambda(t;\mathcal{H})$, and $\lambda(t;\mathcal{H}) \geq \mu(t)$. Consequently no parameter choice makes the intensity after an added event* lower *than before it, nor ever lower than the baseline: a post-deal decrease in a firm's own intensity is inexpressible in any linear Hawkes model with nonnegative kernels, univariate or multivariate, for any (κ, θ).*

Proof. The intensity is additive in history: $\lambda(t;\mathcal{H} \cup \{t_0\}) = \lambda(t;\mathcal{H}) + h(t-t_0) \geq \lambda(t;\mathcal{H})$ since $h \geq 0$; dropping the whole excitation sum gives $\lambda \geq \mu$. The infimum over parameters of the added term is 0 (attained as $\kappa \to 0$), so the family's expressive range for the effect of an own event is $[0, \infty)$: "no effect" is its best imitation of inhibition. The positivity transform $\kappa = e^a$

of Section 5.3 makes the restriction syntactically invisible — the optimizer happily drives a very negative — which is why this proposition must be stated rather than discovered in production. □

Three legal routes remain. Each is developed with its model, validity argument, estimation story, and stability discussion; the decision box at the end commits.

Route (a): log-link self-modulated intensity ("Hawkes without Hawkes")

Move history inside the link:

$$\lambda(t) \;=\; \exp\bigl(\beta^\top x(t) + \gamma^\top r(t)\bigr), \tag{5.11}$$

where $r(t)$ is a vector of *engineered history covariates*: recency indicators and ages (a cooldown flag $\mathbf{1}\{t - \tau_i(t) \le W\}$ with $\tau_i(t)$ the last own-deal time, the gap transform $\log(1 + t - \tau_i(t))$), and decayed event counts

$$H_\theta(t) \;=\; \sum_{t_m<t} e^{-\theta(t-t_m)}, \qquad \mathrm{d}H_\theta = -\theta H_\theta\,\mathrm{d}t + \mathrm{d}N(t), \tag{5.12}$$

at a small grid of *fixed* (or profiled) decays, per source sector in the multivariate case. Negative components of γ are inhibition, positive are excitation: the sign restriction of Proposition 5.15 evaporates because positivity is carried by the exponential, not by the kernel. This is the point-process GLM of Truccolo et al. (2005), and the discrete-time weekly face of it is the repository's current sector model.

Proposition 5.16 (Validity and concavity of the log-link model)**.** *Let each component of r be a left-continuous functional of $\mathcal{H}_{t^-}$ as above. Then (i) λ in Equation (5.11) is predictable, strictly positive, and a.s. finite on the observation window, so the likelihood Equation (2.3) is well defined with no positivity constraints on (β, γ); (ii) $(H_\theta, t - \tau_i(t))$ is a finite-dimensional Markov state, so Equation (5.11) depends on history only through a computable summary; (iii) for fixed decays and windows, the log-likelihood is globally concave in (β, γ).*

Proof. (i) Each feature is adapted and left-continuous (built from events strictly before t), hence predictable; exp makes λ positive; $H_\theta(t) \le N(t^-) < \infty$ a.s. keeps it finite on the window. (ii) is the ODE in Equation (5.12) plus the trivial age dynamics. (iii) The likelihood is $\sum_m (\beta^\top x + \gamma^\top r)(t_m) - \int_0^T \exp(\beta^\top x + \gamma^\top r)(u)\,\mathrm{d}u$: the event sum is linear in the parameters, and the compensator is an integral of exp(affine), convex in the parameters for every u, so its negative is concave; a sum of concave functions is concave. Estimation is a GLM: the entire fragility of Hawkes fitting lived in the decay, and fixing it returns the problem to convex territory. Note what (i) does *not* claim: global-in-time stability of the process. That is Warning 5.18. □

The log-link model is not a retreat from Hawkes; to first order it *contains* it.

Proposition 5.17 (First-order nesting of linear Hawkes)**.** *Fix θ and constant baseline μ, and write $x(t) = \kappa\theta H_\theta(t)/\mu \ge 0$ for the excitation load. The linear Hawkes intensity $\lambda_{\text{lin}} = \mu(1+x)$ and the log-link intensity $\lambda_{\log} = \mu\, e^x$ (i.e. Equation (5.11) with feature H_θ and coefficient $\gamma = \kappa\theta/\mu$) satisfy*

$$0 \;\le\; \log\lambda_{\log}(t) - \log\lambda_{\text{lin}}(t) \;=\; x(t) - \log\bigl(1 + x(t)\bigr) \;\le\; \tfrac{1}{2}\,x(t)^2. \tag{5.13}$$

Thus a decayed-count feature with $\gamma > 0$ reproduces linear-Hawkes excitation to first order in the load, with a second-order overshoot; and $\gamma < 0$ delivers the mirrored soft inhibition that Proposition 5.15 forbids the linear family. The GLM route nests soft versions of both.

Proof. $\log \lambda_{\text{lin}} = \log \mu + \log(1+x)$ and $\log \lambda_{\text{log}} = \log \mu + x$; the elementary bounds $x - \frac{x^2}{2} \leq \log(1+x) \leq x$ for $x \geq 0$ give Equation (5.13). In the fitted regime of this repository the load is small — covariates dominate and excitation contributes decimal points of held-out likelihood (Appendix C) — so the surrogate error is second-order in an already-small quantity. The overshoot has a sign, however: the log-link *amplifies* large excursions that the linear model would meet additively, and that is precisely the seed of the instability below. □

Warning 5.18 ($\rho(G) < 1$ does not stabilize a log-link count model (Mechanism B)). For the linear model, subcriticality is a level-free certificate: it depends only on kernel masses. For log-link feedback it has no such force. In the weekly discrete face $\log m_{s,t} = a_s + \sum_r B_{sr} y_{r,t-1}$ (counts y in the exponent), the deterministic skeleton $m \mapsto \exp(a + Bm)$ has Jacobian $\operatorname{diag}(m^*)B$ at a fixed point m^*: local stability requires $\rho(\operatorname{diag}(m^*)B) < 1$, an *operating-point* condition that scales with the level m^*. A stochastic excursion raises the operating point, the local certificate can silently fail, and the process runs away — explosion is an *absorbing rare event*, invisible to in-sample diagnostics. The rehearsal measured exactly this: a metastable excursion of ∼169,000 events in a configuration whose linear-style spectral radius was nominally far subcritical (Appendix C). The theory agrees from both sides: geometric ergodicity for log-linear Poisson autoregressions holds only under contraction conditions on the feedback coefficients (Fokianos and Tjøstheim, 2011), and the continuous-time stability theory for nonlinear Hawkes requires a Lipschitz link (Brémaud and Massoulié, 1996) — exp is not Lipschitz. Design rules that follow: history enters the exponent only through *bounded or saturating* transforms (EWMA levels, $\log(1+\cdot)$ counts, clipped features), never raw counts; every simulation carries event caps; and the operating-point diagnostic $\rho\Big(\operatorname{diag}(\hat{m}_t)\hat{B}\Big)$ is monitored along simulated paths, not certified once at fit time.

Route (b): rectified additive intensity

Keep the kernel signed and restore positivity by rectification:

$$\lambda(t) \;=\; \Big[\mu(t) + \textstyle\sum_{t_m<t} h^+(t-t_m) - \sum_{t_m<t} h^-(t-t_m)\Big]_+, \qquad h^\pm \geq 0. \tag{5.14}$$

Validity is immediate ($[\cdot]_+$ of a predictable process is predictable and nonnegative), and the mathematics is genuinely available: viewing $\lambda = \phi(\mu + \int h_{\text{signed}}\,\mathrm{d}N)$ with $\phi(x) = x_+$ Lipschitz(1), existence, uniqueness, and stability of a stationary version hold when the total variation of the kernels is subcritical (Brémaud and Massoulié, 1996), and the renewal structure of signed-exponential-kernel processes is worked out in Costa et al. (2020). Maximum likelihood is feasible for the exponential kernel because the times at which the rectifier binds (intensity hits zero, then restarts) are computable in closed form, giving an exact compensator (Bonnet et al., 2021). The costs are structural, not numerical. The likelihood is nonsmooth in the parameters (kinks wherever a zero-crossing time moves), estimation loses the GLM concavity of route (a), and — decisively for us — *expectation and rectification do not commute*: $\mathbb{E}[X]_+ \neq [\mathbb{E}X]_+$ whenever the rectifier binds with positive probability. The mean-intensity Volterra equation $\xi = \mu + h * \xi$, whose closed-form exponential-kernel solution is the engine of the interval-censored MBPP machinery of Chapter 6, therefore has no analogue under Equation (5.14). Since defect (C4) forces much of our fitting through interval-censored likelihoods, route (b) would sever the chapter that keeps timing estimation honest. It stays on the shelf as the escalation for a genuinely signed continuous-time ground kernel — nothing less.

Route (c): move inhibition to the mark factor

The factorization of Theorem 3.7 offers the cleanest exit: keep the *ground* process linear Hawkes over sectors — positive momentum only, which is the economically natural sign at the sector level — and let firm-level inhibition live in the mark factor, where the ranker of Chapter 10 already consumes a cooldown covariate with a negative coefficient. The firm-level intensity is

$$\lambda_i(t) \;=\; \lambda_{s(i)}(t)\, p(i \mid s(i), t, \mathcal{H}_{t^-}), \qquad p(i \mid s,t) = \frac{e^{q_i(t)}}{\sum_{j\in\mathcal{R}_{s,t}} e^{q_j(t)}}, \quad q_i(t) = w^\top z_i(t) + \eta\, c_i(t), \tag{5.15}$$

with $c_i(t) = \mathbf{1}\{t - \tau_i(t) \le W\}$ the cooldown flag and $\eta \le 0$.

Proposition 5.19 (Cooldown in the mark factor is firm-level inhibition)**.** *Under Equation (5.15) with $\eta < 0$: (i) λ_i is predictable and positive, and the joint likelihood factorizes exactly as ground $\times$ mark, so both factors are fitted by their own valid likelihoods; (ii) holding other firms' features fixed, the switch $c_i : 0 \to 1$ multiplies $p(i \mid s,t)$ by a factor in $(e^\eta, 1)$ — a strict decrease — hence λ_i strictly decreases while λ_s is untouched; (iii) the ground layer retains nonnegative kernels, so the cluster representation, the $\rho(G)$ certificate, and the MBPP aggregation theory of Chapter 6 all survive verbatim.*

Proof. (i) is Theorem 3.7 plus predictability of c_i (built from events strictly before t). For (ii), write $p_i(\eta) = e^{q+\eta}/(D + e^{q+\eta})$ with $q = w^\top z_i$ and $D = \sum_{j\neq i} e^{q_j} > 0$: the map $x \mapsto x/(D+x)$ is strictly increasing, so $\eta < 0$ strictly decreases p_i; the ratio $p_i(\eta)/p_i(0) = e^\eta(D + e^q)/(D + e^{q+\eta})$ lies strictly between e^η and 1. (iii) is by construction: no signed object ever touches the ground intensity. What (c) *cannot* express is a firm's own deal suppressing the *sector aggregate*: the mark factor only redistributes a fixed sector flow across firms. If aggregate post-deal suppression is real, it belongs to route (a) at the sector level, or ultimately to route (b). $\square$

Route (c) is the current production architecture (`hawkes_calibration/sector_stability.py` + `hawkes_calibration/sector_ranker.py`), and the rehearsal gives it empirical teeth: the self-inhibition *ordering* recovered through the ranker cooldown was robust under deliberate misspecification (rank correlation 0.937 against regime-B truth, Appendix C), even though the parametric shape of the cooldown was wrong.

Design decision (D5.1 — signed dynamics without signed kernels).

We commit to routes (a) and (c), with division of labor: *(c) owns firm-level inhibition* — the linear sector Hawkes ground layer plus the ranker cooldown covariate, as implemented today — because it preserves the cluster representation, the subcriticality certificate, and the MBPP interval-censored machinery that defect (C4) makes indispensable; *(a) owns sector-level covariate modulation and timing-feature models* — fixed-decay log-link intensities with bounded history features, as in `hawkes_calibration/models/event_block_hawkes.py` — because it keeps concavity and sign freedom at the price of the operating-point stability discipline of Warning 5.18. Route (b), the rectified additive model (Bonnet et al., 2021; Costa et al., 2020), is the recorded escalation if a genuinely signed *continuous-time ground kernel* is ever required. Reversal triggers: escalate to (b) if the time-rescaling and PIT diagnostics of Chapter 7 show persistent post-burst *overprediction of sector aggregates* that route-(a) saturating features cannot absorb (aggregate suppression, which (c) cannot express by Proposition 5.19), or if the decision layer of Chapter 12 demands calibrated signed amplification in continuous time. Restrict (a) — or shift its mass to (c) — if the Mechanism-B operating-point monitor fires in production simulation.

5.7 Frailty versus contagion

A fitted $\hat{\kappa} > 0$ is not proof of contagion. Clustered, overdispersed event flow has two structural explanations: events beget events (Hawkes), or a latent common factor modulates everyone's rate (frailty; the doubly-stochastic alternative). Both produce overdispersion, autocorrelated counts, and cross-sector comovement, and from a single realization of moderate length they are close to observationally equivalent — the point-process face of the state-dependence-versus-heterogeneity problem of Heckman and Borjas (1980). The mechanism of confounding is concrete: if a positively autocorrelated factor $\xi(t)$ multiplies the true baseline and the model omits it, the score of an excitation loading at zero has expectation

$$\mathbb{E}_0\Big[\partial_\gamma \ell\big|_{\gamma=0}\Big] \;\propto\; \int_0^T \mathrm{Cov}\left(H_\theta(t),\, \xi(t)\right) \mathrm{d}t \;>\; 0, \tag{5.16}$$

because the decayed count H_θ is built from past events that themselves loaded on ξ: the fitted excitation absorbs the common factor. The rehearsal calibrated the damage: under a latent AR(0.90) frailty, the freely fitted branching radius came out 2.4× the truth (Appendix C).

The doctrine that follows is layered. *Covariates first* (Duffie et al., 2007): every observable common driver belongs in $\mu_s(t)$; comovement explained by covariates was never contagion, and our ladder runs baseline-first by construction (Chapter 4 precedes this chapter). *Latent-factor escalation* (Duffie et al., 2009): with many sectors, frailty and contagion have different fingerprints — a common factor loads simultaneously across sectors, contagion propagates along the event stream at the kernel's lag structure — so a latent-factor model (EM or particle methods over ξ) is genuinely estimable; it is an escalation beyond this chapter that we have deliberately not built. *Both coexist* (Azizpour et al., 2018): the operational conclusion for corporate default data, after conditioning on covariates, was contagion *and* frailty — so the honest posture is not to pick one but to bound. The posterior parentage diagnostic of Section 5.4 is the sharpest single-sample probe we have: attributed parent–child gaps concentrated at the kernel timescale are consistent with contagion; attributions spread evenly in time indict a common factor.

Protocol 5.20 (Branching estimates are upper bounds on contagion)**.** Every reported $\hat{\kappa}$, $\hat{G}$, or endogenous share is accompanied by: (i) the covariate set of the baseline it was fitted against; (ii) the sentence "upper bound on contagion; unobserved common factors inflate this estimate" with the 2.4× AR(0.90) calibration of Appendix C cited as the measured severity; (iii) the parent–child gap histogram of Section 5.4. No standalone $\hat{\kappa}$ appears in any deliverable.

5.8 Defect declaration (D2.1)

Against the ledger of Table 2.1, for the models of this chapter (linear sector Hawkes; log-link history models; the mark-factor cooldown belongs to Chapter 10).

(C1) Administrative censoring: robust (exact). The likelihood Equation (5.5) integrates the compensator to T; the open window contributes its survival factor $\exp(-\int \lambda)$ exactly. Unlike the Poisson case, that factor is *path-dependent* — it depends on the realized history through the excitation terms — but for exponential kernels it is available in closed form given the history, so nothing is approximated. Within EM, the right-edge exposure correction of Proposition 5.11 is the same fact seen from the cluster side: an observed parent's offspring process really is Poisson observed on $(t_l, T]$. The subtleties of history-dependent survival under *interval* censoring are deferred to Chapter 6.

(C2) Left truncation: corrects under assumptions. All compensator integrals start at entry times, and risk-set indicators Y multiply intensities as in Equation (2.3). The Hawkes-specific residue is the *left edge*: in-window events whose parents predate the window are structurally

misattributed to immigration, biasing $\hat{\mu}$ up and $\hat{G}$ down. Burn-in of several half-lives mitigates; with 90-day-scale kernel mass and a 13.5-year window the residual bias is small, but short per-sector subwindow fits are not licensed (Protocol 5.13).

(C3) Unlabeled exits: robust at the sector layer, vulnerable at the firm layer. Sector counts are complete regardless of zombie firms, so the ground models of this chapter are immune. Any firm-level intensity (the route-(a) block model, the ranker) keeps dead firms at risk: fitted firm hazards are deflated and cooldown effects attenuated, with the documented sign (downward on firm-level intensities). The risk-set sensitivity is the largest in the system (Appendix C); treatments live in Chapter 10.

(C4) Untrusted dates and aggregation: VULNERABLE for θ, with signed statement. Date jitter and binning destroy the information that identifies the timescale: Fisher information in θ collapses once bin width exceeds the kernel half-life, while κ remains estimable from count balance. The bias is not a small perturbation of $\hat{\theta}$ but a loss of identification — the κ–θ ridge — and timing claims from weekly data are vetoed outright. This defect is the reason Chapter 6 exists: the interval-censored MBPP likelihood is the tractable replacement for the exact likelihood this chapter derived, and Chapter 7 quantifies the bucket-width rule (safe at ≤ 1 half-life, destroyed at ≥ 4).

(C5) Stale firm covariates: pointer. The sector baselines of this chapter consume macro and sector-activity covariates, which are never stale (as-of joined); the engineered history covariates of route (a) are point-in-time correct by construction. Firm-level covariates entering z_i or block-model baselines inherit the attenuation and informative-observation pathologies of Chapter 8.

(C6) Out-of-universe deals: robust at the ground layer, pointer at the mark layer. Sector event streams include out-of-universe deals, so both the excitation sources and the counts this chapter models are complete. The mark factor must carry an outside option or overstate every tracked firm's share; that is Chapter 11's treatment.

Code hooks.

Existing. `hawkes_calibration/eventtime/likelihood.py`: exponential-kernel Jacod likelihood with the per-pair Ogata recursions of Proposition 5.7 in numerically stable block-cumsum form, plus analytic gradients. `hawkes_calibration/eventtime/estimate.py`: fixed-decay MLE with the $\alpha = e^a$ positivity transform and delta-method standard errors; note the code parameterizes $h(u) = \alpha e^{-\beta u}$, so $\kappa = \alpha/\beta$, $\theta = \beta$ — migrating to the (κ, θ) convention of this document is a pending hygiene item. `hawkes_calibration/eventtime/simulate.py`: Ogata thinning specialized via Lemma 5.14, with baseline-breakpoint bound refresh. `hawkes_calibration/eventtime/covariates.py`: piecewise-constant exp-link baselines with closed-form compensators. `hawkes_calibration/eventtime/lasso.py`: FISTA-based ℓ_1 recovery of sparse excitation matrices (Hansen et al., 2015). `hawkes_calibration/sector_stability.py`: the current discrete face (weekly log-link count model, nonnegative lag excitation, row-sum constraint $\rho_{\max} = 0.95$, `aggregate_excitation` and `excitation_spectral_radius` diagnostics). `hawkes_calibration/models/event_block_hawkes.py`: route (a) in production form — exp-link event-time intensity with signed recency term ($\rho \geq 0$ on the inhibitory feature) and sector-block excitation, concave by Proposition 5.16. *To be written.* A continuous-time sector Hawkes module (`hawkes_calibration/eventtime/sector_hawkes.py`) implementing Protocol 5.13: EM fitter with posterior-parentage output (Proposition 5.10), right-edge exposure corrections, profile-decay grids, and the $\rho\left(\hat{G}\right)$ enforcement; the exact Dassios–Zhao simulator (Dassios and Zhao, 2013) alongside thinning; and the

Mechanism-B operating-point monitor of Warning 5.18 as a first-class simulation diagnostic.

Chapter 6

Untrusted and Censored Deal Times

6.1 What the dates are, and what we choose to trust

Chapter 2 recorded, as defect (C4), that the event times t_m of Definition 2.1 are not delivered to us. What PitchBook delivers is a recorded date $\hat{t}_m$ produced by an editorial process: announced versus closed dates that differ by days, back-filled records whose dates were reconstructed after the fact (often rounded to a month or quarter boundary), and duplicate rows collapsed by the hygiene rules of Protocol 2.5. Every continuous-time likelihood in Chapters 4 and 5 consumes t_m as if it were exact; this chapter decides what to do when it is not. The thesis is stated once and defended throughout: *the correct response to untrusted positions is to choose the observation resolution honestly, and to use likelihoods that consume only what we trust — counts of events on intervals, never positions within them.* The count is an editorial fact we can audit; the timestamp is an editorial choice we cannot.

We formalize the corruption with three assumptions, kept deliberately weak.

Assumption 6.1 (Bounded date jitter)**.** Each deal m has a true time t_m and a recorded time $\hat{t}_m = t_m + \varepsilon_m$ with $|\varepsilon_m| \leq \delta_m \leq \delta$ for a known trust radius δ. Nothing else is assumed about the errors ε_m: they may be dependent on each other, on the process, and on the marks; they need not be centered. The bound δ is an output of the date audit (Section 6.8), not a modeling choice.

Assumption 6.2 (Recording conventions)**.** The recorded date decomposes as $\hat{t}_m = t_m + b(r_m) + \varepsilon_m$, where $r_m \in \{\text{closed}, \text{announced}, \text{backfilled}\}$ is an observable record type, $b(\cdot)$ is a systematic convention offset with $b(\text{closed}) = 0$, and backfilled records may additionally exhibit *heaping*: rounding to calendar boundaries, visible as excess mass on month and quarter starts. After auditing to a single convention (subtracting an estimated b, flagging heaped records), the residual error satisfies Assumption 6.1 with a type-specific δ_m.

Assumption 6.3 (Count completeness)**.** After the deduplication of Protocol 2.5, every in-window deal of the tracked process appears exactly once in the record. Corruption lives in the *position* $\hat{t}_m$, never in the *multiplicity.* (Thinning of the record — events missing entirely — is a different defect: (C6) for out-of-universe attribution, (C3) for unobserved exits. This chapter is orthogonal to both; see Section 6.8.)

The deliberate weakness of Assumption 6.1 — adversarial, possibly dependent errors — is what makes the induced modeling choice sharp. If the ε_m were i.i.d. with a known density we could deconvolve; we refuse to assume that, because announced-versus-closed offsets are systematic, not noise. Under adversarial bounded jitter the only event-level statements that survive are set-membership statements, and the following proposition is the complete list of what we keep.

Proposition 6.4 (What bounded jitter leaves invariant). *Let $0 = o_0 < o_1 < \cdots < o_J = T$ be a partition into bins $I_j = (o_{j-1}, o_j]$ of width $\Delta = o_j - o_{j-1}$, let $C_j = \#\{m : t_m \in I_j\}$ be the true bin counts and $\hat{C}_j = \#\{m : \hat{t}_m \in I_j\}$ the observed ones. Under Assumptions 6.1 and 6.3:*

(i) every event observed in bin j has its true time in the dilated bin $(o_{j-1} - \delta,\, o_j + \delta]$;

(ii) $\left|\hat{C}_j - C_j\right| \le \#\{m : \min_k |t_m - o_k| \le \delta\}$ restricted to the two boundaries of I_j: only events within δ of a bin boundary can be miscounted, and only into an adjacent bin;

(iii) if the process has locally bounded mean rate $\sup_t \mathbb{E}[\lambda(t)] \le \bar{\lambda}_{\sup}$, then $\mathbb{E}\left|\hat{C}_j - C_j\right| \le 4\delta\,\bar{\lambda}_{\sup}$, so the expected miscounted fraction *per bin is of order $4\delta/\Delta$;*

(iv) counts over unions of consecutive bins are more robust still: transfers across interior boundaries cancel, leaving only the two outer boundaries.

Proof. (i) is immediate from $\left|\hat{t}_m - t_m\right| \le \delta$. (ii): an event with $t_m \in I_j$ and $\hat{t}_m \notin I_j$ requires $\hat{t}_m$ on the far side of o_{j-1} or o_j, hence $|t_m - o_k| \le \delta$ for a boundary of I_j; conversely for events entering I_j. (iii): by (ii) and the mean-measure bound $\mathbb{E}\,\#\{m : t_m \in A\} = \int_A \mathbb{E}[\lambda] \le |A|\,\bar{\lambda}_{\sup}$, the expected number of events within δ of a fixed boundary is at most $2\delta\,\bar{\lambda}_{\sup}$; each bin has two boundaries. Comparing with $\mathbb{E}C_j \asymp \Delta\bar{\lambda}$ gives the fraction. (iv): an event transferred from I_j to I_{j+1} leaves the count of $I_j \cup I_{j+1}$ unchanged. □

The induced choice is now forced. Any inference that reads event *positions* at scales comparable to δ is reading editorial noise; any inference that reads *counts* on bins of width $\Delta \gg \delta$ is reading the process, up to a boundary-transfer error of controlled order $4\delta/\Delta$ that vanishes as the resolution coarsens. We therefore commit to modeling at resolution no finer than the trust radius — in practice $\Delta \ge$ several multiples of δ — at which point *the data is, by construction, interval-censored count data*, and the rest of this chapter builds the likelihoods that consume exactly that and nothing more. One reassurance before the hard work: the mark factor of Equation (1.4) conditions on a deal occurring and compares candidates within a risk set, so it consumes dates only through risk-set membership and coarse ordering; jitter at the scale of days is invisible to it (Chapter 10 quantifies this). It is the ground process — the timing factor — that absorbs (C4), which is why this chapter is about intensities.

6.2 Right censoring with history-dependent intensities

Before interval censoring, we settle right censoring — defect (C1) — for the Hawkes case, because a doubt naturally arises there that does not arise for Poisson models. For an NHPP the censored-window contribution $\exp(-\int \lambda)$ involves a deterministic function, and the argument of Chapter 4 is elementary. For a Hawkes process the intensity is random and feeds on the history: one might worry that the survival factor for the window $(t_{\text{last}}, T]$ secretly depends on events we have not seen, or that the ABGK independent-censoring argument breaks. It does not, and the reason — predictability — is worth stating precisely, because it is the same property that will *fail* to rescue us under interval censoring.

Assumption 6.5 (Independent, noninformative censoring). Let N be a counting process with $\mathcal{F}_t$-intensity $Y(t)\lambda(t;\vartheta)$ as in Definition 2.3. There is an enlarged filtration $\mathcal{G}_t \supseteq \mathcal{F}_t$ carrying the censoring information such that (i) the $\mathcal{G}_t$-intensity of N is still $Y(t)\lambda(t;\vartheta)$; (ii) the censoring indicator $C(t) \in \{0,1\}$ (e.g. $C(t) = \mathbf{1}\{t \le \tau\}$ for a censoring time τ) is $\mathcal{G}_t$-predictable; (iii) the censoring mechanism's factor in the full likelihood does not involve ϑ (Andersen et al., 1993, III.2). Administrative censoring at the fixed date T (June 2024) satisfies all three with $\mathcal{G} = \mathcal{F}$ and $C(t) = \mathbf{1}\{t \le T\}$.

Theorem 6.6 (Censored-window validity for history-dependent intensities)**.** *Under Assumption 6.5, with λ any $\mathcal{G}$-predictable intensity — in particular the Hawkes intensity $\lambda(t) = \mu(t) + \sum_{t_m<t} h(t - t_m)$, which depends on the realized history:*

(i) the observed process $N^o(t) = \int_0^t C(u)\,\mathrm{d}N(u)$ has $\mathcal{G}_t$-intensity $C(t)\,Y(t)\,\lambda(t;\vartheta)$;

(ii) the likelihood of the observed data factorizes as the Jacod likelihood Equation (2.3) evaluated with intensity $CY\lambda$, times a factor free of ϑ; the first factor is therefore a genuine partial likelihood, and a censored spell contributes exactly its survival factor $\exp(-\int_{spell}\lambda)$ — no term dropped, no term invented;

(iii) the score of this partial likelihood is a $\mathcal{G}$-martingale evaluated at T, so the estimation theory of Chapter 3 (unbiased score, information identities, asymptotic normality) applies unchanged.

Proof. (i) $M = N - \int Y\lambda\,\mathrm{d}u$ is a local $\mathcal{G}$-martingale by Assumption 6.5(i). The process C is bounded and $\mathcal{G}$-predictable by (ii), so the stochastic integral $\int_0^t C\,\mathrm{d}M = N^o(t) - \int_0^t CY\lambda\,\mathrm{d}u$ is again a local $\mathcal{G}$-martingale; since $\int CY\lambda\,\mathrm{d}u$ is predictable and nondecreasing, it is the compensator of N^o, i.e. $CY\lambda$ is the intensity of the observed process. Observe what the proof used: boundedness and predictability of C, predictability of λ — *nowhere* that λ is deterministic. History dependence is immaterial because predictability is preserved under enlargement by Assumption 6.5(i); this is the ABGK III.2 argument verbatim. (ii) is the standard likelihood factorization for counting processes with the intensity from (i) (Andersen et al., 1993, II.7, III.2): the full likelihood is the product integral of the observed-jump and survival factors for N^o times the censoring factor, and (iii) removes the latter from the maximization. (iii) follows because the score is $\int_0^T \partial_\vartheta \log\lambda\,(\,\mathrm{d}N^o - CY\lambda\,\mathrm{d}u)$, a stochastic integral of a predictable process against the martingale of (i). □

What, then, is genuinely different from the Poisson case? Only this: the survival factor is now *random* — but it is a functional of the *observed* data alone. The next proposition makes the spec's "depends on the unobserved post-censoring events only through — nothing" precise.

Proposition 6.7 (Locality of the compensator)**.** *Let $\tau \le T$ be the censoring time of Assumption 6.5 and let λ be the Hawkes intensity with causal kernel (h supported on $(0,\infty)$). For every $t \le \tau$, $\lambda(t)$ is measurable with respect to $\mathcal{H}_{t^-}$, the history of events* strictly before *t; consequently the integrated compensator $\Lambda(\tau) = \int_0^\tau \lambda(u)\,\mathrm{d}u$ is measurable with respect to the observed history $\mathcal{H}_\tau$, and the survival factor $\mathrm{e}^{-\Lambda(\tau)}$ is a deterministic functional of the data we possess. Events in (τ,∞) — observed or not — enter neither the event terms nor the compensator of Theorem 6.6(ii).*

Proof. $\lambda(t) = \mu(t) + \sum_{t_m<t} h(t - t_m)$ involves only event times $t_m < t \le \tau$, all of which are uncensored under Assumption 6.3; measurability of the integral follows from joint measurability of the predictable path $u \mapsto \lambda(u)$ on $(0,\tau]$ and Tonelli. The claim is exactly the statement that the compensator up to a stopping time is determined by the pre-stopping-time path — the defining property of predictability, which the Hawkes feedback respects because the kernel is causal. □

Warning 6.8 (Completeness of the pre-censoring record is now load-bearing). For a Poisson model, an event missing from the record costs exactly its own two terms (one jump term, one sliver of compensator); the rest of the likelihood is untouched. For a Hawkes model, a missing past event corrupts the intensity *path* at every later time: the reconstructed $\lambda(t)$ omits a kernel contribution $h(t-t_m)$, so every subsequent event term and the entire survival factor are evaluated at the wrong intensity. The bias has no universal sign — the fitted baseline μ and branching scale κ reapportion the missing excitation — which is precisely why Assumption 6.3

(counts complete, positions untrusted) is the boundary of this chapter's mandate: it protects the history that the intensity feeds on. Where completeness itself fails, we are in (C3)/(C6) territory and no date discipline repairs it.

6.3 Interval censoring for Poisson models: exactness

Coarsening event times to bin counts is interval censoring by construction. For the Poisson rung of the ladder this costs *no approximation at all* — only resolution. We restate the binned-likelihood theorem of Chapter 4 in the form needed here, and prove it by a route chosen deliberately: the same configuration integral, evaluated for a history-dependent intensity in the next section, is exactly where tractability dies.

Proposition 6.9 (Binned NHPP likelihood is exact)**.** *Let N be an NHPP with deterministic, locally integrable intensity $\lambda(t;\vartheta)$ on $(0,T]$, and let $I_1,\dots,I_J$ partition $(0,T]$ with $\Lambda_j(\vartheta) = \int_{I_j}\lambda(u;\vartheta)\,\mathrm{d}u$. Then the bin counts $C_1,\dots,C_J$ are independent with $C_j \sim \mathrm{Poisson}(\Lambda_j)$, and the log-likelihood of the interval-censored data,*

$$\ell_{\mathrm{IC}}(\vartheta) \;=\; \sum_{j=1}^{J}\Big(C_j \log\Lambda_j(\vartheta) - \Lambda_j(\vartheta) - \log C_j!\Big), \tag{6.1}$$

is the exact *likelihood of what was observed — not an approximation to the event-time likelihood but the correct likelihood of the coarsened data. Moreover, if $\lambda(\cdot;\vartheta)$ is constant on each bin (the weekly piecewise-constant models of Chapter 4), the counts are sufficient for ϑ and coarsening loses nothing; in general the loss is confined to the within-bin shape of λ.*

Proof. Start from the density of the ordered event times under the full model: for n events at $0 < t_1 < \cdots < t_n \le T$, $f(t_{1:n}) = \prod_{m=1}^{n}\lambda(t_m)\,e^{-\int_0^T\lambda(u)\,\mathrm{d}u}$. The probability of the count vector $C_{1:J}$ (with $n = \sum_j C_j$) is the integral of f over the configuration region $\mathcal{A}(C_{1:J}) = \{\,0 < t_1 < \cdots < t_n \le T : \#\{m : t_m \in I_j\} = C_j\ \forall j\,\}$. Because λ is deterministic, the integrand is a product of factors each depending on one coordinate only, so the ordered integral factorizes across bins, and within bin j the ordered integral of $\prod\lambda$ over the C_j-dimensional ordered simplex equals $\Lambda_j^{C_j}/C_j!$. Hence $\mathbb{P}(C_{1:J}) = \prod_j\,(\Lambda_j^{C_j}/C_j!)\,e^{-\Lambda_j}$, which is Equation (6.1) and simultaneously the independence and Poisson-marginal claims. Sufficiency under piecewise-constant λ is the factorization criterion applied to Equation (6.1): the likelihood depends on the data only through $C_{1:J}$. □

Corollary 6.10 (For NHPP calibration, untrusted dates cost only resolution)**.** *Under Assumptions 6.1 and 6.3, fitting any NHPP of Chapter 4 on bin counts at resolution $\Delta \gg \delta$ uses an exact likelihood (Proposition 6.9) applied to data whose corruption is bounded by Proposition 6.4(iii). No approximation enters anywhere; the entire price of (C4) is the inability to resolve λ within bins — a price we were going to pay anyway by modeling weekly.*

This is the sense in which the weekly sector GLM of the current two-layer architecture (`hawkes_calibration/sector_stability.py`) is already fully (C4)-compliant: it never consumed positions in the first place. The classical ancestor of likelihoods under grouped and censored observation is Turnbull (1976); what is distinctive in our setting is not grouping per se but the interaction of grouping with *dependence*, which is the next section's subject.

6.4 Interval censoring for Hawkes: where exactness dies

Run the proof of Proposition 6.9 again with the Hawkes intensity. The density of the ordered times is still the Jacod product, so the exact interval-censored likelihood is still a configuration integral:

$$L_{\mathrm{IC}}(\vartheta) \;=\; \int_{\mathcal{A}(C_{1:J})} \prod_{m=1}^{n} \lambda(t_m \mid \mathcal{H}_{t_m^-};\vartheta)\; \exp\Big(-\int_0^T \lambda(u \mid \mathcal{H}_{u^-};\vartheta)\,\mathrm{d}u\Big)\; \mathrm{d}t_1 \cdots \mathrm{d}t_n, \tag{6.2}$$

with $\lambda(t \mid \mathcal{H}_{t^-}) = \mu(t) + \sum_{t_m<t} h(t-t_m)$ and $n = \sum_j C_j$. But now the factorization step fails, for the precise reason the model was chosen: the intensity at t_m depends on the *positions* of all earlier events, including those in earlier bins and those in the same bin. The integrand does not separate across bins, the within-bin integrals are not simplex moments of a fixed function, and Equation (6.2) is an n-dimensional integral — at the scale of this dataset, dimension $\sim 5 \times 10^4$ — over a combinatorially-constrained region, with no product structure to exploit. Data augmentation (imputing positions by MCMC or EM over Equation (6.2)) is possible in principle and is how the Bayesian literature treats small problems (Rasmussen, 2013); at our scale, and for a likelihood that must be refit across sectors, splits, and rehearsal campaigns, it is not a workable primary route. The likelihood requires event positions inside bins *because intensity feeds back on history*; interval censoring withholds exactly that. Something must give, and the honest thing to give up is the randomness of the intensity — replacing it by its mean — rather than the honesty of the resolution.

6.5 The mean-intensity route

This section is the chapter's centerpiece: the *mean behavior Poisson process* (MBPP) of Rizoiu et al. (2022), implemented in `hawkes_calibration/mbpp/`. The idea is to fit, instead of the intractable interval-censored Hawkes likelihood, the exact interval-censored likelihood of the NHPP whose deterministic intensity is the *expected* Hawkes intensity. The parameters (μ, κ, θ) are shared, the first moments are matched exactly, and the price — discarded dependence and overdispersion — is stated, bounded, and paid with open eyes.

6.5.1 The mean intensity and its Volterra equation

Theorem 6.11 (Renewal equation for the mean intensity)**.** *Let N be a Hawkes process on $[0,T]$ with intensity $\lambda(t) = \mu(t) + \int_{(0,t)} h(t-u)\,\mathrm{d}N(u)$, where μ is deterministic, locally bounded, and $h \geq 0$ with $\int_0^\infty h = \kappa < 1$. Then the mean intensity $\bar\lambda(t) := \mathbb{E}[\lambda(t)]$ is finite and satisfies the linear Volterra equation of the second kind*

$$\bar\lambda(t) \;=\; \mu(t) \;+\; \int_0^t h(t-u)\,\bar\lambda(u)\,\mathrm{d}u. \tag{6.3}$$

Proof. Step 1 (finiteness). By the cluster representation of Chapter 5 (Hawkes and Oakes, 1974), N is the superposition of immigrant events arriving with rate μ and, per event, an independent Galton–Watson cascade with offspring mean $\kappa < 1$; the expected total progeny per immigrant is $1/(1-\kappa)$, so $\mathbb{E}N(t) \leq \int_0^t \mu(u)\,\mathrm{d}u \,/\, (1-\kappa) < \infty$, and $\bar\lambda(t) \leq \mu(t) + \kappa\theta\,\mathbb{E}N(t) < \infty$ for the exponential kernel (and similarly for any bounded kernel).

Step 2 (martingale property of the compensator). For any *nonnegative predictable* integrand H, the defining property of the intensity gives the compensator formula $\mathbb{E}\int_0^t H(u)\,\mathrm{d}N(u) = \mathbb{E}\int_0^t H(u)\lambda(u)\,\mathrm{d}u$, both sides possibly infinite (Andersen et al., 1993, II.4); this is the statement $\mathbb{E}[\,\mathrm{d}N(u)] = \bar\lambda(u)\,\mathrm{d}u$ made rigorous. Fix t and apply it to the deterministic (hence predictable) integrand $H(u) = h(t-u)\mathbf{1}\{u < t\}$:

$$\mathbb{E}\int_{(0,t)} h(t-u)\,\mathrm{d}N(u) \;=\; \mathbb{E}\int_0^t h(t-u)\,\lambda(u)\,\mathrm{d}u \;=\; \int_0^t h(t-u)\,\bar\lambda(u)\,\mathrm{d}u,$$

the last step by Tonelli (nonnegative integrand) and Step 1. Taking expectations through $\lambda(t) = \mu(t) + \int_{(0,t)} h(t-u)\,\mathrm{d}N(u)$ yields Equation (6.3). $\square$

Note what was used: only the compensator formula — the same martingale structure that powers every likelihood in this document — plus subcriticality for finiteness. No stationarity, no exponential form, no constant baseline.

Theorem 6.12 (Existence, uniqueness, Neumann series)**.** *Under the hypotheses of Theorem 6.11, Equation (6.3) has exactly one locally bounded solution on* $[0, T]$*, namely*

$$\bar{\lambda} \;=\; \mu + R * \mu, \qquad R \;\coloneqq\; \sum_{n\geq 1} h^{*n}, \qquad \int_0^\infty R \;=\; \frac{\kappa}{1-\kappa}, \tag{6.4}$$

where h^{*n} *is the* n*-fold convolution power. Picard iteration* $\bar{\lambda}_{k+1} = \mu + h * \bar{\lambda}_k$*,* $\bar{\lambda}_0 = \mu$*, converges to it geometrically:* $\sup_{[0,T]} \left|\bar{\lambda}_k - \bar{\lambda}\right| \leq \kappa^{k+1}(1-\kappa)^{-1} \sup_{[0,T]} \mu$*.*

Proof sketch. The map $\Phi(f) = \mu + h * f$ on $(L^\infty[0,T], \|\cdot\|_\infty)$ satisfies $\|\Phi(f) - \Phi(g)\|_\infty \leq \|h\|_{L^1} \|f - g\|_\infty = \kappa \|f - g\|_\infty$, a contraction since $\kappa < 1$; Banach's fixed point theorem gives existence, uniqueness among bounded solutions, and the geometric rate. Unrolling the iteration gives the Neumann series Equation (6.4), whose mass identity follows from $\|h^{*n}\|_{L^1} = \kappa^n$ and geometric summation. The general theory of Volterra equations — including the resolvent formalism, L^1 well-posedness, and what survives when $\kappa \geq 1$ locally — is Gripenberg et al. (1990). When $\kappa \uparrow 1$ the geometric rate degrades as $(1-\kappa)^{-1}$ and at criticality the series diverges: the mean-intensity route inherits exactly the stability boundary of the process itself. □

The resolvent R has a probabilistic reading that connects back to Chapter 5: h^{*n} is the arrival density of n-th generation offspring, so Equation (6.4) says the mean intensity is the immigrant rate plus the expected arrival rate of descendants of all generations — the cluster representation, averaged.

6.5.2 Exponential kernel: closed form

Proposition 6.13 (Closed form for the exponential kernel)**.** *Let* $h(t) = \kappa\theta e^{-\theta t}$ *and* $\mu(t) \equiv \mu$ *constant, with* $\kappa \in (0,1)$*,* $\theta > 0$*. Then*

$$\bar{\lambda}(t) \;=\; \mu \;+\; \frac{\kappa\theta\,\mu}{(1-\kappa)\theta}\Big(1 - e^{-(1-\kappa)\theta\, t}\Big) \;=\; \mu \;+\; \frac{\kappa\,\mu}{1-\kappa}\Big(1 - e^{-(1-\kappa)\theta\, t}\Big), \tag{6.5}$$

so $\bar{\lambda}$ *relaxes from* $\bar{\lambda}(0) = \mu$ *to the stationary level* $\bar{\lambda}(\infty) = \mu/(1-\kappa)$ *at the* effective relaxation rate $r = (1-\kappa)\theta$*. Equivalently, the resolvent is itself exponential,* $R(t) = \kappa\theta\, e^{-(1-\kappa)\theta t}$*, and for* time-varying $\mu(\cdot)$ *the solution is the single convolution* $\bar{\lambda} = \mu + R * \mu$*; the integrated version* $\bar{\Lambda}(t) = \int_0^t \bar{\lambda}$ *obeys the same equation,* $\bar{\Lambda} = \int_0^\cdot \mu + h * \bar{\Lambda}$*.*

Proof. Write $\bar{\lambda}(t) = \mu + g(t)$ with $g(t) = \int_0^t \kappa\theta\, e^{-\theta(t-u)} \bar{\lambda}(u)\,\mathrm{d}u$. The kernel's exponential form makes g differentiable with

$$g'(t) \;=\; \kappa\theta\,\bar{\lambda}(t) \;-\; \theta\, g(t) \;=\; \kappa\theta(\mu + g(t)) - \theta\, g(t) \;=\; \kappa\theta\,\mu \;-\; (1-\kappa)\theta\, g(t),$$

a linear ODE with $g(0) = 0$, solved by $g(t) = \dfrac{\kappa\theta\,\mu}{(1-\kappa)\theta}(1 - e^{-(1-\kappa)\theta t})$; adding μ gives Equation (6.5). *Sanity checks.* At $t = 0$ the excitation has not started: $\bar{\lambda}(0) = \mu$. As $t \to \infty$, $\bar{\lambda} \to \mu + \kappa\mu/(1-\kappa) = \mu/(1-\kappa)$ — the immigrant rate times the mean cluster size, agreeing with the cluster representation and with the stationary Hawkes mean of Chapter 5; and substituting the constant $\mu/(1-\kappa)$ into Equation (6.3) verifies it is the fixed point. For the resolvent claim, either solve $R = h + h * R$ by the same ODE argument or sum the Erlang series $h^{*n}(t) = \kappa^n \theta^n t^{n-1} e^{-\theta t}/(n-1)!$ directly; then $\bar{\lambda} = \mu + R * \mu$ is Equation (6.4). The equation for $\bar{\Lambda}$ follows by integrating Equation (6.3) and Fubini. □

The repository implements exactly this structure (`hawkes_calibration/mbpp/core.py`): the impulse response $R(t) = \kappa\theta e^{(\kappa-1)\theta t}$ is Eq. (14) of Rizoiu et al. (2022); their Theorems 1–2 and Corollary 3 give $\bar{\lambda} = s + R * s$ and $\bar{\Lambda} = S + R * S$ in closed form for every exogenous input $s(\cdot)$ in `hawkes_calibration/mbpp/exogenous.py` (constant, rectangle, piecewise-constant, sinusoidal, impulse trains, and covariate-driven baselines), with a numerical finite-convolution solver for kernels without closed forms (power-law). The relaxation rate $r = (1-\kappa)\theta$ of Proposition 6.13 will reappear twice: as the decay rate of cross-bin count correlation, and as the parameter whose product with the bin width governs the κ–θ ridge.

6.5.3 The MBPP working likelihood, and precisely what it discards

Definition 6.14 (Mean behavior Poisson process and IC likelihood)**.** The MBPP associated with a Hawkes model (μ, h) is the NHPP with deterministic intensity $\bar{\lambda}(\cdot)$ solving Equation (6.3). Its interval-censored log-likelihood on the partition of Proposition 6.4 is the exact Poisson form Equation (6.1) with means $\bar{\Lambda}_j(\vartheta) = \int_{I_j} \bar{\lambda}(u;\vartheta)\,\mathrm{d}u$:

$$\ell_{\mathrm{MBPP}}(\vartheta) \;=\; \sum_{j=1}^{J} \Big(C_j \log \bar{\Lambda}_j(\vartheta) \;-\; \bar{\Lambda}_j(\vartheta) \Big) \;+\; \text{const}, \tag{6.6}$$

maximized over $\vartheta = (\mu, \kappa, \theta, \dots)$ (`hawkes_calibration/mbpp/interval_censored.py`, `fit_mbpp_ic`).

Because precision about approximations is this document's currency, we state exactly what Equation (6.6) gets right and wrong when the data is generated by the Hawkes process itself.

Exact: the first moments. By Theorem 6.11 and the compensator formula, $\mathbb{E}C_j = \mathbb{E}\int_{I_j} \lambda = \bar{\Lambda}_j$ *exactly*, for every j, transient or stationary. Consequently the expected score of Equation (6.6) at the true parameters is $\sum_j (\mathbb{E}C_j/\bar{\Lambda}_j - 1)\partial_\vartheta \bar{\Lambda}_j = 0$: the MBPP maximum-likelihood estimator is an M-estimator with an unbiased estimating equation — a moment-matching estimator dressed as a likelihood — and is consistent under the usual regularity and ergodicity conditions for correctly specified means. Rizoiu et al. (2022, Prop. 8) give the population version: the expected IC log-likelihood is (up to constants) the generalized Kullback–Leibler divergence between the true interval means and the model's interval compensators, minimized exactly at parameters reproducing the true means — identifiability is preserved *in the KL sense*, to the extent that the interval means themselves separate the parameters (they pin μ and κ through levels and transients; how much of θ they retain is Section 6.5.4).

Discarded: the second moments. Two falsehoods are asserted by Equation (6.6). First, that C_j is Poisson — but the Hawkes random compensator $\int_{I_j} \lambda$ has been replaced by its expectation, and the law of total variance charges for it: $\mathrm{Var}(C_j) = \mathbb{E}[(\text{martingale variation})] + \mathrm{Var}\big(\int_{I_j} \lambda\big) + \text{cross terms} > \mathbb{E}C_j$. Second, that $C_1, \dots, C_J$ are independent — but excitation propagates across bin boundaries, and for the exponential kernel the count autocovariance decays geometrically with lag at exactly the AR root $e^{-r\Delta}$, $r = (1-\kappa)\theta$. The next proposition quantifies the first falsehood; the second is smallest precisely when bins are wide relative to the relaxation time $1/r$.

Proposition 6.15 (Hawkes bin counts are overdispersed)**.** *For a stationary Hawkes process with $h = \kappa\theta e^{-\theta t}$, $\kappa \in (0,1)$, constant μ, let $F(\Delta) = \mathrm{Var}(C_j)/\mathbb{E}(C_j)$ be the Fano factor of a bin of width Δ. Then*

$$F(\Delta) \;\longrightarrow\; 1 \quad (\Delta \to 0), \qquad F(\Delta) \;\longrightarrow\; \frac{1}{(1-\kappa)^2} \quad (\Delta \to \infty), \tag{6.7}$$

and $F(\Delta) > 1$ for all $\Delta > 0$: true bin counts are strictly overdispersed relative to the Poisson law asserted by Equation (6.6), by a factor approaching $(1-\kappa)^{-2}$ for wide bins.

Proof sketch. Wide bins. By the cluster representation (Hawkes and Oakes, 1974), N is a superposition of i.i.d. clusters attached to Poisson(μ) immigrants; the total cluster size S is the total progeny of a Galton–Watson process with Poisson(κ) offspring, for which $\mathbb{E}S = (1-\kappa)^{-1}$ and $\operatorname{Var} S = \kappa(1-\kappa)^{-3}$. Cluster durations have finite mean (exponential delays), so as $\Delta \to \infty$ the count over a bin is, up to boundary terms of $O(1)$ against a mean of $O(\Delta)$, a compound Poisson sum $\sum_{i\le K} S_i$ with $K \sim \text{Poisson}(\mu\Delta)$; hence $F \to \mathbb{E}S^2/\mathbb{E}S = \big(\operatorname{Var} S + (\mathbb{E}S)^2\big)/\mathbb{E}S = \kappa(1-\kappa)^{-2} + (1-\kappa)^{-1} = (1-\kappa)^{-2}$. *Narrow bins.* $\mathbb{E}[C_j(C_j-1)] = \iint_{I_j^2} \rho_2 = O(\Delta^2)$ for the (bounded) second-order product density ρ_2 of the stationary process, while $\mathbb{E}C_j = \Theta(\Delta)$, so $F = 1 + O(\Delta) \to 1$. Strict inequality for all Δ follows from positivity of the covariance density of a self-exciting process with nonnegative kernel (Hawkes, 1971; Daley and Vere-Jones, 2003). What breaks if $\kappa \ge 1$: $\mathbb{E}S = \infty$ and the Fano factor diverges — another face of the stability boundary. □

Warning 6.16 (MBPP standard errors are optimistic). The observed-information standard errors of Equation (6.6) (as reported by `fit_mbpp_ic`) price the counts as Poisson and independent. By Proposition 6.15 the true variance is up to $(1-\kappa)^{-2}$ larger per bin — at the rehearsal's $\kappa \approx 0.35$ a ceiling of ≈ 2.4, i.e. naïve standard errors up to $\approx (1-\kappa)^{-1} \approx 1.5\times$ too small — and positive cross-bin correlation compounds the understatement. (This 2.4 is a derived variance ceiling, unrelated to the 2.4× branching-ratio inflation under latent frailty recorded in Appendix C.) Remedies, in committed order: (i) sandwich (robust M-estimation) covariance, $\mathcal{I}^{-1} V \mathcal{I}^{-1}$ with V the empirical score variance — valid because the estimating equation is unbiased regardless of the working variance (van der Vaart, 1998, Ch. 5); (ii) a negative-binomial working likelihood with variance $\bar\Lambda_j + \bar\Lambda_j^2/\nu$ when the dispersion diagnostic of Chapter 7 (Pearson statistic, `hawkes_calibration/mbpp/gof.py`) exceeds the predicted Fano band; (iii) the Bayesian route of `hawkes_calibration/mbpp/bayes.py`, whose posterior exhibits the κ–θ ridge as a correlation instead of hiding it in a scalar. Never report the naïve SEs alone.

When is the mean-intensity replacement good? Three regimes, all of which hold for our intended use. (i) *Aggregation over many low-feedback units:* sector-level counts superpose many firms' sparse streams; superpositions of many sparse, weakly dependent point processes are close to Poisson (Daley and Vere-Jones, 2003, Ch. 9), so the Poisson working law is close to true even before the mean replacement. (ii) *Subcritical, moderate κ:* the entire second-moment error is bounded by the Fano ceiling $(1-\kappa)^{-2}$, which is $O(1)$ and known; the fitted κ on this data lives near 0.3–0.4, far from the boundary. (iii) *Bin width at or beyond the kernel half-life:* the neglected cross-bin dependence has AR root $e^{-r\Delta}$, small once $\Delta \gtrsim 1/r$; and within-bin position information — the thing MBPP cannot use — was already destroyed by (C4). The regime where MBPP is *bad* is the mirror image: a single high-κ stream at fine resolution, where one should be fitting the event-time likelihood of Chapter 5 instead — on verified dates.

6.5.4 What survives aggregation: κ yes, θ no

The stationary interval mean is $\bar\Lambda_j = \mu\Delta/(1-\kappa)$: it contains κ and *no trace of θ*. All θ-information in counts therefore lives in temporal structure — transients of Equation (6.5) and cross-bin autocovariances — and both enter only through the relaxation root $e^{-r\Delta}$. The rehearsal campaigns made this quantitative (inherited results, Appendix C): the Fisher information for θ carried by counts decays like $\Delta^2 e^{-2(1-\kappa)\theta\Delta}$ — exponentially in the bin width — so binning does not merely blur the timescale, it erases it. At daily bins the misspecified regime-B world ($\theta = \ln 2/7$ per day) was recovered unbiased; at weekly bins ($\approx$ one half-life) essentially nothing was lost; at 28-day bins ($\approx$ 4 half-lives) the θ estimate was biased low by $\sim 20\%$ *with a deceptively small spread* — the signature of an unidentified direction collapsing onto a ridge — while κ, read from first moments, stayed within 0.28–0.38 of its true 0.35

at every width. Hence the standing rule, inherited as a veto: *bucket width* $\leq$ *one kernel half-life is safe for* θ*;* $\geq$ *four half-lives destroys timing;* κ *and covariate effects survive any width.* The Bayesian implementation (`hawkes_calibration/mbpp/bayes.py`) is the honest reporting vehicle on the ridge: a flat-prior posterior on (κ, θ) stays wide along θ and tight along κ, and a θ prior from a verified-date subset tightens it legitimately.

Warning 6.17 (Never claim timing precision finer than the honest resolution). A θ fitted from weekly counts is a ridge coordinate, not a measurement. Design consequence, binding on every downstream consumer (Chapter 12): no model output may assert within-week timing structure — expected response times, decay half-lives, "momentum windows" — unless θ was estimated from a verified-date subset at event-time resolution, and the honest resolution of the input data is reported next to any timing claim. The corresponding failure mode is seductive precisely because the fit *converges*: the optimizer happily returns a θ with a small naïve SE while sliding along the ridge.

6.6 Single untrusted dates as one-bin censoring

The machinery above treats wholesale aggregation. The same logic handles retail corruption: a *single* deal whose date is trusted only to $\pm\delta_m$ is an event observed in the one-off interval $I_m = (\hat{t}_m - \delta_m, \hat{t}_m + \delta_m]$, i.e. a private one-bin censoring. Mixed records — verified dates for some deals, windows for others — then call for a mixed likelihood.

Proposition 6.18 (Exact mixed likelihood, Poisson case)**.** *Let N be an NHPP with deterministic intensity λ, and suppose the record consists of (a) verified events at exact times $\{t_m\}_{m\in\mathcal{V}}$ and (b) untrusted events known only to lie in windows $\{I_m\}_{m\in\mathcal{U}}$, one event per window, where the windows are pairwise disjoint and contain no verified events (overlapping windows are first merged into a single interval carrying its total count, reducing to Proposition 6.9 locally), and (c) the assertion, by Assumption 6.3, that no further events occurred in $(0, T]$. The exact log-likelihood of this record is*

$$\ell_{\mathrm{mix}}(\vartheta) \;=\; \sum_{m\in\mathcal{V}} \log\lambda(t_m;\vartheta) \;+\; \sum_{m\in\mathcal{U}} \log\Lambda(I_m;\vartheta) \;-\; \int_0^T \lambda(u;\vartheta)\,\mathrm{d}u, \qquad \Lambda(I) = \int_I \lambda. \tag{6.8}$$

Proof. The restrictions of an NHPP to disjoint sets are independent. Partition $(0, T]$ into the windows $\{I_m\}_{m\in\mathcal{U}}$ and their complement B. On B the observation is exact: points at $\{t_m\}_{m\in\mathcal{V}}$ and none elsewhere, with density $\prod_{\mathcal{V}} \lambda(t_m)\, e^{-\Lambda(B)}$. On each I_m the observation is the event $\{N(I_m) = 1\}$, with probability $\Lambda(I_m)e^{-\Lambda(I_m)}$. Multiplying and collecting exponents (the Λ's add to $\int_0^T \lambda$ over the partition) gives Equation (6.8). Every factor is exact; no within-window position was used. $\square$

So for Poisson calibration, mixed trust costs nothing but the windows' internal resolution: verified deals keep their full event terms, untrusted deals degrade gracefully to interval terms, and the survival factor is common. For Hawkes the by-now-familiar obstacle returns — the intensity after an untrusted event depends on where in I_m it actually happened — and the exact mixed likelihood is again a configuration integral over $\prod_{m\in\mathcal{U}} I_m$. The committed practical recipe, flagged as an approximation, mirrors the MBPP construction locally: keep exact terms where dates are verified, and push untrusted events through their *expected* kernel contribution. Concretely, define the smeared intensity

$$\tilde{\lambda}(t;\vartheta) \;=\; \mu(t) \;+\; \sum_{m\in\mathcal{V}:\, t_m<t} h(t - t_m) \;+\; \sum_{m\in\mathcal{U}} \frac{1}{|I_m|} \int_{I_m} h(t-u)\,\mathbf{1}\{u < t\}\,\mathrm{d}u, \tag{6.9}$$

(the untrusted event's excitation averaged uniformly over its window, with only the past part of the window contributing), and maximize Equation (6.8) with $\tilde{\lambda}$ in place of λ. This is exact as $\kappa \to 0$ (Poisson limit), errs only in the untrusted events' second-order feedback terms, and must be validated by the protocol of Section 6.7 before use — the same discipline as the MBPP itself. When untrusted events are dense in some sector-period, the recipe degenerates gracefully: merge that period's windows into bins and the untrusted sum in Equation (6.8) becomes the MBPP likelihood Equation (6.6) restricted to that period.

6.7 The validation loop

Every approximation admitted in this chapter — the mean-intensity replacement, the mixed-likelihood smearing, the choice of resolution — is validated by the same closed loop, which the repository already implements for the MBPP (`hawkes_calibration/mbpp/ic_simulate.py`) and which is hereby made the standing acceptance test.

Protocol 6.19 (Simulate exactly, censor, refit, compare)**.**

(a) *Simulate exactly* from the continuous-time Hawkes model at the fitted (or stress) parameters: the cluster/branching construction of Hawkes and Oakes (1974) (`simulate_separable_hawkes`), or exact simulation by Dassios and Zhao (2013), or Ogata thinning (Ogata, 1981; Lewis and Shedler, 1979) — never from the MBPP itself, which would assume the conclusion.

(b) *Censor as production does*: collapse event times to the production partition with `interval_censor`; optionally inject adversarial jitter at the audited δ before binning, to certify Proposition 6.4 empirically.

(c) *Refit with the production likelihood* (`fit_mbpp_ic` and variants; the mixed recipe of Section 6.6 where applicable).

(d) *Compare*: (1) parameter recovery — κ must be recovered within Monte Carlo bands at every candidate resolution; θ only at resolutions the ridge rule permits, and its failure at coarse bins must *reproduce* the known low-bias signature, not surprise us; (2) SE honesty — the ratio of Monte Carlo spread to reported SE is the empirical dispersion factor, to be compared against the Fano prediction of Proposition 6.15 and repaired per Warning 6.16; (3) fit diagnostics — Pearson dispersion and PIT-style count calibration (Czado et al., 2009) via `hawkes_calibration/mbpp/gof.py`, with the event-time time-rescaling check (Brown et al., 2002) reserved for verified-date subsets (Chapter 7).

(e) *Gate*: a resolution/likelihood pair is admitted for real-data use only after passing (d) in the synthetic world at matched sample sizes — the transfer convention of Section 2.5.

6.8 Defect declaration and commitments

Per Table 2.1 and decision D2.1, this chapter *owns (C4)* and relates to the rest of the ledger as follows. *(C1) corrected here*: Theorem 6.6 and Proposition 6.7 establish that right censoring costs nothing for history-dependent intensities when open spells are kept, closing the Hawkes case left open by Warning 2.4. *(C2) respected mechanically*: every integral and every bin in this chapter starts at the entry time V_i where firm-level objects are involved; the selection content of (C2) is owned by Chapter 11. *(C3) vulnerable, out of scope*: nothing here repairs contaminated risk sets — sector-level count likelihoods are largely insulated (sectors do not die), but any firm-level use of these tools inherits (C3) with the deflation bias documented in Chapter 2, owned by Chapters 4, 9 and 10. *(C5) orthogonal*: covariates enter the MBPP through the exogenous input $s(\cdot)$ (`hawkes_calibration/mbpp/exogenous.py`); their staleness is Chapter 8's problem and its remedies compose with everything here. *(C6) robust at sector level*: sector deal counts include out-of-universe deals, so sector-level MBPP consumes complete counts; firm-attributable applications inherit (C6) as thinning, in tension with Assumption 6.3, and are owned by Chapter 11.

Design decision (D6.1 — the honest-resolution rule).

The observation resolution for continuous-time ground-process modeling on PitchBook 2011–2024 is *weekly*. Daily resolution is permitted only for subsets whose closed dates the date audit certifies with $\delta \leq 1$ day; back-filled and announced-only records never qualify. Rationale: Proposition 6.4 (boundary-transfer error $\sim 4\delta/\Delta$), Assumption 6.2 (announced-vs-closed spreads of days, heaping in backfills), and the ridge rule of Section 6.5.4 (weekly $\approx$ one half-life of the regime-B kernel: safe for κ, marginal for θ). Reversal trigger: a data vintage whose audit certifies $\delta \leq 1$ day for a sector's majority of deals moves that sector to daily; conversely, evidence of heaping beyond month boundaries (quarter-level backfill) forces that segment to monthly.

Design decision (D6.2 — MBPP is the committed aggregation likelihood).

Any sector-level continuous-time Hawkes model fitted on binned counts uses the MBPP interval-censored likelihood Equation (6.6) (`fit_mbpp_ic` and variants), with standard errors repaired per Warning 6.16 (sandwich by default; negative-binomial or Bayesian escalation on a failed dispersion check), and with every fit gated by Protocol 6.19. Rationale: exact first moments (Theorem 6.11), KL-sense identifiability (Rizoiu et al., 2022), closed forms for the exponential kernel (Proposition 6.13), and a bounded, quantified price (Proposition 6.15) — against an exact alternative Equation (6.2) that is computationally out of reach. Reversal trigger: if dispersion diagnostics persistently exceed the Fano ceiling $(1-\hat{\kappa})^{-2}$ (pointing at latent-frailty variance the MBPP cannot absorb), the sector-level count model reverts to the discrete autoregressive count family of Chapter 9 (Fokianos et al., 2009), which models the conditional variance directly.

Design decision (D6.3 — event-time likelihoods reserved for verified dates).

The exact event-time likelihood of Chapter 5 (and the time-rescaling diagnostics that need it) may be fitted only on date-audit-verified subsets; θ estimated anywhere else is prior-constrained from such a subset or fixed (decay banks). Mixed records use Equation (6.8): exactly for Poisson (Proposition 6.18), with the flagged smeared-intensity recipe Equation (6.9) for Hawkes. Rationale: Warning 6.17 — timing claims must trace to data that contains timing. Reversal trigger: none for the principle; the boundary of "verified" moves with the date audit.

Code hooks.

The mathematics of this chapter maps onto `hawkes_calibration/mbpp/` as follows. *`core.py`*: the MBPP object — kernels, the resolvent/impulse response $R(t) = \kappa\theta e^{(\kappa-1)\theta t}$ of Proposition 6.13, closed-form and numeric solutions of Equation (6.3), interval compensators $\bar{\Lambda}_j$, and the Proposition-6/7 compensator bounds used for forecasting. *`interval_censored.py`*: the IC likelihood Equation (6.6) (`ic_ll`), the SSE/Bregman alternative, and the fitting family `fit_mbpp_ic / _multi / _covariates / _sumexp / _excitation`, plus `forecast_counts`. *`exogenous.py`*: the exogenous inputs $s(\cdot)$ — piecewise-constant, covariate-driven, impulse, LHPP — through which Chapter 8's covariates and Chapter 4's baselines enter. *`ic_simulate.py`*: the branching simulator and `interval_censor`, powering Protocol 6.19. *`gof.py`*: Pearson/dispersion diagnostics for counts and time-rescaling residuals for verified-date subsets; *`bayes.py`*: the posterior treatment of the κ–θ ridge. *To be written*: `hawkes_calibration/data/date_audit.py` — the audit behind Assumptions 6.1 and 6.2: classify each deal date as closed/announced/backfilled, estimate convention offsets $b(r)$, detect month/quarter heaping, emit per-event trust radii δ_m and the verified-date flags consumed by D6.1/D6.3 — and the sandwich-covariance wrapper for `fit_mbpp_ic` required by Warning 6.16.

Chapter 7

Goodness of Fit, Model Comparison, and Calibration Discipline

7.1 The evaluation constitution

Every preceding chapter builds an estimator; this chapter fixes the court in which all of them are tried. Its deliverables are not models but rules: a definition of "calibrated" for each of the four targets of Chapter 1, the test batteries that operationalize those definitions, the comparison ladder that decides which model family is promoted, and the reporting discipline that keeps every number attached to the caveats it needs. We call this the *evaluation constitution* because it is meant to be hard to amend: metrics chosen after seeing results are not metrics, and the repository's synthetic rehearsals (Appendix C) demonstrated repeatedly that the difference between a defensible claim and an embarrassing one was almost never the model — it was whether the evaluation respected the risk set, the temporal split, and the identifiability limits of the parameter being quoted.

The word "calibrated" means something different for each target, and conflating the meanings is the failure mode Section 1.2 warns against.

Definition 7.1 (Calibration, per target)**.** Fix an evaluation era and the point-in-time discipline of Protocol 2.5.

(i) *T2 (timing).* An intensity or count model is calibrated if it survives the time-rescaling battery of Protocol 7.3 at the finest resolution its event times honestly support, *and* the count-calibration battery of Protocol 7.8 (PIT uniformity, dispersion near 1, reliability per sector-week) at weekly resolution. Both are required: rescaling sees timing structure that counts hide, and PIT sees marginal-distribution failures that rescaling averages out.

(ii) *T3 (horizon events).* A horizon classifier is calibrated if its predicted probabilities match conditional frequencies: $\mathbb{P}(\text{event within } \Delta \mid \hat{p} = p) = p$ across the support of $\hat{p}$, assessed by reliability diagram, calibration intercept and slope. Discrimination (PR-AUC, lift at K) is necessary but never sufficient; recalibration maps (Platt, 1999; Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005) are legal but are model components, fitted on the validation era only.

(iii) *T1 (identity).* A ranker is judged by ranking metrics (recall at K, MRR, computed on the declared risk set) *and* by held-out conditional likelihood (nats per deal of the mark factor). Each alone is gameable: recall at K is blind to the probabilities attached to the ranks it gets right, and conditional NLL can improve while the tail ordering that sourcing consumes inverts. Both must be reported; neither may be traded silently for the other.

(iv) *T4 (marks).* Mark models are scored by proper scoring rules per mark (Section 7.3), with the same temporal split.

> **Design decision (D7.1 — the evaluation constitution).**
> The definitions in Definition 7.1, the protocols Protocols 7.3, 7.8, 7.14 and 7.15, and the ladder of Section 7.4 are the only admissible evaluation for models in this repository. A new metric enters the constitution only together with a new consuming action, per decision D1.1 of Chapter 1; a metric may never be added because a model happens to do well on it. Rationale: post hoc metric selection is the cheapest way to fool ourselves, and the rehearsal campaigns showed that every family looks good on *some* score. Reversal trigger: a change in the decision layer (Chapter 12) that adds or retires a consuming action.

7.2 Continuous-time goodness of fit

7.2.1 Time rescaling in practice

The instrument for event-time fit is the time-rescaling theorem, proved in Chapter 3 from Watanabe's characterization and restated here in the form the diagnostics use (Brown et al., 2002; Ogata, 1988; Daley and Vere-Jones, 2003).

Theorem 7.2 (Time rescaling, evaluation form)**.** *Let N be a point process on $[0,T]$ with event times $t_1 < t_2 < \cdots < t_n$, adapted to $\{\mathcal{F}_t\}$, admitting a strictly positive conditional intensity $\lambda(t)$ whose compensator $\Lambda(t) = \int_0^t \lambda(u)\,\mathrm{d}u$ is a.s. finite and continuous. Then the rescaled times $\tau_m = \Lambda(t_m)$ form a unit-rate Poisson process on $[0, \Lambda(T)]$; equivalently, with $\tau_0 = 0$, the gaps*

$$\xi_m \;=\; \tau_m - \tau_{m-1} \;\sim\; \mathrm{Exp}(1) \quad \textit{i.i.d.}, \qquad u_m \;=\; 1 - e^{-\xi_m} \;\sim\; U(0,1) \quad \textit{i.i.d.} \tag{7.1}$$

Proof. Since $\lambda > 0$, Λ is continuous and strictly increasing, so $s \mapsto \Lambda^{-1}(s)$ is a well-defined family of $\{\mathcal{F}_t\}$-stopping times. Set $\tilde{N}(s) = N(\Lambda^{-1}(s))$ and $\tilde{\mathcal{F}}_s = \mathcal{F}_{\Lambda^{-1}(s)}$. Because $N(t) - \Lambda(t)$ is a local martingale (Chapter 2, Equation (2.2)), optional stopping at $\Lambda^{-1}(s)$ shows $\tilde{N}(s) - s$ is a local $\{\tilde{\mathcal{F}}_s\}$-martingale: $\tilde{N}$ is a counting process with deterministic continuous compensator s, hence a unit-rate Poisson process by Watanabe's characterization (Chapter 3). The gap and uniform statements are the standard exponential-spacings and probability-integral facts. The proof fails exactly where its hypotheses do: if λ vanishes on at-risk gaps, rescale over at-risk time only; and if the t_m are jittered (defect (C4)), the theorem still holds for the *true* times but the computed τ_m use corrupted ones — the test then interrogates model and jitter jointly (Section 7.7). $\square$

In practice Λ is the *fitted* compensator $\hat{\Lambda}$, and three details separate a usable diagnostic from a misleading one: the open final interval, the null distribution under estimated parameters, and localization.

Protocol 7.3 (Time-rescaling battery)**.**

(a) *Residuals.* Compute ξ_m from Equation (7.1) with the fitted compensator, integrating over at-risk time only, per sector and pooled (`hawkes_calibration/mbpp/gof.py`, `time_rescaling_residuals`).

(b) *The last interval is censored.* The final gap $\hat{\Lambda}(T) - \tau_n$ is a right-censored Exp(1) draw by memorylessness. It is either excluded from the KS sample or included as a censored observation in a likelihood-based check; it is *never* treated as a complete gap. Treating it as complete injects one draw that is stochastically too small per unit — across many sectors this manufactures a spurious excess of short gaps, i.e. phantom clustering.

(c) *KS and QQ.* Test the complete gaps against Exp(1) (KS, `ks_test_exp1`) and plot QQ coordinates (`qq_exp1`) with envelope bands. Report the statistic per sector and pooled.

(d) *Localization.* A rejection is the beginning, not the end: identify *which* sectors, which calendar windows, and which covariate strata carry the misfit, using the stratified residuals of Equation (7.2) — this is Ogata's residual-analysis program (Ogata, 1988).

(e) *Defect exposure.* Under (C4) this battery is run only at the resolution the dates honestly support (daily-resolved deals in the event-time track; never sub-week on jittered dates); under (C3) the fitted $\hat{\lambda}$ absorbs dead exposure and the test inherits the optimism of Warning 7.16.

Warning 7.4 (Estimated parameters make standard bands optimistic for the model). The KS null distribution assumes a fixed hypothesized law. When the compensator is fitted on the same data, $\hat{\Lambda}$ adapts to the sample, the KS statistic is stochastically smaller than under a fixed truth, and standard p-values are conservative: a model can *pass* while wrong. The honest null is simulation-based: simulate from the fitted model, refit, recompute the statistic, and compare the observed value to this refit-in-loop distribution. The envelope machinery of Protocol 7.5 provides the harness; held-out-era residuals (fit on train, rescale on test) are the cheap partial fix and are the repository default.

7.2.2 Martingale residuals over strata

Time rescaling compresses the whole fit into one distributional statement; martingale residuals decompose it. For any predictable stratum g — a set of (unit, time) pairs selected by sector, calendar window, or covariate values known at t^- — define

$$R(g) \;=\; \sum_m \mathbf{1}\{(e_m, t_m) \in g\} \;-\; \sum_e \int_0^T \mathbf{1}\{(e,u) \in g\}\, Y_e(u)\, \hat{\lambda}_e(u)\,\mathrm{d}u, \qquad Z(g) \;=\; \frac{R(g)}{\sqrt{\hat{\Lambda}(g)}}, \tag{7.2}$$

where $\hat{\Lambda}(g)$ is the integral term. Under a correctly specified model $R(g)$ is the terminal value of a mean-zero martingale with predictable variation $\Lambda(g)$, so $Z(g)$ is approximately standard normal when $\hat{\Lambda}(g)$ is large (Andersen et al., 1993). The design use is *systematic lack-of-fit localization*: compute $Z(g)$ over a fixed grid of strata — sector $\times$ calendar quarter, staleness-age bands of $a_i(t)$, macro-covariate terciles — and read the sign pattern. A block of positive Z across all sectors in one era is a missing calendar covariate; positive Z concentrated at high staleness age is the (C5) signature that Chapter 8 predicts; one wild cell is usually data hygiene, not model failure. Isolated $|Z| > 2$ values among dozens of strata are expected; structure is what indicts the model.

7.2.3 Simulation-based envelope tests

Likelihood-based diagnostics ask whether the data look probable under the model; envelope tests ask whether the model's own sample paths look like the data. They are the only GOF instrument that sees *functionals we did not fit.*

Protocol 7.5 (Envelope battery).

(a) Simulate B (default 999) replicates of the evaluation era from the fitted model using the simulators of Chapter 5: Ogata thinning (Ogata, 1981), the cluster representation (Hawkes and Oakes, 1974), or exact exponential-kernel simulation (Dassios and Zhao, 2013); pass each replicate through the same observation layer (aggregation, at-risk masks) as the real data.

(b) Compute, per replicate and for the data, a fixed pre-registered statistic vector: weekly-count autocorrelations at lags 1–8, the dispersion index of Proposition 7.7, the coefficient of variation of inter-event gaps, the maximum weekly count, and the sector-level count cross-correlations.

(c) Reject a statistic at (pointwise) level $\alpha \approx 2/(B+1)$ if the observed value falls outside the central simulated band; report all statistics together and treat the multiplicity honestly — one excursion among a dozen statistics at $B = 999$ is weak evidence.

(d) *Defect exposure.* Simulating through the observation layer is mandatory: envelopes from the latent (uncorrupted) model test the wrong null. (C1)/(C4) are reproduced by construction; (C3) cannot be reproduced (we do not know who is dead) and the envelope inherits the same contaminated risk set as the fit.

The double-use caveat of Warning 7.4 applies verbatim: parameters estimated from the data pull the envelope toward the data, so envelope non-rejection is weak; envelope *rejection* is strong and localizes what the model cannot reproduce (typically: the real data's overdispersion or its cross-sector comovement).

7.3 Count-level goodness of fit

At weekly resolution the model emits a predictive distribution $F_{s,w}$ for each sector-week count $Y_{s,w}$, and the calibration question becomes: are the observed counts distributed as the model says? The continuous-data answer — the probability integral transform $F(Y) \sim U(0,1)$ — fails for discrete data: $F(Y)$ is a discrete random variable, stochastically larger than uniform, and naive PIT histograms read miscalibration where there is none, most severely in sparse cells where $\mathbb{P}(Y = 0)$ is large. Two standard repairs exist (Smith, 1985; Brockwell, 2007; Czado et al., 2009).

Proposition 7.6 (Randomized PIT)**.** *Let Y have cumulative distribution F on $\mathbb{N}_0$ with $p_y = F(y) - F(y-1)$, and let $V \sim U(0,1)$ independent of Y. Then*

$$U \;=\; F(Y-1) + V\,p_Y \tag{7.3}$$

satisfies $U \sim U(0,1)$.

Proof. Fix $u \in (0,1)$ and let y^* be the unique value with $F(y^*-1) < u \le F(y^*)$. Conditioning on $Y = y$: if $F(y) \le u$ (i.e. $y < y^*$) then $U \le u$ with probability 1; if $y > y^*$ then $F(y-1) \ge u$ and $U \le u$ has probability 0; if $y = y^*$ then $\mathbb{P}(U \le u \mid Y = y^*) = (u - F(y^*-1))/p_{y^*}$. Summing,

$$\mathbb{P}(U \le u) = \sum_{y<y^*} p_y + p_{y^*} \cdot \frac{u - F(y^*-1)}{p_{y^*}} = F(y^*-1) + u - F(y^*-1) = u. \qquad \square$$

Under a correctly calibrated sequence of predictive distributions the $U_{s,w}$ are i.i.d. uniform and the whole continuous toolkit (KS, histograms, drift plots) applies. The price is randomization: two analysts get two dashboards. The non-randomized alternative of Czado et al. (2009)

replaces each draw by its conditional expectation: with $a = F_{s,w}(y_{s,w} - 1)$, $b = F_{s,w}(y_{s,w})$, define the conditional PIT $c(u; a, b) = \min\{\max\{(u - a)/(b - a), 0\}, 1\}$ and the aggregate $\bar{F}(u) = \frac{1}{n} \sum_{s,w} c(u; a_{s,w}, b_{s,w})$; calibration implies $\mathbb{E}\,\bar{F}(u) = u$, and the histogram of increments of $\bar{F}$ is the deterministic analogue of the PIT histogram. Repository rule: the non-randomized version drives dashboards and regressions (reproducible), the randomized version drives formal tests (exact uniformity).

Proposition 7.7 (Dispersion index and its sampling distribution)**.** *Let* $Y_i \sim \text{Poisson}(\mu_i)$ *independently,* $i = 1, \dots, n$*, and*

$$\hat{\phi} \;=\; \frac{1}{n-p} \sum_{i=1}^{n} \frac{(Y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}, \tag{7.4}$$

with p *fitted parameters. With known means* ($p = 0$, $\hat{\mu}_i = \mu_i$)*:* $\mathbb{E}\,\hat{\phi} = 1$ *and* $\operatorname{Var}(\hat{\phi}) = n^{-2} \sum_i (2 + 1/\mu_i)$*, so that if* $n^{-1} \sum_i 1/\mu_i \to \bar{m} < \infty$*,*

$$\sqrt{n}\,(\hat{\phi} - 1) \;\xrightarrow{d}\; \mathcal{N}(0,\; 2 + \bar{m}).$$

Estimated means reduce the degrees of freedom, whence the divisor $n - p$*.*

Proof. For $Y \sim \text{Poisson}(\mu)$ the central moments are $m_2 = \mu$, $m_4 = \mu + 3\mu^2$. Hence $\mathbb{E}[(Y - \mu)^2/\mu] = 1$ and $\operatorname{Var}[(Y - \mu)^2/\mu] = (m_4 - m_2^2)/\mu^2 = (\mu + 2\mu^2)/\mu^2 = 2 + 1/\mu$. Independence gives the variance sum; the Lindeberg CLT (bounded $1/\mu_i$ suffices) gives normality. The proof shows what practitioners forget: the null variance is *not* $2/n$ — the $1/\mu_i$ term dominates in sparse cells, and sector-weeks with $\mu_i \lesssim 1$ (common in this data) widen the honest acceptance band. Judging $\hat{\phi}$ against $1 \pm 2\sqrt{2/(n-p)}$ in a sparse panel manufactures false overdispersion alarms. □

Overdispersion ($\hat{\phi} > 1$) is the escalation trigger of the model ladder — it says history or heterogeneity is missing — but it is *never* evidence for excitation specifically, because frailty and contagion produce it identically (Chapter 5, Section 7.5).

Protocol 7.8 (Count-calibration battery)**.** For every count model, on the evaluation era: (a) non-randomized PIT diagram pooled and per sector; (b) randomized-PIT KS test, seeds logged; (c) dispersion Equation (7.4) with the honest band of Proposition 7.7, reported *alongside every count NLL*; (d) reliability per sector-week: bin cells by predicted mean decile, compare observed to predicted bin means with Poisson error bars; (e) PIT drift over calendar time, plotted — drift is the cheapest detector of vendor-coverage change and of (C5) staleness confounding. Defect exposure: complete sector counts make this battery robust to (C6) at sector level; (C3) does not bite sector counts (sectors do not die); (C4) is absorbed by construction — weekly counts are the honest resolution.

Scoring rules for counts. A scoring rule $S(F, y)$ assigns a loss to forecasting F when y materializes; it is *proper* if $\mathbb{E}_{Y \sim G}\, S(G, Y) \le \mathbb{E}_{Y \sim G}\, S(F, Y)$ for all F, and *strictly proper* if equality forces $F = G$ (Gneiting and Raftery, 2007). Propriety is not decoration: an improper score rewards hedged or sharpened forecasts and corrupts every comparison built on it. We use two:

$$\text{logS}(F, y) \;=\; -\log f(y), \qquad \text{RPS}(F, y) \;=\; \sum_{k=0}^{\infty} \big(F(k) - \mathbf{1}\{y \le k\}\big)^2. \tag{7.5}$$

Proposition 7.9 (Strict propriety of the log score)**.** *On the class of probability mass functions with common support,* logS *is strictly proper.*

Proof. Let g be the true pmf and f any competitor with the same support. Then

$$\mathbb{E}_g[-\log f(Y)] - \mathbb{E}_g[-\log g(Y)] \;=\; \sum_y g(y) \log \frac{g(y)}{f(y)} \;=\; \mathrm{KL}(g \,\|\, f) \;\geq\; 0,$$

with equality iff $f = g$, by Jensen's inequality applied to the strictly concave logarithm ($\sum g \log(f/g) \leq \log \sum f = 0$, strict unless f/g is constant). Hence the expected score is uniquely minimized at the truth. If supports differ the score can be infinite — which is a feature: a model that assigns zero mass to an observed count is infinitely wrong, and the repository treats $f(y) = 0$ on observed data as a bug, not an outlier. □

The ranked probability score is proper on distributions with finite mean (Gneiting and Raftery, 2007; Czado et al., 2009) but not local: it rewards mass *near* the outcome. The two scores are complementary — logS is the likelihood and dominates the ladder (Section 7.4); RPS is distance-sensitive and robust to a single mis-tailed cell — and both are reported, logS as primary, RPS as the sanity companion.

7.4 Model comparison

7.4.1 The held-out NLL ladder

The primary comparison metric of the repository is held-out negative log likelihood, in two units: *nats per event* for event-time models ($-\ell/n$ over test-era events) and *nats per sector-week cell* for count models ($-\frac{1}{|\text{cells}|}\sum_{s,w} \log \hat{f}_{s,w}(y_{s,w})$). Every model is reported as a difference against three named baselines fitted on the same train era: *random* (homogeneous Poisson, one global rate), *historical-mean* (per-sector constant rate), and *seasonal NHPP* (the covariate/seasonal Poisson model of Chapter 4). A model that has not been run against all three has not been evaluated.

Warning 7.10 (Units are not exchangeable)*.* Nats per event and nats per cell are different currencies: they condition on different information and normalize by different denominators. No number in one unit may be compared to a number in the other. Comparing an event-time model to a count model requires evaluating both at a common, honest resolution — in practice, aggregating the event-time model's predictive distribution to weekly cells and paying the aggregation cost explicitly (Chapter 6).

Design decision (D7.2 — NLL ladder with the covariates-dominate prior).

Held-out NLL against the three named baselines is the primary promotion metric for T2 models, with Protocol 7.15 governing the split. The standing prior, inherited from the rehearsal campaigns (Appendix C), is that *covariates dominate and excitation is decimal points*: moving from the seasonal baseline to the covariate NHPP was worth +3.5/+5.7/+1.5 nats across the three rehearsal worlds, while adding weekly excitation contributed +0.04/−1.15/+0.53. Consequently, any claim that excitation matters must clear the seasonal NHPP *and* the covariate NHPP under rolling-origin evaluation, not merely improve on a weak baseline. Reversal trigger: excitation contributing ≥ 0.5 nats per cell over the covariate NHPP on real data in two consecutive campaigns retires the prior.

AIC and BIC have a narrow legal scope here: quick screening *within* a family, on the train era, when models differ by a few regular interior parameters. Three limits keep them off headlines. First, they are in-sample devices, and our risk is out-of-era: a temporal regime shift is invisible to them. Second, their penalty calculus assumes regular asymptotics, which fail exactly at the comparison we care most about — excitation present versus absent — because $\kappa = 0$ is a boundary point, where even the effective parameter count is wrong (below). Third, under misspecification neither estimates what its derivation promises. Held-out NLL is the arbiter; information criteria are a screen.

7.4.2 Testing for excitation at the boundary

The natural test of "is there any excitation?" is the likelihood ratio between the covariate NHPP ($\kappa = 0$) and the exponential-Hawkes extension ($\kappa \geq 0$). But $\kappa = 0$ lies on the boundary of the parameter space — negative branching is not in the model — so the classical χ^2_1 asymptotics fail (Self and Liang, 1987).

Theorem 7.11 (Chi-bar-square limit for one bounded parameter)**.** *Let the model have parameter (κ, ψ) with $\kappa \geq 0$ on the boundary at the null, ψ interior nuisance parameters, and the decay θ* fixed *(a decay bank, Chapter 5). Assume the LAN regularity conditions of Self and Liang (1987): consistency, a nonsingular Fisher information $\mathcal{I}$ at the true value $(0, \psi_0)$, and differentiability in quadratic mean in a neighborhood intersected with the cone $\{\kappa \geq 0\}$. Then the likelihood-ratio statistic $\lambda_n = 2\big(\sup_{\kappa \geq 0, \psi} \ell - \sup_\psi \ell|_{\kappa=0}\big)$ satisfies*

$$\lambda_n \xrightarrow{d} \tfrac{1}{2}\,\chi^2_0 + \tfrac{1}{2}\,\chi^2_1, \qquad i.e. \qquad \mathbb{P}(\lambda_n \leq x) \to \tfrac{1}{2} + \tfrac{1}{2}\,\mathbb{P}(\chi^2_1 \leq x), \tag{7.6}$$

where χ^2_0 is the point mass at zero.

Proof. Profile out ψ: under LAN, uniformly over $\kappa = O(n^{-1/2})$,

$$\ell(\kappa) - \ell(0) \;=\; \kappa\,\tilde{S}_n \;-\; \tfrac{1}{2}\,\kappa^2\,\tilde{\mathcal{I}}_n \;+\; o_p(1),$$

where $\tilde{S}_n$ is the efficient score for κ (the score residualized on the nuisance scores) and $\tilde{\mathcal{I}}_n$ the efficient information, with $Z_n := \tilde{S}_n/\tilde{\mathcal{I}}_n^{1/2} \xrightarrow{d} \mathcal{N}(0,1)$ at the null. Maximizing the quadratic over the half-line $\kappa \geq 0$ gives $\hat{\kappa} = \max(\tilde{S}_n, 0)/\tilde{\mathcal{I}}_n$ and

$$\lambda_n \;=\; \big(\max(Z_n, 0)\big)^2 + o_p(1).$$

For $Z \sim \mathcal{N}(0,1)$: $\mathbb{P}(\max(Z,0)^2 \leq x) = \mathbb{P}(Z \leq 0) + \mathbb{P}(0 < Z \leq \sqrt{x}) = \frac{1}{2} + \frac{1}{2}\mathbb{P}(\chi^2_1 \leq x)$. The mixture is exactly the geometry: with probability $\frac{1}{2}$ the unconstrained optimum wants negative excitation, the constrained optimum sits at the boundary, and the LR is zero. When more than one parameter is bounded (e.g. a nonnegative excitation matrix), the mixture weights become the probabilities that the projected Gaussian lands on each face of the cone (Self and Liang, 1987), and closed forms disappear — simulate instead. □

Practical rule. The p-value for the boundary LRT is $p = \frac{1}{2}\,\mathbb{P}(\chi^2_1 \geq \lambda_{\text{obs}})$ — halve the naive p-value; the 5% critical value is 2.71, not 3.84. A team using χ^2_1 tables is running a conservative test and will under-detect excitation; the direction of this error is at least safe, which is why the mistake survives unnoticed.

Warning 7.12 (When the decay is free, the mixture is wrong). Under $H_0 : \kappa = 0$ the decay θ is unidentified — every θ yields the same null model — so if θ is estimated, the LR is effectively a supremum over the decay bank and is stochastically larger than Equation (7.6); the 50:50 rule then *over*-detects excitation. Two legal responses: (i) test per fixed decay-bank member (each obeys Theorem 7.11) with a multiplicity correction across the bank; (ii) parametric bootstrap — simulate from the fitted null NHPP, refit both models, and compare λ_{obs} to the simulated null distribution, reusing the envelope harness of Protocol 7.5. Separately: Wald standard errors from the inverse Hessian are invalid for any parameter whose estimate sits at a boundary (fitted zeros in nonnegative excitation matrices accumulate there); report profile-likelihood intervals or held-out NLL instead.

7.5 Identifiability diagnostics

A model comparison is only as meaningful as the parameters it compares are identified. Three diagnostics are standing obligations for the Hawkes layer.

The κ–θ ridge. Define the profile likelihood $\ell_p(\kappa) = \max_{\theta,\psi} \ell(\kappa, \theta, \psi)$ and its two-dimensional analogue over (κ, θ). On event-time data the profile is curved in both directions at sufficient volume; on aggregated counts it is not, and the reason is elementary.

Proposition 7.13 (Stationary counts pin κ and hide θ)**.** *For a stationary exponential Hawkes process with baseline μ and kernel $h(\tau) = \kappa\theta e^{-\theta\tau}$, $\kappa < 1$, the mean bucket count over width Δ is $\Xi = \mu\,\Delta/(1-\kappa)$. Hence*

$$\frac{\partial \Xi}{\partial \kappa} = \frac{\mu\,\Delta}{(1-\kappa)^2} > 0, \qquad \frac{\partial \Xi}{\partial \theta} = 0 :$$

first moments of counts identify κ and carry zero *information about θ. All θ-information in counts lives in the temporal covariance structure, whose sensitivity to θ collapses at rate $\Delta^2 e^{-2\theta(1-\kappa)\Delta}$ as the bucket widens (Chapter 6).*

Proof. The stationary mean intensity solves the first-moment equation $\bar{\xi} = \mu + \bar{\xi} \int_0^\infty h(\tau)\,\mathrm{d}\tau = \mu + \kappa\,\bar{\xi}$ (Chapter 5), so $\bar{\xi} = \mu/(1-\kappa)$, and $\Xi = \bar{\xi}\Delta$ does not involve θ. Differentiation is immediate. The covariance statement is Chapter 6's Fisher-information computation, restated here because it is the operative rule: the exponential factor beats any polynomial, so no amount of clever estimation recovers a timescale from buckets much wider than the kernel's half-life. □

The consequences were measured (Appendix C): θ estimated from 28-day buckets came in biased low by ∼20% with a deceptively tight spread — the signature of an unidentified direction collapsing onto a ridge — while κ was stable at every width. The standing rule, restated from Chapter 6: bucket width at most one kernel half-life is safe for timing; four or more half-lives destroys it; when in the destroyed regime, fix θ to a decay bank and report κ only. Operationally, every Hawkes fit ships its profile slices: if ℓ_p varies by less than 0.5 nats across an order of magnitude in θ, the fit *declares* θ unidentified and the reporting layer refuses to print a decay estimate.

Frailty versus contagion: the upper-bound rule. Overdispersion and clustering are produced identically by contagion (past deals raise future intensity) and by frailty (persistent latent heterogeneity) (Duffie et al., 2009; Azizpour et al., 2018); a fitted Hawkes model channels both into $\hat{\kappa}$. The rehearsal quantified the masquerade: under a latent AR(0.90) frailty factor the fitted effective excitation radius came out 2.4× the truth (Appendix C). Since additional structure (shared covariates, factor models) can only remove apparent excitation, every fitted branching ratio in this repository is reported as an *upper bound* on contagion — the sentence "$\hat{\kappa} = 0.3$" is banned; the sentence "excitation plus unmodeled common factors account for at most $\hat{\kappa} = 0.3$" is the template (Chapter 5 develops the decomposition attempts and their limits).

Parameter recovery as a shipping gate.

Protocol 7.14 (Parameter-recovery rehearsal)**.** No new model family ships until it passes, on synthetic data, all of:

(a) *Gradient check.* Analytic gradients of the likelihood agree with finite differences at relative error 10^{-6}–10^{-8} at random interior points — the repository's standing discipline (Appendix C); a family whose gradients do not check does not proceed to fitting.

(b) *Recovery.* Simulate ≥ 20 datasets from the family at parameter values matched to fitted real-data scale, pass them through the synthetic observation layer with defects (C1)–(C6) active (Section 2.5), fit with the production code path, and verify: parameters claimed interpretable are recovered with bias small against their sampling spread; parameters known unidentified under the observation scheme (e.g. θ under coarse aggregation, Proposition 7.13) are *declared* non-recoverable, and the declaration is part of the family's documentation.

(c) *Coverage.* Whatever interval procedure the family advertises (profile, bootstrap, posterior) achieves approximately nominal coverage in the rehearsal.

(d) *Archival.* The rehearsal is an `experiments/` script with logged seed and config, its results filed under the conventions of Protocol 7.15(d).

A family failing (b) for some parameter may still ship *for prediction*, but that parameter is barred from interpretation until a passing rehearsal exists. Defect exposure: the gate is exactly as honest as the observation layer's reproduction of (C1)–(C6); (C3)'s unlabeled exits are simulated, which is the one place we can rehearse a defect we cannot observe.

7.6 The temporal evaluation protocol

All of the above presumes an evaluation era. Choosing it is not bookkeeping: with 13.5 years of data spanning a zero-rate boom (2021), a rate-shock bust (2022–2023), and a partial normalization (2024), the split determines which regime the headline numbers describe.

Protocol 7.15 (Temporal evaluation)**.**

(a) *Development split (fixed).* One split serves all model development: train on data through 2019-12-31; validate (model selection, hyperparameters, calibration maps) on 2020-01-01–2021-12-31; test on 2022-01-01–2024-06-30, touched only when a family is frozen. No result computed on the test era before freezing may influence any modeling choice.

(b) *Rolling-origin evaluation (headline).* Frozen families are reported by rolling origin: expanding-window refits at quarterly steps across the test era, each evaluated on the following quarter; the headline number is the mean across origins with its range. Rolling origins convert one number into a small distribution and are the only defense against a verdict that is an artifact of a single split point.

(c) *No post-split information, anywhere.* Every transform — imputation, scaling, feature construction, probability recalibration — obeys Protocol 2.5: fitted on train (calibration maps on validation), applied frozen downstream. Covariate support is checked: test-era covariate ranges are reported next to every NLL, because a log-linear intensity extrapolated outside the train support is exponential extrapolation.

(d) *Reproducibility.* Every evaluation run logs its seed, config hash, split boundaries, and code version; randomized-PIT draws and bootstrap resamples are seeded; results are machine-readable files under `results/`, one per experiment, following the `experiments/` conventions.

(e) *Defect exposure.* (C1) is respected by construction (the test era ends at the administrative censoring time and open spells carry their survival factors); (C2)/(C3) exposure is inherited from the risk sets used, and every reported metric names its risk-set rule (Section 7.7).

Design decision (D7.3 — default temporal split).

The default development split is train $\leq$ 2019, validation 2020–2021, test 2022–June 2024, as in Protocol 7.15(a). Rationale: the validation era contains a violent regime change (COVID shock, then the 2021 boom), so model selection is stress-tested against nonstationarity rather than sheltered from it; the test era contains the bust and normalization, which is the regime a deployed model faces next; and train ends before any of it, so no family is selected on the regime it will be scored on. The known risk is that 2021–2023 spans a structural break, so fixed-split verdicts may be regime artifacts — which is precisely why headlines come from rolling origins, not from the fixed split. Reversal trigger: if rolling-origin family orderings disagree with fixed-split validation orderings in two consecutive campaigns, the fixed split is replaced by a multi-split (nested rolling) development scheme and this box is rewritten.

7.7 Reporting discipline and defect exposure

Design decision (D7.4 — no number without its caveat).

The following pairings are mandatory in every report, dashboard, README table, and paper draft produced from this repository:

(i) no hazard or intensity estimate without its *risk-set rule* (who was at risk, from when, closed by what);

(ii) no branching ratio without its *upper-bound caveat* (Section 7.5);

(iii) no probability without its *calibration plot* (reliability diagram or PIT, per Definition 7.1);

(iv) no ranking metric without its *risk-set definition* (the top-5 rate 0.14 vs 0.29 observed-vs-oracle gap of Appendix C is the standing proof that the risk set, not the model, can be the largest term);

(v) no count NLL without its *dispersion* printed alongside (Protocol 7.8).

Rationale: each rule is the fossil of a specific rehearsal failure in which the naked number was misleading and its caveat reversed the conclusion. Reversal trigger: none anticipated; amendments add pairings, they do not remove them.

Defect declaration (per decision D2.1). The “models” of this chapter are the tests themselves, and they are corrupted by the same six defects of Table 2.1 they are meant to police. (C1) *robust*: administrative censoring is handled exactly — the open final rescaled interval is treated as censored (Protocol 7.3b) and censored cells enter count likelihoods with their survival factors. (C2) *robust at sector level, conditional at firm level*: all firm-level GOF statements are statements about the entered (post-first-deal) universe; the protocols do not correct the selection, they inherit Chapter 11's scope. (C3) *vulnerable, optimistic* — see Warning 7.16. (C4) *corrects by resolution*: the batteries are run at the honest resolution only — weekly PIT always; event-time rescaling only on the daily-resolved track — so date jitter is absorbed rather than tested against; the cost is that sub-weekly misfit is invisible by design. (C5) *vulnerable, detectable*: stale covariates push lack of fit into the staleness-age strata of Equation (7.2) and into calendar PIT drift; the diagnostics detect the damage but cannot attribute it without Chapter 8's machinery. (C6) *robust for the ground process* (sector counts are complete), *thinned for the mark factor* (ranker metrics run on tracked risk sets with the outside option of Chapter 11).

Warning 7.16 (GOF cannot detect dead exposure). Under (C3), dead firms held at risk deflate every fitted firm-level hazard — and the *same* contaminated risk set is used at evaluation time, so the deflated model is scored against the corruption that produced it: held-out NLL, time-rescaling, and reliability can all pass while the hazard is wrong for every living firm. Goodness of fit is a consistency check between model and observed data, not between model and truth. This is why risk-set truthfulness has its own diagnostics (Chapters 4 and 10), why the exit-classifier AUC of 0.877 was insufficient to repair it (Appendix C), and why rule (i) of decision D7.4 exists: the risk-set rule *is* part of the estimate.

Code hooks.

Existing: `hawkes_calibration/mbpp/gof.py` implements `time_rescaling_residuals`, `ks_test_exp1` (numpy-only KS), `qq_exp1`, `poisson_pearson_residuals`, and `dispersion` — the cores of Protocols 7.3 and 7.8. To be written: `hawkes_calibration/eval/`, the evaluation harness that makes the constitution executable: `hawkes_calibration/eval/ladders.py` (held-out NLL versus the three named baselines, nats per event and per cell, decision D7.2); `hawkes_calibration/eval/pit.py` (randomized and non-randomized PIT, reliability binning, PIT-drift series); `hawkes_calibration/eval/rescaling.py` (wraps `mbpp.gof` with censored-final-gap handling, stratified martingale residuals Equation (7.2), bootstrap bands per Warning 7.4); `hawkes_calibration/eval/envelope.py` (simulation envelopes over `hawkes_calibration/eventtime/simulate.py` and `hawkes_calibration/mbpp/ic_simulate.py`, doubling as the bootstrap null for Warning 7.12); `hawkes_calibration/eval/boundary.py` (chi-bar LRT, Theorem 7.11); `hawkes_calibration/eval/rolling.py` (the rolling-origin driver of Protocol 7.15). Conventions: every campaign is an `experiments/expNN_name.py` script emitting seeded, config-stamped JSON to `results/`; the recovery gate of Protocol 7.14 reuses `synthetic/` world generation; profile-likelihood slices and the gradient check attach to the fitters in `hawkes_calibration/sector_stability.py`, `hawkes_calibration/eventtime/estimate.py`, and `hawkes_calibration/mbpp/`.

Part III

The Covariate Problem

Chapter 8

The Stale-Covariate Problem

8.1 Why staleness owns a chapter

Defect (C5) of Chapter 2 is easy to state and easy to underestimate. The firm covariates that every model in this document consumes — financials, headcount, capitalization — are not observed as processes. PitchBook records them when, and only when, the firm itself transacts: each deal row carries the covariates as reported around that deal, and between deals the pipeline carries the last recorded value forward (Equation (2.1), implemented in `synthetic/loaders.py` and to be mirrored by every real-data loader per Protocol 2.5). Two facts make this more than a measurement-error nuisance. First, the error is not noise but *drift*: the gap between the last recorded value and the current truth grows with the time since the firm's last deal. Second — and this is the point the whole chapter turns on — the observation times are event times *of the process being modeled*. The firms that have gone longest without a deal have the stalest covariates, and they are exactly the low-intensity firms. Staleness is informative: the age of a firm's covariates is itself evidence about the firm.

The rehearsal campaigns justify the chapter's rank. Design decision D2.1 of Chapter 2 records that the largest errors in the synthetic rehearsal came not from wrong model families but from unstated exposure to (C3) and (C5); the covariate-noise campaign (exp22, Appendix C) measured the mechanics — structure erodes before prediction does — and the standing verdict that *covariates dominate, excitation is decimal points* means the covariate channel is where the predictive mass lives. A defect that silently corrupts the dominant channel, differentially across the risk set, is one of this document's two reasons to exist.

The chapter proceeds in the document's standard order: formalize (Section 8.2), prove the damage in a tractable case (Section 8.3), then commit to a remedy ladder — R1, staleness-aware observed-data modeling, committed and always on (Section 8.4); R2, supervised drift modeling with regression calibration (Section 8.5); R3, an explicit covariate state space (Section 8.6); R4, joint modeling, adopted only if a defined trigger fires (Section 8.7) — and close with the bucket-model face of the same mathematics, the rehearsal evidence, and the defect declaration (Sections 8.8 and 8.9).

8.2 Formalization

Throughout, i ranges over tracked firms, N_i is firm i's deal counting process, $\mathcal{F}_t$ is the full filtration of Definition 2.1 *enlarged with the latent covariate paths*, and $\mathcal{G}_t \subset \mathcal{F}_t$ is the *observed* filtration: the internal history of the marked point process, the published macro paths, and the covariate values recorded at past deals — everything a production pipeline can know at t, nothing it cannot.

Assumption 8.1 (Latent covariate dynamics)**.** Each tracked firm carries a latent càdlàg covariate process $x_i(t) \in \mathbb{R}^q$, partitioned into *mechanical* components — deterministic functionals of the firm's own observed event history and marks (raised-to-date, deal count, time-stamped last-round descriptors) — and *drifting* components (revenue, EBITDA, headcount, latent valuation). As a working model, the drifting components follow a mean-reverting Ornstein–Uhlenbeck diffusion (Uhlenbeck and Ornstein, 1930) with sector- and macro-driven mean,

$$\mathrm{d}x_i(t) \;=\; -\Theta\,(x_i(t) - m_i(t))\,\mathrm{d}t \;+\; \Sigma_W^{1/2}\,\mathrm{d}W_i(t), \qquad m_i(t) \;=\; m_{s(i)} + \Gamma^\top x^{\mathrm{mac}}(t), \tag{8.1}$$

with $\Theta = \mathrm{diag}(\vartheta_1, \dots, \vartheta_q)$, $\vartheta_k > 0$ (the mean-reversion rates ϑ are unrelated to the excitation decay θ of Chapter 5), Σ_W diagonal, and W_i idiosyncratic across firms given sector and macro. Where stationarity is invoked, coordinate k has stationary variance $\sigma^2_{X,k} = \Sigma_{W,kk}/(2\vartheta_k)$. Only the qualitative content — mean reversion toward an observable anchor, conditional variance growing in elapsed time — is load-bearing; the Gaussian linear form is what makes the propositions below exact.

Read concretely, Equation (8.1) says: a firm's revenue and headcount track an anchor set by its sector's trajectory and the macro state — rate cuts and sector booms lift the anchor, contractions lower it — while ϑ_k sets the memory of coordinate k, with covariate half-life $\log 2/\vartheta_k$. Nothing in the data pins these half-lives a priori — headcount is plausibly slow, revenue faster, reported EBITDA noisier still — and estimating them is exactly what the drift dataset of Section 8.5 is for. The OU form is chosen because it is the tractable member of its qualitative class: it is the Gaussian Markov process with mean reversion, it closes the filtering recursions of Section 8.6 in closed form, and it makes every distortion formula in Section 8.3 exact rather than first-order. When the working model fails — heavy-tailed revenue jumps, regime-switching growth — the exact constants below degrade to first-order approximations, but the monotone conclusions (attenuation increasing in age, variance increasing in age, survival tilt downward) survive, because they use only the qualitative content.

Assumption 8.2 (LOCF observation scheme)**.** The data delivers, for each firm and each t past its entry V_i,

$$\tilde{x}_i(t) \;=\; x_i(\tau_i(t)), \qquad \tau_i(t) \;=\; \sup\{t_m < t : i_m = i\}, \qquad a_i(t) \;=\; t - \tau_i(t), \tag{8.2}$$

with the strict inequality of Protocol 2.5(b): the value used at t comes from deals *strictly before* t, so that $\tilde{x}_i$ and the staleness age a_i are left-continuous in t. At its own deal times the firm's drifting covariates are observed (as reported in the deal row); Section 8.6 adds reporting noise. For mechanical components LOCF is *exact* — they are step functions that only move at own deals — so (C5) binds on the drifting components alone. This split is the basis of the column classification in Appendix B and of decision D8.3 below.

Proposition 8.3 (The observation scheme is internal and informative)**.** *Under Assumptions 8.1 and 8.2:*

(i) Each observation time of firm i's covariates is a jump time of N_i, hence a stopping time of the modeled filtration $\{\mathcal{F}_t\}$; the processes $\tau_i(t)$, $a_i(t)$, and $\tilde{x}_i(t)$ are left-continuous and adapted to $\{\mathcal{G}_t\}$, hence $\mathcal{G}$-predictable — they are legitimate intensity covariates.

(ii) The staleness age is a deterministic functional of the event history: $\{a_i(t) > u\} = \{N_i([t-u, t)) = 0\}$; it is the backward recurrence time of the modeled process.

(iii) The scheme is informative*: whenever the true intensity λ_i depends on x_i, the conditional law of $x_i(t)$ given $(\tilde{x}_i(t), a_i(t))$ differs from the law obtained from the free dynamics Equation (8.1) alone, because $\mathbb{P}(a_i(t) > u \mid \mathcal{F}^X_t \vee \mathcal{F}_\tau) = \exp\big(-\int_{t-u}^{t} Y_i(v)\,\lambda_i(v)\,\mathrm{d}v\big)$ depends on the latent covariate path over the gap.*

Proof. (i) Jump times of an adapted counting process are stopping times; between jumps τ_i is frozen and a_i grows deterministically, and the strict-inequality convention in Equation (8.2) makes all three processes left-continuous adapted, hence predictable. (ii) is Equation (8.2) read backwards. (iii) The displayed survival probability is the standard conditional-inhomogeneous-Poisson formula given the intensity path; if λ_i is x_i-measurable and nonconstant in it, the event $\{a_i(t) > u\}$ has different probability under different covariate paths, so by Bayes the conditional law of the path given the gap is tilted (Lemma 8.8 computes the tilt). □

In the Kalbfleisch–Prentice taxonomy of time-dependent covariates (Kalbfleisch and Prentice, 2002, Ch. 6), the macro series are *external*: they evolve regardless of the deal process, and survival-type probability statements conditioning on their future paths are legitimate. The pair $(\tilde{x}_i, a_i)$ is *internal*: it is a functional of the firm's own event history and marks. The consequence, inherited verbatim from the internal-covariate discussion of Chapter 5, is that intensities conditioned on $(\tilde{x}_i, a_i)$ retain their interpretation, while plug-in survival functionals ("probability of no deal in two years at today's covariates") do not — the covariate would not stay at today's value. One-step prediction is clean; multi-step statements must simulate the system jointly, covariate observation process included.

How much information does the age itself carry? The rehearsal gives a sharp, inherited answer (Appendix C): in the model-ladder campaign a recency-only ranker — score firms by how recently they raised — performed *worse than the popularity baseline*, because in a cooldown world recent activity is anti-signal at short range; yet the committed choice models carry an explicit gap covariate precisely because the gap is informative — with the measured caveat that its fitted functional form is DGP-specific (on the misspecified regime the gap weight tracked the generator's 30-day inhibition, not the cooldown it was designed around). The age is neither ignorable nor monotone in effect: it interacts with everything else, which is why Section 8.4 insists on interactions and why Section 8.8 argues trees are the natural estimator of that surface. The lesson to carry forward: a is simultaneously a nuisance index (how corrupted is $\tilde{x}$) and a signal (how long has the market declined to fund this firm), and a correct treatment must let the data express both roles at once.

8.3 What staleness does to a log-linear intensity

Fix one firm, drop the subscript, and let the true intensity be log-linear in the current latent covariates,

$$\lambda(t) \;=\; \exp(\beta^\top x(t)), \qquad x(t) \;=\; \tilde{x}(t) + \varepsilon(t), \tag{8.3}$$

where $\varepsilon(t) = x(t) - x(\tau)$ is the drift accumulated since the last observation. (A baseline and external covariates can be carried along untouched; they are omitted for legibility.) We fit with $\tilde{x}$ in place of x. Everything in this section is a statement about what that substitution does.

8.3.1 Attenuation in the Gaussian-drift case

Two benchmark drift laws bracket the mechanism. The first switches off mean reversion and the survival information; it isolates the pure variance effect.

Proposition 8.4 (Independent Gaussian drift: multiplicative distortion)**.** *Suppose that, given $(\tilde{x}, a) = (\tilde{x}, a)$ and under the free drift law (survival information ignored), $\varepsilon(t) \sim \mathcal{N}(0, \Sigma(a))$ independent of $\tilde{x}$ — the $\Theta \to 0$ (Brownian) limit of Equation (8.1), where $\Sigma(a) = \Sigma_W\, a$. Then the intensity implied for the observed data is*

$$\tilde{\lambda}(t) \;=\; \mathbb{E}\big[\exp(\beta^\top x(t)) \mid \tilde{x}, a\big] \;=\; \exp\Big(\beta^\top \tilde{x} \;+\; \tfrac{1}{2}\,\beta^\top \Sigma(a) \beta\Big) : \tag{8.4}$$

a staleness-dependent multiplicative distortion $\exp\{\frac{1}{2}\beta^\top\Sigma(a)\beta\} \geq 1$, *increasing in* a, *with the coefficient on* $\tilde{x}$ *unbiased at fixed* a.

Proof. $\mathbb{E}[\exp(\beta^\top\varepsilon) \mid \tilde{x}, a] = \exp(\frac{1}{2}\beta^\top\Sigma(a)\beta)$ is the Gaussian moment generating function; independence lets it factor out of $\exp(\beta^\top\tilde{x})$. □

Remark 8.5. The structure "truth = observed + independent error" is Berkson-type, and Equation (8.4) shows its signature: no slope attenuation *conditional on* a, but an omitted a-dependent term. Convexity of exp makes uncertain firms look *more* intense on average — the variance term rewards staleness — which is one half of the ranking distortion of Proposition 8.9. The classical-error benchmark, $\tilde{x} = x + U$ with $U \perp x$, gives the textbook attenuation instead: $\mathbb{E}[x \mid \tilde{x}] = \mu + \Sigma_X(\Sigma_X + \Sigma_U)^{-1}(\tilde{x} - \mu)$, so the induced log-intensity is linear in $\tilde{x}$ with slope $\Sigma_X(\Sigma_X + \Sigma_U)^{-1}\beta$ — in the scalar case $\beta\,\sigma_X^2/(\sigma_X^2 + \sigma_U^2)$ (Prentice, 1982; Carroll et al., 2006). The exp22 rehearsal campaign realizes exactly this benchmark (Section 8.8).

The second benchmark restores mean reversion; it produces genuine, staleness-graded attenuation and, as a corollary, the functional form of remedy R1.

Proposition 8.6 (Stationary OU drift: staleness-graded attenuation)**.** *Let the drifting covariates follow Equation* (8.1) *with constant mean* m, *diagonal* Θ, *stationary initial condition, and survival information ignored. Then, given* $(\tilde{x}, a)$,

$$x(t) \mid \tilde{x}, a \;\sim\; \mathcal{N}\Big(m + e^{-\Theta a}(\tilde{x} - m),\; V(a)\Big), \qquad V(a)_{kk} = \sigma_{X,k}^2(1 - e^{-2\vartheta_k a}), \tag{8.5}$$

and the observed-data intensity is

$$\tilde{\lambda}(t) \;=\; \exp\Big(\underbrace{(e^{-\Theta a}\beta)^\top\tilde{x}}_{\textit{attenuated signal}} \;+\; \underbrace{\beta^\top(I - e^{-\Theta a})m + \tfrac{1}{2}\,\beta^\top V(a)\beta}_{=\,g_0(a),\ \textit{staleness offset}}\Big). \tag{8.6}$$

At fixed staleness the coefficient on $\tilde{x}_k$ *is* $e^{-\vartheta_k a}\beta_k$*: an attenuation factor* $e^{-\vartheta_k a} \in (0, 1)$, *decreasing in the age, equal to the regression-calibration factor* $\mathrm{Cov}(x_k(t), \tilde{x}_k)/\operatorname{Var}(\tilde{x}_k)$.

Proof. Variation of constants on Equation (8.1) gives $x(t) = m + e^{-\Theta a}(x(\tau) - m) + \int_\tau^t e^{-\Theta(t-u)}\Sigma_W^{1/2}\,\mathrm{d}W(u)$; the stochastic integral is Gaussian, independent of $x(\tau)$, with the stated variance (compute $\int_0^a e^{-2\vartheta_k u}\Sigma_{W,kk}\,\mathrm{d}u$ and use $\sigma_{X,k}^2 = \Sigma_{W,kk}/2\vartheta_k$), which proves Equation (8.5). Then Equation (8.6) is the Gaussian mgf as in Proposition 8.4. For the last claim, stationarity makes $(x(\tau), x(t))$ jointly Gaussian with $\operatorname{Var}(\tilde{x}_k) = \sigma_{X,k}^2$ and $\mathrm{Cov}(x_k(t), \tilde{x}_k) = \sigma_{X,k}^2 e^{-\vartheta_k a}$, so the ratio is $e^{-\vartheta_k a}$; equivalently, time reversibility of the stationary OU lets one write $\tilde{x}_k = m_k + e^{-\vartheta_k a}(x_k(t) - m_k) + U_k$ with $U_k \perp x_k(t)$ — the classical shrink-plus-noise surrogate structure of Prentice (1982), with $e^{-\vartheta_k a}$ playing the role of $\sigma_X^2/(\sigma_X^2 + \sigma_U^2)$. □

Corollary 8.7 (Pooled fit: risk-set-weighted attenuation plus an informativeness term)**.** *Take the scalar case with centered covariate* ($m = 0$)*, and fit the misspecified model* $\lambda = \exp(c + b\,\tilde{x})$ *across the risk set while the truth is Equation* (8.6)*. In the small-signal (first-order) regime, the quasi-likelihood fit is the exposure-weighted least-squares projection of the true log-intensity onto* $(1, \tilde{x})$*, so the estimand is*

$$b \;=\; \frac{\mathrm{Cov}_w(\tilde{x},\; e^{-\vartheta a}\,\tilde{x})}{\operatorname{Var}_w(\tilde{x})}\,\beta \;+\; \frac{\mathrm{Cov}_w(\tilde{x},\; g_0(a))}{\operatorname{Var}_w(\tilde{x})}, \tag{8.7}$$

with weights w *proportional to exposure. If* $a \perp \tilde{x}$ *across the risk set,* $b = \mathbb{E}_w[e^{-\vartheta a}]\,\beta$*: pure attenuation by the average factor. Informativeness breaks the independence with a definite* *sign: if the intensity is increasing in $\beta^\top x$ with $\beta > 0$, then a is stochastically decreasing in $\tilde{x}$ (better last-seen firms raise sooner), so for the increasing offset g_0 the second term is ≤ 0 — the omitted staleness structure biases the fitted slope* further *toward zero, beyond the average attenuation.*

Proof. Linearizing the log-likelihood Equation (2.3) in the small-signal regime reduces the score equations for (c, b) to the normal equations of exposure-weighted least squares of the true log-intensity $e^{-\vartheta a}\beta\tilde{x} + g_0(a)$ on $(1, \tilde{x})$; Equation (8.7) is the resulting projection coefficient. Under independence the first ratio is $\mathbb{E}_w[e^{-\vartheta a}]$ and the second covariance vanishes. For the sign claim: a stochastically decreasing in $\tilde{x}$ means $\mathbb{E}[g_0(a) \mid \tilde{x}]$ is nonincreasing in $\tilde{x}$ for any increasing g_0, hence $\mathrm{Cov}_w(\tilde{x}, g_0(a)) \le 0$. □

This corollary is the precise version of the chapter's slogan: staleness does not merely add noise, it attenuates the fitted covariate effects by a factor tied to the age distribution of the risk set, and the informativeness of the ages adds a signed omitted-variable term on top. Both effects disappear from a model that conditions on the age — which is remedy R1.

8.3.2 The informativeness twist: distortion is differential across the risk set

Propositions 8.4 and 8.6 deliberately ignored one piece of information: the firm has *not* raised during $(\tau, t]$. By Proposition 8.3(iii) that silence is evidence. The following lemma pins its direction; to stay tractable it summarizes the intensity over the gap through the endpoint signal, which is adequate for the slowly-varying covariates at issue.

Lemma 8.8 (Survival tilt)**.** *Let $S = \beta^\top x(t)$ have conditional density $p(s)$ given $(\tilde{x}, a)$ under the free drift law, and suppose the integrated intensity over the gap is $\int_\tau^t \lambda(u)\,\mathrm{d}u = a\,\Lambda(S)$ with Λ strictly increasing (e.g. covariates approximately constant over the gap, $\Lambda(s) = e^s$). Then the conditional density given additionally $\{$no event in $(\tau, t]\}$ is*

$$p_{\mathrm{surv}}(s) \;\propto\; p(s)\, e^{-a\Lambda(s)},$$

which is dominated by p in the likelihood-ratio order; consequently, for every increasing h with finite means, $\mathbb{E}[h(S) \mid \tilde{x}, a, \text{no event}] \le \mathbb{E}[h(S) \mid \tilde{x}, a]$, strictly if $a > 0$, $\beta \ne 0$ and S is nondegenerate.

Proof. Bayes gives the displayed density; the ratio $p_{\mathrm{surv}}/p \propto e^{-a\Lambda(s)}$ is strictly decreasing in s, which is likelihood-ratio dominance, and likelihood-ratio dominance implies the expectation inequality for increasing h (Chebyshev's correlation inequality applied to the decreasing weight $e^{-a\Lambda(s)}$ and increasing h). When the monotone-summary assumption fails — strongly oscillating paths — the pointwise ordering can fail locally, but the mechanism survives: the tilt $\exp(-\int\lambda)$ always downweights high-intensity *paths.* □

Proposition 8.9 (Sign of the ranking bias under naive LOCF plug-in)**.** *Rank firms at time t by the naive plug-in score $\tilde{s}_i = \beta^\top\tilde{x}_i(t)$, while the correct observed-data score (the log of the projected intensity, Proposition 8.12) is $s_i^* = \log\mathbb{E}[\exp(\beta^\top x_i(t)) \mid \mathcal{G}_{t-}]$. Under Assumption 8.1 with stationary drift and Lemma 8.8:*

- *(i)* Conditional on the truth*: the score error is $\tilde{s}_i - \beta^\top x_i(t) = -\beta^\top\varepsilon_i(t)$; a firm whose covariates deteriorated in the β direction since its last deal ($\beta^\top\varepsilon_i < 0$) is over-ranked, one that improved is under-ranked, with error magnitude growing in the drift, hence stochastically in a_i.*

- *(ii)* Conditional on the data*: by Proposition 8.6 and Lemma 8.8,*

$$s_i^* \;\le\; \beta^\top m + e^{-\vartheta a_i}\,\beta^\top(\tilde{x}_i - m) + \tfrac{1}{2}\beta^\top V(a_i)\beta,$$

so for firms whose last-seen signal was above the sector mean ($\beta^\top \tilde{x}_i > \beta^\top m$), the naive score exceeds the correct one by at least $(1 - e^{-\vartheta a_i})\,\beta^\top(\tilde{x}_i - m) - \frac{1}{2}\beta^\top V(a_i)\beta$, a quantity that is positive and increasing in a_i once $\beta^\top(\tilde{x}_i - m)$ clears the variance term, and the survival tilt of Lemma 8.8 strictly enlarges the gap. Symmetrically, below-mean stale firms are pushed up by mean reversion and down by the tilt: their net bias is smaller and can take either sign.

(iii) Across the risk set*: a concentrates on low-intensity firms (Proposition 8.3), so the distortion is differential — largest exactly where the naive score trusts old good news. The characteristic failure is the* stale champion*: a firm that looked excellent at its last round years ago and has been silent since is systematically over-ranked by LOCF plug-in, because its signal should have been shrunk toward the prior and tilted down for the silence, and was neither.*

Proof. (i) is arithmetic on Equation (8.3). (ii): the unconditional (no-tilt) projection is Equation (8.6), whose log is the displayed bound; Lemma 8.8 applied to $h = \exp$ shows the survival-conditioned expectation is strictly smaller. Subtracting from $\tilde{s}_i = \beta^\top \tilde{x}_i$ gives the stated gap. (iii) combines (ii) with Proposition 8.3(iii). □

Example 8.10 (The size of the inversion)**.** Scalar covariate, $m = 0$, $\sigma_X = 1$, $\beta = 1$, covariate half-life two years ($\vartheta = \log 2/2 \approx 0.347$ per year). Firm A last raised four years ago looking excellent, $\tilde{x}_A = +2$; firm B raised a quarter ago looking merely good, $\tilde{x}_B = +1$. Naive LOCF plug-in scores A far above B (2 vs 1). The corrected log-scores of Equation (8.6), before any survival tilt: for A, $e^{-1.386} \cdot 2 + \frac{1}{2}(1 - e^{-2.77}) \approx 0.50 + 0.47 = 0.97$; for B, $e^{-0.087} \cdot 1 + \frac{1}{2}(1 - e^{-0.173}) \approx 0.92 + 0.08 = 1.00$. The correction alone *reverses* the ranking, and the survival tilt of Lemma 8.8 — four years of silence versus one quarter — pushes A down further. At the risk-set scale this is not an edge case: with inter-deal gaps of one to four years and covariate half-lives of the same order, the attenuation factor $e^{-\vartheta a}$ ranges over $(0.25, 1)$ *within a single risk set*, so a model fitting one common slope is wrong for most of its members at once.

Warning 8.11 (Look-ahead through deal-time covariates is fatal)*.* The covariates in deal row m are measured *at or after* the deal: revenue as reported in the round's diligence, raised-to-date *including* the round, valuation — the outcome itself. Using row m's covariates as features for predicting deal m, or any as-of join keyed on reference date rather than availability date, imports the label into the features. This is the internal-covariate "price sticker" in its most concrete form: the covariate value exists *because* the event happened. The failure mode is silent — metrics inflate in backtest and evaporate in deployment — and no downstream remedy can undo it. Policy: features at t use deals strictly before t (Protocol 2.5); every loader ships a look-ahead audit that verifies no feature path changes value at the timestamp of the label event it feeds (Section 8.9, code hooks). The synthetic loader already enforces the strict inequality; the real-data loader must inherit it verbatim.

8.4 Remedy R1: staleness-aware observed-data modeling

The first remedy is not an attempt to undo the defect but to model what we actually observe. Condition on the pair $(\tilde{x}, a)$ and fit

$$\lambda_i(t) \;=\; \exp\Big(\beta^\top \tilde{x}_i(t) \;+\; g(a_i(t)) \;+\; \tilde{x}_i(t)^\top\, \Gamma_g\, \phi(a_i(t)) \;+\; \delta^\top z_i(t)\Big), \tag{8.8}$$

with g flexible (a spline or step function on an age grid with knots at the empirical staleness quantiles), ϕ a low-dimensional basis in the age whose interaction with $\tilde{x}$ lets the covariate *slope* vary with staleness, and z_i the other engineered history covariates of Definition 2.2(iii). By Proposition 8.3(i) every term is $\mathcal{G}$-predictable, so Equation (8.8) is a well-specified intensity model *for the observed filtration* and the likelihood Equation (2.3) applies with its full martingale estimation theory (Andersen et al., 1993; Andersen and Gill, 1982). The deep point is that this is not a heuristic patch:

Proposition 8.12 (Projection: the observed-data intensity is a genuine intensity)**.** *Let N be a counting process adapted to a subfiltration $\{\mathcal{G}_t\} \subseteq \{\mathcal{F}_t\}$ and let λ be its $\mathcal{F}$-intensity. Then the $\mathcal{G}$-predictable projection*

$$\lambda^{\mathcal{G}}(t) \;=\; \mathbb{E}[\lambda(t) \mid \mathcal{G}_{t-}]$$

is an intensity of N with respect to $\{\mathcal{G}_t\}$: $N(t) - \int_0^t \lambda^{\mathcal{G}}(u)\,\mathrm{d}u$ is a $\mathcal{G}$-local martingale. Consequently the best-fitting element of a rich class of functions of $(\tilde{x}, a, \mathcal{H}_{t-})$ estimates the projection of the true structural intensity onto the available information — fitting Equation (8.8) is estimating a well-defined object, not compensating an error (Daley and Vere-Jones, 2008, Ch. 14) (Andersen et al., 1993, II.4).

Proof. For any bounded nonnegative $\mathcal{G}$-predictable C (which is also $\mathcal{F}$-predictable), the $\mathcal{F}$-intensity property gives $\mathbb{E}\int_0^\infty C\,\mathrm{d}N = \mathbb{E}\int_0^\infty C(u)\lambda(u)\,\mathrm{d}u$. Tower on $\mathcal{G}_{u-}$: since $C(u)$ is $\mathcal{G}_{u-}$-measurable, $\mathbb{E}[C(u)\lambda(u)] = \mathbb{E}[C(u)\,\mathbb{E}(\lambda(u) \mid \mathcal{G}_{u-})] = \mathbb{E}[C(u)\lambda^{\mathcal{G}}(u)]$, whence $\mathbb{E}\int C\,\mathrm{d}N = \mathbb{E}\int C\lambda^{\mathcal{G}}$. This identity for all such C is the defining property of a $\mathcal{G}$-intensity (martingale characterization via the monotone class theorem; the technical identification of $\mathbb{E}[\,\cdot \mid \mathcal{G}_{t-}]$ with the predictable projection is Daley and Vere-Jones, 2008, Ch. 14). □

Corollary 8.13 (R1 is exactly specified in the OU world)**.** *Under Assumption 8.1 with stationary drift and the monotone-summary approximation of Lemma 8.8, the projected intensity is $\exp\{(e^{-\Theta a}\beta)^\top \tilde{x} + g_0(a) - d(\beta^\top \tilde{x}, a)\}$ with g_0 as in Equation (8.6) and $d \geq 0$ the survival-tilt correction, a smooth function of the scalar signal and the age. The class Equation (8.8) contains the first two terms exactly (the interaction basis reproduces $e^{-\vartheta_k a}\beta_k$) and approximates the third within the same $(\tilde{x}^\top\gamma, a)$ interaction structure. R1 is therefore not merely valid but — in the working model — essentially well specified.*

Two practical points before the caveats. First, the repository already implements degenerate cases of Equation (8.8): the risk-set ranker of `hawkes_calibration/sector_ranker.py` carries a months-since-last-funding (cooldown) covariate — a parametric g inside the choice model — and the synthetic loader emits it point-in-time (`log1p_months_since_last_funding` in `CHOICE_FEATURES`). D8.1 is the commitment that this stops being a per-model accident: the pair $(\tilde{x}, a)$, with interactions where the model class allows them, becomes mandatory plumbing in every stack. Second, an identifiability warning that doubles as an interpretation warning: the fitted g absorbs *everything* that varies with the gap — drift-variance inflation (Proposition 8.4), mean-reversion offsets (Proposition 8.6), the survival tilt, *and* genuine duration dependence of the hazard itself (post-round cooldown, runway mechanics, the renewal structure of Chapter 5). These are observationally confounded in g by construction: the projection does not distinguish "covariates have drifted" from "the hazard truly depends on time since last round." For prediction under the observed filtration this is harmless — the projection is the target (Proposition 8.12) — but it means g has no structural reading either, and separating the channels requires the state-space machinery of Section 8.6, which supplies the drift-variance component of g from fitted dynamics and leaves the remainder as an honest estimate of duration dependence.

What R1 gives up is exactly what the projection destroys. The fitted β in Equation (8.8) is a property of the *observation scheme composed with* the structural model: it mixes the true covariate effect with mean reversion, drift variance, and the informativeness tilt. It must not be read as "the effect of revenue on funding intensity," used for counterfactuals ("if this firm's revenue were 20% higher. . . ” — the projection does not move that way), or extrapolated to a vendor with a different update policy. And because $(\tilde{x}, a)$ is internal, multi-horizon functionals require joint simulation of events and the LOCF update process, never plug-in survival formulas (Section 8.2). Within those limits, R1 is committed, cheap, and always on — decision D8.1.

8.5 Remedy R2: supervised drift modeling and regression calibration

R1 conditions on staleness; R2 tries to *reverse* it. The key observation is that the dataset labels its own drift: at every deal the firm's true drifting covariates are re-observed. Every consecutive pair of own deals $(\tau_m, \tau_{m'})$ of the same firm therefore yields a supervised example

$$\underbrace{\left(x_i(\tau_m),\ a = \tau_{m'} - \tau_m,\ \{x^{\text{mac}}_{\text{pub}}(u)\}_{u \in (\tau_m, \tau_{m'}]},\ \text{sector activity}\right)}_{\text{inputs}} \longmapsto \underbrace{x_i(\tau_{m'})}_{\text{target}},$$

a *drift dataset* on which a conditional model $\hat{m}(\tilde{x}, a, \cdot)$ with predictive variance $\hat{V}(\tilde{x}, a, \cdot)$ can be fit — linear-Gaussian (which is exactly the OU regression of Equation (8.5), making R2 and R3 two estimators of one object) or gradient-boosted with a variance head. The input design follows Equation (8.1): the macro path enters through summaries of the anchor movement over the gap — cumulative rate change, CPI and unemployment changes between the two deal dates, as-of joined on publish dates per Protocol 2.5(a) — and sector activity through trailing deal counts, so that the linear-Gaussian fit is literally a discretized regression estimate of $(\Theta, \Gamma, m_s, \Sigma_W)$, one consecutive-pair likelihood term per row. The gradient-boosted variant keeps the same inputs and relaxes the functional form; it is admitted only if it beats the linear-Gaussian model on *temporally* held-out pairs (pairs whose second deal falls after the training era, per Protocol 2.5(e)) on both mean error and variance calibration — a drift model whose $\hat{V}$ is uncalibrated poisons Equation (8.9) through the exponent. Imputation then replaces the stale value by $\hat{x}_i(t) = \hat{m}(\tilde{x}_i(t), a_i(t), \text{macro path})$, and the intensity uses regression calibration:

Proposition 8.14 (Plug-in of the conditional mean, with variance correction)**.** *Suppose $x(t) \mid \mathcal{G}_{t-} \sim \mathcal{N}(\hat{x}(t), V(t))$. Then the projected intensity of Proposition 8.12 is* exactly

$$\lambda^{\mathcal{G}}(t) \;=\; \exp\!\Big(\beta^\top \hat{x}(t) + \tfrac{1}{2}\,\beta^\top V(t)\,\beta\Big). \tag{8.9}$$

For a general conditional law with mean $\hat{x}$ and variance V, $\log \lambda^{\mathcal{G}} = \beta^\top \hat{x} + \frac{1}{2}\beta^\top V \beta + O(\|\beta\|^3)$ *by the cumulant expansion: plugging in the conditional* mean *makes the first-order term exact, and the variance factor* $\exp(\frac{1}{2}\beta^\top V \beta)$ *absorbs the second order. Plugging in the raw $\tilde{x}$, or a mode or median, leaves a first-order error. This is regression calibration (Carroll et al., 2006, Ch. 4), and Equation (8.9) is the log-linear analogue of the induced-hazard formula of Prentice (1982) — including his caveat, which here is Lemma 8.8: the conditioning $\mathcal{G}_{t-}$ includes the survival information, so the honest $\hat{x}$ is the* tilted *mean, not the free-drift mean.*

Proof. The Gaussian case is the mgf again. In general, $\log \mathbb{E}[e^{\beta^\top x}] = \sum_{r \geq 1} c_r(\beta)/r!$ with first and second cumulants $\beta^\top \hat{x}$ and $\beta^\top V \beta$, the remainder collecting third and higher cumulants of order $\|\beta\|^3$. □

Proposition 8.15 (Selection: the drift dataset is drift conditional on success)**.** *A training pair exists only if a second deal occurred. Under the monotone-summary approximation of Lemma 8.8 with $\Lambda(s) = e^s$, the drift law represented in pairs with gap a is tilted relative to the free law by $e^s\, e^{-ae^s}$ (event density at the endpoint), while the population the imputation* *serves — firms still waiting at age a — is tilted by e^{-ae^s} alone. The ratio of the two tilts is e^s, increasing in $s = \beta^\top x(\tau + a)$, so the pair-conditioned law dominates the survivor-conditioned law in the likelihood-ratio order:*

$$\mathbb{E}[\beta^\top x(\tau+a) \mid \textit{pair with gap } a,\ \tilde{x}] \ > \ \mathbb{E}[\beta^\top x(\tau+a) \mid \textit{no event by } \tau+a,\ \tilde{x}].$$

A drift model fit naively on pairs therefore learns drift conditional on funding success *and systematically over-predicts the β-direction state of the silent firms it is deployed on — optimistic imputation, which re-creates the stale-champion over-ranking of Proposition 8.9 through the back door.*

Proof. The next-event density at gap a given the endpoint summary is $e^s e^{-ae^s}$ (hazard times survival), the no-event probability is e^{-ae^s}; both statements are the conditional-Poisson formulas of Proposition 8.3(iii). Their ratio e^s is increasing, giving likelihood-ratio dominance of the pair law over the survivor law, and dominance implies the strict mean inequality for the nondegenerate increasing functional s. □

Three remedies for the selection, in the order they will be tried. *Censoring-aware drift estimation*: treat pair generation as a survival problem — every firm silent since τ_m contributes a right-censored drift spell $(x_i(\tau_m), a_i(T), \text{no endpoint})$, and the drift model is fit by a likelihood that includes the survival factor, i.e. a small joint model of drift and event (this is R4's machinery in miniature, and the honest default). *Inverse-probability weighting*: weight each pair by the inverse of the estimated probability that a firm with those inputs produces a second deal within the observed gap, taken from the discrete-hazard bucket model of Chapter 9; valid under the bucket model's own correctness, cheap, and diagnosable. *Exogenous restriction*: fit drift only for covariates whose evolution is plausibly exogenous to the funding process on the gap timescale — headcount and revenue drift with the business; raised-to-date, deal counts and last-round descriptors are *mechanical*, move only at events, and must never be imputed (LOCF is exact for them, Assumption 8.2). The mechanical-vs-drifting classification of every PitchBook column is maintained in Appendix B and is decision D8.3; Table 8.1 fixes the scheme on the representative columns, including the third class that Warning 8.11 makes necessary: columns that are *outcomes* of the deal they ride on.

8.6 Remedy R3: an explicit covariate state space

R2's linear-Gaussian special case deserves its own treatment, because it is simultaneously the honest uncertainty model and the supplier of the $\Sigma(a)$, $V(a)$ objects that R1 and R2 consume. Model the drifting components per firm by Equation (8.1) — OU toward a sector mean with macro loadings — observed at own deals with reporting noise: $y_{i,m} = x_i(\tau_m) + \eta_m$, $\eta_m \sim \mathcal{N}(0, R)$ (PitchBook financials are reported figures, not audited truth; $R = 0$ recovers exact observation).

Lemma 8.16 (OU prediction and deal-time update)**.** *Under Equation (8.1) with constant mean m (time-varying $m_i(u)$ adds the deterministic integral $\int_\tau^t e^{-\Theta(t-u)}\Theta\, m_i(u)\,\mathrm{d}u$, computable from the published macro path), the exact recursions for the conditional law $x_i(t) \mid \mathcal{G}_t \sim \mathcal{N}(\hat{x}_i(t), P_i(t))$ are:*

$$\begin{aligned} \textit{predict over } a = t - \tau: \quad \hat{x}_i(t) \ &= \ m + e^{-\Theta a}(\hat{x}_i(\tau) - m), \\ P_i(t) \ &= \ e^{-\Theta a} P_i(\tau)\, e^{-\Theta a} + V(a), \qquad V(a)_{kk} = \sigma^2_{X,k}(1 - e^{-2\vartheta_k a}), \end{aligned} \tag{8.10}$$

$$\textit{update at an own deal with report } y: \quad K = P\,(P+R)^{-1}, \qquad \hat{x}^+ = \hat{x} + K\,(y - \hat{x}), \qquad P^+ = (I-K)\,P. \tag{8.11}$$

Representative columns	**Class**	**Staleness treatment**
`total_raised_to_date`; deal count to date; `last_financing_date/_size/_deal_type`; `vc_round` progression; firm age	mechanical	LOCF exact by construction; recomputed point-in-time from the event log; imputation forbidden (D8.3)
`revenue`, `ebitda`, `gross_profit`, `net_income`, `number_of_employees`, growth since last deal; valuation between rounds	drifting	carried with a always (D8.1); eligible for R2 imputation with variance; modeled by the R3 state space
`deal_size`, `pre_money_valuation`, `post_valuation`, up/down/flat flags of the *current* round	outcome-at-event	usable as covariates only strictly after their deal; as predictors of their own deal they are look-ahead (Warning 8.11); as targets they belong to T4
`business_status`	drifting, laggy	exit-proxy territory of (C3); point-in-time only, never applied retroactively

Table 8.1: The column classification of decision D8.3 on representative PitchBook columns; the full dictionary lives in Appendix B. "Mechanical" columns are deterministic functionals of the observed event history, for which LOCF is exact; "drifting" columns are the carriers of defect (C5).

With $R = 0$ the update collapses to $\hat{x}^+ = y$, $P^+ = 0$, and the filter between deals is pure prediction: $\hat{x}_i(t)$ is exactly the shrunk mean of Equation (8.5) and $P_i(t) = V(a_i(t))$ is the explicit $\Sigma(a)$ that Propositions 8.4 and 8.6 postulated.

Proof. The predict step is Equation (8.5) applied to the current posterior (linear-Gaussian closure under the OU transition); the update step is the standard Kalman gain for a linear observation with Gaussian noise (Kalman, 1960; Harvey, 1989). The $R = 0$ limit is immediate. □

The state-space parameters $(\Theta, \Sigma_W, m_s, \Gamma, R)$ are estimable by exact Gaussian likelihood on the drift dataset of Section 8.5 — with the same selection caveat, treated by the same censoring-aware likelihood: Proposition 8.15 applies to $\hat{\Theta}$ too, and ignoring it biases mean-reversion estimates toward the reversion experienced by successful firms. Estimation must pool: the median tracked firm contributes a handful of deals, far too few to identify per-firm dynamics, so (Θ, Σ_W) are shared at sector level and Γ globally, fitted by summing pair likelihoods across firms; the drifting covariates enter on log scales (log revenue, log headcount), where the Gaussian working model is tenable and where the loaders already live, with missing and undisclosed reports handled by the rules of Appendix B, never ad hoc. Firm-level heterogeneity beyond the pooled parameters is precisely the shared random effect that R4 would add — another sense in which the ladder is nested rather than branching. What R3 buys beyond R2: (i) $V(a)$ in closed form, feeding R1's offset g_0 and R2's variance correction Equation (8.9) instead of leaving them as free spline coefficients; (ii) calibrated per-firm uncertainty that grows honestly to the stationary variance as $a \to \infty$ — the model *knows* it knows nothing about a firm silent for a decade, which is precisely the epistemic behavior the ranking layer needs; (iii) a principled bridge to the doubly-stochastic intensity tradition — default intensities driven by stochastic covariates with fitted dynamics — whose viability at scale is established in credit risk (Duffie et al., 2007). The cost is the linear-Gaussian straitjacket, which is why R3 enters the ladder as the variance model *inside* R2 rather than as a replacement for the gradient-boosted drift mean.

8.7 Remedy R4: joint modeling, gated by a measured trigger

The statistically complete treatment is a joint model of the longitudinal covariates and the event intensity with shared random effects (Rizopoulos, 2012; Tsiatis and Davidian, 2004): latent per-firm effects b_i drive both the covariate trajectory ($x_i(t) = \mu(t) + \Phi(t)b_i + \epsilon_i(t)$) and the intensity ($\lambda_i(t) = \exp(\beta^\top x_i(t) + \alpha^\top b_i + \cdots)$), and the likelihood integrates b_i out over both data streams jointly. This is the construction that makes the informative observation scheme part of the likelihood rather than a correction: the two-stage shortcut — fit the longitudinal model, plug fitted trajectories into the intensity — was shown biased precisely when the event process informs the covariate observations (Tsiatis et al., 1995), which is our Proposition 8.3(iii) verbatim, and the bias direction is the one Proposition 8.15 derives: trajectories fitted to success-selected observations are optimistic for the silent. The joint likelihood repairs this by making the no-event stretches pay their survival factor inside the same integral that fits the trajectory — R2's censoring-aware drift estimation and R3's censoring-aware state-space likelihood are exactly the first two terms of this construction, which is why the ladder converges to R4 rather than being overturned by it. What R4 adds beyond them is the shared random effect: persistent firm-level frailty entering both streams, so that a firm's covariate *level and trajectory shape* inform its intensity beyond the current imputed value (Tsiatis and Davidian, 2004; Rizopoulos, 2012). The price is steep: numerical integration over random effects for every firm ($\sim$7,000 tracked firms $\times$ quadrature nodes $\times$ optimizer iterations), delicate identifiability between α, β and the drift parameters — the frailty-vs-observation confounding of Appendix C's branching-inflation verdict returns here in longitudinal form — and a model class fragile enough that its failure modes are harder to audit than the defect it fixes. R4 is therefore *escalation, not default*, and it is gated:

Protocol 8.17 (Attenuation-gap test, the R4 trigger)**.** On the synthetic rehearsal world with drifting firm covariates, LOCF observation, and known structural β: (i) run the committed pipeline R1 + R2 (with R3's variance); (ii) measure the attenuation ratio $\hat{r}$ = slope of the fitted log-intensity scores regressed on the oracle scores $\beta^\top x_i(t)$ over the risk set, and the top-K ranking gap between fitted and oracle covariates; (iii) the trigger fires if the 95% interval for $\hat{r}$ excludes $[0.9, 1.1]$ *and* the oracle-covariate ablation attributes a material top-K loss to residual staleness (not to (C3), which is measured separately per Chapter 2), in two consecutive evaluation campaigns. Only a fired trigger authorizes the R4 build; the same harness then has to show R4 closing the measured gap on rehearsal data before any real-data use.

8.8 The bucket-model face, and what the rehearsal measured

Chapter 9 attacks the horizon target T3 with gradient boosting on a bucketed panel whose feature rows read, in words: "it has been a since this firm's last deal, and these are its last-known covariates." That is $(\tilde{x}, a)$ — and the trees' recursive partitioning of the joint feature space fits arbitrary piecewise-constant surfaces in $(\tilde{x}, a)$, i.e. nonparametric estimates of exactly the projected intensity that Corollary 8.13 derived: offset $g_0(a)$, age-attenuated slopes $e^{-\Theta a}\beta$, and the survival-tilt correction, none of which need to be prespecified (Friedman, 2001; Chen and Guestrin, 2016). Gradient boosting on $(\tilde{x}, a)$ *is* R1, implemented nonparametrically. This is the principled reason the discrete-time stack tolerates staleness better than a rigid log-linear form fitted without age terms: the rigid form suffers the pooled attenuation and signed omitted-variable bias of Corollary 8.7, while the trees simply learn the interaction surface. It is also a plausible component of the fired B1 test — the parity XGBoost beating the linear conditional logit by +0.22 nats and +9pp top-5 on regime A (Appendix C) — though the rehearsal has not yet isolated how much of that gap is staleness interaction versus other nonlinearity; the isolation run is scheduled below.

The rehearsal evidence on the attenuation mechanics themselves comes from the covariate-noise campaign exp22, catalogued in Appendix C and inherited here without re-derivation: relative Gaussian noise on the covariates at fit and prediction time — the *classical-error benchmark* of Section 8.3.1, not yet informative staleness — attenuated the fitted choice weights by $\times 0.88$ at 20% noise ($\times 0.93$ at 15%) and eroded the stage-1 β correlation from 0.67 to 0.53, while prediction was nearly untouched (stage-1 covariate lift ≈ 1.0 nats throughout; top-5 moved $0.446 \to 0.439$). Two lessons transfer. First, the attenuation arithmetic of Proposition 8.6 is not hypothetical — it fires at the predicted order of magnitude on data of our shape, and it hits *interpretation* (structural readings of β, cross-regime transfer) long before it hits ranking or likelihood. Second, exp22's noise was exogenous and uninformative — independent of the risk set — so it measures the benchmark, not the twist: the differential, informative distortion of Proposition 8.9 is precisely what exp22 could *not* see. The planned follow-up (same harness, drifting covariates with LOCF ages emitted by the loader, ranking bias reported by staleness decile) closes that gap and doubles as the standing harness for Protocol 8.17.

Because aggregate metrics demonstrably dilute exactly the damage this chapter is about — exp22's headline prediction numbers barely moved while the fitted structure decayed — the chapter installs a standing reporting discipline:

Protocol 8.18 (Staleness accounting in every evaluation)**.** Every model evaluation report in the repository includes: (a) the distribution of a over the evaluation risk sets (quantiles, and the share exceeding one and two estimated covariate half-lives once R3 supplies $\hat{\Theta}$); (b) the headline metrics stratified by staleness quartile of the scored firm; (c) for ranking deliverables, the share of top-K selections whose drifting covariates are older than one year, reported next to the hit rate. Rationale: Proposition 8.9 localizes the damage in the stale strata, so a model can win on average while being systematically wrong about the firms it is most confidently promoting from old information — and the consuming action (a diligence short list) is exposed to exactly those firms. The stratified report is the cheap standing detector of residual (C5) damage and the attribution input to Protocol 8.17(iii).

8.9 Defect declaration and commitments

Per decision D2.1, the models of this chapter declare their position against the ledger of Table 2.1. **(C5)** is *owned here*: R1 corrects the fitting bias exactly in the OU working model (Corollary 8.13) and validly in general (Proposition 8.12); R2/R3 additionally correct the ranking bias under a correct, selection-aware drift model; residual vulnerability is the survival-tilt approximation and drift-model error, monitored by Protocol 8.17. **(C1)**: robust — all constructions are point-in-time and censored spells enter the drift likelihood as censored, per Section 8.5; dropping them would reimport Proposition 8.15 with the sign derived there. **(C2)**: robust by construction — $(\tilde{x}, a)$ is defined only after first observation; pre-entry firms are the newcomer problem of Chapter 11. **(C3)**: *vulnerable, with a signed interaction that must be stated.* An unrecorded-dead firm accumulates maximal staleness with frozen covariates: (C3) keeps it in the risk set, (C5) keeps its last good numbers on its face. The two defects compound on the same firms — the stale champion of Proposition 8.9 is, in the worst case, dead — and both corrupt the risk set's information state in the same direction: overstated scores for long-silent firms, hence jointly worse for ranking than either alone. The staleness terms $g(a)$ partially absorb (C3) (large age predicts death as well as drift), which improves ranking but permanently confounds "quiet" with "gone": no structural reading of g is permitted while (C3) is open. **(C4)**: robust — staleness ages are measured in weeks-to-years and deal-date jitter is days; aggregation does not touch the LOCF map. **(C6)**: out-of-universe firms are the $a \to \infty$ limit — no covariates at all — and are handled by the outside-option construction of

Chapter 11, which this chapter's g should be checked against for continuity (a firm silent for a decade should not score very differently from an unseen firm).

Design decision (D8.1 — staleness age is a mandatory feature).

From this document forward, every firm-level model in every stack — event-time intensities, the risk-set ranker, the bucketed panel, mark regressions — includes the staleness age $a_i(t)$ (and, where the model class permits, its interactions with $\tilde{x}_i$) among its features. Rationale: a is free, predictable (Proposition 8.3), and carries the offset and attenuation structure of Proposition 8.6; omitting it produces the signed biases of Corollary 8.7 and Proposition 8.9. Reversal trigger: none — a fitted model may drive its age terms to zero, but the feature may not be withheld from it.

Design decision (D8.2 — the remedy ladder).

Committed sequence: R1 (staleness-aware observed-data modeling, Equation (8.8)) now and always on; R2 (drift dataset, supervised drift model, regression calibration per Proposition 8.14, selection handled censoring-aware first) next quarter; R3 (OU/Kalman state layer, Lemma 8.16) folded in as R2's variance model rather than a separate stack; R4 (joint model) only if Protocol 8.17 fires. Rationale: the ladder is cost-ordered, each rung consumes the previous rung's outputs, and the rehearsal evidence says prediction is initially robust (exp22) — so the expensive rungs must prove their marginal value on the measured gap, not on principle. Reversal trigger: a fired gap test promotes R4; a gap test passing twice consecutively with R1 alone demotes R2/R3 to research status.

Design decision (D8.3 — mechanical vs. drifting column classification).

Every PitchBook covariate column is classified in the data dictionary (Appendix B) as *mechanical* (deterministic functional of the observed event history: raised-to-date, deal counts, last-round descriptors, ownership accumulations — LOCF exact, imputation forbidden) or *drifting* (revenue, EBITDA, net income, employees, growth rates, latent valuation — LOCF stale, carried with a, eligible for R2/R3 treatment). The classification is maintained under vendor schema changes and consumed by the loaders and the drift-dataset builder. Rationale: Assumption 8.2 shows (C5) binds only on drifting columns, and imputing a mechanical column is not an approximation but an error. Reversal trigger: none; reclassification of individual columns requires a data-dictionary change with review.

Code hooks.

Planned module `hawkes_calibration/covstate/`: `pairs.py` — the drift-dataset builder (consecutive own-deal pairs with gap, macro path and sector-activity summaries; right-censored spells emitted with flags per Section 8.5; selection diagnostics from Proposition 8.15); `ou.py` — OU transition and $V(a)$ in closed form, Kalman predict/update of Lemma 8.16, censoring-aware Gaussian likelihood for $(\Theta, \Sigma_W, m_s, \Gamma, R)$;

`impute.py` — the imputation API returning $(\hat{x}_i(t), V_i(t))$ and the calibrated intensity factor $\exp(\frac{1}{2}\beta^\top V\beta)$ of Equation (8.9). Loaders: `synthetic/loaders.py` already enforces strictly-prior LOCF and carries a months-since-last-funding feature; it and the real-data adapter `hawkes_calibration/data/pitchbook.py` must emit the pair $(\tilde{x}_i, a_i)$ explicitly for every stack (D8.1), keyed to the mechanical/drifting classification of Appendix B (D8.3). Audits (`hawkes_calibration/data/audits.py`): the look-ahead shift test of Warning 8.11 — no feature path may change value at the timestamp of the label event it feeds — plus a staleness-age consistency check (a recomputed from the event list must match the emitted feature). The gap-test harness of Protocol 8.17 extends the exp22 campaign driver with LOCF-age worlds and staleness-decile ranking reports.

Part IV

Discrete-Time and Ranking Alternatives

Chapter 9

Temporal Buckets and Gradient Boosting

9.1 The horizon target and the bucketed panel

Target T3 of Chapter 1 asks the question the sourcing workflow actually poses: will firm i have a deal within the next Δ? Formally, the object is the fixed-horizon event probability $p_i(t, \Delta)$ of Equation (1.3), consumed by binary, capacity-constrained actions — enter diligence now, keep on the watch list, deprioritize. The natural attack is supervised learning on a bucketed panel: turn the event history into rows "firm i at time t, did it raise within Δ," and hand the rows to a learner that is good at nonlinearity. This chapter commits to exactly that attack — with gradient boosting as the learner — but insists on doing it as statistics. The bucketed panel is not a machine-learning convenience sitting outside the point-process theory of Part II; it *is* a discrete-time hazard model, with a precise link to the conditional intensity (Theorem 9.4), a censoring theory (Section 9.3), a sampling theory (Section 9.5), and a dependence structure that invalidates the i.i.d. habits that usually travel with the tooling (Section 9.4). Every labeling shortcut that folklore ML treats as plumbing — what to do with windows that cross the end of the data, whether a firm that died quietly is a negative, whether sliding the window weekly "adds data" — is a statistical decision with a signed bias, and this chapter makes each one explicitly.

Definition 9.1 (Bucketed panel)**.** Fix a horizon $\Delta > 0$ and an evaluation grid of bucket start times

$$\mathcal{T}_\Delta \;=\; \{\, t_b = t_0 + b\,\Delta \;:\; b = 0, 1, \ldots, B_\Delta - 1 \,\} \subset [0, T], \tag{9.1}$$

aligned to the weekly calendar of Chapter 2. The panel for horizon Δ is the row set $\mathcal{P}_\Delta = \{(i, t_b) : i \in \mathcal{I},\; Y_i(t_b) = 1\}$ — one row per tracked firm at risk at each bucket start — with binary label

$$y_{i,b} \;=\; \mathbf{1}\{\, N_i(t_b + \Delta) - N_i(t_b) \;\geq\; 1 \,\}, \tag{9.2}$$

and feature vector $z_{i,b}$ assembled at t_b under Protocol 2.5:

(i) the LOCF firm covariates $\tilde{x}_i(t_b)$ of Equation (2.1);

(ii) the staleness age $a_i(t_b) = t_b - \tau_i(t_b)$ — "it has been a since the last deal" — carried *always*, as mandated by design decision D8.1 of Chapter 8;

(iii) engineered history: deal counts $N_i(t_b) - N_i(t_b - w)$ over trailing windows $w \in \{26, 52, 104\}$ weeks, last round type and size (with an undisclosed flag), raised-to-date;

(iv) sector activity summaries: trailing sector deal counts and their changes, computed from the *complete* sector event stream (including out-of-universe deals, whose counts are trustworthy even when their firms are not);

(v) as-of macro $x^{\mathrm{mac}}_{\mathrm{pub}}(t_b)$, joined on publication date.

Negative rows are all at-risk firm-buckets with no deal in the window; there is no other source of negatives. The committed horizon set is $\Delta \in \{26, 52, 104\}$ weeks.

The label Equation (9.2) deserves one immediate remark: it depends on the event history only through a bucket *count*, never through a within-bucket position. This is deliberate. Defect (C4) of Chapter 2 says interval counts are trusted while positions are not; the panel is built on exactly the trusted functional, and Corollary 9.5 below prices what that choice costs.

Defect declaration (per D2.1). Against the ledger of Table 2.1: **(C1)** corrected — buckets whose window crosses T are flagged censored and either dropped or retained with the exposure-offset likelihood of Section 9.3; they are never labeled negative. **(C2)** robust by construction — no row exists before V_i; the deeper selection (the tracked universe is conditioned on having raised at least once) restricts the estimand to tracked firms and is owned by Chapter 11. **(C3)** vulnerable with signed bias — unlabeled deaths sit in the negative class; mitigations and the residual downward bias are stated in Section 9.3. **(C4)** robust by design — labels are invariant to within-bucket date jitter (only events within jitter distance of a bucket boundary can flip a label), at the price of zero information about within-bucket timing, a price this chapter chooses deliberately (Corollary 9.5). **(C5)** vulnerable, mitigated — LOCF covariates enter only alongside their staleness age, and the tree learner can represent the interaction "informativeness of $\tilde{x}$ decays with a" that Chapter 8 shows is the first-order correction; residual attenuation remains and is inherited from Chapter 8. **(C6)** robust for the stated estimand — labels are own-firm events of tracked firms, and out-of-universe deals enter only through complete sector counts; the panel makes no claim about untracked firms.

Design decision (D9.1 — the bucketed panel is a hazard dataset).

The T3 panel is built exactly as Definition 9.1: calendar-aligned disjoint buckets per horizon, labels from bucket counts only, features under Protocol 2.5 with the staleness age mandatory, censored buckets flagged rather than zero-labeled, and no rows before first observation. Rationale: each clause is the discrete-time image of a continuous-time censoring or truncation rule (Theorem 9.4 and Proposition 9.7); violating any one of them injects a bias whose sign we can already state, which is worse than a variance problem because more data will not fix it. Reversal trigger: none for the labeling rules; the feature list may grow freely under Protocol 2.5.

Design decision (D9.2 — horizon set {6M, 1Y, 2Y}).

We commit to $\Delta \in \{26, 52, 104\}$ weeks and build one model per horizon. Rationale: the three consuming actions named in Chapter 1 (diligence now; pipeline within a year; deprioritize if nothing plausible in two years) fix the horizons — the target is chosen by the action, not by the data; and per-horizon models keep the cloglog consistency of Theorem 9.4 exact. Reversal trigger: if the decision layer of Chapter 12 starts consuming horizon *curves* $\Delta \mapsto p_i(t, \Delta)$ rather than three scalars, the per-Δ design yields to the single-model AFT alternative of Section 9.8.

9.2 The bucket model is a hazard model

We now prove the claim that organizes the chapter: the panel of Definition 9.1 is a discrete-time hazard model, and the bucket probability is an exact functional of the same conditional intensity that Part II estimates in continuous time. Nothing about this is asymptotic; the identity holds bucket by bucket.

Lemma 9.2 (Waiting-time hazard of the bucket)**.** *Fix a bucket (i, b) with $Y_i(t_b) = 1$ and let $W_{i,b} = \inf\{u > 0 : N_i(t_b + u) > N_i(t_b)\}$ be the waiting time to firm i's next deal. Suppose the conditional law of $W_{i,b}$ given $\mathcal{F}_{t_b}$ is absolutely continuous on $(0, \infty)$ with hazard function $r_{i,b}(u)$. Then*

$$p_{i,b} \;:=\; \mathbb{P}\big(y_{i,b} = 1 \mid \mathcal{F}_{t_b}\big) \;=\; 1 - \exp\big(-\Lambda_{i,b}\big), \qquad \Lambda_{i,b} \;:=\; \int_0^{\Delta} r_{i,b}(u)\,\mathrm{d}u, \tag{9.3}$$

and the hazard is the predictable projection of the conditional intensity onto the no-event paths: $r_{i,b}(u) = \mathbb{E}\big[\lambda_i(t_b + u) \mid \mathcal{F}_{t_b}, W_{i,b} > u\big]$.

Proof. The first display is the hazard representation of an absolutely continuous survival law, $\mathbb{P}(W > \Delta \mid \mathcal{F}_{t_b}) = \exp(-\int_0^{\Delta} r)$, applied to $\{y_{i,b} = 0\} = \{W_{i,b} > \Delta\}$. For the second, the innovation theorem (Andersen et al., 1993, II.4.2) states that the intensity of N_i with respect to the coarser filtration generated by $\mathcal{F}_{t_b}$ and the post-t_b own-event history is the conditional expectation of λ_i given that filtration; on the event $\{W_{i,b} > u\}$ the post-t_b own-event history is trivial, and the hazard of the first point of a counting process equals its intensity before the first point. □

The lemma already says something practical: the bucket probability depends on the future only through an *integrated* intensity, with the unobserved evolution of covariates and competing events averaged out along the no-event paths. What the panel model estimates is therefore not an exotic new object; it is the compensator increment of Part II, coarsened. To get the classical grouped-data form we add the proportional-hazards structure.

Assumption 9.3 (Grouped proportional hazards within buckets)**.** For every bucket (i, b) there is a baseline $\lambda_{0,b}(u) \geq 0$, common across firms, and a coefficient vector β such that $r_{i,b}(u) = \lambda_{0,b}(u)\, e^{\beta^\top z_{i,b}}$ for $u \in (0, \Delta]$: covariates are frozen at their bucket-start values, and their proportional effect on the no-event intensity holds throughout the bucket. Piecewise-constant intensity ($\lambda_{0,b}(u)$ constant in u) is the special case used for exposure offsets below.

Theorem 9.4 (Hazard link)**.** *Let the panel be built as in Definition 9.1 and let $h_{i,b} = p_{i,b}$ denote the discrete-time hazard Equation (9.3).*

(i) (Exact cloglog / grouped PH.) *Under Assumption 9.3,*

$$\mathbb{P}(y_{i,b} = 1 \mid z_{i,b}) = 1 - \exp\big(-\Lambda_{0,b}\, e^{\beta^\top z_{i,b}}\big), \qquad \operatorname{cloglog} p_{i,b} \;:=\; \log\big(-\log(1 - p_{i,b})\big) = \alpha_b + \beta^\top z_{i,b}, \tag{9.4}$$

with $\alpha_b = \log \Lambda_{0,b}$, $\Lambda_{0,b} = \int_0^{\Delta} \lambda_{0,b}$, and β identical *to the continuous-time coefficients (Prentice and Gloeckler, 1978; Allison, 1982). The Bernoulli likelihood of the panel is the exact likelihood of the point process observed through the coarsening "at least one event per bucket": a bucket classifier estimates a deterministic coarsening $\lambda \mapsto (\Lambda_{i,b})_b$ of the same intensity object as Part II.*

(ii) (Continuum limit.) *Drop Assumption 9.3. If $u \mapsto r_{i,b}(u)$ is right-continuous at 0 then $h_{i,b}/\Delta \to r_{i,b}(0^+) = \mathbb{E}[\lambda_i(t_b) \mid \mathcal{F}_{t_b}]$ as $\Delta \downarrow 0$; and for a fixed window $[t, t + v]$ partitioned*

into buckets of mesh $\Delta \to 0$, with hazards refreshed at each bucket start and bounded above,

$$\prod_b (1 - h_{i,b}) \;\longrightarrow\; \exp\Big(- \int_t^{t+v} \lambda_i(u)\, \mathrm{d}u \Big) \quad \textit{along no-event paths:}$$

the product-limit of the discrete hazards recovers the continuous-time survival factor, so the discrete model converges to the continuous intensity model, not merely to an approximation of it.

(iii) (Logit versus cloglog.) *Writing $\Lambda = \Lambda_{i,b}$ for the bucket's hazard mass,*

$$\operatorname{logit} p \;=\; \log\left(e^{\Lambda} - 1\right) \;=\; \operatorname{cloglog} p + d(\Lambda), \qquad d(\Lambda) = \log \frac{e^{\Lambda} - 1}{\Lambda} = \frac{\Lambda}{2} + \frac{\Lambda^2}{24} - \frac{\Lambda^4}{2880} + \cdots, \tag{9.5}$$

so for small per-bucket probabilities the log-odds equal the log-hazard-mass up to $O(\Lambda)$: a logistic-link panel model estimates the same object as the cloglog model to first order. The divergence is material at large p: $d = 0.026$ nats at $p = 0.05$, 0.114 at $p = 0.2$, 0.367 at $p = 0.5$.

Proof. (i) Substituting Assumption 9.3 into Equation (9.3) gives $\Lambda_{i,b} = e^{\beta^\top z_{i,b}} \Lambda_{0,b}$ and hence both displays of Equation (9.4), since $\operatorname{cloglog}(1 - e^{-\Lambda}) = \log \Lambda$. For the likelihood claim: the coarsened data for bucket (i, b) is the binary $y_{i,b}$, whose conditional law given $\mathcal{F}_{t_b}$ is $\mathrm{Bernoulli}(p_{i,b})$ by Lemma 9.2; over disjoint buckets the joint likelihood factorizes by the tower property into exactly the product of these Bernoulli terms, which is the person-period likelihood of Allison (1982). No approximation enters at any step. (ii) $h = 1 - \exp(-\int_0^\Delta r)$ with $\int_0^\Delta r = \Delta\, r(0^+) + o(\Delta)$ by right-continuity, and $1 - e^{-x} = x + O(x^2)$, give $h/\Delta \to r(0^+)$; the identification of $r(0^+)$ is Lemma 9.2. For the product limit, $\log \prod_b (1 - h_b) = \sum_b \log(1 - h_b) = -\sum_b h_b + O\big(\sum_b h_b^2\big)$; along a no-event path each refreshed hazard mass satisfies $h_b = \int_{t_b}^{t_b + \Delta} \lambda_i(u)\, \mathrm{d}u + o(\Delta)$, so $\sum_b h_b \to \int_t^{t+v} \lambda_i$, while $\sum_b h_b^2 \le (\max_b h_b) \sum_b h_b \to 0$ because $\max_b h_b = O(\Delta)$ under the boundedness assumption. (iii) $p = 1 - e^{-\Lambda}$ gives $p/(1-p) = e^{\Lambda} - 1$, hence the first equality. For the expansion, $(e^{\Lambda} - 1)/\Lambda = e^{\Lambda/2} \sinh(\Lambda/2)/(\Lambda/2)$, so $d(\Lambda) = \Lambda/2 + \log\big(\sinh y / y\big)$ with $y = \Lambda/2$, and $\log(\sinh y/y) = y^2/6 - y^4/180 + O(y^6)$ yields the stated series. The numerical values follow from $\Lambda = -\log(1-p)$. □

Part (iii) has a sharp practical reading. When per-bucket probabilities are small — short horizons, ordinary firms — the logistic link is the cloglog link plus a distortion of order the probability itself, and a logistic learner (including XGBoost with its native `binary:logistic` objective) estimates a hazard-consistent object. When per-bucket probabilities are large — the two-year horizon on recently active firms — the gap $d(\Lambda)$ is covariate-dependent, so it is *not* absorbed by an intercept: logistic coefficients cease to be transferable across horizons, while cloglog coefficients remain the same β at every Δ by Equation (9.4). This is the precise sense in which cloglog, not logit, is the aggregation-coherent link, and it is why the calibration layer of Section 9.7 is mandatory rather than cosmetic: it repairs, on the validation era, exactly the link distortion we can bound here.

Corollary 9.5 (Choosing Δ is choosing the aggregation defect)**.** *Any two conditional intensities with equal bucket masses $(\Lambda_{i,b})_{i,b}$ induce the same panel law; consequently the panel Fisher information is identically zero in every parameter direction that moves λ only within buckets. In particular, timescale parameters faster than Δ (a kernel decay θ with half-life $\ll \Delta$) are unidentifiable from the panel.*

Proof. The panel likelihood in Theorem 9.4(i) depends on the underlying process only through $(\Lambda_{i,b})$; a parameter direction that leaves all bucket masses fixed leaves the likelihood fixed, so the score in that direction vanishes identically and so does its variance. □

This is the κ–θ ridge of Chapter 6 returned as a design dial. There, weekly binning was an observation defect (C4) that destroyed the decay while sparing the branching scale — bucketing at or below one kernel half-life was safe, four half-lives destroyed timing (Appendix C). Here the same mathematics appears with the sign of agency reversed: T3 does not *want* within-horizon timing, so we choose Δ to integrate it out, and Corollary 9.5 states exactly what has been discarded. The defect is not incurred by negligence; it is purchased, at a known price, because the consuming action is horizon-bound.

9.3 Censoring and eligibility

Three eligibility rules determine which (i, t_b) pairs become rows and what their labels may say. Each is the discrete image of a continuous-time censoring statement, and each has a signed bias when violated.

Right censoring at the window end (C1). Define the exposure fraction of bucket (i, b) as $e_{i,b} = \min\{1, (T - t_b)/\Delta\}$ (rows require $t_b < T$, so $e_{i,b} \in (0, 1]$). A bucket with $e_{i,b} < 1$ — any window crossing June 2024 — is *censored*: the label Equation (9.2) is not computable from data, because the window's unobserved remainder may contain the event.

Warning 9.6. Labeling censored buckets $y = 0$ is not a conservative convention; it is a biased estimator with a known sign. It converts "unobserved" into "negative" and deflates every fitted hazard, most strongly in the most recent calendar eras — which are exactly the eras closest to deployment and the eras whose macro covariates differ most from the training bulk, so the bias masquerades as an era effect and contaminates the macro coefficients. This is the bucketed instance of Warning 2.4.

Proposition 9.7 (Censored-bucket bias)**.** *Suppose bucket* (i, b) *has hazard mass* $\Lambda_{i,b} > 0$ *distributed with constant within-bucket intensity, and exposure* $e = e_{i,b} < 1$*. The naive label* $\tilde{y}_{i,b} = \mathbf{1}\{$*event observed in* $[t_b, T)\}$ *satisfies* $\mathbb{E}[\tilde{y}_{i,b} \mid \mathcal{F}_{t_b}] = 1 - e^{-e\Lambda_{i,b}} < 1 - e^{-\Lambda_{i,b}} = p_{i,b}$*: on the cloglog scale the fitted linear predictor is shifted by* $\log e < 0$*, i.e. hazards are biased down by exactly the unobserved exposure fraction.*

Proof. With constant within-bucket intensity the observed sub-window carries hazard mass $e\Lambda_{i,b}$, so the observed-event probability is $1 - e^{-e\Lambda_{i,b}}$ by Lemma 9.2 applied to the sub-window; $\operatorname{cloglog}(1 - e^{-e\Lambda}) = \log \Lambda + \log e$. □

Two legal treatments follow. *Drop:* for the pure classification pipeline, delete rows with $e_{i,b} < 1$. This restores unbiasedness — administrative censoring at fixed T is independent by construction (Chapter 2, defect C1) — at the cost of discarding the most recent eras; the cost grows with the horizon, and at $\Delta = 104$ weeks it amounts to the final two years of bucket starts. *Keep with an exposure offset:* retain censored rows with the survival-style Bernoulli likelihood implied by Proposition 9.7,

$$\ell \;=\; \sum_{(i,b)} \Big[\, \tilde{y}_{i,b} \log \Big(1 - \exp\big(-e_{i,b}\, e^{\eta_{i,b}}\big)\Big) \;-\; (1 - \tilde{y}_{i,b})\, e_{i,b}\, e^{\eta_{i,b}} \,\Big], \tag{9.6}$$

where $\eta_{i,b}$ is the cloglog linear predictor (or boosted margin) and $\log e_{i,b}$ enters as a fixed offset. Under within-bucket constant intensity this is exact; it is the grouped-data analogue of keeping open spells with their survival factor. The classification stack commits to the drop rule for simplicity, and Section 9.8 names the offset likelihood and AFT as the routes that recover the discarded eras.

Entry (C2). No row is emitted for $t_b < V_i$: a firm contributes person-periods only after its first observation. This is delayed entry handled mechanically by the at-risk indicator, exactly as in Equation (2.3); emitting earlier rows would populate the panel with firms observed *because* they later raised, a truncation bias with the opposite sign to Warning 9.6.

Unlabeled exits (C3). The negative class is contaminated: a firm that quietly died before t_b has $Y_i(t_b) = 1$ in the naive risk set and contributes an eternal string of structural negatives. The consequences and their measured severity are inherited from Chapter 2 and Appendix C: hazards deflate by the contamination fraction, and covariates correlated with staleness acquire *inverted* weight, because the Bayes-optimal response to a pool containing zombies is to bet against staleness. Three mitigations are committed, in order of force. First, *exposure windows*: a row (i, t_b) requires observed activity (a deal, or a point-in-time exit proxy refresh) within the trailing E years. Because the rule is a predictable function of the observed history, it is exactly independent censoring; its benefit is not repairing the mechanism but capping each dead firm's phantom exposure at E years. Second, an explicit *active model* — a point-in-time classifier of "still alive" — used as a feature, not as a hard filter: the rehearsal showed a discriminative exit model (AUC 0.877) is *insufficient* to reconstruct truthful risk sets, so it may inform but must not silently redefine the estimand. Third, the staleness age itself partially absorbs the defect: dead firms accumulate enormous a, and a flexible learner routes their predicted probability toward zero through the a feature — which is simultaneously why the fitted a effect is *not* interpretable as the true renewal profile (it mixes the genuine gap dependence with the contamination). The residual bias after all three mitigations is signed: fitted hazards remain biased *downward*, most severely at large staleness ages, and every probability this chapter emits for a long-quiet firm is to be read as a lower bound in that direction.

Protocol 9.8 (Panel construction and audits)**.** The panel builder emits, per horizon: the grid Equation (9.1); rows filtered by $V_i \leq t_b < T$, the exposure-window rule with recorded E, and $Y_i(t_b) = 1$; labels Equation (9.2) for fully exposed buckets and censor flags with $e_{i,b}$ otherwise; features exactly as Definition 9.1 under Protocol 2.5, with the exp26 prohibition on end-of-sample snapshot columns (the companies-table `last_*` fields, totals, employees) enforced by schema. Audits, run on every build: (a) no row precedes its firm's entry; (b) no label consults data past T; (c) censor-flag retention — the count of censored rows must equal the count implied by the grid and T; (d) as-of-join look-ahead test on macro features; (e) cross-horizon coherence — empirical event rates must be nondecreasing in Δ on shared origins; (f) base rate and row count by era, published next to every fit.

9.4 Dependence: what the rows share

The panel looks like a supervised-learning table and is not an i.i.d. sample. Rows share firms across buckets: the same latent quality that Chapter 8 models as a persistent state makes $y_{i,b}$ and $y_{i,b'}$ positively dependent for all pairs of buckets of the same firm. Rows share time within buckets: macro regimes and sector cycles are common shocks, and the covariates-dominate verdict of Appendix C is precisely the statement that this shared structure carries most of the signal. Both dependencies survive conditioning on the features whenever the features are incomplete — which defect (C5) guarantees they are. The effective sample size is therefore far smaller than the row count, nominal i.i.d. standard errors are fictions, and shuffled cross-validation is leakage by construction (rows from the same firm and era land on both sides of the fold).

Warning 9.9. Sliding the bucket window (origins every $s < \Delta$ weeks) does not create data. Each realized deal then appears in $\lceil \Delta/s \rceil$ positive labels, and overlapping windows re-count the same survival exposure: the resulting objective is not the likelihood of any process, and any resampling scheme that treats the rows as exchangeable understates uncertainty by a factor up to $\sqrt{\Delta/s}$. Sliding windows are pseudo-replication, not augmentation.

Design decision (D9.3 — disjoint buckets for training, sliding for evaluation only).

Training panels use disjoint (non-overlapping) buckets per horizon, as in Equation (9.1); the likelihood factorization of Theorem 9.4(i) is exact only then. Phase-shifted disjoint grids (origins offset by $s < \Delta$) may be built as robustness replicates but are never pooled into one training set. Sliding origins *are* permitted for evaluation curves — lift@K or calibration as a function of the decision date — because evaluation is description, not estimation; their inference uses the blocked resampling below. Rationale: disjointness trades a constant factor of rows for an exact likelihood and honest uncertainty; the rehearsal's era-blocked evaluations showed the between-era variance dominates anyway, so the discarded overlap bought little. Reversal trigger: if the $\Delta = 104$ panel's bucket count per firm (≈ 6 over the 2011–2024 window) proves too coarse for stable fitting, revisit pooled phase-shifted grids with firm-and-era clustered inference priced in.

Uncertainty for every panel-derived quantity — metrics, calibration curves, coefficient contrasts — is computed by resampling that respects both sharing directions: firms are resampled as whole clusters (all rows of a firm move together) and eras as contiguous blocks (all rows of a calendar block move together), following the temporal split and blocked-inference protocol fixed in Chapter 7. The split itself is temporal: training era, validation era (model selection, early stopping, calibration), test era touched once. This is the same discipline the exp26 benchmark harness already implements — temporal early stopping strictly inside the training era, a single touch of the test era — and the panel stack reuses that procedure under the calendar-era boundaries fixed by Chapter 7.

9.5 Class imbalance and negative sampling

Positives are rare: most at-risk firm-buckets contain no deal, with the base rate rising in Δ. Rarity has two costs — a statistical one for the parametric benchmarks and a computational one for the full panel — and both have exact treatments; neither is an excuse for folklore reweighting.

For the linear cloglog and logistic benchmarks that every boosted model must beat, rare-event finite-sample bias is real: the MLE intercept is biased in small-event-count regimes and separation becomes possible in sparse eras. The corrections are standard and committed: the King and Zeng (2001) rare-events adjustment for logistic fits, and Firth (1993) penalization whenever any era-by-feature cell is sparse enough to threaten separation.

For scale, the committed device is case-control subsampling of negatives, with its exact correction.

Proposition 9.10 (Subsampling offset). *Retain every positive row and each negative row independently with probability $r \in (0, 1]$, with retention independent of z given y; let $S = 1$ denote retention. If $p(z) = \mathbb{P}(y = 1 \mid z)$, then*

$$\operatorname{logit} \mathbb{P}(y = 1 \mid z, S = 1) \;=\; \operatorname{logit} p(z) \;-\; \log r. \tag{9.7}$$

Hence if the population model is logistic with predictor $\alpha + f(z)$, the retained sample follows a logistic model with the same f and intercept $\alpha - \log r$: fitting on the subsample with the fixed *offset* $-\log r$ *in the predictor (equivalently, adding* $\log r$ *to every fitted margin at prediction time) recovers the population model exactly.*

Proof. By Bayes, $\mathbb{P}(y = 1 \mid z, S = 1) = p(z) \cdot 1 / \left[p(z) \cdot 1 + (1 - p(z))\, r\right]$, so the retained odds are $p(z)/\left[(1 - p(z))\, r\right]$, i.e. the population odds divided by r; taking logs gives Equation (9.7). The second claim follows because the shift is constant in z and the logistic family is closed under intercept shifts. □

Two remarks bound the result's reach. First, the exactness is a property of the *logistic* link: the retained-sample law under a cloglog population model is not a cloglog model with shifted intercept, so subsampling lives in the logistic pipeline, where Theorem 9.4(iii) already prices the link discrepancy, and the calibration layer absorbs the residual. Second, the same logic underwrites the ranking stack: estimating a conditional logit on a *sampled subset of alternatives* is consistent when the sampling protocol satisfies the positive-conditioning property, by exactly the same Bayes correction (McFadden, 1978) — the sampled-softmax training of Chapter 10 and Equation (9.7) are one result in two costumes: a known sampling scheme induces a computable likelihood correction, so sampling is free of bias whenever it is honestly recorded.

For the boosted trees themselves the folklore alternative is `scale_pos_weight`: multiply positive-class gradients by a constant w. We reject it for probability work. In the unregularized, infinite-capacity limit weighting shifts every logit by $\log w$ uniformly — harmless. In an actual boosted tree, leaf values are $-G/(H + \lambda_{\text{reg}})$ (Section 9.6) and splits are accepted against γ and `min_child_weight` thresholds on H: reweighting rescales G and H heterogeneously across leaves, so the induced distortion is *not* a constant shift, and no single intercept correction undoes it. Subsampling with the offset of Proposition 9.10, followed by post-hoc calibration on the *unsubsampled* validation era, achieves the same variance-per-cost benefit with a distortion we can state and remove.

Design decision (D9.4 — subsample negatives, correct exactly, calibrate after).

Training panels may subsample negatives at a recorded rate r per era-stratum; positives are never subsampled. Corrections are mandatory and layered: the $\log r$ margin correction of Proposition 9.10 at prediction time, then the calibration map of Section 9.7 fit on the full (unsubsampled) validation era. `scale_pos_weight` and other loss reweightings are banned in the probability pipeline (permitted in pure-ranking experiments, where only order matters). Rationale: Proposition 9.10 makes subsampling exactly correctable; reweighting is not. Reversal trigger: none; this is a mathematical, not empirical, commitment.

9.6 XGBoost as the committed learner

The learner attached to the panel is gradient-boosted trees in the XGBoost formulation (Chen and Guestrin, 2016), i.e. the functional gradient view of Friedman (2001) with a second-order objective and explicit regularization. We derive the fitting rule once, because two of its internals (the Hessian denominators and the default directions) are load-bearing for our data.

The model after m rounds is $F_m(z) = \sum_{k \le m} f_k(z)$, each f_k a regression tree with T leaves, leaf weights $w \in \mathbb{R}^T$, and leaf-assignment map q. Round m solves

$$\min_f \; \sum_n l\big(y_n,\; F_{m-1}(z_n) + f(z_n)\big) \;+\; \gamma T \;+\; \tfrac{1}{2}\lambda_{\text{reg}}\, \|w\|^2 \,, \tag{9.8}$$

with l the per-row loss — for us `binary:logistic`, $l(y, F) = -yF + \log(1 + e^F)$, or the cloglog-exposure loss implied by Equation (9.6) as a custom objective.

Proposition 9.11 (Second-order boosting step)**.** *Let* $g_n = \partial_F\, l(y_n, F)\big|_{F_{m-1}(z_n)}$ *and* $h_n = \partial_F^2\, l(y_n, F)\big|_{F_{m-1}(z_n)}$*, and for a fixed tree structure* q *let* $I_j = \{n : q(z_n) = j\}$*,* $G_j = \sum_{n\in I_j} g_n$*,* $H_j = \sum_{n\in I_j} h_n$*. To second order in* f*, the objective Equation (9.8) is minimized over leaf weights by*

$$w_j^* \;=\; -\,\frac{G_j}{H_j + \lambda_{\mathrm{reg}}}, \qquad \textit{with optimal value} \quad -\tfrac{1}{2}\sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda_{\mathrm{reg}}} \;+\; \gamma T,$$

and the gain of splitting a leaf with statistics (G, H) *into* (G_L, H_L) *and* (G_R, H_R) *is*

$$\mathrm{Gain} \;=\; \tfrac{1}{2}\left[\frac{G_L^2}{H_L + \lambda_{\mathrm{reg}}} + \frac{G_R^2}{H_R + \lambda_{\mathrm{reg}}} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda_{\mathrm{reg}}}\right] - \gamma. \tag{9.9}$$

For the logistic loss, $g_n = \sigma(F_{m-1}(z_n)) - y_n$ *and* $h_n = \sigma(F_{m-1}(z_n))(1 - \sigma(F_{m-1}(z_n)))$*.*

Proof. Taylor-expanding each loss term to second order, $l(y_n, F_{m-1} + f) \approx l(y_n, F_{m-1}) + g_n f(z_n) + \frac{1}{2} h_n f(z_n)^2$, and summing within leaves (where $f \equiv w_j$) turns Equation (9.8) into $\sum_j [\, G_j w_j + \frac{1}{2}(H_j + \lambda_{\mathrm{reg}}) w_j^2\,] + \gamma T$ up to a constant: a separable quadratic, minimized at w_j^* with the stated value. The gain formula is the difference of optimal values before and after the split, with γ charged for the extra leaf (Chen and Guestrin, 2016). □

Why trees earn their keep here. Four reasons, one of them a measured verdict. First, *the staleness interaction is the model*: Chapter 8 establishes that the predictive content of a LOCF covariate decays with its age, so the truth has the form $f(\tilde{x}, a)$ with effect modification in a — exactly what axis-aligned splits represent cheaply and what a linear cloglog must hand-engineer term by term. Second, PitchBook columns are incorrigibly mixed-type: round-type categories, flags, sizes spanning orders of magnitude, and pervasive missingness; XGBoost's sparsity-aware default directions (Chen and Guestrin, 2016) learn, per split, which way missing values go — and since missingness here is informative (undisclosed size is a signal, not an accident), learning the direction is modeling, not imputation. Third, nonlinearity in the engineered history (trailing counts, raised-to-date) is expected and cheap to capture. Fourth, and decisively: the B1 benchmark verdict of Appendix C. At strict feature parity — the same six features, the same choice sets, the same temporal split — XGBoost beat the linear conditional logit by +0.22 nats and +9pp top-5 on regime A. The linear-utility restriction is thereby empirically falsified on the identity target even in a clean six-feature world; on the panel, whose feature set is wider and dirtier, we do not expect linearity to fare better. The linear cloglog GLM remains in the harness as the benchmark every boosted fit must beat, but it is the challenger, not the champion.

Hyperparameters that matter. Depth (`max_depth`) bounds interaction order; the rehearsal's gains appeared already at depth 3–5, and deeper trees mostly buy overfitting to firm identity. `min_child_weight` thresholds the leaf Hessian mass H_j; under the logistic loss $h_n = p_n(1 - p_n)$, so the threshold is an effective-count guard that, under class imbalance, mechanically prevents leaves fit to a handful of positives — it is the imbalance safeguard that *is* compatible with calibration, unlike loss reweighting. Row subsampling per boosting round reduces variance but should sample firms as clusters when feasible, for the reasons of Section 9.4; column subsampling decorrelates trees in the presence of the many collinear trailing-window counts. The number of trees is chosen by early stopping on the temporal validation era — never by shuffled cross-validation — following the exp26 protocol (grid over depth and learning rate, early stop on the validation tail, refit at the chosen round count, one shot at test).

Monotonicity constraints. XGBoost accepts per-feature monotone constraints, and economics suggests candidates: probability nondecreasing in trailing deal counts, nonincreasing in the policy rate. The tempting constraint — intensity decreasing in staleness age a — we explicitly decline to impose. The true gap dependence is not monotone: renewal-style maturation pushes hazard *up* over the first year-plus of a gap before frailty heterogeneity pulls the population hazard down (Vaupel et al., 1979; Aalen et al., 2008), and the apparent steep decline at large a is partly the (C3) contamination artifact of Section 9.3, not economics. Forcing monotonicity would hard-code the artifact. Constraints are therefore used as robustness probes — fit with and without, compare test loss and reliability — and imposed only where the constrained fit is not measurably worse.

9.7 Probability calibration

The decision layer of Chapter 12 consumes the panel's outputs as probabilities: expected shortlist yield $\sum_{i\in\text{list}} \hat{p}_i$, thresholds set by expected utility, horizon-qualified pipeline pacing. A model whose ranking is superb but whose probabilities run hot by a factor of two poisons every one of those computations, which is why calibration here is mandatory infrastructure, not a finishing touch. Three distortions are anticipated and named: the logit–cloglog gap of Theorem 9.4(iii), residual subsampling distortion after the $\log r$ correction of Proposition 9.10, and the boosted-margin distortion documented by Niculescu-Mizil and Caruana (2005), who find boosted trees among the models that benefit most from post-hoc mapping.

The committed instruments are the two standard ones. *Platt scaling* (Platt, 1999): fit $c(m) = \sigma(Am + B)$ on validation margins by logistic regression — two parameters, stable on modest validation eras, and well matched to distortions that are smooth and sigmoidal in the margin. *Isotonic regression* (Zadrozny and Elkan, 2002): the nonparametric monotone map fit by pooled adjacent violators — strictly more flexible, needs materially more validation data, and prone to piecewise-constant artifacts in the tails where our decisions live. Guidance follows Niculescu-Mizil and Caruana (2005): Platt as default; isotonic when the validation reliability curve is visibly non-sigmoidal *and* the validation era supplies enough positives per bin to estimate it — per-horizon, the 26-week panel is the most likely to qualify, the 104-week panel the least.

Protocol 9.12 (Calibration and its audit)**.** (a) The calibration map is fit on the validation era only, after the $\log r$ subsampling correction, and is part of the model artifact: no downstream consumer ever sees raw margins. (b) Reliability diagrams (binned $\hat{p}$ versus empirical frequency, with firm-cluster, era-block resampled bands) are produced per horizon and per era, before and after calibration. (c) Scores: the Brier score with its reliability–resolution–uncertainty decomposition, and the log score; both are proper, so no calibration map can game them in expectation (Gneiting and Raftery, 2007). (d) Drift: calibration is re-audited on each new era; a reliability shift with stable discrimination triggers recalibration, not refitting.

Design decision (D9.5 — calibration is mandatory).

Every deployed horizon model carries a calibration map fit per Protocol 9.12; uncalibrated scores are internal quantities. Rationale: the consuming actions are probability arithmetic (Chapter 12), and the three distortion sources above are known and repairable at the cost of a validation-era fit. Reversal trigger: none. If a future learner is verified reliably calibrated end-to-end, the map degenerates to the identity, at zero cost.

9.8 Survival-aware boosting alternatives

The per-Δ classifier design pays two visible costs: censored buckets are dropped (Section 9.3), losing the most recent eras precisely at the long horizons; and the three horizon models are fit separately, so nothing enforces coherence of $\hat{p}_i(t, 26) \leq \hat{p}_i(t, 52) \leq \hat{p}_i(t, 104)$ beyond the audit in Protocol 9.8. Both costs are removed by boosting a survival model instead of a classifier.

XGBoost–AFT (Barnwal et al., 2022) boosts an accelerated failure time model, $\log W_i = \mu(z_i) + \sigma\varepsilon$, with ε standard normal, logistic, or extreme-value, and μ the boosted ensemble. Its likelihood consumes exactly the data the classifier discards: an observed waiting time contributes a density term, a bucket that crosses T contributes a right-censored survival term $\mathbb{P}(W > T - t_b)$, and grouped observations contribute interval-censored terms $\mathbb{P}(l < W \leq u)$ — censored rows are retained natively, no offsets required. One fitted model produces the whole curve $\Delta \mapsto \hat{\mathbb{P}}(W \leq \Delta)$, monotone in Δ by construction, serving all three horizons and any future one. *Gradient-boosted Cox* (Pölsterl, 2020) boosts the partial likelihood instead: it inherits the proportional-hazards reading of Theorem 9.4(i) and needs no error-distribution choice, but it delivers relative risks; absolute horizon probabilities require a Breslow-type baseline estimate on top, at which point its calibration story is no simpler than the classifier's.

The costs run the other way. AFT buys its coherent curve with a global parametric assumption on ε: a misspecified error shape miscalibrates *every* horizon in a correlated way, and repairing it per horizon reintroduces exactly the per-Δ calibration maps the design was meant to avoid. AFT also consumes exact waiting times, thereby re-inheriting the date-jitter component of (C4) that the bucket design is immune to by construction. And the surrounding tooling — imbalance-aware sampling theory (Proposition 9.10 is logistic-specific), calibration instruments, evaluation conventions — is mature for binary classification and improvised for boosted AFT.

Design decision (D9.6 — per-Δ classifiers now, AFT as the benchmarked alternative).

The committed T3 stack is the per-Δ, cloglog-consistent classifier pipeline of this chapter. XGBoost–AFT is maintained in the benchmark harness on the identical feature set and splits, evaluated on the same per-horizon probabilities $\hat{\mathbb{P}}(W \leq \Delta)$ after its own calibration audit. Rationale: exact censoring and sampling theory, mature calibration, and (C4) robustness today outweigh the coherent curve and censored-row retention. Reversal trigger: if AFT beats the calibrated per-Δ classifiers on test log score at two of three horizons in two consecutive evaluation campaigns — or if D9.2's reversal fires and the decision layer demands curves — AFT is promoted.

The repository's prior art is acknowledged and mined: experiment `exp16` already implements a discrete-time logistic-hazard model with Kaplan–Meier calibration reference and a pairwise ranking-loss hybrid on synthetic firms — the panel stack of this chapter is that sketch industrialized, with the labeling, censoring, and sampling theory made explicit.

9.9 Evaluation, and the bridge to ranking

Evaluation follows the temporal split of Chapter 7, reuses the exp26 harness conventions (selection on the validation era, one shot at test), and reports per horizon and per era. The headline metrics are chosen to proxy the consuming action, per Equation (1.1). *PR-AUC*, not ROC-AUC, is the discrimination summary: under heavy imbalance ROC-AUC is inflated by the ocean of easy negatives, while precision–recall tracks the operating region the sourcing action lives in; because PR-AUC depends on the base rate, it is reported next to the era's base rate and never compared across eras or horizons without that context. *Lift at K* — the event rate in the top-K list divided by the base rate, at K set to realistic diligence capacities — is the closest offline proxy to sourcing value. *Reliability and proper scores* per Protocol 9.12 complete the set: discrimination without calibration is half a model here. All uncertainty is firm-clustered, era-blocked (Section 9.4).

Sorting firms by $\hat{p}_i(t, \Delta)$ is a ranking, and the T1 machinery of Chapter 10 is close enough that the relationship must be stated rather than left to intuition. The two targets differ in conditioning: the ranker estimates $\mathbb{P}(i \mid \text{deal in } s, t, \mathcal{H})$ — identity given that a deal happens, timing canceled — while the horizon model estimates a marginal that mixes identity with timing mass; under the two-layer factorization of Chapter 1, heuristically $p_i(t, \Delta) \approx 1 - \exp(-\int_t^{t+\Delta} \Lambda_s(u)\, \pi_i(u)\, \mathrm{d}u)$, so a mid-ranked firm in a hot sector can out-probability a top-ranked firm in a cold one. They differ in universe: the ranker competes firms within a sector risk set and must own the outside option (C6); the horizon model scores each tracked firm absolutely, no competition and no outside option. And they differ in unit: recall@K is computed against realized deals and is calibration-free; lift@K on the panel is computed on all at-risk rows and consumes calibrated levels. The two are complements, not rivals: the sector-conditional ranker is the defensible object when timing is untrusted (Chapter 1), the firm-level horizon model is the object the diligence trigger consumes, and the composed continuous-time stack furnishes a third estimate of the same $p_i(t, \Delta)$ by integrating the fitted ground intensity against the mark factor. Comparing the boosted panel's $\hat{p}_i(t, \Delta)$ with that composed structural estimate on identical origins is the standing cross-stack audit demanded by design decision D1.2: when the two stacks disagree systematically, one of them is wrong about something nameable — and every diagnostic in this chapter exists to name it.

Code hooks.

Panel layer (to be written): `hawkes_calibration/panel/builder.py` — grids, at-risk filtering, labels, exposure fractions $e_{i,b}$, censor flags, and feature assembly per Definition 9.1 under Protocol 2.5; `hawkes_calibration/panel/audits.py` — the audit battery of Protocol 9.8 (entry consistency, censor-flag retention, look-ahead tests, cross-horizon monotonicity, base rates by era). *Model layer* (to be written): `hawkes_calibration/models/xgb_horizon.py` — per-Δ XGBoost with the D9.4 subsampling-offset correction, an optional cloglog-exposure custom objective implementing Equation (9.6), monotone-constraint configuration, and the calibration map of Protocol 9.12 as part of the persisted artifact; `hawkes_calibration/models/xgb_aft.py` — the D9.6 AFT benchmark on identical features and splits. *Harness*: the temporal-split, validation-tail early-stopping, grid-then-one-test-shot protocol is inherited from `experiments/exp26_xgb_benchmark.py` and must not be reimplemented; `experiments/exp16_survival.py` (discrete-time logistic hazard, KM calibration, ranking-loss hybrid) is the prior art this chapter supersedes. Linear benchmarks (cloglog GLM with Firth (1993) penalization) live beside the boosted models in the same harness so that every gain is measured, as in the fired B1 test, at feature parity.

Chapter 10

Ranking Objectives: Who Gets Funded Next

10.1 The metric is the ranking

Chapter 1 argued that the first answer to "what is valuable to know" is an identity, not a time: given finite diligence capacity, which K firms should be looked at now because they are most likely to be the next to raise. The consuming action is a short list, refreshed on a cadence; nothing in that action consumes an arrival time. The target is therefore the conditional distribution of the next deal's identity, already fixed as Equation (1.2),

$$p(i \mid s,t) \;=\; \mathbb{P}\big(\text{next deal in sector } s \text{ goes to firm } i \,\big|\, \text{a deal occurs, } \mathcal{F}_{t-}\big), \qquad i \in \mathcal{R}_s(t), \tag{10.1}$$

where $\mathcal{R}_s(t) \subseteq \mathcal{I}$ is the risk set of tracked firms at risk in sector s at t^-. This chapter's thesis is a sharpening of that framing: when T1 is the valuable thing, *the likelihood of the full marked process is not the metric — the ranking is.* The full likelihood rewards timing accuracy that no action consumes and that the data defects of Chapter 2 corrupt; the object to be scored is the ordered list, and the object to be fitted is the conditional law Equation (10.1), whose own likelihood — the conditional NLL of the mark factor in Theorem 3.7 — is the proper training loss precisely because it is a likelihood *of the ranking target and of nothing else.*

The bridge between intensity models and Equation (10.1) is the competing-intensities identity. Chapter 3 states it as part of the ground–mark factorization; here we give the operational proof, because every robustness claim in this chapter is a corollary of the *form* of the identity — a ratio.

Assumption 10.1 (Firm-level intensity system)**.** Fix a sector s and work conditionally on $\mathcal{F}_{t-}$. Each firm $i \in \mathcal{R}_s(t)$ has a counting process N_i with $\mathcal{F}$-conditional intensity $\lambda_i(t) \geq 0$ (Definition 2.3), the intensities are locally integrable, no two firms jump at the same instant almost surely, and the sector ground process $N_s = \sum_i N_i$ has intensity $\lambda_{\mathrm{g}}(t) = \sum_{j \in \mathcal{R}_s(t)} \lambda_j(t) \in (0, \infty)$.

Proposition 10.2 (Competing intensities)**.** *Under Assumption 10.1:*

(i) (Exact form, at the event.) *For almost every t, the conditional distribution of the identity of an event occurring at t is*

$$\mathbb{P}\big(\textit{identity} = i \mid \textit{event at } t,\ \mathcal{F}_{t-}\big) \;=\; \frac{\lambda_i(t)}{\sum_{j \in \mathcal{R}_s(t)} \lambda_j(t)}. \tag{10.2}$$

(ii) (Next-event form.) *Fix a decision time τ and suppose in addition that on the no-event continuation after τ the intensities are $\mathcal{F}_\tau$-measurable functions $u \mapsto \lambda_i^\tau(u)$ (true for*

internal-history models with exogenous covariate paths frozen at τ). Then the next event time T has conditional density $\lambda_{\mathrm{g}}^{\tau}(u)\exp(-\int_{\tau}^{u}\lambda_{\mathrm{g}}^{\tau})$ and, jointly with the identity J,

$$\mathbb{P}(J=i,\ T\in \mathrm{d}u \mid \mathcal{F}_\tau) \;=\; \lambda_i^{\tau}(u)\,\exp\Bigl(-\int_{\tau}^{u}\lambda_{\mathrm{g}}^{\tau}(v)\,\mathrm{d}v\Bigr)\,\mathrm{d}u, \qquad \mathbb{P}(J=i \mid T=u,\mathcal{F}_\tau) \;=\; \frac{\lambda_i^{\tau}(u)}{\lambda_{\mathrm{g}}^{\tau}(u)}. \tag{10.3}$$

Proof. (i) View the system as a single marked point process on $[0,T]\times\mathcal{R}_s(t)$ with intensity kernel $\lambda(t,\{i\})=\lambda_i(t)$; the absence of common jumps makes this kernel well defined, and superposition gives the ground intensity $\lambda_{\mathrm{g}}=\sum_j\lambda_j$. The ground–mark factorization of Theorem 3.7 writes the kernel as $\lambda(t,\{i\})=\lambda_{\mathrm{g}}(t)\,p(i\mid t,\mathcal{F}_{t-})$, where $p(\cdot\mid t,\mathcal{F}_{t-})$ *is*, by construction, the conditional mark distribution given an event at t (Daley and Vere-Jones, 2003, Ch. 6). Identifying the two expressions for the kernel and dividing by $\lambda_{\mathrm{g}}(t)>0$ yields Equation (10.2).

(ii) The operational derivation. Partition $(\tau,u]$ into n intervals of mesh $\delta=(u-\tau)/n$. Given $\mathcal{F}_\tau$ and the frozen intensity functions, the probability that no firm jumps in $(v,v+\delta]$, given no jump before v, is $1-\lambda_{\mathrm{g}}^{\tau}(v)\delta+o(\delta)$, and these no-jump factors multiply across subintervals because, in the absence of events, the conditional jump probabilities depend only on the deterministic functions λ_j^{τ} — this is the independent-increment approximation, exact in the limit $\delta\to0$ by the product-integral representation of the survivor function (Andersen et al., 1993). Hence $\mathbb{P}(T>u\mid\mathcal{F}_\tau)=\exp(-\int_\tau^u\lambda_{\mathrm{g}}^{\tau})$. The event $\{T\in(u,u+\delta],\ J=i\}$ requires no jump on $(\tau,u]$ and a jump of firm i in $(u,u+\delta]$, of probability $\lambda_i^{\tau}(u)\delta+o(\delta)$; multiplying and letting $\delta\to0$ gives the joint density in Equation (10.3). Dividing by the marginal density of T — the same expression with λ_i^{τ} replaced by $\lambda_{\mathrm{g}}^{\tau}$ — gives the ratio. When exogenous covariates evolve stochastically after τ, part (ii) holds with the intensities replaced by their conditional expectations along the no-event continuation; part (i) is unaffected, and as $u\downarrow\tau$ the two parts agree, so ranking *at the decision time* inherits the exact form (i). $\square$

Corollary 10.3 (Every intensity model is a ranker)**.** *Any firm-level intensity model $\{\lambda_i\}$ induces a ranker: order $\mathcal{R}_s(t)$ by $\lambda_i(t)$, equivalently by the conditional probability Equation (10.2), of which $\lambda_i(t)$ is a monotone transform. Under the model, this ordering is optimal for every list metric considered in this chapter (Lemma 10.9). Conversely — and this is the direction the chapter exploits — the ratio Equation (10.2) can be modeled* directly*, without ever committing to a timing model for λ_{g}.*

The ratio form is the load-bearing observation. Any factor common to the whole risk set — the sector's level of activity, the calendar week's deal flow, a macro regime, an unmodeled burst — appears in every λ_j and cancels. The next section turns this cancellation into a likelihood with a proof of exactly what it buys.

10.2 The conditional-choice mark factor

The repository's mark factor models the ratio directly. Each candidate $i\in\mathcal{R}_s(t)$ receives a score built from point-in-time features $z_i(t)$ (firm covariates with staleness ages, Chapter 8; engineered history summaries, Definition 2.2) and a cooldown indicator $C_i(t)=\mathbf{1}\{\text{firm } i \text{ funded in the last } K_c \text{ weeks}\}$:

$$q_i(t) \;=\; (w_0+u_s)^{\top}z_i(t) \;+\; \eta_s\,C_i(t), \qquad \eta_s\le 0, \tag{10.4}$$

with a global weight vector w_0, sector deviations u_s, and sector-specific cooldown coefficients η_s. The conditional choice probability is the softmax over the *current* risk set,

$$p(i\mid s,t) \;=\; \frac{\exp q_i(t)}{\sum_{j\in\mathcal{R}_s(t)}\exp q_j(t)}, \tag{10.5}$$

and the training loss is the conditional negative log-likelihood over observed events (t_m, s_m, i_m),

$$-\ell_c(\vartheta) \;=\; \sum_m \Big[\operatorname{logsumexp}_{j \in \mathcal{R}_{s_m}(t_m)} q_j(t_m) \;-\; q_{i_m}(t_m) \Big]. \tag{10.6}$$

The normalization over $\mathcal{R}_s(t)$ rather than over a fixed candidate list is the crucial design choice: firms enter the risk set by becoming active and leave it by the exposure rules of Chapter 9, and the model is well defined for every risk-set size. Algebraically, Equations (10.5) and (10.6) are McFadden's conditional logit (McFadden, 1974; Train, 2009); they are simultaneously, term for term, the factors of Cox's partial likelihood for a proportional-intensity model (Cox, 1972, 1975). The two literatures prove the same theorem twice, and we will lean on both.

Proposition 10.4 (Baseline cancellation)**.** *Suppose the firm intensities have the multiplicative form*

$$\lambda_i(t) \;=\; \lambda_0(s,t)\, \exp q_i(t), \qquad i \in \mathcal{R}_s(t), \tag{10.7}$$

where $\lambda_0(s,t) \geq 0$ is any *sector–time baseline common to the risk set: unknown, unmodeled, nonstationary, dependent on the whole history, or misspecified in any way whatsoever. Then the conditional identity distribution given an event is Equation (10.5), free of λ_0. Consequently the conditional likelihood Equation (10.6) involves no timing model, and any deformation of the time axis or of the baseline that acts on the risk set as a common factor — weekly aggregation, jittered or back-filled dates that preserve the assignment of events to risk-set snapshots, bursts, regime shifts, unmodeled excitation in the ground process — leaves Equation (10.6) unchanged.*

Proof. Substitute Equation (10.7) into Equation (10.2):

$$\frac{\lambda_i(t)}{\sum_j \lambda_j(t)} = \frac{\lambda_0(s,t)e^{q_i(t)}}{\lambda_0(s,t)\sum_j e^{q_j(t)}} = \frac{e^{q_i(t)}}{\sum_{j \in \mathcal{R}_s(t)} e^{q_j(t)}}.$$

The baseline cancels because it is common; nothing about its form was used. For the aggregation claim: replacing t_m by its containing bin changes which snapshot $(\mathcal{R}_s, z, C)$ the event is matched to, but if features and risk sets are constructed at bin resolution from pre-bin information (Protocol 2.5), the conditional law of the winner given the bin's snapshot is again Equation (10.5) with the bin-level baseline cancelled — interval censoring corrupts λ_0, which does not appear. What does *not* cancel is instructive: a firm-specific multiplicative frailty $\lambda_i = \lambda_0 \xi_i e^{q_i}$ does not cancel (it is not common) and is absorbed into the score as an omitted variable; and a baseline that differs *within* the risk set — e.g. stage-specific deal clocks inside one sector — breaks the common-factor premise and must be moved into q_i as a covariate or handled by stratifying the risk set. □

Proposition 10.5 (Robustness of the conditional MLE)**.** *Let events $m = 1, \ldots, M$ carry risk sets and features constructed point-in-time, with $i_m \in \mathcal{R}_{s_m}(t_m)$. Suppose (a) the conditional model is correctly specified: there exists ϑ^* such that $\mathbb{P}(i_m = i \mid \text{event } m, \mathcal{F}_{t_m^-}) = p(i \mid s_m, t_m; \vartheta^*)$ of Equation (10.5); (b) features are uniformly bounded, ϑ^* is identified and interior to a compact parameter set; (c) the averaged conditional information $\bar{\mathcal{I}}_M = M^{-1} \sum_m \mathbb{E}[V_m]$, with V_m the conditional covariance of the score gradient under Equation (10.5), converges to a positive-definite limit. Then the maximizer $\hat{\vartheta}$ of ℓ_c is consistent and asymptotically normal,* with no assumption on the ground process*: the law of (t_m, s_m) may be arbitrarily misspecified, dependent, aggregated, or simply never modeled.*

Proof. Let $\mathcal{G}_m = \mathcal{F}_{t_m}$ be the event-indexed filtration. The m-th score contribution at ϑ^* is

$$U_m \;=\; \nabla q_{i_m}(t_m) \;-\; \sum_{j \in \mathcal{R}_{s_m}(t_m)} p(j \mid s_m, t_m; \vartheta^*)\, \nabla q_j(t_m),$$

and by (a), $\mathbb{E}[U_m \mid \mathcal{G}_{m-1}, \text{event } m \text{ at } (t_m, s_m)] = \sum_i p_i^* \nabla q_i - \sum_j p_j^* \nabla q_j = 0$: the score is a martingale difference array with respect to $\{\mathcal{G}_m\}$ *conditionally on the event stream, whatever its law.* Boundedness (b) and nondegeneracy (c) then give a law of large numbers and a martingale central limit theorem for the score, and consistency plus asymptotic normality of $\hat{\vartheta}$ follow by the standard argument — this is precisely Cox's partial-likelihood heuristic (Cox, 1975) made rigorous by counting-process martingale theory in Andersen and Gill (1982); the conditional-logit sampling theory of McFadden (1974) gives the same conclusion when events are treated as independent choice occasions, and the martingale version shows independence is not needed. The proof fails exactly where it should: if the risk set is wrong (defect (C3): members retained who cannot win, or competitors omitted), the softmax in U_m is normalized over the wrong set and the conditional mean of the score is no longer zero at ϑ^* — the estimand silently changes to the conditional law *given the contaminated risk set.* Robustness to timing is not robustness to membership. □

Propositions 10.4 and 10.5 together are the precise version of the claim previewed in Section 1.3: the mark factor needs no timing model, survives the (C4) aggregation and untrusted-date defect wholesale, and is insulated from ground-process misspecification — the three failure modes that Chapter 6 shows corrupt every timing estimate on current data. That is why T1 is the most defensible deliverable on present data quality, and why the ranking layer, not the Hawkes layer, is the repository's headline deliverable while the timing stack matures.

10.3 Positive-only labels: manufacturing the comparison

A supervised-classification framing of T1 needs a negative class, and the data has none (Chapter 2): we observe fundings, never rejections. No row anywhere says "firm i sought capital at t and was declined," so per-firm labels are positive-only, and any attempt to manufacture firm-level negatives by declaring non-events ("firm i did not raise in week t") imports the timing model through the back door — the probability of a non-event depends on λ_0, which we just proved we do not need and cannot trust. The conditional-choice construction solves the label problem structurally: each funding event manufactures a valid comparison — the funded firm against its risk-set peers at the same sector-time — and Proposition 10.5 says these comparisons identify the score parameters without a single explicit negative. This is the same device by which Cox regression learns hazard ratios from cases and risk sets alone, and the connection to case-control methodology is exact: the risk set is the control pool, sampled at the case's own event time.

The practical obstacle is scale: real sector risk sets contain hundreds to thousands of firms, and Equation (10.6) sums over all of them per event. Sampling the risk set fixes this without changing the estimand.

Theorem 10.6 (Consistency of the sampled softmax)**.** *For event m, let a subset $D_m \subseteq \mathcal{R}_{s_m}(t_m)$ with $i_m \in D_m$ be drawn by a sampling protocol $\pi(D \mid i, \mathcal{R})$ — the probability of drawing D given that the winner is i — satisfying the* uniform conditioning property*: $\pi(D \mid i) = \pi(D \mid j) > 0$ for all $i, j \in D$. Then, under the model Equation (10.5), the conditional law of the winner given D_m is the softmax restricted to D_m,*

$$\mathbb{P}(i_m = i \mid D_m, \mathcal{F}_{t_m^-}) \;=\; \frac{\exp q_i(t_m)}{\sum_{j \in D_m} \exp q_j(t_m)}, \tag{10.8}$$

so maximizing the subsampled conditional likelihood built from Equation (10.8) is maximum likelihood in a correctly specified model and inherits the consistency of Proposition 10.5. In particular, uniform sampling of J non-winners from $\mathcal{R} \setminus \{i_m\}$, with the winner always retained, *satisfies the property, and no correction term is needed. For a general protocol, replacing q_j by $q_j + \log \pi(D \mid j)$ restores Equation (10.8) (McFadden, 1978; Train, 2009).*

Proof. By Bayes' rule on the joint draw (winner first, then the sampled set given the winner),

$$\mathbb{P}(i_m = i \mid D_m = D) \;=\; \frac{p(i \mid s_m, t_m)\, \pi(D \mid i)}{\sum_{j \in D} p(j \mid s_m, t_m)\, \pi(D \mid j)} \;=\; \frac{e^{q_i + \log \pi(D|i)}}{\sum_{j \in D} e^{q_j + \log \pi(D|j)}},$$

where the full-risk-set normalizer of $p(\cdot \mid s_m, t_m)$ cancels between numerator and denominator — the same ratio trick as Proposition 10.4, now applied to the sampling design. Under uniform conditioning the $\log \pi$ terms are equal across $j \in D$ and cancel as a common shift of the scores, giving Equation (10.8). Uniform sampling of J alternatives yields $\pi(D \mid i) = \binom{|\mathcal{R}|-1}{J}^{-1}$ for every $i \in D$, which is constant. The subsampled events (i_m, D_m) then follow a correctly specified conditional logit over D_m, and Proposition 10.5 applies verbatim with $\mathcal{R}$ replaced by D. What is lost is efficiency, not consistency: the conditional information per event shrinks with J, so J is a compute–variance dial, not a bias dial. □

10.4 Cooldown: inhibition carried by the mark factor

Chapter 5 establishes that a nonnegative Hawkes kernel cannot express inhibition and enumerates the three legal routes: (a) rectified additive intensities, (b) exp-link log-intensity models, and (c) inhibition carried by the mark factor. The cooldown term of Equation (10.4) is route (c). A firm that has just raised is, for a while, less likely to raise again — not because the sector cooled, but because that firm left the market for capital. Encoding this as a negative-coefficient covariate in the score gives, by Proposition 10.4, a multiplicative suppression of the firm's *share*: while $C_i = 1$, firm i's conditional probability is scaled by $e^{\eta_s} < 1$ relative to its own baseline share, renormalized within the risk set. The ground layer is untouched — sector excitation stays positive and the linear Hawkes/MBPP mean-field structure of Chapters 5 and 6 survives, which is exactly why route (c) was chosen as the default (the positivity link and the expectation operator do not commute for routes (a) and (b), breaking the closed-form machinery).

The economic interpretation is direct: η_s is the log of the factor by which a recent raise suppresses a firm's odds of being the next funded firm in its sector, and the constraint $\eta_s \leq 0$ (enforced as a bound in `fit_startup_ranker`) encodes the sign as a modeling assumption rather than a hope. The rehearsal evidence says the device works even under misspecification: in regime B — where the generator's inhibition is a continuous 30-day-half-life kernel, not a binary window — the fitted mark-factor cooldown recovered the generator's self-inhibition *ordering* across firms with rank correlation 0.937 (Appendix C). The binary window $C_i(t)$ with default $K_c = 26$ weeks is, of course, a one-column basis for a decaying curve; the natural refinements — an age-since-last-deal feature with learned shape, or a few staggered windows — belong to the gradient-boosted scorer of the next section, which can learn the shape without a parametric commitment.

10.5 Gradient-boosted scores in the same likelihood

The linear score Equation (10.4) is a restriction, and it has been falsified: the B1 rehearsal test fired, with a feature-parity XGBoost ranker beating the linear conditional logit by +0.22 nats of held-out conditional NLL and +9 points of top-5 hit rate on regime A (Appendix C). The design response is not to abandon the conditional-choice likelihood but to change the score family inside it: replace $q_i(t)$ by $f(z_i(t), C_i(t), a_i(t), \dots)$ with f an additive tree ensemble (Friedman, 2001; Chen and Guestrin, 2016), keeping the softmax over the current risk set and the loss Equation (10.6) exactly as they are. Training requires only the per-candidate gradient and Hessian of the grouped-softmax loss, which is where the "listwise" machinery of the learning-to-rank literature collapses into three lines of calculus.

Lemma 10.7 (Grouped-softmax gradients)**.** *Fix one event group m with candidate scores $\{q_j\}_{j\in\mathcal{R}_m}$, softmax probabilities p_j, and one-hot label $y_j = \mathbf{1}\{j = i_m\}$. The per-event loss $\ell_m = \operatorname{logsumexp}_j q_j - q_{i_m}$ has*

$$\frac{\partial \ell_m}{\partial q_j} = p_j - y_j, \qquad \frac{\partial^2 \ell_m}{\partial q_j \partial q_k} = p_j \mathbf{1}\{j = k\} - p_j p_k, \tag{10.9}$$

so the diagonal Hessian entries are $h_j = p_j(1 - p_j)$. The full Hessian $\operatorname{diag}(p) - pp^\top$ is positive semidefinite (it is the covariance matrix of the candidate-indicator vector under p), so ℓ_m is convex in the scores and the diagonal is a legitimate curvature surrogate.

Proof. Differentiate logsumexp: $\partial_{q_j} \operatorname{logsumexp} = p_j$; subtracting $\partial_{q_j} q_{i_m} = y_j$ gives the gradient. Differentiating p_j once more gives $\partial_{q_k} p_j = p_j \mathbf{1}\{j = k\} - p_j p_k$. For a vector v, $v^\top(\operatorname{diag}(p) - pp^\top)v = \mathbb{E}_p[v_J^2] - (\mathbb{E}_p[v_J])^2 = \operatorname{Var}_p(v_J) \geq 0$. □

Implementation is a custom objective for XGBoost: candidates of one event form one group; per boosting round, compute the softmax within each group from the current margin scores, emit $g_j = p_j - y_j$ and $h_j = p_j(1 - p_j)$ per candidate row, and let the tree learner take the regularized Newton step in function space (Chen and Guestrin, 2016). Dropping the off-diagonal Hessian terms is the standard diagonal approximation; by convexity of ℓ_m it costs step quality, never correctness of the stationary point. Note what this objective is and is not: it *is* the exact conditional NLL of Equation (10.6) — the listwise cross-entropy per event group — so the fitted trees produce genuine conditional probabilities through the softmax; it is *not* a pairwise surrogate in the RankNet family (Burges et al., 2005), which we treat separately in Section 10.7. Sampled groups per Theorem 10.6 keep the objective correct at scale.

Warning 10.8 (The robustness belongs to the family, not the score). Every robustness statement of Section 10.2 — baseline cancellation, timing-free consistency, survival of aggregation — is a property of the conditional-choice *likelihood family*: any scorer, linear or boosted, normalized over the current risk set inherits them. None of them certify the *linear* score, which the B1 verdict falsified, and none of them certify a boosted scorer trained on a different loss (a pairwise surrogate, a pointwise classifier): change the likelihood family and the cancellation proofs no longer apply. When this chapter says "the ranker is robust to (C1) and (C4)," it means the grouped-softmax likelihood is; claims about any particular fitted score function are empirical and live in Appendix C.

10.6 Multiple deals per sector-week: Plackett–Luce

Weekly resolution creates ties: several deals land in the same sector-week cell and share a risk-set snapshot. The current implementation treats them as repeated independent softmax draws over the same risk set. Survival analysts will recognize the situation instantly: tied event times in a Cox partial likelihood, where the repeated-softmax treatment is exactly Breslow's approximation (Breslow, 1974), the exact treatment is the sequential-conditioning likelihood, and Efron's correction sits between them (Efron, 1977; Therneau and Grambsch, 2000). The discrete-choice literature derives the exact object independently as the Plackett–Luce model (Plackett, 1975; Luce, 1959).

The derivation is a corollary of Proposition 10.2. Let d deals occur in cell (s, t) with winners $i_1, \ldots, i_d$ in sequence. After the k-th deal, firm i_k ceases to compete for the remaining within-cell deals (it has just raised; within one week its continued presence in the denominator is a modeling error, not a modeling choice). Applying Equation (10.2) to each successive deal over the shrinking risk set gives the sequential softmax without replacement:

$$\mathbb{P}(i_1,\ldots,i_d \mid s,t,d \text{ deals}) \;=\; \prod_{k=1}^{d} \frac{\exp q_{i_k}(t)}{\sum_{j\in\mathcal{R}_s(t)\setminus\{i_1,\ldots,i_{k-1}\}} \exp q_j(t)}, \tag{10.10}$$

which is the Plackett–Luce likelihood of the ordered top-d under score utilities. The repeated-softmax approximation replaces every denominator by the full-risk-set sum, so its per-cell log-likelihood error is

$$\Delta\ell_{s,t} \;=\; \sum_{k=2}^{d} -\log\left(1-P_{k-1}\right), \qquad P_{k-1} \;=\; \sum_{l<k} p(i_l \mid s,t), \tag{10.11}$$

the cumulative softmax mass of already-funded firms. Two consequences. First, the sign is systematic: the approximation understates every later-deal probability, so it penalizes exactly the concentrated, confident score profiles the ranker is supposed to produce, biasing fitted scores toward diffuseness. Second, the size is quantifiable: to first order $\Delta\ell_{s,t} \approx \binom{d}{2}\bar{p}$, with $\bar{p}$ the typical winner probability — negligible when risk sets are large and diffuse ($\bar{p} \sim 10^{-2}$, $d \le 2$), material when many deals hit a small or concentrated risk set in one week ($d \ge 3$ with $\bar{p} \sim 10^{-1}$ costs tenths of a nat per cell, the same order as the entire B1 improvement). Since same-week multiplicity is common in hot sectors — precisely the cells the business cares about — the exact likelihood is worth its implementation cost. One honest complication: Equation (10.10) conditions on the within-cell *order*, and (C4) says within-week order is exactly what the dates cannot support. The clean treatment is the exchangeable-order PL: average the log-likelihood over random permutations of the cell's winners (exact summation over $d!$ orders for the typical $d \le 3$, sampled permutations beyond), which removes the dependence on untrusted ordering at the price of a small, controlled Monte Carlo cost.

Design decision (D10.2 — Plackett–Luce in the next ranker revision).

The next revision of the mark factor replaces repeated softmax with the Plackett–Luce likelihood Equation (10.10), with within-cell order handled by exact averaging over permutations for $d \le 3$ and sampled permutations beyond, and with cooldown state updated between within-cell draws in simulation (as `simulate_marked_paths` already does). Rationale: Equation (10.11) gives a systematic, concentration-punishing bias whose magnitude in hot cells reaches the scale of the B1 gain; the Cox-ties literature says the exact method dominates Breslow whenever ties are heavy (Therneau and Grambsch, 2000). Reversal trigger: if on real data the measured NLL and recall@K differences between PL and repeated softmax are inside the era-blocked bootstrap intervals of Protocol 10.10 for two consecutive evaluation campaigns, revert to repeated softmax for simplicity and record the equivalence.

10.7 Optimizing the list directly

If the consumed object is only the top-K list, one can ask whether the likelihood should be bypassed entirely in favor of a loss that targets the list metric — the learning-to-rank program: pairwise RankNet (Burges et al., 2005), listwise LambdaMART with its λ-gradients aimed at NDCG (Burges, 2010; Järvelin and Kekäläinen, 2002). The temptation deserves a precise answer, and the answer has two halves: a lemma that says likelihood training is already list-optimal *under the model*, and an empirical clause that says surrogates can win only through misspecification — which B1 proved is real.

Lemma 10.9 (Top-K optimality of the conditional probabilities)**.** *Fix a sector-time* (s, τ) *and let* p_i, $i \in \mathcal{R}_s(\tau)$, *be the conditional identity distribution of the next deal. For a list* $L \subseteq \mathcal{R}_s(\tau)$ *with* $|L| = K$, *the expected recall@K against the next deal is* $\mathbb{E}[\mathbf{1}\{winner \in L\}] = \sum_{i \in L} p_i$, *and it is maximized by any* L *consisting of* K *largest-*p_i *candidates. For an ordered list* σ, *the expected reciprocal rank* $\sum_{r \geq 1} p_{\sigma(r)}/r$ *is maximized by sorting* p *in descending order. Hence, under the model, producing the top-*K *by conditional probability is the optimal list policy for both headline metrics, and NDCG with flat gains and any decreasing discount is maximized by the same sort.*

Proof. Recall: the objective $\sum_{i \in L} p_i$ is a set function maximized by exchange: if $i \in L$, $i' \notin L$, and $p_{i'} > p_i$, swapping i' for i strictly increases the sum; iterating terminates at a set of K largest probabilities. MRR: the discounts $1/r$ are strictly decreasing; if an ordering places $p_{\sigma(r)} < p_{\sigma(r')}$ with $r < r'$, transposing them changes the objective by $(p_{\sigma(r')} - p_{\sigma(r)})(1/r - 1/r') > 0$; absence of improving transpositions is exactly the descending sort — the rearrangement inequality. The NDCG claim is the same argument with $1/r$ replaced by the discount sequence. Under multiple deals per window the argument applies to each successive deal with the PL-updated distribution; ranking by the current p_i remains optimal for the first hit, which is what recall@K scores. □

The lemma settles the division of labor. A correctly specified, likelihood-trained conditional model followed by a sort cannot be beaten in expectation on recall@K, MRR, or NDCG; training on a surrogate can help only by trading bias in regions the likelihood cares about for accuracy in the top of the list when the score family cannot represent the truth. That trade is real (B1), but it has a cost we refuse to pay in the primary model: a LambdaMART scorer emits uncalibrated ordinal scores, while the decision layer of Chapter 12 consumes *probabilities* — conditional identity probabilities composed with ground intensities to produce statements like "firm i is on the list *and* a deal is likely within the quarter" (T1 composed with T2/T3), and the conditional NLL is the strictly proper score for that object (Gneiting and Raftery, 2007). A scorer that cannot feed the composition is a dead end in this architecture regardless of its list metrics.

Design decision (D10.3 — LambdaMART as challenger only).

LambdaMART (and pairwise RankNet as its diagnostic little sibling) is maintained as a benchmarked *challenger* in the evaluation harness, never as the primary model. Rationale: Lemma 10.9 says it can only win through score-family misspecification, which the boosted grouped-softmax primary already addresses inside the likelihood family; and its output does not feed the probability-consuming decision layer. Reversal trigger: if LambdaMART beats the likelihood-trained boosted ranker on pooled recall@5 by more than the era-blocked bootstrap interval for two consecutive campaigns, *and* a post-hoc calibration overlay (Platt, 1999; Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005) on its scores passes the calibration checks of Chapter 7 at the event-group level, it is promoted to primary and this section is rewritten around the surrogate-first architecture.

10.8 The recall at K protocol

Everything above concerns fitting; what follows is the committed evaluation contract — the repository's headline metric, stated fully so that no future implementation "simplifies" it into something unauditable.

Protocol 10.10 (Rolling recall@K and MRR for the mark factor)**.**

(a) *Origins.* Evaluation origins τ run on a weekly grid over the held-out era. At each τ and each sector s, freeze the risk set $\mathcal{R}_s(\tau)$ under the versioned risk-set definition (item (g)), compute point-in-time scores, and emit the full ordering of $\mathcal{R}_s(\tau)$; the top-K prefixes are the lists $L_{s,\tau,K}$, $K \in \{1, 5, 10\}$.

(b) *Window and hits.* The evaluation window is the next W weeks, $(\tau, \tau + W]$; we commit to next-W-weeks rather than next-K'-deals so that empty windows are informative and the metric has a fixed temporal meaning. W is matched to the consuming action's refresh horizon: default $W = 4$ (a monthly sourcing look-ahead), with $W = 13$ reported as a robustness column. A deal in the window scores a *hit* at level K iff its firm was on $L_{s,\tau,K}$ — membership at τ, not at deal time.

(c) *Denominator and newcomers.* The denominator counts deals in the window whose firm was in $\mathcal{R}_s(\tau)$. Deals to firms outside every frozen risk set at τ — first appearances and out-of-universe deals — are *excluded from the denominator and reported separately* as the newcomer share, under the newcomer-vs-entrant discipline of Chapter 11. Silently scoring newcomer deals as misses (or dropping them without reporting) is forbidden: the first deflates the metric for a failure the mark factor cannot commit, the second hides the (C6) exposure.

(d) *Metrics.* Report recall@K for $K \in \{1, 5, 10\}$ and MRR (reciprocal rank of the funded firm in the full frozen ordering), per sector and pooled (deal-weighted), aggregated over origins rolling-origin style. Alongside, *always*, the held-out conditional NLL of Equation (10.6) on the same events: the list metrics say whether the sort is right, the NLL says whether the probabilities feeding the decision layer are right, and Lemma 10.9 is the license to demand both from one model.

(e) *Uncertainty.* Era-blocked bootstrap: resample contiguous blocks of evaluation origins (blocks aligned with the macro-era segmentation of Chapter 7) with replacement, recompute pooled metrics, report percentile intervals. Origin-level i.i.d. resampling is forbidden — overlapping windows and regime persistence make it anti-conservative.

(f) *Baselines.* Every report carries the same metrics for the random ranker over the same frozen risk sets (chance floor) and, on synthetic data, the oracle ranker (ceiling); the rehearsal reference values are in Table 10.1.

(g) *Frozen, versioned risk sets.* The risk-set definition — entry rule, exposure/exit rule from Chapter 9, sector assignment — is frozen per evaluation campaign and identified by a version string recorded with every metric row; changing the rule starts a new metric series rather than overwriting one. Rationale: risk-set truthfulness is the single largest sensitivity in the system — rehearsal top-5 was 0.14 on observed (contaminated) risk sets against 0.29 under oracle membership (Appendix C) — so an unversioned membership rule would make every metric movement uninterpretable.

Design decision (D10.1 — the T1 deliverable and its two metrics).

The primary T1 deliverable is the sector-conditional risk-set ranker: gradient-boosted scores inside the grouped-softmax conditional likelihood Equation (10.6) (per Lemma 10.7), with cooldown and staleness-age features, sampled risk sets per Theorem 10.6 where scale requires, upgraded to Plackett–Luce per D10.2. It is evaluated by Protocol 10.10 *and*

Held-out metric	fitted	oracle (ceiling)	random (floor)
top-1 hit rate	16.7%	21.2%	2.9%
top-5 hit rate	49.2%	53.7%	14.3%
MRR	0.328	0.371	0.118

Table 10.1: Rehearsal reference numbers for the mark-factor ranker (Appendix C): the fitted ranker recovers most of the oracle's performance, and the oracle itself is far from 1 — the residual is irreducible softmax stochasticity, not model error. These values calibrate expectations for real-data campaigns; they are priors and vetoes, never substitutes (Section 2.5).

held-out conditional NLL — both, always; a model improving one while degrading the other is investigated, not shipped. Rationale: the competing-intensities identity makes the conditional model the exact T1 target (Proposition 10.2); the cancellation and consistency results make it the defect-robust factor (Propositions 10.4 and 10.5); the B1 verdict selects boosted over linear scores within the family; and Lemma 10.9 guarantees likelihood training plus sorting is list-optimal under the model while the NLL keeps the probabilities honest for Chapter 12. Reversal triggers: if the boosted scorer fails to beat the linear conditional logit on real data (both metrics, two campaigns), demote to the linear score for interpretability; if the challenger clause of D10.3 fires, the primary itself is reconsidered.

10.9 Defect declaration

Per design decision D2.1, the mark factor's position against the ledger of Table 2.1, with the proofs above doing the work.

(C1) Administrative censoring: robust. The conditional likelihood Equation (10.6) is a sum over observed events only; no survival factor appears because timing is conditioned away (Proposition 10.4), and under independent censoring the risk set at each observed event is uncorrupted. Open spells cost the ranker nothing — there is simply no term.

(C2) Left truncation: robust for the conditional estimand, with a scope caveat. Comparisons are made within risk sets of firms that have entered observation; the estimand is conditional on membership, so delayed entry does not bias it. What (C2) forbids is re-reading scores as statements about the untracked population — that extension belongs to Chapter 11.

(C3) Unlabeled exits: vulnerable — the binding defect. Dead firms retained in $\mathcal{R}_s(t)$ occupy denominator mass and top-K slots; Proposition 10.5's proof shows the estimand silently becomes the conditional law given the contaminated pool. The rehearsal quantified the cost as the largest sensitivity in the whole system (top-5 0.14 observed vs 0.29 oracle; an exit classifier with AUC 0.877 was not sufficient to close the gap, Appendix C). Mitigations are the exposure rules of Chapter 9 (trailing-activity windows closing spells point-in-time) plus staleness features that let the scorer learn to demote ghost-like profiles. One inherited caution must be stated: the rehearsal's inverted-weight result shows that the Bayes-optimal score weighting *given a contaminated pool* differs from the oracle weighting — the fitted model partially adapts to contamination, so cleaning the risk set changes the optimal parameters, and oracle-world or cleaned-world coefficients must never be transplanted onto contaminated risk sets or vice versa (Appendix C). Risk-set definition and fitted model are a matched pair; Protocol 10.10(g) versions them together.

(C4) Untrusted dates and aggregation: robust, by proof. Proposition 10.4: binning and jitter deform $\lambda_0(s,t)$, which cancels; the residual exposure is only through snapshot assignment

(a deal jittered across a week boundary meets a one-week-shifted risk set and covariates) and through within-cell order, which D10.2's exchangeable-order Plackett–Luce removes.

(C5) Stale covariates: vulnerable, with signed bias, remedies inherited. Stale z_i is classical covariate measurement error inside a choice model: score coefficients attenuate toward zero, the softmax flattens, lists diffuse, and recall@K degrades toward the random floor — a bias against the model, never in its favor. The remedies of Chapter 8 (staleness ages as first-class features, covariate-state layer where it pays) apply verbatim to the ranker's feature block, and the boosted scorer can learn staleness-dependent discounting that the linear score cannot.

(C6) Out-of-universe deals: vulnerable without the outside option. Deals from untracked firms mean the true choice set exceeds $\mathcal{R}_s(t)$; a mark factor without an outside option overstates every tracked firm's probability by the factor $1/(1-\pi_{\text{new}})$. The treatment — the newcomer branch with its exact separation of binary and choice likelihoods — is Chapter 11's; this chapter's protocol already quarantines the exposure by excluding newcomer deals from the recall denominator and reporting their share (Protocol 10.10(c)).

Code hooks.

Existing. `hawkes_calibration/sector_ranker.py`: `fit_startup_ranker` implements Equations (10.4) to (10.6) with sector deviations, the $\eta_s \leq 0$ bound of Section 10.4 (`constrain_cooldown_negative`), and dynamic risk sets via `candidate_set` and `cooldown_vector`; `ranker_predict_proba` exposes the conditional probabilities; `evaluate_ranker` computes held-out conditional NLL, MRR, and top-k hit rates and is the seed of the harness below. `hawkes_calibration/models/hawkes_ranker.py` carries the regime-B rehearsal ranker behind the cooldown result of Section 10.4.

To be written. (i) The grouped-softmax XGBoost objective module (`hawkes_calibration/ranking/grouped_softmax_objective.py`): per-group softmax from margin scores, per-candidate (g_j, h_j) of Equation (10.9), group boundaries from event ids, uniform risk-set sampling per Theorem 10.6 with the winner retained. (ii) The Plackett–Luce upgrade of D10.2 inside `hawkes_calibration/sector_ranker.py` (sequential denominators, exchangeable-order averaging for $d \leq 3$). (iii) The evaluation harness (`hawkes_calibration/eval/ranking.py`) implementing Protocol 10.10 verbatim: weekly origins, frozen versioned risk sets (`riskset_version` recorded per metric row), W-week windows, newcomer-share reporting, era-blocked bootstrap, and the random/oracle baseline columns of Table 10.1. (iv) The LambdaMART challenger of D10.3 as a harness plug-in, never as a dependency of the decision layer.

Part V

The Open Universe

Chapter 11

Deals from Companies We Cannot See

11.1 Two populations, one deal stream

Every model developed so far has quietly assumed that we know who could receive the next deal. This chapter is about the two ways that assumption fails. First, deals arrive from firms that are simply not in the tracked universe: the deal row exists, carries a date and a sector, and is flagged `company_universe = 'Out of Universe'`, but the firm has no companies-table row, no covariates, and no history — defect (C6) of Chapter 2. Second, deals arrive from firms that will be in the universe tomorrow but are invisible today: a firm enters observation *by raising*, so the tracked population is selected on the very event we model — defect (C2). Neither failure is small. In the synthetic rehearsal, roughly 27% of all deals went to firms outside the current risk set, and the share sat at 23–25% in held-out test windows (Appendix C). A mark factor that ignores this mass overstates every tracked firm's probability by a factor of roughly 1.3–1.4; a ground model that ignores it cannot forecast sector totals honestly. The remedy of this chapter is structural, not cosmetic: an explicit outside option in the mark factor, an external component in the ground process, a model-free audit of the unseen mass, and an entry model for the population side of (C2).

We first fix what is and is not observed, in the notation of Definition 2.1.

Definition 11.1 (Tracked and untracked populations; observed stream)**.** The firm universe splits as $\mathcal{I} \cup \mathcal{I}^\dagger$: the *tracked* universe $\mathcal{I}$ (a companies-table row, covariates $\tilde{x}_i$, an event history N_i) and the *untracked* remainder $\mathcal{I}^\dagger$ (no row, no covariates, no attributable history). The observed deal stream is

$$\{(t_m,\, s_m,\, \iota_m)\}_{m\geq 1}, \qquad \iota_m \;=\; \begin{cases} i_m, & i_m \in \mathcal{I}, \\ \text{`Out of Universe'}, & i_m \in \mathcal{I}^\dagger, \end{cases} \tag{11.1}$$

together with deal marks m_m: the time and sector of every deal are recorded, but firm identity is recorded only on $\mathcal{I}$. Operationally, a deal is *untracked* if its universe flag says so *or* if its firm makes its first-ever appearance in the data at t_m (no prior attributable history existed at t_m^-).

Assumption 11.2 (Complete sector stream)**.** Every deal row carries a deal time and a sector label; no deal is dropped from the stream for being untracked. Sector labels are correct up to the small mislabel noise documented in Appendix B, whose treatment (hygiene rules; the group-level variant of Section 11.3) is orthogonal to universe membership.

Assumption 11.2 is a statement about the vendor's delivery format, and it is the load-bearing asymmetry of this chapter: sector-level ground counts are *complete* — every deal has a sector — while firm-level attribution is complete only on $\mathcal{I}$. The consequence deserves to be stated as a result, because it silently decides which layers of the architecture (C6) can hurt.

Proposition 11.3 (Attribution asymmetry)**.** *Under Assumption 11.2, with the exact factorization of Theorem 3.7:*

(i) The ground factor — the likelihood of times and sectors, with sector intensities $\Lambda_s(t)$ built from sector-level histories and exogenous covariates — is a functional of fully observed data. Every ground-process estimand and estimator is invariant to (C6): the ground layer is immune.

(ii) The mark factor is not. Let $\pi_{s,t} = \mathbb{P}(i_{next} \notin \mathcal{R}_s(t) \mid deal\ in\ s\ at\ t, \mathcal{F}_{t^-})$ be the probability that the next deal in s goes outside the tracked risk set. A mark factor fitted on tracked-attributed deals only estimates the conditional law $\mathbb{P}(i \mid deal, i \in \mathcal{R}_s(t))$, and for every tracked firm i,

$$\mathbb{P}(i \mid \text{deal in } s, t) \;=\; (1 - \pi_{s,t})\, \mathbb{P}(i \mid \text{deal in } s, t,\ i \in \mathcal{R}_s(t)). \tag{11.2}$$

Reading the restricted model's output as an absolute probability overstates every tracked firm by the factor $(1 - \pi_{s,t})^{-1}$.

(iii) Any firm-level intensity assembled from the two layers, $\lambda_i(t) = \Lambda_{s(i)}(t)\, \mathbb{P}(i \mid \text{deal})$, inherits the inflation of (ii) multiplicatively, as do all horizon probabilities (target T3) built from it.

Proof. (i) By Theorem 3.7 the ground factor of the log-likelihood is $\sum_m \log \Lambda_{s_m}(t_m) - \sum_s \int_0^T \Lambda_s(u)\,\mathrm{d}u$, which depends on the data only through the pairs (t_m, s_m) and the exogenous covariate paths. By Assumption 11.2 these are observed for every deal, untracked or not; hence the ground likelihood, its maximizer, and any functional of the fitted Λ_s are unchanged by the loss of firm identities. (The caveat — that Λ_s's history term must itself be built from *all* deals — is exactly the histories rule of Section 11.4; violate it and immunity is forfeited.) (ii) The event $\{i\} \subseteq \{i \in \mathcal{R}_s(t)\}$ for $i \in \mathcal{R}_s(t)$, so the chain rule gives $\mathbb{P}(i) = \mathbb{P}(i \in \mathcal{R})\, \mathbb{P}(i \mid i \in \mathcal{R}) = (1 - \pi_{s,t})\mathbb{P}(i \mid i \in \mathcal{R})$. A likelihood evaluated only on deals with $i_m \in \mathcal{R}$ is, term by term, the likelihood of the conditional law given $\mathcal{R}$-membership; its MLE targets that conditional law, not the absolute one. (iii) Multiply (ii) by $\Lambda_s(t)$; the marking theorem for marked point processes (Daley and Vere-Jones, 2003, Ch. 6) identifies $\Lambda_s(t)\mathbb{P}(i \mid \text{deal})$ as firm i's intensity, and the factor $(1 - \pi_{s,t})^{-1}$ carries through every monotone functional of λ_i, including $p_i(t, \Delta)$. □

At the rehearsal's test-window shares, $\pi \approx 0.23$–0.25, the inflation in Equation (11.2) is 30–33% on every tracked firm — uniform enough to leave within-sector *rankings* intact (which is why Chapter 10 could defer the issue) but fatal to any absolute probability, any cross-sector comparison in which $\pi_{s,t}$ differs by sector, and any sector total decomposed by firm. Proposition 11.3 is also one more argument for the two-layer architecture of Theorem 3.7: the defect ledger of Table 2.1 assigns (C6) to the mark layer *alone*, so the timing deliverable (T2) survives on sector-complete counts while the identity deliverable (T1) is repaired where it is broken — in the mark factor, by the outside option of Section 11.3.

11.2 Newcomers are not entrants

Before building the repair, one distinction must be carved into the chapter, because every downstream interpretation depends on it.

Warning 11.4 (Newcomer ≠ entrant). The *newcomer probability* $\pi_{s,t}$ is the probability that the next deal's firm is not in the current tracked risk set — equivalently, that the deal's firm has no attributable prior appearance in the data. This is an *observable*: every deal either matches a tracked history or it does not, so $\pi_{s,t}$ can be estimated, calibrated, and audited. The *entrant probability* — the probability that a genuinely new economic entity enters the market — is *not identified* from this data without an entry model and auxiliary coverage information, because a first appearance conflates two channels: economic entry (a firm's first raise ever) and first-time *tracking* of an old firm (vendor coverage catching up). The two channels are an unidentified mixture: in the rehearsal, untracked incumbents accounted for roughly 12% of out-of-universe deals — a nontrivial slice of the $\sim 27\%$ newcomer share — and separating the channels with oracle labels changed the likelihood by only +0.003 nats (Appendix C). The conflation therefore costs almost nothing in fit and cannot be detected by fit; what it costs, permanently, is every compositional claim: "entry accelerated in 2021," "new company formation shifted to fintech," "the entrant cohort of 2020 is larger than 2019's" are all unsupported unless the coverage channel is modeled or measured. Policy: in code, documentation, and reporting, this repository says *newcomer*, never *entrant*, for the observable; *entrant* is reserved for the entry-model constructs of Section 11.6, always with their identification caveats attached.

The warning has a converse worth stating: because the two channels are indistinguishable in likelihood, a single outside option that absorbs both loses essentially nothing as a *predictive* device. The architecture below therefore models the newcomer mass as one bucket and defers the decomposition to the entry model and the coverage audit, which are the only places it can honestly live.

11.3 The mark factor with an outside option

11.3.1 The extended risk set

The repair to the mark factor is a one-row change with exact semantics. Extend each sector's risk set with an *outside option* $\Diamond_s$:

$$\mathcal{R}_s^+(t) \;=\; \mathcal{R}_s(t) \cup \{\Diamond_s\}, \tag{11.3}$$

and give the outside option a score built from *context* covariates only — nothing firm-level exists on that branch by construction:

$$q_{\Diamond_s}(t) \;=\; v^\top g_{s,t}, \qquad g_{s,t} = \big[\text{trailing newcomer share, sector deal volume, macro as-ofs, vintage/era effects, 1} \tag{11.4}$$

while tracked firms keep their ranker scores $q_i(t) = \beta^\top z_i(t)$ from Chapter 10. The mark distribution is the softmax over the extended set,

$$\mathbb{P}\big(m = a \mid \text{deal in } s, t,\ \mathcal{F}_{t^-}\big) \;=\; \frac{e^{q_a(t)}}{e^{q_{\Diamond_s}(t)} + \sum_{j\in\mathcal{R}_s(t)} e^{q_j(t)}}, \qquad a \in \mathcal{R}_s^+(t), \tag{11.5}$$

so that $\pi_{s,t} = \mathbb{P}(\Diamond_s \mid \cdot)$ is produced by the same softmax that produces firm probabilities. The next proposition says this is not a heuristic patch but the *correct* conditional model of the observable mark when the next deal may go outside $\mathcal{I}$.

Proposition 11.5 (The collapsed mark model is exact)**.** *Let $P^*(\cdot \mid s, t, \mathcal{F}_{t^-})$ be the true next-deal firm distribution on $\mathcal{I} \cup \mathcal{I}^\dagger$, and define the collapsed mark $\tilde{m} = i$ if $i \in \mathcal{R}_s(t)$ and $\tilde{m} = \Diamond_s$ otherwise. Then:*

- *(i) $\tilde{m}$ is fully observed for* every *deal, including untracked ones; the extended-softmax likelihood built from Equation (11.5) is the exact mark-factor likelihood of the collapsed marked process — no approximation is introduced by the collapse, and no deal is dropped.*

(ii) The family Equation (11.5) is well specified for the collapsed law: for any categorical distribution with full support on $\mathcal{R}_s^+(t)$ there exist scores realizing it (logits equal log-probabilities up to an additive constant), and the parameterization adds only the outside score Equation (11.4); identification is up to the usual location shift, with the outside option playing the role of the outside good's alternative-specific constant (McFadden, 1974; Train, 2009).

(iii) Estimation is unchanged: the log-likelihood is the standard conditional-choice form of Chapter 10 with one extra row per choice set, and its MLE targets the KL projection of the true collapsed law onto the family. In particular the tracked-firm probabilities it outputs are absolute *— the inflation of Equation (11.2) is removed by construction, since $\mathbb{P}(i \mid \cdot) = (1 - \pi_{s,t}) \cdot \mathrm{softmax}_{\mathcal{R}_s(t)}(q)_i$ holds identically in the extended model.*

Proof. (i) The collapse map $a \mapsto \tilde{m}$ is a deterministic, measurable coarsening of the mark space. For a tracked deal we observe i_m and hence $\tilde{m}_m$; for an untracked deal we observe precisely the event $\{i_m \notin \mathcal{R}_{s_m}(t_m)\}$, which *is* $\tilde{m}_m = \Diamond_{s_m}$. The likelihood of a coarsened mark is the pushforward of P^* under the collapse, and Equation (11.5) is a density for exactly that pushforward; nothing unobservable enters. What is lost is only resolution *within* $\mathcal{I}^\dagger$ and among out-of-pool firms — which the data does not carry anyway. (ii) Given target probabilities $(p_a)_{a\in\mathcal{R}^+}$ with $p_a > 0$, set $q_a = \log p_a + c$: the softmax returns (p_a) for any c, and conversely the softmax determines scores up to c — the standard bijection between positive categorical laws and score differences. Restricting $q_\Diamond$ to the span of $g_{s,t}$ makes the family parametric in the usual sense; whether the *true* $\pi_{s,t}$ lies in that span is an empirical question answered by the calibration protocol below, not by the algebra. (iii) Adding one alternative to a conditional-logit choice set changes neither the objective's form, nor its concavity in the linear-in-features parameterization, nor the softmax gradient; misspecified-model MLE converges to the KL projection under the usual regularity. The displayed identity is the chain rule applied inside the softmax: conditioning Equation (11.5) on $\{a \neq \Diamond_s\}$ cancels the outside term from the denominator. □

Two estimation remarks. First, at newcomer shares of 23–27% the outside option is not a rare event, so the intercept bias of rare-event logistic estimation (King and Zeng, 2001) is minor; if per-sector shares in thin sectors fall to a few percent, the Firth correction (Firth, 1993) costs one line and removes the concern. Second, the context features in Equation (11.4) are exactly the covariates that made the newcomer share *calibratable over time* in the rehearsal: with the enriched context specification, quarterly newcomer-share forecasts calibrated to within ±2pp on test quarters, the standing acceptance target (Appendix C). The trailing newcomer share must be computed on an expanding or trailing window strictly before t (Protocol 2.5), and the vintage/era terms exist to absorb the coverage drift of Section 11.7 rather than let it contaminate the macro loadings.

11.3.2 The architecture is a two-stage classifier, recognized

The committed factorization — ground process for sector timing, mark factor for firm-given-sector with an outside option — should be recognized as an instance of a familiar design pattern: *predict the group first, then the member within the group, with an outlier class.*

Proposition 11.6 (Two-stage equivalence). *Given a deal at t, let $\Lambda(t) = \sum_s \Lambda_s(t)$ be the superposed ground intensity. Then for any tracked firm i with sector $s(i)$,*

$$\mathbb{P}(i \mid \text{deal at } t, \mathcal{F}_{t^-}) \;=\; \underbrace{\frac{\Lambda_{s(i)}(t)}{\Lambda(t)}}_{\text{stage 1: sector}} \;\times\; \underbrace{\frac{e^{q_i(t)}}{e^{q_{\Diamond_{s(i)}}(t)} + \sum_{j\in\mathcal{R}_{s(i)}(t)} e^{q_j(t)}}}_{\text{stage 2: firm or outlier}}, \tag{11.6}$$

and the residual mass $\sum_s \frac{\Lambda_s(t)}{\Lambda(t)}\pi_{s,t}$ *is the probability the deal goes outside the tracked universe altogether. Thus the two-layer model* is *the two-stage sector→firm classifier with a per-sector outlier bucket; conversely, any such two-stage classifier with strictly positive stage probabilities defines a valid mark distribution on* $\bigcup_s \mathcal{R}_s^+(t)$*.*

Proof. By the marking theorem for the superposition of sector streams (Daley and Vere-Jones, 2003, Ch. 6), conditional on a deal at t the sector is distributed as $\Lambda_s(t)/\Lambda(t)$; multiplying by the conditional firm law Equation (11.5) is the chain rule. The converse direction is immediate: nonnegative stage probabilities that sum to one compose, by the same chain rule, into a categorical law on the disjoint union of extended risk sets. □

The equivalence earns its display for one practical reason: it makes the *group-level variant* a first-class alternative rather than an architectural upheaval. Sector labels carry mislabel noise (Assumption 11.2; Appendix B), and at the finest granularity a mislabeled deal both pollutes one sector's ground counts and lands in the wrong choice set. Collapsing sectors into industry groups is another measurable coarsening of the mark — the same quotient construction as Proposition 11.5 — so stage 1 may equally predict the industry *group* and stage 2 rank firms within the group's (larger) risk set, with one outlier bucket per group. Mislabels that stay within a group then cancel entirely. The trade is variance for bias: coarser groups mean larger softmax denominators and weaker sector-specific context in Equation (11.4). The decision between granularities belongs to the goodness-of-fit machinery of Chapter 7 applied per level; the mathematics is level-agnostic, provided — and this is a hard constraint — ground and mark layers use the *same* partition, or Equation (11.6) silently stops being a probability distribution.

11.3.3 Calibration of the outside option is a deliverable

With the outside option in place, the newcomer share stops being a nuisance and becomes a first-class model output: $\pi_{s,t}$ is a predicted probability, realized outcomes are observed for every deal, and the pair supports exactly the calibration discipline of Chapter 7.

Protocol 11.7 (Reliability of the outside option)**.** On every evaluation window: (a) per quarter and per sector, compare the average predicted $\pi_{s,t}$ over deals against the realized newcomer share, reporting the mean absolute error (rehearsal reference: 3–4pp with context covariates, 6–9pp for a constant; Appendix C); (b) pool deals into bins by predicted π and report a reliability diagram with binomial bands; (c) report the same by era, so that coverage drift (Section 11.7) is visible as an era-structured miscalibration rather than averaged away. Failures of (a) with stable context covariates are escalated to the coverage audit (Protocol 11.12) *before* any model change.

Design decision (D11.1 — outside option in the next ranker revision).

The next revision of the mark-factor ranker extends every choice set with the outside-option row Equation (11.3)–Equation (11.4): context features on the $\Diamond_s$ row only, firm features on tracked rows only, one shared softmax. This extends `hawkes_calibration/sector_ranker.py`'s risk-set construction and mirrors the synthetic loader's `build_choice_sets(newcomer_context=True)` path, with `experiments/exp28_newcomer_designs.py` (designs N1–N4) as prior art. Rationale: Proposition 11.3 (the restricted model's probabilities are inflated by $(1-\pi)^{-1}$), Proposition 11.5 (the extension is exact and costs nothing in estimation), and the measured calibration gain of context covariates. Reversal trigger: if on real data the realized newcomer share falls below 2% in every sector and era *and* Protocol 11.7 shows a constant share calibrates within 1pp, the outside option may be frozen to an intercept-only score — removed it is not, since Equation (11.2) holds at any $\pi > 0$.

11.4 The ground process: the external component and the histories rule

11.4.1 Decomposing the immigrant intensity

On the ground layer, (C6) and (C2) appear not as missing labels but as a component of intensity that no tracked-firm bookkeeping can generate. Write the sector ground intensity in Hawkes form (Chapter 5) and split the baseline:

$$\Lambda_s(t) \;=\; \underbrace{\mu_s^{\mathrm{tr}}(t) + \mu_s^{\mathrm{ext}}(t)}_{\mu_s(t)} \;+\; \sum_{m:\, t_m < t} h_{ss_m}(t - t_m), \tag{11.7}$$

where μ_s^{tr} carries the part of the immigrant intensity attributable to the tracked population's state and μ_s^{ext} is the *external component*: deal flow from firms we cannot see, driven by the context covariates $g_{s,t}$ of Equation (11.4) (and, once Section 11.6 exists, by cohort-based newcomer supply). The decomposition is not identified by the ground layer alone — both components are baselines — but it is pinned by the mark layer through the marking theorem:

Corollary 11.8 (Newcomer substream intensity)**.** *If the outside-option probability $\pi_{s,t}$ of Equation (11.5) is predictable, then the newcomer deals of sector s form a point process with conditional intensity $\pi_{s,t}\,\Lambda_s(t)$, and the tracked-attributed deals one with intensity $(1-\pi_{s,t})\,\Lambda_s(t)$. The sector total decomposes exactly as*

$$\Lambda_s(t) \;=\; \underbrace{(1 - \pi_{s,t})\Lambda_s(t)}_{\textit{tracked}} \;+\; \underbrace{\pi_{s,t}\Lambda_s(t)}_{\textit{newcomer}}, \tag{11.8}$$

which is the position-dependent thinning of the ground process by the mark factor (Daley and Vere-Jones, 2003, Ch. 5).

Equation (11.8) is the honest version of a sector forecast: a predicted quarterly total *with its newcomer fraction attached*, rather than a tracked-only total silently presented as the market. It is also the Hawkes–Oakes immigrant/offspring split (Hawkes and Oakes, 1974) read at the mark level: the newcomer substream is the immigrant channel of the branching representation, arriving from outside the tracked system's genealogy. One rehearsal finding tempers the interpretation: when the two substreams were fitted separately, the *newcomer* stream carried the larger apparent excitation radius (0.30 against 0.08–0.11 for repeats; Appendix C) — entry waves are where latent common factors live, so apparent "excitation" on the newcomer stream must be read as clustering of unknown origin, in line with the frailty-inflation verdict of Chapter 5 (Appendix C: fitted branching ratios are upper bounds).

11.4.2 The histories rule, and the price of breaking it

The excitation term in Equation (11.7) raises the operational question with the sharpest teeth in this chapter: *which deals enter the history sum?* Untracked deals are real events; they move the sector's state and excite future deal flow exactly as tracked deals do. The rule is therefore:

Design decision (D11.2 — histories include all deals).

Every history functional in every fitter — lagged sector counts in the weekly GLM, exponential-kernel sums in the event-time fitters, trailing volume covariates, cooldown clocks at sector level — is computed from the *complete* deal stream, untracked deals included. Only *firm-level attribution* (risk-set membership, firm histories N_i, firm covariates) is restricted to $\mathcal{I}$. Rationale: Proposition 11.9 below — dropping untracked deals from histories understates excitation by a factor of about the tracked share and distorts the decay estimate — and Proposition 11.3(i), whose immunity claim is conditional on exactly this rule. Reversal trigger: none; this is an accounting identity about what an event is, not a tunable choice.

The bias when the rule is broken can be derived exactly in the linear case. Model the loss of untracked deals as independent thinning: each event is retained (tracked) with probability $q = 1 - p$ independently of everything else — optimistic, since real coverage is not independent of firm type, but sufficient to sign and size the effect.

Proposition 11.9 (Thinning bias in the branching ratio)**.** *Let N be a stationary linear Hawkes process with immigrant rate μ, kernel h with branching ratio $\kappa = \int_0^\infty h < 1$, and let N^q be its independent thinning with retention $q = 1 - p$ (Daley and Vere-Jones, 2003, Ch. 11). Then:*

(i) (Rates) N^q has mean rate $\bar{\lambda}^q = q\,\mu/(1-\kappa)$: thinning scales the mean by q exactly.

(ii) (First generation) In the Hawkes–Oakes cluster representation (Hawkes and Oakes, 1974), the expected number of retained *direct offspring of a retained event is $q\kappa = (1-p)\kappa$. Any estimator that identifies branching by attributing observed events to observed parents — the declustering/EM estimand (Veen and Schoenberg, 2008) — therefore targets first-generation mass scaled by $(1-p)$.*

(iii) (All generations) The expected number of retained descendants (all generations) of a retained event is $q\kappa/(1-\kappa)$; matching the total-progeny identity $\kappa_{\text{app}}/(1-\kappa_{\text{app}}) = q\kappa/(1-\kappa)$ gives

$$\kappa_{\text{app}} = \frac{(1-p)\,\kappa}{1-p\kappa} \in \big[(1-p)\kappa,\ \kappa\big), \tag{11.9}$$

again $(1-p)\kappa$ to first order in κ. In both readings the apparent branching ratio is understated, and the missing excitation mass reappears as an inflated apparent baseline: $\mu_{\text{app}} = \bar{\lambda}^q(1-\kappa_{\text{app}}) = q\mu/(1-p\kappa) > q\mu$.

Proof. (i) The stationary mean of a subcritical linear Hawkes solves $\bar{\lambda} = \mu + \kappa\bar{\lambda}$, giving $\mu/(1-\kappa)$ (sum the geometric cascade of generations, $\mu\sum_{k\geq 0}\kappa^k$). Independent thinning multiplies the first moment measure by q (Daley and Vere-Jones, 2003, Ch. 11). (ii) In the cluster representation each event, independently of everything else, produces direct offspring as an inhomogeneous Poisson process with mean κ. The retention marks are i.i.d. Bernoulli(q), independent of the genealogy and of each other; by Wald's identity the expected number of retained direct offspring is $q\kappa$, and conditioning on the parent's own retention changes nothing by independence of the marks. (iii) Generation-k descendants of a fixed event have expected count κ^k; each is retained with probability q independently, so the expected retained progeny is $q\sum_{k\geq 1}\kappa^k = q\kappa/(1-\kappa)$. Solving the displayed identity for κ_{app} gives Equation (11.9); monotonicity in q and the bounds follow by inspection ($q = 1$ recovers κ). The baseline claim is (i) combined with $\mu_{\text{app}} = \bar{\lambda}^q(1-\kappa_{\text{app}})$ and $1-\kappa_{\text{app}} = (1-\kappa)/(1-p\kappa)$. □

Two remarks complete the picture. First, the *decay* is distorted alongside the scale: a retained grandchild whose intermediate parent was dropped presents to the fitter as a direct child at the *sum* of two kernel lags, so the apparent kernel is contaminated by convolution powers h^{*k} — heavier-tailed than h — and a fitted exponential decay θ is dragged toward slower values. Since Chapter 7 already shows θ is the fragile parameter (the κ–θ ridge), history thinning attacks the model exactly where it is weakest. Second, the repository's actual failure mode is the *mixed* case — sector counts complete (Assumption 11.2) but the history term built from tracked deals only, as happens whenever a fitter joins deals to the companies table and drops non-matches. Then the excitation covariate is a q-scaled, noisy version of the true excitation state: an errors-in-variables regression that attenuates the excitation coefficient and reroutes the missing mass into the baseline and into any covariate correlated with activity — the same understatement, arrived at by a different door. The covariates-dominate verdict of Appendix C already leaves excitation only decimal points of likelihood to live on; Proposition 11.9 says that breaking the histories rule takes a $(1-p)$-sized bite out of precisely those decimal points.

11.5 Unseen mass: the model-free audit

Everything so far trusts the model to measure the outside world through $\pi_{s,t}$. Good practice demands an estimate of the unseen mass that does not pass through the model at all, so that model and audit can disagree informatively. The classical tool is exactly fitted to our question: *what is the probability that the next deal comes from a firm we have not seen before?*

Fix a sector and a window containing N deals, and regard the firm identities as draws $X_1, \dots, X_N$ from an unknown distribution (p_i) over the (countable, partly invisible) firm population. Let N_r be the number of firms appearing exactly r times in the window, and let the *missing mass* $M_0 = \sum_i p_i \mathbf{1}\{i \notin \{X_1, \dots, X_N\}\}$ be the probability mass of firms not seen at all. The Good–Turing estimator (Good, 1953) is

$$\widehat{M}_0 \;=\; \frac{N_1}{N}, \tag{11.10}$$

the fraction of deals going to firms seen exactly once.

Proposition 11.10 (Good–Turing is nearly unbiased)**.** *If $X_1, \dots, X_N$ are i.i.d. (exchangeability suffices), then*

$$\mathbb{E}[\widehat{M}_0] \;=\; \mathbb{E}[M_0^{(N-1)}] \qquad \textit{and} \qquad 0 \;\le\; \mathbb{E}[\widehat{M}_0] - \mathbb{E}[M_0^{(N)}] \;=\; \sum_i p_i^2 (1-p_i)^{N-1} \;\le\; \tfrac{1}{N}\big(1-\tfrac{1}{N}\big)^{N-1} \;\le\; \tfrac{1}{2N},$$

where $M_0^{(n)}$ is the missing mass of a sample of size n: the estimator is unbiased for the missing mass of a sample one draw smaller, hence biased upward for $M_0^{(N)}$ by at most $O(1/N)$.

Proof. Leave one out. By exchangeability,

$$\mathbb{E}\Big[\frac{N_1}{N}\Big] = \frac{1}{N}\sum_{n=1}^{N} \mathbb{P}(X_n \text{ is a singleton in } X_{1:N}) = \mathbb{P}\big(X_N \notin \{X_1, \dots, X_{N-1}\}\big),$$

since X_N is a singleton in the full sample precisely when its firm is absent from the other $N-1$ draws. Conditioning on the first $N-1$ draws, that probability is $\mathbb{E}[\sum_i p_i \mathbf{1}\{i \notin X_{1:N-1}\}] = \mathbb{E}[M_0^{(N-1)}]$, which proves the identity; the estimator answers "will the *next* deal come from an unseen firm" almost by definition. For the bias, $\mathbb{E}[M_0^{(n)}] = \sum_i p_i (1-p_i)^n$, so

$$\mathbb{E}[\widehat{M}_0] - \mathbb{E}[M_0^{(N)}] = \sum_i p_i \big[(1-p_i)^{N-1} - (1-p_i)^N\big] = \sum_i p_i^2 (1-p_i)^{N-1} \le \max_{x\in[0,1]} x(1-x)^{N-1} \sum_i p_i,$$

and the maximum is attained at $x = 1/N$ with value $\frac{1}{N}(1-\frac{1}{N})^{N-1} \le \frac{1}{2N}$ for $N \ge 2$ (and $\to (eN)^{-1}$). When exchangeability fails the identity fails with it: this is the assumption to watch, not the algebra. □

The proof makes the operative assumption explicit: exchangeability of deals *within the window*. On this data that is violated by everything this document models — excitation, macro dynamics, coverage drift — so the estimator is applied on *rolling windows per sector*, short enough that within-window dynamics are second-order, long enough that N_1 is not noise; the smoothing practice of Gale and Sampson (1995) applies when small counts make the raw frequencies-of-frequencies ragged. Two further caveats. The window convention must match the functional: N_1/N on a trailing window estimates the window-relative unseen mass, which is the right comparator for a model probability evaluated under the same convention — first-ever appearance versus first-in-window appearance are different events, and the audit must fix one and compute both sides accordingly. And GT estimates a *probability mass*, not a population count. For the count question — how many fundable firms are we not seeing? — the Chao lower bound (Chao, 1984) on the number of distinct firms S active in the window is

$$\widehat{S}_{\text{Chao}} \;=\; S_{\text{obs}} + \frac{N_1^2}{2N_2} \;\le\; S \quad \text{(in expectation, as a lower bound)}, \tag{11.11}$$

valid as a floor precisely under the abundance heterogeneity that is certain to hold here (firms differ wildly in deal propensity), which is the honest direction: the invisible pool is *at least* this large.

Protocol 11.11 (Good–Turing versus the model)**.** Standing diagnostic, run with every evaluation campaign and reported alongside Protocol 11.7: per sector and rolling window, (a) compute $\widehat{M_0} = N_1/N$ and $\widehat{S}_{\text{Chao}}$ from the raw deal stream — no model involved; (b) compute the model's average predicted outside-option mass over the same deals under the same window convention; (c) report the paired series and their gap. Agreement within the binomial noise band is a nontrivial consistency check (two estimators of one functional, one parametric and one not); a drifting gap with stable context covariates is escalated to the coverage audit of Protocol 11.12 before any refit.

Design decision (D11.3 — unseen-mass audit ships with the evaluation harness).
The GT/Chao audit of Protocol 11.11 is implemented once, in the shared evaluation layer (planned `hawkes_calibration/eval/unseen_mass.py`), and run by default in every campaign — not as an occasional notebook. Rationale: it is the only estimate of the outside world that does not share assumptions with the model it audits, and it is nearly free (counting singletons). Reversal trigger: none anticipated; if window choice proves contentious, the audit gains a sensitivity panel over window lengths rather than being dropped.

11.6 Modeling entry: cohorts on the Lexis diagram

Defect (C2) has appeared so far as a missing-mass problem. Its population face is different: the tracked universe is *selected on having raised*, so the data is silent about the reservoir of founded-but-not-yet-funded firms from which tomorrow's newcomers arrive. An *entrant-profile module* models what can be modeled about that reservoir: the intensity of *first observed rounds*, organized by founding cohort.

The natural coordinate system is the Lexis diagram (Keiding, 1990): calendar time t on one axis, firm age $a = t - c$ (time since founding, cohort c) on the other; firms travel along unit-slope lines, and a first observed round at age a is a point on that line — observed only if it falls inside the data window (firms whose first raise postdates June 2024 are invisible entirely: right truncation of entry, the mirror image of (C1)). Because the number of not-yet-funded firms in a cohort is unobserved, only the *aggregate* first-round intensity is identified, and that is what the module fits:

$$\lambda_{s,c}^{\text{entry}}(t) \;=\; \exp\bigl(\alpha_s + \phi_{\text{era}(c)} + f_s(t-c) + \beta^\top x_{\text{pub}}^{\text{mac}}(t)\bigr), \qquad \mu_s^{\text{ext}}(t) \;\supseteq\; \sum_{c\le t} \lambda_{s,c}^{\text{entry}}(t), \quad (11.12)$$

with f_s an age profile (entry hazards peak in the first years after founding), ϕ era-of-founding effects (the dry-powder vintages), and macro as-ofs on the calendar axis. Founding dates come from the companies table — available, with (C4)-grade noise, for firms once they are tracked — so the cohort structure is estimable from each firm's *first appearance*, the one moment at which entry-relevant features are observed.

What the module buys is exactly two things. First, *honest sector totals*: with Equation (11.12) feeding μ_s^{ext}, the decomposition Equation (11.8) becomes a forecast of incumbent *plus* newcomer intensity with the newcomer part driven by cohort supply rather than extrapolated share — the difference matters whenever era effects move (a 2021-vintage cohort bulge produces newcomer flow for years). Second, *cohort-based forecasts of newcomer supply*: questions like "how much first-round flow should sector s expect next year given the founding activity of 2022–2024" become answerable, with the era effects carrying the dry-powder cycle.

What it cannot buy must be stated with equal clarity: *firm-level prediction for unseen firms*. A firm with no covariates admits no firm-level score; no entry model changes that. The honest statement of the module's output is a *sector-level newcomer intensity* together with a mark distribution over *observable-at-entry* features — stage, geography, sector, and the deal-time profile measured at first appearance — learned from tracked firms at their first rounds. That learned distribution is a *selection* distribution (features conditional on having entered), which supports predicting the profile of the next entrant; it says nothing about the latent population of firms that have not raised and may never (Warning 11.4 applies in full: the "entry" intensity Equation (11.12) is estimated from first *appearances* and inherits the coverage channel unless the audit of Section 11.7 clears it). The rehearsal adds a caution: newcomer utility was dominated by the alternative-specific constant and the context/pool-size terms rather than by entrant-profile quality (Appendix C: ASC 6.86 on regime A against observable-profile weights an order of magnitude smaller) — effort belongs in $g_{s,t}$ and the cohort structure, not in rich profile models.

Design decision (D11.4 — a planned `entrants/` module, scoped by identification).

The repository plans an `hawkes_calibration/entrants/` module implementing Equation (11.12): cohort–age–era first-round intensities per sector, an observable-at-entry mark distribution estimated at first appearances, and the coupling of both into μ_s^{ext} and the outside-option features $g_{s,t}$. The module's API exposes sector-level newcomer intensity and entrant-profile distributions only — it deliberately has no per-firm scoring surface, so the identification limit is enforced by the interface rather than by discipline. Rationale: the honest-totals and cohort-supply gains above; the interface constraint is Warning 11.4 made structural. Reversal trigger: acquisition of auxiliary coverage data (a second vendor, or vendor backfill metadata) that identifies the entrant-versus-untracked split would justify promoting the module to a genuine entry model with compositional claims.

11.7 Two-sided universe drift

One final structural fact cuts across everything above: the universe boundary itself moves, from both sides. Firms cross into $\mathcal{I}$ as PitchBook's coverage expands, and coverage has expanded materially over 2011–2024 — tracking improves, backfill happens, whole segments (regions, stages) arrive in waves. The untracked share is therefore time-varying *for reasons that are not market dynamics*: a declining `off_universe_share` is what improving coverage looks like, and it is also exactly what a genuine shift toward incumbent-dominated deal flow would look like. Every quantity in this chapter — $\pi_{s,t}$, the GT mass, the cohort intensities, the era effects of Equation (11.12) — confounds the two explanations unless coverage is audited first.

Protocol 11.12 (Coverage audit by era)**.** The loader reports `off_universe_share` by sector and by era (fixed calendar blocks) as a first-class output, on synthetic and real data alike (Chapter 2 already requires the aggregate; this protocol requires the era breakdown). Any trend in the series is treated as a coverage artifact *candidate* until examined: check for step changes aligned with vendor events rather than market events, for divergence between the untracked share and the GT newcomer mass (Protocol 11.11) — which respond differently to backfill — and for era-localized failures of Protocol 11.7.

Design decision (D11.5 — coverage audit precedes entry claims).

No claim about entry dynamics — trends in newcomer share, cohort sizes, era effects in Equation (11.12) — is reported, internally or externally, until Protocol 11.12 has been run on the relevant eras and the trend has survived it. Rationale: the drifting-share tripwire is the one diagnostic that doubles as a data-quality monitor; its most likely cause on this vendor's history is coverage, not the market, and Warning 11.4 already establishes that the data cannot arbitrate by itself. Reversal trigger: none; this is an ordering constraint on analysis, and it costs one audit per campaign.

11.8 Defect declarations and the ranking interface

Per Table 2.1 and decision D2.1, this chapter *owns* (C2) and (C6), and its models relate to the ledger as follows. *(C6)*: the ground layer is robust under Assumption 11.2 and the histories rule (Proposition 11.3); the extended mark factor *corrects* the defect exactly at the collapsed-mark resolution (Proposition 11.5); fitters that break the histories rule are vulnerable with signed bias — excitation understated by a factor $\approx (1-p)$, baseline inflated, decay dragged slow (Proposition 11.9). *(C2)*: the outside option absorbs its deal-stream face; the entry module models its population face at sector level under the stated identification limits; compositional claims remain vulnerable permanently absent auxiliary coverage data (Warning 11.4), and the vulnerability is accepted and fenced by D11.4/D11.5 rather than assumed away. *(C1)*: robust — all windows and spells here inherit the censoring treatment of Chapters 2 and 6; right truncation of entry is handled by construction in Equation (11.12). *(C3)*: not corrected here, and the interaction matters: unlabeled dead firms *deflate* tracked-firm probabilities by padding the softmax denominator, while a missing outside option *inflates* them (Equation (11.2)); the two biases can partially cancel in aggregate metrics, and this document forbids reading such cancellation as health — the two defects are diagnosed separately (Chapter 10 for (C3), Protocol 11.7 here). *(C4)*: sector mislabels perturb Assumption 11.2 marginally and motivate the group-level variant of Section 11.3.2; date jitter is immaterial to shares at weekly resolution.

(C5): the outside-option score Equation (11.4) uses only context covariates, which are never stale; the tracked rows inherit the treatment of Chapter 8 unchanged.

The interface with the ranking metrics of Chapter 10 must be fixed explicitly, because the outside option changes what recall can mean. Newcomer deals have no tracked target to recall: including them in the denominator of recall@K mechanically caps the metric at $1-\pi$ and makes scores incomparable across eras with different coverage. The coordinated protocol is therefore: recall@K and MRR are computed on tracked-attributed deals only, exactly as Chapter 10 specifies; the newcomer mass is reported *alongside* as the calibrated $\pi_{s,t}$ output under Protocol 11.7; and every ranking report carries both numbers, so that a model cannot buy apparent recall by silently wishing the outside world away, nor be penalized for deals no tracked-universe model could have named.

Code hooks.

Mark factor. `hawkes_calibration/sector_ranker.py` gains the extended risk set Equation (11.3): one outside-option row per choice set carrying the context features Equation (11.4), mirroring `synthetic/loaders.py`'s `build_choice_sets(newcomer_context=True)`; prior art and reference fits in `experiments/exp28_newcomer_designs.py` (N1 context score, N2 entrant moments, N3 two-step, N4 immigrant/offspring split). *Ground process.* The histories rule (D11.2) binds `hawkes_calibration/sector_stability.py`, the event-time fitters in `hawkes_calibration/eventtime/`, and the interval-censored tools in `hawkes_calibration/mbpp/`: history terms consume the complete deal stream; the external baseline component of Equation (11.7) enters as covariates until `hawkes_calibration/entrants/` exists. *Audits.* `hawkes_calibration/eval/unseen_mass.py` (to be written) implements Protocol 11.11 (Good–Turing mass, Chao floor, model comparison) and the reliability report of Protocol 11.7; `hawkes_calibration/data/audits.py` extends its universe-share check to the by-era breakdown of Protocol 11.12. *Entry.* The planned `hawkes_calibration/entrants/` module implements the cohort–age–era intensities Equation (11.12) and the observable-at-entry mark distribution, exposing sector-level outputs only (D11.4).

Part VI

Synthesis

Chapter 12

From Model Output to Investment Decisions

12.1 Closing the loop with Chapter 1

Chapter 1 defined worth by the value-of-information identity Equation (1.1): a model output matters only through the action it improves. The intervening chapters built the outputs — a sector-level ground intensity (Chapters 4 to 6), a firm-level conditional ranker with an outside option (Chapters 10 and 11), horizon event probabilities (Chapter 9), all under the evaluation constitution of Chapter 7 and the covariate honesty of Chapter 8. This chapter specifies the *decision layer*: the small set of composed, consumable deliverables, the decision rules that consume them, and the feedback loop that measures whether the whole apparatus is earning its keep. The layer is deliberately thin — it contains no new statistical estimation, only composition, thresholds, and reporting — because every quantity it consumes arrives already calibrated and already uncertainty-qualified; anything thicker would smuggle modeling into a place with no likelihood to check it.

12.2 The three deliverables

12.2.1 The sourcing list (T1 + T3)

The headline deliverable is a per-sector ranked list, refreshed weekly: for each sector s at time τ, the top-K tracked firms by conditional next-deal probability $\hat{p}(i \mid s, \tau)$ (Chapter 10), each annotated with its horizon probabilities $\hat{p}_i(\tau, \Delta)$ for $\Delta \in \{6\text{M}, 1\text{Y}, 2\text{Y}\}$ (Chapter 9), its staleness age $a_i(\tau)$ and last-known covariates (honesty about the information state, Chapter 8), and the sector's calibrated newcomer share $\hat{p}(\Diamond_s)$ (Chapter 11). The two probability columns answer different questions and must not be merged: the ranker column is *relative* (who, given a deal), the horizon column is *absolute* (whether, within Δ). A firm can top the ranker while all horizon probabilities are low because the sector itself is quiet; the pairing is exactly what tells the desk to watch rather than act.

Proposition 12.1 (Optimality of the top-K sourcing rule)**.** *Suppose diligence capacity allows opening at most K files in sector s, each opened file on firm i yields expected utility $b \cdot \mathbf{1}\{i \text{ raises within } \Delta\}$ minus cost c, with b, c constant across firms, and outcomes are as modeled. Then the expected-utility-maximizing selection is the top-K firms by $\hat{p}_i(\tau, \Delta)$, and opening the k-th file is worthwhile iff $\hat{p}_{(k)}(\tau, \Delta) \geq c/b$.*

Proof. Expected utility of a selection A, $|A| \leq K$, is $\sum_{i \in A} \left(b\, \hat{p}_i(\tau, \Delta) - c\right)$, additive in members; it is maximized by taking the firms with the largest positive terms, which are the top-ranked ones by $\hat{p}_i$, truncated at K or at the threshold $\hat{p}_i \geq c/b$, whichever binds first. The threshold form is the classical cost–benefit decision boundary for calibrated probabilities; the argument fails exactly when $\hat{p}$ is uncalibrated, which is why Chapter 9 makes calibration mandatory, and when utilities interact across firms (portfolio effects), which is out of scope by Section 12.3. □

The ranking-only analogue — expected recall of the next deals is maximized by ranking on conditional probability — was proved as the top-K optimality lemma of Chapter 10. Together the two results say: *under the model, the list is the decision*; everything else is threshold arithmetic supplied by the desk's own c/b.

12.2.2 Sector pacing (T2)

For capital-deployment pacing the deliverable is the predictive distribution of sector deal counts over the next 4–13 weeks: mean paths with predictive intervals, produced by simulating the fitted ground process forward (Chapters 5 and 6 simulators; scenario or bootstrapped macro paths per Chapter 2), with dispersion honestly propagated (negative-binomial noise where adopted, Chapter 4). Every pacing chart carries two inherited caveats as fixed footnotes: the fitted branching mass is an upper bound on contagion (Chapter 5), and timing resolution is bounded below by the honest resolution of Chapter 6 — no intra-week timing claims.

12.2.3 Deal-terms context (T4)

Mark regressions — round size, up/down/flat probability, valuation step conditional on a deal — contextualize a candidate: the same gradient-boosting stack as Chapter 9 applied to deal-level targets from Appendix B, with staleness features and the undisclosed-size selection handled per the data dictionary's imputation rules (`deal_size_status` is itself informative and enters as a feature, never silently imputed away). Terms models are context, not gates: they refine the utility b in Proposition 12.1, firm by firm, when the desk wants probability-weighted expected terms rather than a flat b.

12.3 Scope: what the decision layer refuses to do

Design decision (D12.1 — no silent portfolio optimization).

The decision layer outputs ranked lists, calibrated probabilities, and predictive count distributions. It does *not* output buy/pass verdicts, position sizes, or fund-level allocations. Rationale: portfolio-level utility involves correlations, capacity, and mandate constraints that live outside the data this document models; any such optimization must be a separate, owned system consuming our calibrated outputs. Reversal trigger: a stated fund-level objective with its own evaluation protocol.

Design decision (D12.2 — probabilities travel with their caveats).

Every number leaving the system carries: the model version, the evaluation-era metrics of Chapter 7, the risk-set definition (Chapter 10), the staleness summary of the inputs (Chapter 8), and the newcomer share (Chapter 11). The reporting discipline of Chapter 7 — no hazard without its risk-set rule, no branching ratio without its upper-bound caveat, no newcomer probability sold as an entrant probability — binds the decision layer most of all, because it is the layer other people quote. Reversal trigger: none; this is charter.

12.4 Measuring the value of information

Offline metrics are proxies. The loop closes only by measuring Equation (1.1) on the workflow itself.

Protocol 12.2 (VoI measurement)**.** (a) *Shadow deployment*: for at least one full quarter, sourcing lists are produced and frozen weekly but not shown to the desk; realized deals score them retrospectively (recall@K, lift over the desk's contemporaneous shortlists). (b) *Assisted period*: lists shown; the desk logs which opened files originated from the list vs. other channels. (c) The VoI estimate is the difference in hit rates per opened file between assisted and baseline periods, era-blocked as in Chapter 7; secular market shifts are partially controlled by the shadow-period comparison. (d) Attribution is reported with its confounds stated — this is an observational estimate, not a randomized one; if the desk's capacity permits, randomize *which sectors* get assisted lists per quarter for a cleaner contrast.

12.5 Monitoring, drift, and retraining cadence

Models decay: coverage drifts (Chapter 11), macro regimes shift, and the 2021–2023 funding cycle sits inside our evaluation eras as a standing warning. The cadence is quarterly re-evaluation against the rolling-origin protocol of Chapter 7, with three tripwires, each mapped to its owning chapter: (i) calibration drift on horizon probabilities beyond the tolerance band of the reliability protocol (Chapter 9) triggers recalibration before retraining; (ii) a trend in `off_universe_share` or in the Good–Turing missing mass (Chapter 11) triggers the coverage audit before any modeling response; (iii) attenuation-gap growth in covariate effects (Chapter 8's gap test) escalates the staleness remedy ladder. Retraining without a fired tripwire is discouraged: it churns the desk's mental model of the lists for no measured gain.

Code hooks.

The decision layer is a reporting module, not a modeling module: `hawkes_calibration/decision/` (to be written) composes existing outputs — `ranker` scores from `hawkes_calibration/sector_ranker.py` (extended with the outside option), horizon probabilities from `hawkes_calibration/models/xgb_horizon.py` (planned, Chapter 9), simulated count fans from the ground-process simulators, and terms context — into versioned weekly artifacts (per-sector JSON + rendered sheets). The VoI protocol needs a tiny `hawkes_calibration/decision/log.py` for desk-interaction logging. Tripwire metrics reuse the `hawkes_calibration/eval/` harness of Chapter 7; nothing in this layer may compute a probability itself.

Chapter 13

Implementation Roadmap

13.1 Reading the roadmap

This chapter converts the document's decisions into ordered engineering work on `hawkes_calibration/`. Each phase lists its modules (existing, extended, or new), the chapter that owns the mathematics, and the *gate* that must pass before the phase is called done. Gates are inherited from the evaluation constitution of Chapter 7: every new fitter passes a parameter-recovery rehearsal on synthetic data generated from its own family (and a misspecified stress world where one exists), every likelihood passes analytic-vs-numeric gradient checks at the tolerances already standard in the repository (Appendix C), and every data transformation passes its audit. Phases are ordered by the dependency structure of the mathematics, not by expected impact; the impact ordering is already encoded in the standing verdicts (covariates first, risk-set truthfulness above all, excitation last).

13.2 Phase 0 — the real-data spine

Owns: Chapters 2, 6 and 11. Everything else blocks on this.

P0.1 `hawkes_calibration/data/pitchbook.py`: the adapter from raw PitchBook deals/companies/investors extracts (2011–June 2024) to the fitter-ready arrays of `synthetic/loaders.py` — weekly sector counts, as-of macro matrix, LOCF firm features *with staleness ages* (Chapter 8, D8.1), at-risk masks, event list with universe flags. The synthetic loader's array contract is frozen as the interface; Appendix B is the column contract.

P0.2 `hawkes_calibration/data/audits.py`: the defect-ledger validation suite of Chapter 2 — open-spell retention, entry-date consistency, as-of join look-ahead test, duplicate-collapse idempotence, coverage-share report by era (Chapter 11), date-quality audit separating announced/closed/backfilled dates (Chapter 6).

P0.3 Honest-resolution decision executed: the date audit fixes the modeling resolution (expected: weekly) per D6.1.

Gate: audits green on the real extract; a data note recording coverage share, date-quality composition, and staleness-age distributions by sector — the empirical face of Table 2.1.

13.3 Phase 1 — baselines that must be beaten

Owns: Chapters 4, 8 and 9 (R1), Chapter 10 (linear).

P1.1 Sector ground baseline: seasonal + macro NHPP via the binned GLM equivalence (Chapter 4), then the negative-binomial noise upgrade (D4.2); held-out NLL ladder vs. random and historical-mean baselines established as the standing leaderboard (Chapter 7).

P1.2 `hawkes_calibration/panel/`: the bucketed panel builder of Chapter 9 — disjoint buckets per horizon $\Delta \in \{26, 52, 104\}$ weeks, censoring-aware labels, point-in-time features including staleness ages and interactions (R1), leakage audits.

P1.3 `hawkes_calibration/models/xgb_horizon.py`: per-horizon XGBoost classifiers with temporal early stopping and mandatory post-hoc calibration (D9.5); reliability reports into `hawkes_calibration/eval/`.

P1.4 Ranker on real data: the existing linear risk-set ranker (`hawkes_calibration/sector_ranker.py`) with staleness features, evaluated by the recall@K protocol of Chapter 10 — this is the first real sourcing-list prototype.

Gate: the covariates-dominate verdict re-established (or refuted) on real data — nats-ladder note comparing seasonal, macro, and history-augmented sector models; horizon models calibrated within tolerance; recall@K baseline numbers frozen for all later comparisons.

13.4 Phase 2 — the committed architecture

Owns: Chapters 5, 6, 10 and 11.

P2.1 Sector-level continuous-time Hawkes at honest resolution via the MBPP likelihood (`hawkes_calibration/mbpp/` already implements the machinery; wire it to the real arrays); excitation reported with the upper-bound protocol; the κ–θ ridge diagnostics of Chapter 7 standing.

P2.2 Grouped-softmax XGBoost objective (Chapter 10's gradient/Hessian derivation) so the ranker's score function gains interactions while keeping the conditional-likelihood family — the response to the fired B1 test.

P2.3 Outside option $\lozenge_s$ in the risk set (Chapter 11, D11.x): newcomer share becomes a calibrated output; recall@K denominators split tracked vs. newcomer per protocol; Good–Turing and coverage audits wired into `hawkes_calibration/eval/`.

P2.4 `hawkes_calibration/eval/`: the consolidated harness of Chapter 7 — NLL ladders, PIT/dispersion, time-rescaling, rolling-origin driver, recall@K/MRR with era-blocked uncertainty — replacing per-experiment ad hoc scripts.

P2.5 `hawkes_calibration/decision/`: the composition layer of Chapter 12 (weekly artifacts, caveat propagation, VoI logging).

Gate: end-to-end weekly run on real data producing the three deliverables of Chapter 12; a full evaluation campaign note in the style the repository already uses, with every metric attached to its protocol.

13.5 Phase 3 — escalations, each behind its trigger

Owns: Chapter 8 (R2–R4), Chapter 10 (PL), Chapter 11 (entrants), Chapter 5 (signed kernels, frailty).

P3.1 `hawkes_calibration/covstate/`: the supervised drift dataset from consecutive-deal pairs and the imputation API (R2), then the OU/Kalman state layer supplying $\Sigma(a)$ (R3); joint modeling (R4) strictly behind the attenuation-gap trigger of Chapter 8.

P3.2 Plackett–Luce likelihood for multi-deal sector-weeks (D10.x); LambdaMART challenger benchmarked, not adopted, unless it beats the likelihood family on the consumed metric two campaigns running.

P3.3 `hawkes_calibration/entrants/`: cohort entry intensities on the Lexis diagram (Chapter 11), unlocking honest sector-total decompositions.

P3.4 Signed continuous-time kernels (route (b) of Chapter 5) and latent-frailty models: only if a consuming question requires a structural continuous-time inhibition or contagion claim that routes (a)/(c) cannot express — each with its own recovery rehearsal first.

Gate: per item, the trigger that justified it, measured before and after.

13.6 Decision registry

The numbered decisions (D$x.y$ boxes) of Chapters 1, 2 and 4 to 12 form the registry against which pull requests are reviewed: a change that contradicts a decision box must either be rejected or arrive together with a documented reversal — trigger fired, evidence attached, box amended in this document. The document itself is versioned with the code; Appendix C appends new campaign verdicts rather than overwriting old ones, so the evidentiary trail stays auditable.

Code hooks.

New top-level modules by phase: `hawkes_calibration/data/` (P0), `hawkes_calibration/panel/` and `hawkes_calibration/models/xgb_horizon.py` (P1), `hawkes_calibration/eval/` and `hawkes_calibration/decision/` and the grouped-softmax objective (P2), `hawkes_calibration/covstate/` and `hawkes_calibration/entrants/` (P3). Existing modules extended in place: `hawkes_calibration/sector_ranker.py` (staleness features, outside option, PL), `hawkes_calibration/sector_stability.py` (NB noise), `hawkes_calibration/mbpp/` (real-data wiring). The synthetic world (`synthetic/`) remains the rehearsal stage for every gate.

Appendix A

Notation

The table below collects the symbols used throughout the document, in the conventions fixed by Chapter 2 and `macros.tex`. Chapter references point to the definition site.

Symbol	Meaning	Defined
$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$	filtered probability space; $\mathcal{F}_t$ holds the marked-process history and covariate paths	Chapter 2
$\mathcal{H}_{t-}$	internal history strictly before t	Chapter 2
t_m, s_m, i_m, m_m	time, sector, firm, and marks of the m-th deal	Chapter 2
$\mathcal{S}$, $\mathcal{I}$, $\mathcal{I}^\dagger$	sector set; tracked universe; untracked remainder	Chapter 2
$N_e(t)$	counting process for index e (sector, firm, transition)	Chapter 2
$\lambda_e(t)$	conditional intensity of N_e	Chapter 2
$\Lambda_e(t)$	compensator $\int_0^t Y_e \lambda_e \, \mathrm{d}u$	Chapter 3
$\bar{\lambda}$, $\xi(t)$	mean intensity $\mathbb{E}[\lambda(t)]$ (MBPP)	Chapter 6
$Y_e(t)$	at-risk indicator (observed, eligible at t^-)	Chapter 2
$M_e(t)$	compensated martingale $N_e - \Lambda_e$	Chapter 2
$\mu_s(t)$	baseline (immigrant) intensity of sector s	Chapter 5
$h(t) = \kappa\theta e^{-\theta t}$	exponential excitation kernel	Chapter 5
κ	branching ratio (kernel mass $\int h$)	Chapter 5
θ	kernel decay rate; half-life $\ln 2/\theta$	Chapter 5
$G[s, r]$	aggregated excitation matrix $r \to s$; stability $\rho(G) < 1$	Chapter 5
$\rho_{\max}$	row-sum stability budget (default 0.95)	Chapter 5
$x^{\mathrm{mac}}(t)$	macro covariates (FactSet), published with lag	Chapter 2
$x_i(t)$, $\tilde{x}_i(t)$	true (latent) and LOCF-observed firm covariates	Chapter 2
$\tau_i(t)$	time of firm i's last deal before t	Chapter 2
$a_i(t) = t - \tau_i(t)$	staleness age of firm i's covariates	Chapter 2
β, γ	covariate loadings (baseline; history features)	Chapters 4 and 5
$z_i(t)$	firm-level ranker features	Chapter 10
$q_i(t)$	ranker score of firm i	Chapter 10
$\mathcal{R}[s, t]$	sector risk set (live candidates at t)	Chapter 10
$\Diamond_s$	outside option (newcomer/untracked) in sector s's risk set	Chapter 11
V_i	entry (left-truncation) time of firm i	Chapter 2
T	end of observation window (June 2024)	Chapter 2
Δ	bucket horizon (26/52/104 weeks)	Chapter 9
$p_i(t, \Delta)$	horizon event probability $\mathbb{P}(N_i(t+\Delta) - N_i(t) \geq 1 \mid \mathcal{F}_t)$	Chapter 1
$y_{i,b}$	bucket label for firm i, bucket b	Chapter 9

Symbol	**Meaning**	**Defined**
$\ell(\vartheta)$, L	log-likelihood; likelihood	Chapter 3
ϑ, $\hat{\vartheta}$	generic parameter vector; its MLE	Chapter 3
$U(\vartheta)$, $\mathcal{I}(\vartheta)$	score; Fisher information	Chapter 3
(C1)–(C6)	the defect ledger	Table 2.1
T1–T4	the four targets (identity, timing, horizon, marks)	Chapter 1
R1–R4	staleness remedy ladder	Chapter 8
D$x.y$	numbered design decisions (registry: Section 13.6)	throughout

Appendix B

Data Dictionary: PitchBook and FactSet Columns

This appendix is the column-level contract between the raw data and the loader of Section 13.2. It reflects the column memory already encoded in the repository's observation layer (`synthetic/observe.py`, `synthetic/macro.py`, `synthetic/loaders.py`), which mirrors the production PitchBook/FactSet extracts; the real adapter must treat any divergence between this dictionary and a delivered extract as an audit failure, not something to patch silently.

Deals table (one row per financing event)

Column	Notes for modeling
`deal_id`	primary key; duplicates exist upstream — the dedup rule (same company within a few days, jittered size) runs first (Protocol 2.5).
`company_id`	foreign key into companies; may reference a firm with *no* companies row when out-of-universe (C6).
`deal_date`	the untrusted timestamp of (C4): announced vs. closed conventions and backfill jitter; enters models only at the honest resolution of Chapter 6.
`primary_industry_sector` / `_group` / `_code`	sector taxonomy; small mislabel rate — group-level rollups are the robustness fallback (Chapter 11).
`company_universe`	`'Venture Capital'` vs. `'Out of Universe'`; the (C6) flag; coverage share by era is a standing audit.
`deal_size`, `deal_size_status`	size with `Actual/Estimated/Undisclosed` status; undisclosedness is informative — always a feature, never silently imputed.
`pre_money_valuation`, `post_valuation`, `post_valuation_status`	frequently jointly missing; missingness indicator enters T4 models.
`vc_round`, `vc_round_up_down_flat`	round sequence and direction; up/down/flat is a T4 target and a T1 feature.
`deal_type`, `deal_type_2`, `deal_type_3`, `deal_class`	deal taxonomy; defines which events count as "funding rounds" for each model's event definition (an explicit loader config, audited).

Column	Notes for modeling
raised_to_date, total_raised_to_date, total_invested_capital, total_invested_equity, debt, total_new_debt, total_debt, total_investor_ownership, pct_acquired, type_of_stock, deal_status, sellers	capitalization block; *mechanically* event-linked columns (Chapter 8's classification): they update by construction at deals, so their staleness is definitional, not drift.
revenue, ebitda, gross_profit, net_income, revenue_growth_since_last_deal, number_of_employees	firm fundamentals as reported around the deal; the *drifting* class of Chapter 8 — prime candidates for the R2 drift model.
valuation_cash_flow, valuation_ebitda, valuation_revenue, deal_size_cash_flow, deal_size_ebitda, deal_size_revenue	ratio block; derived — recompute from parents rather than trusting, audit inconsistencies.
year_founded, hq_location, company_country, hq_global_region, hq_global_sub_region	static descriptors; founding year anchors cohort entry models (Chapter 11).

Companies table (one row per tracked firm)

Column	Notes for modeling
company_id, company_name, description, website	identity block.
financing_status, business_status, ownership_status, company_universe	status block; business_status is the only exit signal and it is coarse and laggy — the (C3) discussion of Chapter 2 governs its use; it may close spells only through a point-in-time rule.
year_founded, hq_*, primary_industry_sector, other_industries	static descriptors.

Column	Notes for modeling
`last_financing_date`, `last_financing_size`, `last_financing_valuation`, `last_financing_deal_type`, `last_known_valuation_date`, `last_known_valuation_deal_type`, `total_money_raised`, `employees`	snapshot summaries of the latest deal — the companies table is itself LOCF by construction; never join it as if current (the (C5) warning in table form).

Investors table

`company_id`, `investor_id`, `investorname`, `investorstatus`, `investorsince`: the firm–investor link with first-investment dates. Used for engineered features (investor count, investor quality proxies, syndicate recency), all as-of `investorsince`.

FactSet macro feed

Long format `(series_code, reference_date, publish_date, value)`. Series: `FFUNDT` (Fed Funds target, daily step path), `FFUND` (effective Fed Funds, daily), `CPI_RAW` (CPI level, monthly, ∼14-day publication lag), `CPI_YOY` (derived YoY inflation), `RUNEMP` (unemployment rate, monthly). Joins are as-of on `publish_date` only (Protocol 2.5); reference-date joins are look-ahead and fail the audit. Derived transforms (changes, regime dummies, real-rate proxies) are computed inside the loader so their point-in-time correctness is testable in one place.

Appendix C

Synthetic-Rehearsal Results Carried into This Document

This appendix is the evidentiary record behind every "inherited result" cited in the main text. All numbers come from the repository's synthetic rehearsal campaigns (`experiments/`, `EXPERIMENTS.md`, `docs/`); none are real-data results. The transfer convention is fixed here once:

Protocol C.1 (Transfer convention)**.** Rehearsal results set *priors and vetoes* for real-data work — they tell us which failure modes are live, which parameters are identifiable at which resolutions, and which designs were falsified in a world where the truth was known. They never substitute for real-data evaluation, and no rehearsal number may be quoted externally as a property of the market. New campaigns append to this appendix; existing entries are never overwritten (Section 13.6).

The rehearsal world

Twelve sectors; 8,000 firms of which 7,031 tracked; 520 weeks with `TRAIN_END` at week 416; ∼49k deals; ∼27% newcomer share overall (23–25% in test windows, and ∼12% of untracked deals from untracked *incumbents*); risk sets of ∼64 candidates plus the newcomer option. Regime A: well-specified — generated by the two-stage weekly model family we fit (26-week cooldown in the ranker). Regime B: misspecified stress — a daily firm-level block Hawkes with self-inhibition (30-day half-life), sector peer contagion (7-day half-life, true $\theta = \ln 2/7 \approx 0.099$/day), and latent firm quality; the weekly two-stage model is deliberately not the DGP.

Standing verdicts

Verdict	Evidence	Binds
Covariates dominate; excitation is decimal points.	Seasonal → NHPP covariate upgrades carried +3.5/+5.7/+1.5 nats across arms; adding weekly excitation +0.04/−1.15/+0.53; event-time timing terms +0.09/+1.35/+0.53 with the event-time arm the best calibrated (dispersion ≈ 1.0).	Model-investment order (Chapters 4 and 7); Phase 1 before Phase 2 (Chapter 13).

Verdict	Evidence	Binds
Risk-set truthfulness is the largest sensitivity in the system.	Observed vs. oracle risk sets: top-5 hit rate 0.14 → 0.29; an exit classifier with AUC 0.877 was insufficient to repair the risk set; with a contaminated pool the Bayes-optimal candidate weighting inverts the oracle weighting.	(C3) treatment everywhere (Chapters 2, 9 and 10).
κ survives aggregation; θ does not (the κ–θ ridge).	Bucket width ≤ 1 kernel half-life: timing recoverable; ≥ 4 half-lives: Fisher information in θ collapses; branching mass remains estimable.	Honest resolution (Chapter 6); no fine-timing claims (Chapters 7 and 12).
Fitted branching ratios are upper bounds on contagion.	A latent AR(0.90) frailty factor inflated the fitted effective radius $\sim$2.4× (0.15 true vs. 0.355 fitted).	Reporting protocol (Chapters 5 and 12).
The B1 test fired: linear utility is falsified on identity.	A feature-parity XGBoost beat the linear conditional logit by +0.22 nats and +9pp top-5 on regime A.	Grouped-softmax boosted ranker (Chapter 10); tree stack legitimacy (Chapter 9).
"Newcomer" must never be sold as "entrant."	The conflation cost only +0.003 nats in likelihood — invisible to fit — but blocks compositional claims permanently ($\sim$12% of untracked deals were untracked incumbents).	Language and reporting (Chapters 11 and 12).
Log-link count models are not stabilized by $\rho(G) < 1$.	Stability is an operating-point condition $\rho(\mathrm{diag}(m^*)B) < 1$; explosion is an absorbing rare event — one unconstrained fit reached $\rho \approx 3.7$ and a simulated metastability episode produced 169k events from a 50k-event world at radius 0.35.	Mechanism-B warning and the $\rho_{\max} = 0.95$ constraint (Chapter 5).
Self-inhibition is recoverable through the mark factor.	The ranker's cooldown route recovered the true self-inhibition ordering with rank correlation 0.937 under regime B misspecification.	Route (c) commitment (Chapters 5 and 10).

Verdict	**Evidence**	**Binds**
Analytical likelihoods first; learned solvers only as accelerators.	PINO forward-solver accuracy 0.55%/1.52%/2.03% at $M = 3/5/8$ dimensions at $\sim$3.3 ms/solve (a valid accelerator); the amortized *inverse* direction plateaued at 38%/49% recovery and was rejected.	No learned component where a likelihood exists (Chapters 7 and 13).

Reference numbers

Ranker validation (held-out weeks, 5 seeds; fitted / oracle / random): top-1 16.7% / 21.2% / 2.9%; top-5 49.2% / 53.7% / 14.3%; top-10 69.2% / 71.6% / 28.6%; MRR 0.328 / 0.371 / 0.118 — the fitted ranker recovers $\sim$80–92% of the oracle ceiling, 3–6$\times$ chance. Campaign arms (regime A): event NLL 3.346 vs. random 4.011, top-5 0.446; regime B: 3.796 vs. 4.172, top-5 0.316. Average sector concentration (ASC) 6.86 (A), 4.7 (B). Numerical hygiene standards inherited by every new fitter: analytic-vs-numeric gradient agreement 1.3×10^{-8} (survival), 1.8×10^{-6} (block Hawkes), $\sim 10^{-7}$ (event-time); time-rescaling KS statistics 0.04–0.06 on well-specified fits.

Recovery and robustness reference numbers

Aggregation (κ–θ) ridge campaign (true $\theta = \ln 2/7 \approx 0.099$/day): at 1-day bins $\hat{\theta} = 0.112 \pm 0.030$ (unbiased); at 28-day bins $\hat{\theta} = 0.081 \pm 0.014$ — biased low $\sim$20% with a deceptively tight spread; κ recovered within 0.28–0.38 around its true 0.35 at every tested width. *Covariate-noise campaign (exp22)*: relative Gaussian noise on covariates at fit and prediction time attenuated fitted choice weights by $\times$0.88 at 20% noise ($\times$0.93 at 15%), eroded stage-1 β correlation from 0.67 to 0.53, while prediction was nearly untouched (covariate lift $\approx$ 1.0 nats throughout; top-5 0.446 $\to$ 0.439) — structure erodes before prediction does. *Overdispersion*: quasi-Poisson dispersion 2.72 on the regime-B weekly sector fit (variance $\approx 3\times$ the Poisson prediction). *Newcomer substream*: fitted separately, the newcomer (immigrant) stream carried an apparent excitation radius of 0.30 against 0.08–0.11 for the repeat-deal stream — entry waves harbor the latent common factors, reinforcing the upper-bound rule. *Newcomer-share calibration*: with the enriched context ASC, quarterly newcomer-share forecasts met the standing $\pm$2pp acceptance target on test quarters.

Warning C.2. Several rehearsal constants are world-specific (cooldown windows, half-lives, risk-set sizes, newcomer shares) and must be re-estimated, not assumed, on the 2011–2024 PitchBook data. What transfers is the *shape* of each verdict — which defects bind, which parameters are fragile, which designs were falsified — per Protocol C.1.

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